



DesignWare® Cores MIPI UniPro V2.0 Controller

Databook

DWC MIPI UNIPRO V2.0 CONTROLLER - Product Code: F575-0

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Revision History

The following table provides a summary of changes made to this databook.

Version	Date	Description
1.02a-lca00	May 2022	Updated ■ “Standards Compliance” ■ “DWC_mipi_unipro_20 Features” ■ “DME Primitive to Signal and Attribute Mapping” Added ■ “PA_INIT and Error Recovery During PMC When ResetMode is HS_Mode” ■ “Link Startup in High Speed” ■ “Ports Added for Link Startup in High-Speed Mode”
1.01a-lca00	December 2021	Updated ■ “DWC_mipi_unipro_20 Features” ■ “Link Optimization” ■ “L2 TX Buffer Size Calculation” ■ “L2 RX Buffer Size Calculation” ■ “Layer 2 Data Transfer Basics” ■ “RX Data Interface” ■ “Retained State Information” ■ “UniPro Capabilities” Added ■ “Layer 2 Credit Unit Mode” ■ “Flow Control Registers Mapped to VS Registers” ■ “ResetMode Selection”
1.00a-lca00	May 2021	Initial Release

Preface

Databook Organization

The chapters of this databook are organized as follows:

- [Chapter 1, “Product Overview”](#) describes the overall architecture, and lists the product features, deliverables, and general requirements.
- [Chapter 2, “Architecture”](#) describes the DWC_mipi_unipro_20 architecture and its main blocks.
- [Chapter 3, “Parameter Descriptions”](#) describes DWC_mipi_unipro_20 configuration options and parameters.
- [Chapter 4, “Signal Descriptions”](#) provides the pinout and describes the I/O for DWC_mipi_unipro_20.
- [Chapter 5, “Register Descriptions”](#) describes the function of DWC_mipi_unipro_20 control and status registers.
- [Chapter 6, “Application Description”](#) describes the usage of DWC_mipi_unipro_20 with some examples.
- [Appendix A, “Area and Power”](#) provides an estimation of DWC_mipi_unipro_20 area and power values.
- [Appendix B, “Differences from the Specification”](#) describes differences between the DWC_mipi_unipro_20 and the MIPI UniPro Specification.
- [Appendix C, “Synchronizer Methods”](#) describes the synchronizer methods that are used in DWC_mipi_unipro_20 to cross clock boundaries.
- [“Glossary”](#) explains the terms used in this databook.

Reference Documentation

- MIPI® Alliance Standard for UniProSM, version 1.8
- MIPI® Alliance Standard for UniProSM, Draft version 2.0 r10
- MIPI® Alliance Standard for M-PHY, version 5.0
- AMBA APB3 Protocol Specification v1.0

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 - c. Complete the mandatory fields that are marked with an asterisk and click **Save**.
Make sure to include the following:
 - **Product L1:** DesignWare Cores
 - **Product L2:** MIPI Controller
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 - For simulation issues, include the timestamp of any signals or locations in waveforms that are not understood
 - Attach any debug files you created.
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1

Product Overview

This chapter introduces the Synopsys DesignWare® Cores MIPI UniPro V2.0 Controller (DWC_mipi_unipro_20) features and deliverables. It contains the following sections:

- [“Product Description”](#) on page 14
- [“DWC_mipi_unipro_20 Features”](#) on page 14
- [“Deliverables”](#) on page 15
- [“Application Scenarios”](#) on page 16

1.1 Product Description

The `DWC_mipi_unipro_20` is a MIPI UniPro compliant interface module. It provides the capability to control the UniPro link over MIPI M-PHY modules from one or more generic applications. Because of its generic nature, it is capable of transporting any kind of data between applications like the camera, display, and the memory devices on the same physical link.

1.1.1 Applications

The following are the typical applications built with `DWC_mipi_unipro_20`:

- Camera devices
- Display devices
- Memory devices
- Processor devices
- Debug interfaces (for example, Gigabit Trace)

1.1.2 Standards Compliance

The `DWC_mipi_unipro_20` conforms to the following standards:

- MIPI® Alliance Standard for UniProSM, version 1.8
- MIPI® Alliance Standard for UniProSM, Draft version 2.0 r10
- MIPI® Alliance Standard for M-PHY, version 5.0
- AMBA APB3 Protocol Specification v1.0



Attention

For more information about the required license for required version support, see *MIPI UniPro Licenses* section in the *DesignWare Cores MIPI UniPro 2.0 Installation Guide*.

1.2 DWC_mipi_unipro_20 Features

The `DWC_mipi_unipro_20` supports the following features:

- MIPI® M-PHY support with RMMI compliant interface
- Configurable Device Descriptor Block (DDB) settings
- Generic interface for the applications (CPort for data transfer and APB3/native for configuration)
- Single C-Port signal interface
- RMMI symbol bus data width: 80-bit (fixed)
- Automatic flow control between the local and the peer layer 2 buffers (layer 2 flow control)
- Reliable transmissions through error detection and automatic retransmission
- Support for single traffic class: TC0

- Frame prioritization using pre-emption
- Well-structured clock system (clock domains separated using FIFO structures)
- Fixed number of RX and TX M-PHY modules or lanes (two in each direction for RMMI symbol bus data width set to 80-bit)
- Scalable maximum bandwidth (based on the number of lane configuration)
- 128 bits system data interface width
- Minimum restrictions on application clock frequency
- Support for test-modes like scantest
- Configurable number of layer 4 test features
- Generation and check of layer 1.5 test pattern
- Access to vendor specific M-PHY attributes
- Support for low-power flow
 - Advance low-power flow with UPF methodology support
 - Two power rail flow (TPF) support
- Support for UniPro version 2.0
 - Up to High speed G5 support
- Support for General Purpose Input and Output ports (up to 96 ports)
- Support for the override of M-PHY capabilities communicated in PACP_CAP.ind, PACP_EXT1_CAP.ind, and PACP_EXT2_CAP.ind capability packets
- Support for DLCreditUnit128_Mode
- Link Startup in High Speed
- Power Mode Change (PMC) Recovery and PA_INIT in High Speed

1.2.1 Unsupported Features and Exceptions

The unsupported features and exceptions are as follows:

- The DWC_mipi_unipro_20 does not support MIPI D-PHY.
- Traffic Class TC1
- Non-multiplexed CPort

1.3 Deliverables

The DWC_mipi_unipro_20 requires the Synopsys coreConsultant tool to install, unpack, and configure the controller. The license file required to install, unpack, and configure the DWC_mipi_unipro_20 is delivered separately.

The DWC_mipi_unipro_20 image includes the following:

- Verilog RTL source code

- Synthesis scripts for Synopsys Design Compiler
- Verification testbench and test configurations derived from the parameter choices made during controller configuration
- DWC_mipi_unipro_20 ASIC synthesis, design-for-test, timing analysis, formal verification, and timing information

1.4 Application Scenarios

UniPro is a standardized high-speed serial interface that can be used for multiple applications. Its capability to transport the data reliably at high speeds including the flow control and the bi-directional information exchange makes it suitable for image based and inter-processor based applications.

In addition, the switching devices are part of the application scope for UniPro, for example, bridging between CSI-2 or DSI, and the UniPro-based interfaces like CSI-3.

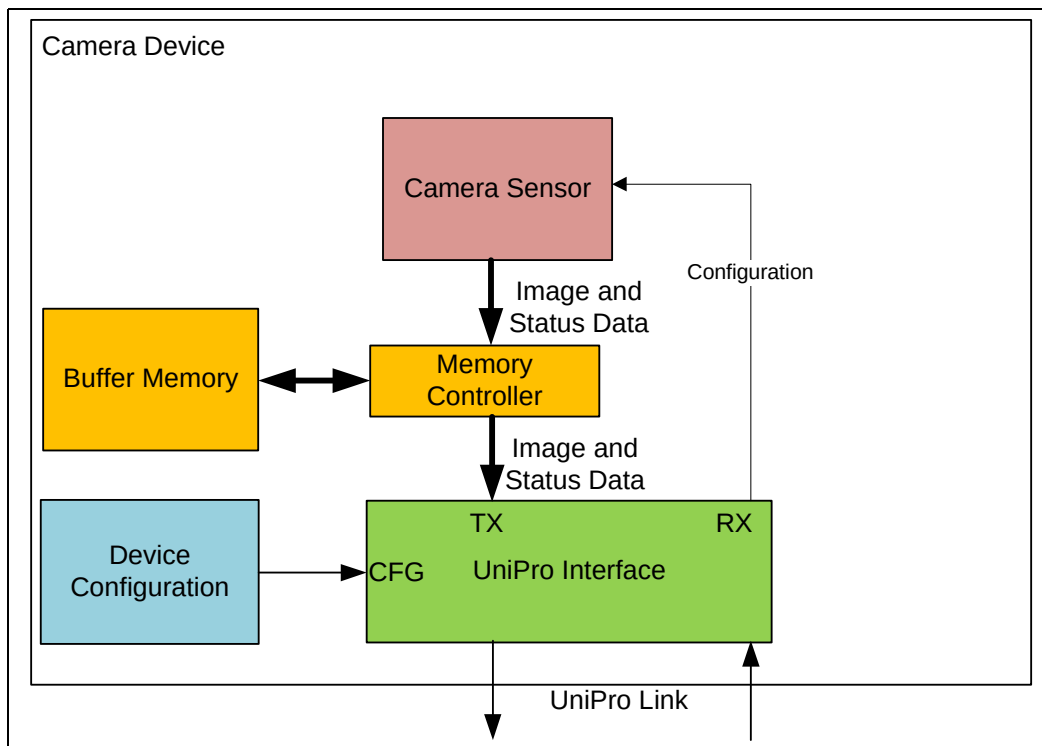
The following sub-sections describe some example application scenarios.

1.4.1 Camera Application

The main requirement of the camera applications is the outgoing bandwidth of the interface. The image data is streamed through a buffering mechanism (because UniPro may introduce some latency) from the sensor to the UniPro interface. The receive port of the link is mainly used for the configuration purposes.

Camera devices may have an unequally distributed number of data lanes, because the output bandwidth requirement is much higher than the input bandwidth.

[Figure 1-1](#) shows the application of UniPro in a camera.

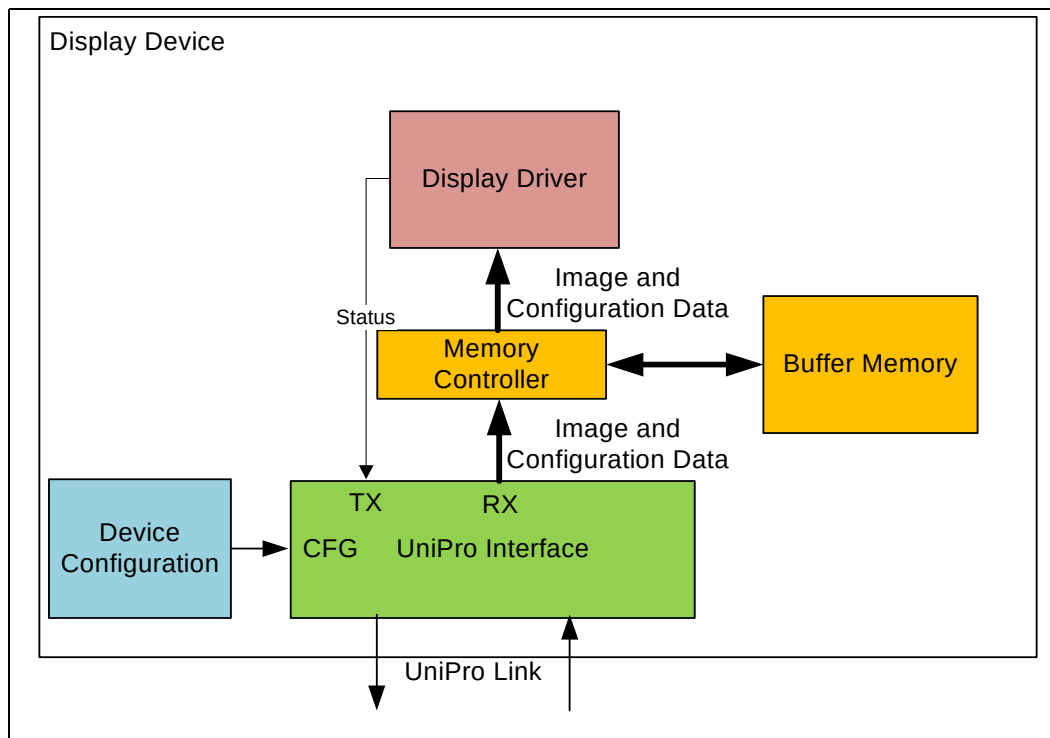
Figure 1-1 Camera Application

1.4.2 Display Application

Unlike camera applications, display devices need a much higher bandwidth on the input direction of the interface. Image data is transferred over MIPI UniPro into an internal or external memory of the display device. Because the built-in error detection and resend mechanism of MIPI UniPro can generate problems for transferring real-time synchronization, the display system must synchronize image data. One option is to transfer the raw data to the display device while controlling the display timing generation on its own. Alternatively, a higher link bandwidth and additional synchronization buffers allow infrequent retransmissions on the link without disturbing continuous image synchronization.

For display applications, the output port of the UniPro link is used for returning the status information back to the video source.

Figure 1-2 shows the application of UniPro in a display device.

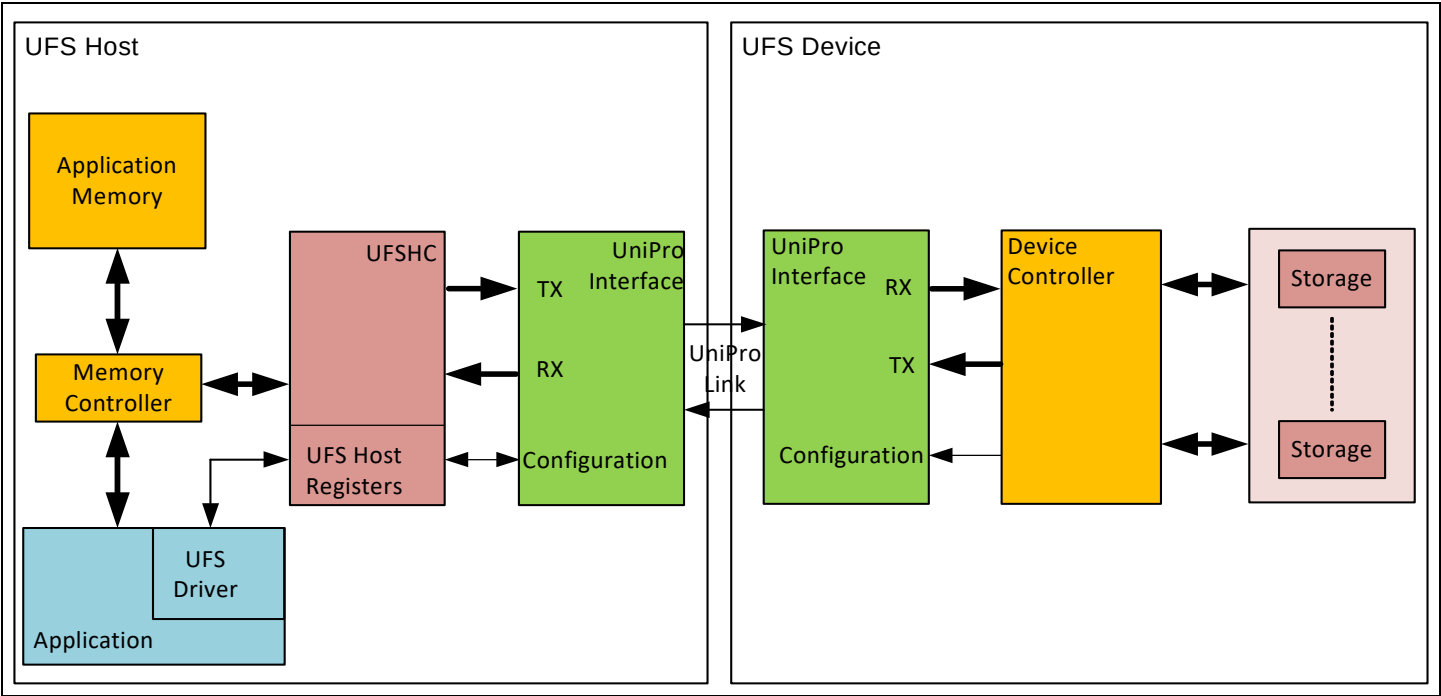
Figure 1-2 Display Application

1.4.3 Flash Storage Application

UniPro is required to be used in Universal Flash Storage (UFS) solutions. In this application, bandwidth requirement on both input and output directions is high and similar. The UFS Host controller fetches the data from the application memory to be stored in a flash memory and transfer it to the UFS device through the UniPro link. Similarly, when the application wants to fetch the data from the flash memory, it instructs the UFS Host controller which reads the data blocks from the flash storage. Thus, the data transfer occurs in both directions. UniPro ensure loss-less data transfer by taking care of retransmissions in case of errors, avoids overflow/underflow of buffers through flow control, and so on. The latencies and overheads introduced by UniPro do not have much impact on the application throughput.

Figure 1-3 shows a typical UFS-based storage system.

Figure 1-3 UFS Application



2

Architecture

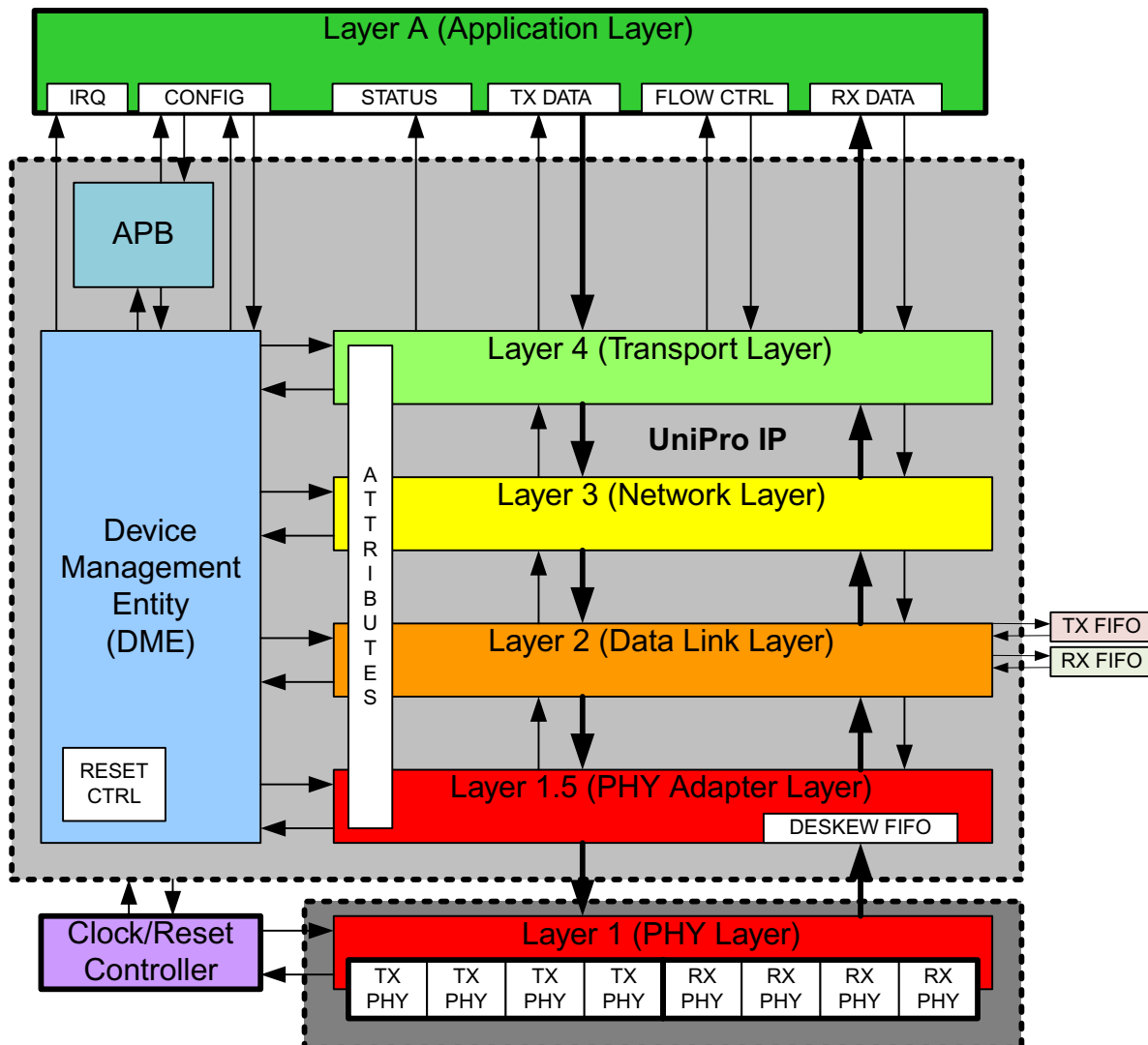
This chapter describes the DWC_mipi_unipro_20 architecture. It contains the following sections:

- [“Structure of DWC_mipi_unipro_20” on page 22](#)
- [“Application Layer Interface” on page 22](#)
- [“Data Write Accesses” on page 24](#)
- [“Data Read Accesses” on page 26](#)
- [“End-to-End Flow Control Support” on page 27](#)
- [“Status Interface” on page 28](#)
- [“Attribute Mapping” on page 29](#)
- [“DME Control Interface” on page 33](#)
- [“Additional Indication Interface” on page 35](#)
- [“Interrupt Interface” on page 36](#)
- [“Configuration Accesses” on page 37](#)
- [“Layer 2 Memory Interface” on page 43](#)
- [“Clock and Reset Interfaces” on page 44](#)
- [“M-PHY Interface” on page 44](#)
- [“Layer 4 Test Feature” on page 62](#)
- [“Layer 1.5 Test Mode” on page 63](#)
- [“Advance Low-Power Flow with UPF Methodology Support” on page 64](#)
- [“General Purpose Input and Output Ports” on page 77](#)
- [“Ports Added for Link Startup in High-Speed Mode” on page 78](#)

2.1 Structure of DWC_mipi_unipro_20

The DWC_mipi_unipro_20 controller uses the layered structure as defined in the MIPI UniPro Specification. All the layers explicitly perform the required functions allowing easy adoptions in case of specification changes or additions. Figure 2-1 shows the general structure of the DWC_mipi_unipro_20 and its interfaces.

Figure 2-1 DWC_mipi_unipro_20 Structure



2.2 Application Layer Interface

The interface to the application layer has the following parts:

- Data Write
- Data Read

- Configuration
- Flow Control
- Status
- Interrupt

While the configuration and interrupt interface connect to Data Management Entity (DME), all other interfaces transport the information between the application layer and transport layer. The data and configuration ports use a very general form of data exchange.

Figure 2-2 Application Layer Interface

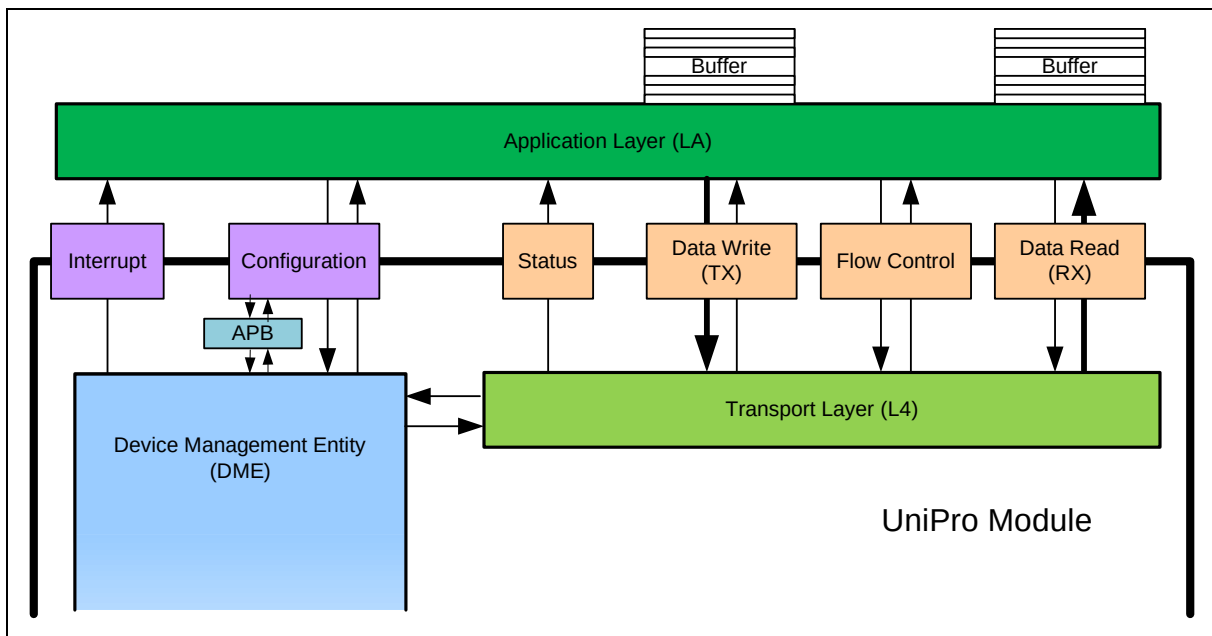


Figure 2-2 shows all the ports between the application layer and the `DWC_mipi_unipro_20`.

All interfaces between the application layer and `DWC_mipi_unipro_20` are synchronous to the `CG_Sysclk` signal. But the APB interface can either use `CG_Sysclk` or `PClk`, based on the configuration. For more information about the clock structure, see the *Clock System* section of the *DesignWare Cores MIPI UniPro 2.0 User Guide*.

2.2.1 CPort Mapping

All data accesses to `DWC_mipi_unipro_20` are handled by specification through CPorts. These CPorts are virtual ports for one or more applications and define the source and sink of information on both sides of the UniPro link.

`DWC_mipi_unipro_20` groups all virtual CPorts to a single interface where the respective CPort ID is selected through a CPortID address bus. The application layer arbitrates between all applications in the system for accessing the `DWC_mipi_unipro_20`. The transport layer stores several attributes related to individual CPorts in its internal attributes and selects the required settings depending on the requested CPortID.

Figure 2-4 shows sample write accesses to several different CPorts in a row. A different CPortID indicates the beginning of a new fragment to the transport layer.

2.2.2 Connection Management

Connection Management (CM) defines the handling and setup of individual connections between two UniPro devices. One or more applications need to take care of setting up all connection related attributes in both the local and the peer devices because UniPro 2.0 cannot handle the connections automatically. For more information about accessing the local and peer attributes, see [“Configuration Accesses”](#) on page 37.

Connections are generated between exactly one CPort on each side of the link. If the connection is properly set up, the DWC_mipi_unipro_20 takes care of handling the flow-control between these two CPorts and the translation of all required communication information automatically.

To enable a connection, program the following attributes in both local and peer devices:

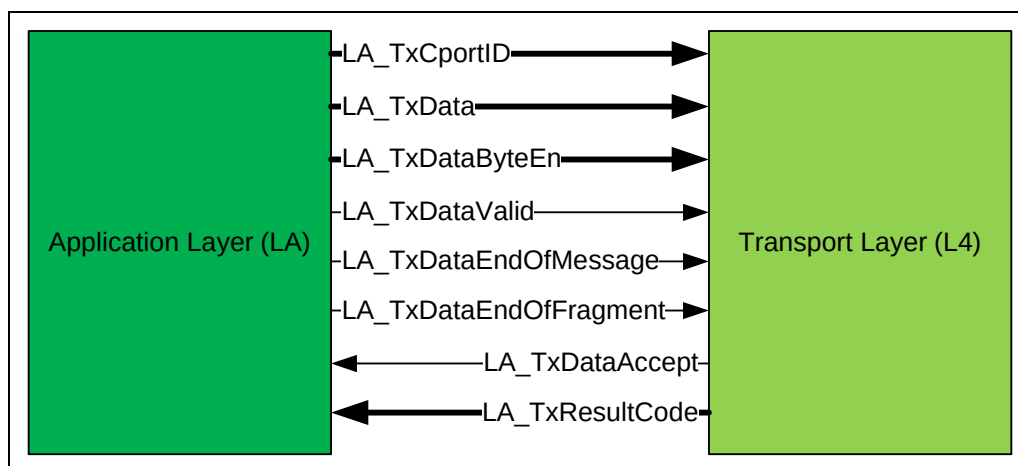
- T_PeerCPortID: Defines the receiving CPort of the peer device
- T_PeerDeviceID: Defines the device ID of the peer device
- T_TrafficClass: Defines whether the messages transported on this connection use TC0 or TC1
- T_CPortFlags: Enables or disables the end-to-end flow control (and controlled segment dropping) for this connection
- T_TxTokenValue: Defines the size of a credit token sent to the peer device (in bytes)
- T_RxTokenValue: Defines the size of a credit token received from the peer device (in bytes)
- T_ConnectionState: Activates the connection, if set to 1

For more information about the connection management, see [“CPort Connection Management”](#) on page 556.

2.3 Data Write Accesses

You can perform the data write accesses on a dedicated interface between the application layer and the transport layer. The interface uses a general structure to allow easy system integration of the module.

Figure 2-3 Data Write (TX) Interface



Pass the data to layer 4 over a parallel data bus using a separate byte enable bus to support any size of fragments. Define the width of the CPort data bus during system integration using the P_DWTX parameter. Select the CPort ID through the CPortID address bus. Because the number of CPorts is configurable as well, define the width of the CPort address bus using the P_CA parameter.

The Data Write interface signals are described in [“Data Write Interface Signals”](#) on page 104.

A valid flag (LA_TxDataValid) indicates the availability of the new data. An additional end-of-message flag indicates that the transport layer must mark the current access as final fragment of a message.

The transport layer can throttle the data throughput by de-asserting the LA_TxDataAccept flag to indicate that it cannot process the data at the moment. This signal stays asserted if the application keeps at least two bytes inactive in this cycle (two bits of LA_TxDataByteEn = 0). This signal is de-asserted

- To insert layer 4 and layer 3 header information (two bytes in total) every time a new segment is created, for example, after a CPort change, or sending a maximum sized segment.
- If layer 2 cannot accept the data, for example, when the layer 2 buffers are full.

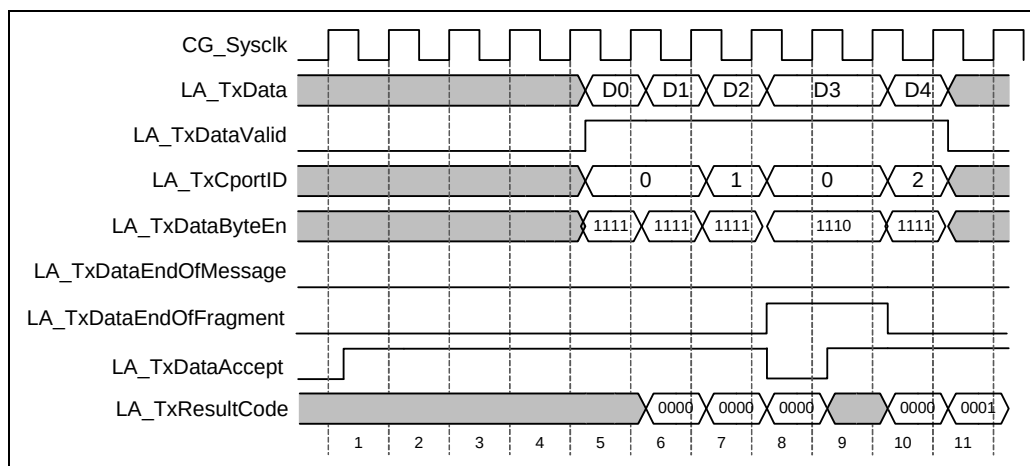
The LA_TxResultCode signal returns the status of the write access back to the application in the next clock.

Like all interfaces between the application layer and the DWC_mipi_unipro_20, the Data Write interface is fully synchronous to the CG_Sysclk clock. If the data is not processed with this clock inside the application layer, a synchronization stage must be added inside the application layer to perform proper clock domain crossing of data.

Some signals are defined related to specific parameters selected during the implementation of the module. These parameters are marked with a “P_” prefix and are described in [“Parameter Descriptions”](#) on page 81.

[Figure 2-4](#) shows a timing example for write accesses to three different CPorts, 0 to 2. The transport layer processes the data only when both the LA_TxDataValid and LA_TxDataAccept signals are active in the same cycle.

Figure 2-4 Data Write Access



[Figure 2-4](#) shows one invalid access to a non-connected CPort (CPort 2). If the access does not target a valid CPort, the LA_TxResultCode indicates this status (0001) back to the application.

At least one byte enable must be active when the LA_TxDataValid signal is set. The only exception is the last cycle of a message, where LA_TxDataEndOfMessage is active.

Data provided at the LA_TxData signal must be kept stable when the DWC_mipi_unipro_20 does not accept the data (LA_TxDataAccept = 0).

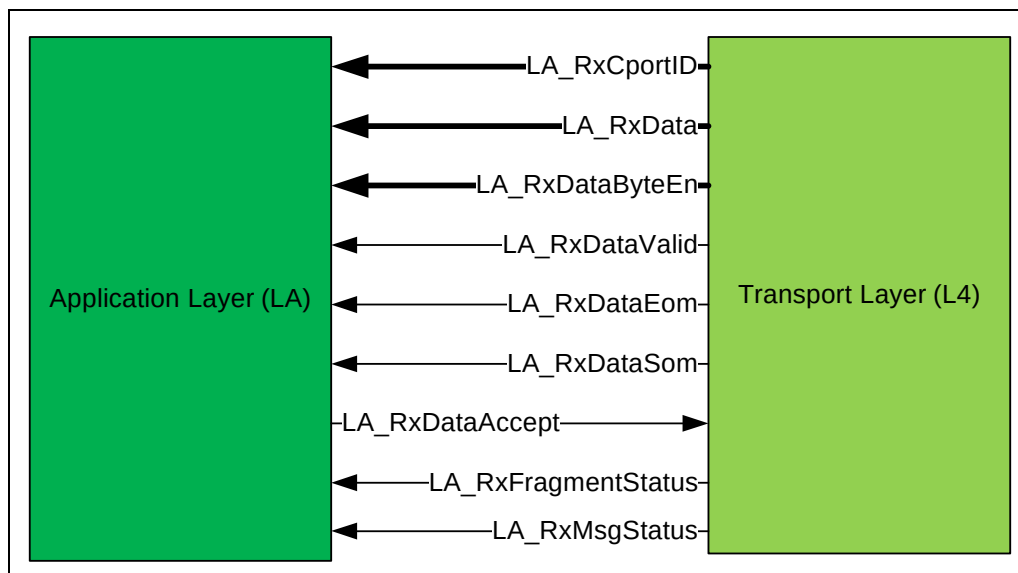
The LA_TxDataEndOfFragment signal can be used by the application to force the generation of a new fragment or segment inside layer 4. This function is useful if the application requires to partition the messages into specific sizes (for example, 256 bytes). If this signal is not used, DWC_mipi_unipro_20 generates 272 byte sized segments automatically.

2.4 Data Read Accesses

Performing the read accesses is similar to performing the write accesses. The transport layer controls the accesses, while the application layer can throttle the data throughput.

Figure 2-5 shows all signals of the Data Read interface.

Figure 2-5 Data Read (RX) Interface



The Data Read interface signals are described in “[Data Read Interface Signals](#)” on page 108.

The transport layer uses the LA_RxDataValid signal to indicate the availability of new data to the application layer. The LA_RxCportID indicates the CPort, which is targeted by the frame. Byte enable shows the byte lanes of the LA_RxData bus that carry the valid data and needs to be processed by the application layer.

The DWC_mipi_unipro_20 is capable of delivering the new received data of a frame in every cycle. By de-asserting the LA_RxDataAccept signal, the application layer can control the data bandwidth by itself if the CPort Safety Valve (CSV) is disabled.

The LA_RxDataEom signal flags that the valid least significant byte (LSB) of the current access cycle carries the last byte of the message. Similarly the LA_RxDataSom signal flags the start of a new message for the corresponding CPort.

If a fragment is dropped by the transport layer, the LA_RxFragmentStatus signal shows this occurrence for every cycle where the data is sent to that specific CPort and the following cycle. For the data dropped through Controlled Segment Dropping (CSD), the first corrupted data cycle shows a value of 01b. For every

following cycle where the data is still dropped, a value of 11b informs the application that current data as well as the data in the previous cycle are corrupted. In the first cycle when the data is valid again, this signal shows a value of 10b to indicate that the current data is valid again.

If the data is dropped because of the CPort Safety Valve, only values of 00b for correct data and 10b for invalid data in the previous cycle are indicated, because DWC_mipi_unipro_20 needs one cycle to determine that the data is not accepted. The LA_RxMsgStatus signal is activated at the following cycle after the first occurrence of the corrupted data and kept active in each valid cycle until the next end-of-message flag is received.

The transport layer activates its valid signal (LA_RxDataValid) to indicate that the new data is received and is ready to be transported to the application layer. By asserting its LA_RxDataAccept signal, the application layer indicates the capability of processing data to DWC_mipi_unipro_20. If this signal is not activated, DWC_mipi_unipro_20 does not provide the data to the application.

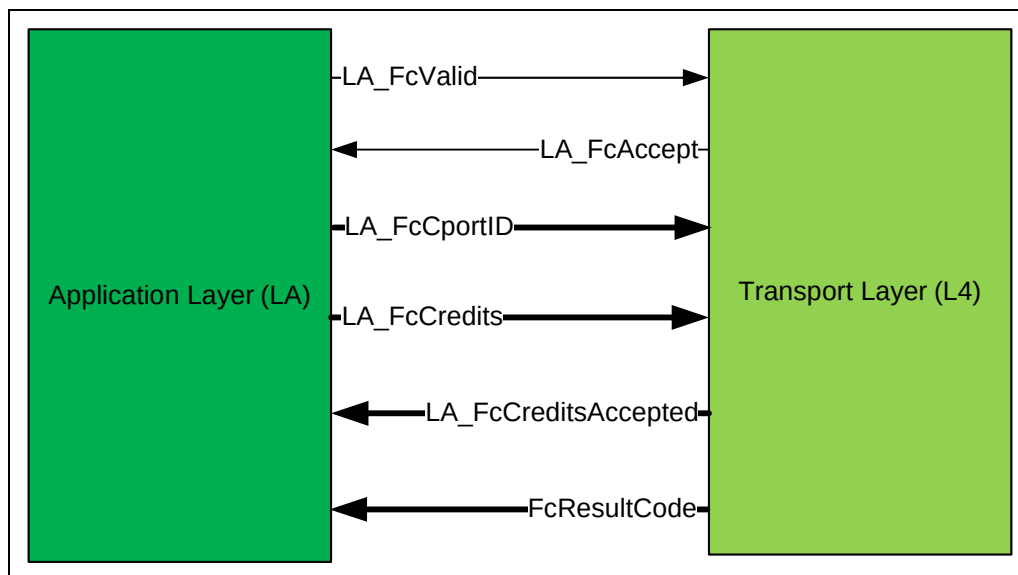
2.5 End-to-End Flow Control Support

The end-to-end flow control mechanism ensures that the transmitting CPort never sends more data over the link than can be absorbed by the buffer of the destination CPort. This feature can be disabled or enabled for each CPort.

The receiver initializes the end-to-end flow control for each CPort in use by providing a certain amount of credits to the transport layer through its Flow Control interface.

Figure 2-6 shows all signals of the Flow Control interface.

Figure 2-6 Flow Control Interface



The Flow Control interface signals are described in “Flow Control Interface Signals” on page 112.

The LA_FcResultCode signal returns the status of the current credit access to the application.

DWC_mipi_unipro_20 divides the number of credits granted by the application layer into smaller equally sized portions called tokens. An attribute of the corresponding CPort defines the size of a token (both RX and TX). The UniPro module informs the remote TX about this free buffer space by sending a Flow Control

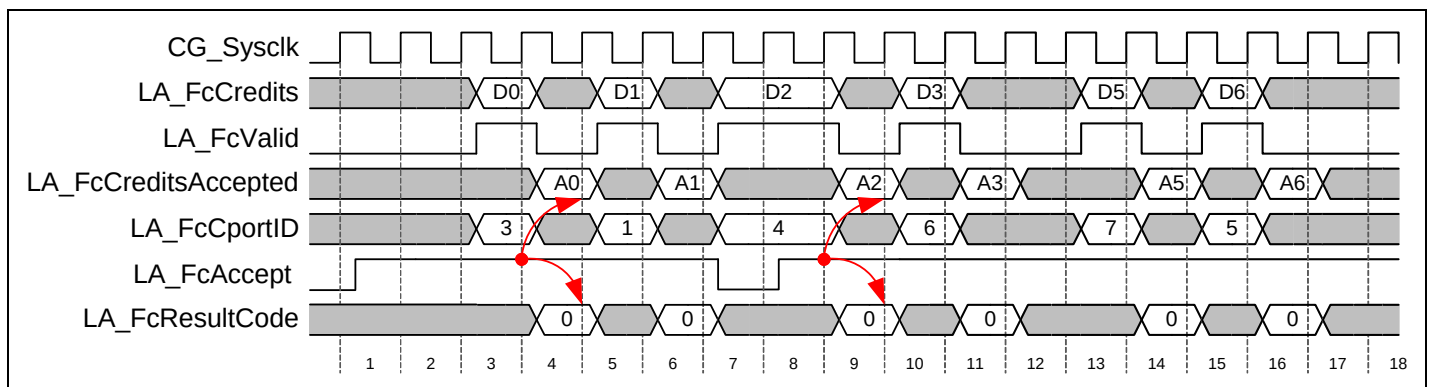
Token (FCT) as part of the layer 4 information in a data frame. After receiving an FCT, the remote TX knows that the receiver has space equal to the configured token size or a multiple of that size, if several tokens are received already. For more information about setting up End-to-End flow control, see [“Setting up End-to-End Flow Control”](#) on page 560.

**Note**

The Flow Control interface processes accesses sequentially. Pipelined operation is not supported. In other words, the application should sample the result of credits accepted by the `DWC_mipi_unipro_20` (on `LA_FcResultCode`) before it provides further credits (on `LA_FcCredits`).

The remote TX accumulates the bytes equivalent to the number of tokens received and decrements this value by the number of bytes sent to the CPort to keep track of the remaining buffer space. [Figure 2-7](#) shows some sample flow control accesses.

Figure 2-7 Flow Control Accesses

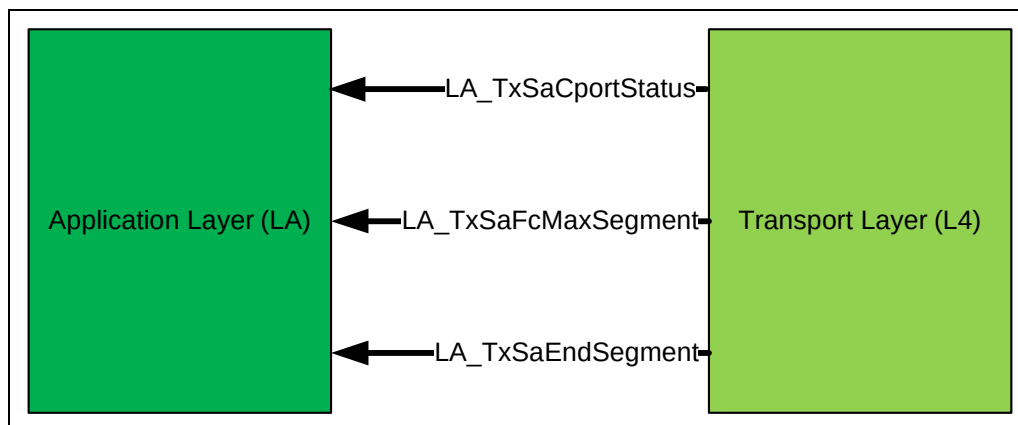


2.6 Status Interface

The transport layer uses a special interface to inform the application layer about its status. The Status interface signals are described in [“Status Interface Signals”](#) on page 115.

[Figure 2-8](#) shows all signals of the Status interface.

Figure 2-8 Status Interface



For each CPort, two bits of the LA_TxSaCportStatus bus provides the status of the CPort back to the application layer. Only if the corresponding bits are set to 2'b00, the application layer is allowed to send data to the corresponding CPort.

The LA_TxSaFcMaxSegment bus indicates if the respective CPort has received enough credits to send a maximum sized segment to the connected remote CPort.

Each bit of LA_TxSaEndSegment shows that the layer 4 collected the data for a maximum segment for the respective CPort. Layer 2 sends this segment as one frame to the peer device. This bit is only valid for a single cycle. An application may use this signal to trigger checking of the LA_TxSaFcMaxSegment signal if enough buffer space is available on the peer device.

2.7 Attribute Mapping

All UniPro attributes are mapped to the address space of the DWC_mipi_unipro_20 as described in the MIPI Specification. The lower 16 bits of the address mark the target layer for the access. The upper byte of the layer address defines the layer (0x00 for layer 1, 0x15 for layer 1.5, and so on). The lower part of the layer address marks the target attribute inside this layer.



Note

The description given for the attribute mapping is applicable for the direct native configuration interface. When the controller is configured for APB configuration interface, the same is applicable for programming the SEL and AID fields of the APB_Command register.

The upper part of the LA_AttrID bus called Select has a special meaning for layer 1 and 4. Because these two layers have multiple instances for each M-PHY module or CPort, the Select bus defines the target M-PHY or CPort.

Figure 2-9 Attribute Address Structure

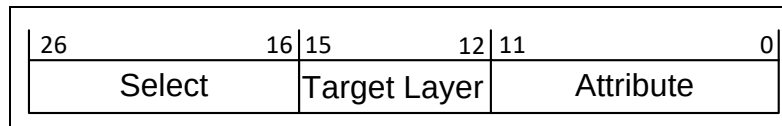


Table 2-1 describes the address space of the module.

Table 2-1 UniPro Attribute Space

Lower 16-bit Address	Target Layer	Select	Description
0x0000 - 0x00FF	L1	■ 0 to 3 TX ■ 4 to 7 RX	Address space M-PHY for layer 1 (M-PHY) specific attributes The Select bits determine the logical M-PHY module.
0x0100 - 0x14FF	None	None	Reserved
0x1500 - 0x15FF	L1.5	None	Address space for the layer 1.5 specific attributes
0x1600 - 0x1FFF	None	None	Reserved
0x2000 - 0x20FF	L2	None	Address space for the layer 2 specific attributes
0x2100 - 0x2FFF	None	None	Reserved

Table 2-1 UniPro Attribute Space (Continued)

Lower 16-bit Address	Target Layer	Select	Description
0x3000 - 0x30FF	L3	None	Address space for the layer 3 specific attributes
0x3100 - 0x3FFF	None	None	Reserved
0x4000 - 0x40FF	L4	0 to 2047	Address space for the layer 4 specific attributes The Select bits determine the CPort number or L4 test feature.
0x4100 - 0x4FFF	None	None	Reserved
0x5000 - 0x50FF	DME	None	Address space for the DME specific attributes
0x5100 - 0x7FFF	None	None	Reserved
0x8000 - 0x80FF	IP	■ 0 to 3 TX ■ 4 to 7 RX	IP related configuration attribute space for the implementation specific M-PHY attributes
0x8100 - 0x81FF	IP	■ 0 to 3 TX ■ 4 to 7 RX	IP related configuration attribute space for the common M-PHY attributes
0x8200 - 0xCFFF	None	None	Reserved
0xD000 - 0xDFFF	IP	None	IP related configuration attribute space for the implementation specific DME attributes
0xE000 - 0xFFFF	None	None	Reserved

Accessing an attribute at address 0xA1 inside layer 2 results in an AttrID address of 0x000_20A1. The Select bits of the address bus are not checked if the target layer does not use these bits, so accessing the same attribute on address 0xFFF_20A1 is also valid.

For accessing the attributes of the M-PHY modules, the Select bits should be set to the target M-PHY module. The TX M-PHY modules are mapped to select 0 to 3 and the RX M-PHY modules are mapped to select 4 to 7. If less than four modules are available in any direction, the upper select indices are not mapped and return an error when accessed. For example, accessing the M-PHY at index 3, if only two TX M-PHY modules are available, returns the error code BAD_INDEX. The same error is flagged, if an available but unconnected PHY lane is accessed after the link startup because the DWC_mipi_unipro_20 indicates that the power may be cut from these PHY modules.

The CPort L4 and test features are addressed using the standard binary addressing. The CPort 0 uses Select bits 0x000 while CPort 2047 is selected by using Select bits 0x7FF.

[Table 2-2](#) shows the mapping of the Select part of the AttrID address to the individual parts of layers 1 and 4.

Table 2-2 Select Bit Translation Examples

Select Bits	Layer	Type	Description
0x000	L1	binary	Target logical TX M-PHY module 0
0x003	L1	binary	Target logical TX M-PHY module 3
0x005	L1	binary	Target logical RX M-PHY module 1
0x000	L4	binary	Target CPort 0
0x001	L4	binary	Target CPort 1
0x003	L4	binary	Target CPort 3
0x100	L4	binary	Target CPort 256
0x7FF	L4	binary	Target CPort 2047

The M-PHY layer attributes always target the logical M-PHY encoded in the upper part of the AttrID bus. Before a link startup is performed, the logical lane map is not defined and PHY accesses target the corresponding physical lane. This access method allows the configuration of critical attributes such as DriverPolarity before link startup.

Avoid accesses to M-PHY attributes during the link startup sequence as they target either a physical or logical lane depending on the current state of the sequence.

Direct write accesses to the M-PHY attributes affects only the shadow registers of the attribute and need to be confirmed by writing the attribute VS_MphyCfgUpdt.

When the application needs to set any M-PHY attribute (standard or vendor specific), it should use DME_SET command and initiate a UniPro power mode change later. The UniPro power mode change ensures that the full set of attributes are consistent and synchronizes the two devices over the M-PHY link. The power mode change procedure performs the configuration update procedure during an active burst and terminates the burst afterwards to allow correct activation of the programmed attributes.

After an application initiates the power mode change, it must wait for the power mode indication before changing any M-PHY attributes.

Accesses to non-power mode related M-PHY attributes (for example, slew rate setting) should be performed before starting the link. If the application needs to re-program the M-PHY attributes while the link is active, it needs to ensure to handle the M-PHY related timing constraints (for example, SaveConfigTime) on its own, ideally by ending all data transmissions and bringing the M-PHY to a SAVE state.

The DWC_mipi_unipro_20-specific configuration and status attributes start from address 0x8000.

From address 0x8000 to 0x80FF, all M-PHY (layer 1) attributes are mapped again (same offsets as in 0x0000 to 0x00FF). However, accessing the M-PHY attributes on this address allows access to otherwise protected attributes (for example, direct access to TxGear or similar attributes). Use this option only for debugging purposes because direct access to these attributes may cause severe problems in the UniPro protocol and may corrupt the link integrity.

Additionally, attributes that are not available can be accessed through this address range. These attributes may be implementation-specific or vendor-specific attributes and any access inside this address range activates the MP_TX_ExtAcc or MP_RX_ExtAcc signal during the complete access.

Address 0x8100 to 0x81ff have same use as address range 0x8000 to 0x80FF. During this address range access, UniPro drives MP_ExtAcc to be high. This can be used to map common M-PHY attribute bank.

**Note**

To understand DWC_mipi_unipro_20 register programming and to help speed up the initial start-up of your system with the configured DWC_mipi_unipro_20 RTL, you can use the script **tools/log2prog.pl**. For more information, see the *DWC_mipi_unipro_20 Programming Sequence Tool* section of the *DesignWare Cores MIPI UniPro 2.0 User Guide*.

2.7.1 Accessing the Implementation-Specific Attributes

2.7.1.1 DME and Vendor-Specific Attributes

The DME attributes can be accessed using byte 0xD0 as upper byte and the addresses as the offset. For example, use address 0xD081 to access the interrupt status.

- All attributes are marked as either DME primitive related (with the DME_ prefix) or vendor-specific (with the VS_ prefix) attributes. These attributes are implementation-specific.
- All DME related attributes are separated into requests, confirmations, and indications. Requests are write only, while confirmations and indications can only be read.
- All DME request related attributes provide the confirmation defined in the MIPI UniPro Specification by reading the attribute on the same address.
- To activate a request any defined bit of the corresponding request attribute needs to be set to 1. If multiple bits are defined for a request, only one bit is activated at a time. Writing 0 to a defined bit has no effect.
- All reserved bits are set to 0 for write accesses and are assumed as *don't care* when reading out the attribute.
- After the reset, all readable attributes are set to 0.
- All indications are cleared after the read.

2.7.2 Register Memory Map

The DWC_mipi_unipro_20 registers are addressable on 32-bit boundaries. Each unused bit or address location is reserved for future use and read back as 0.

Table 2-3 summarizes the register memory map for the DWC_mipi_unipro_20 when APB interface is enabled. The access type RO stands for Read Only bit, WO stands for Write Only bit, and RW stands for Read/Write bit.

**Note**

Unless indicated, the default values (Value after Reset) are in decimal.

Table 2-3 DWC_mipi_unipro_20 Memory Map when APB Interface is Enabled

Register Name	Address Offset	Access Type	Description	Value after Reset
APB Indirect Access Registers				
APB_Command	0x00	RW	Controls the access to the DME attributes when using the APB interface	0
APB_Results	0x04	RO	Indicates the result of the command given in the APB_Command register	0
APB_Data	0x08	RW	Contains the data read during GET_attribute access or the data to be written during SET_attribute access	0
APB_InterruptEnable	0x10	RW	Activates individual interrupt sources when 1 is written into corresponding bit	0
APB_InterruptStatus	0x14	RO	Shows the occurrence of any of the defined interrupt sources independent of the interrupt enable settings	0

2.8 DME Control Interface

All DME control related primitives of the MIPI UniPro Specification are mapped to individual signals or attribute accesses of DWC_mipi_unipro_20. A system using software to control DWC_mipi_unipro_20 can use standard attribute accesses to control the module and interrupts to react on indications of DWC_mipi_unipro_20. In this case, all unused request signals must be tied to 0.

Hardware based applications can use separate signals to ease the controlling of DWC_mipi_unipro_20 and handling of external events.

[Table 2-4](#) shows the translation of all UniPro primitives to the individual signals or attributes.

Table 2-4 DME Primitive to Signal and Attribute Mapping

UniPro Primitive	Signal	Interrupt	Attribute
DME_POWERON.req()	optional -> not implemented yet		
DME_POWERON.cnf_L()	optional -> not implemented yet		
DME_POWEROFF.req()	optional -> not implemented yet		
DME_POWEROFF.cnf_L()	optional -> not implemented yet		
DME_ENABLE.req()	LA_EnableReq	No	DME_LayerEnable
DME_ENABLE.cnf_L()	LA_EnableCnf	No	DME_LayerEnable
DME_RESET.req()	■ LA_ResetReqWarm ■ LA_ResetReqCold ■ LA_ResetModeReq	No	DME_Reset

Table 2-4 DME Primitive to Signal and Attribute Mapping (Continued)

UniPro Primitive	Signal	Interrupt	Attribute
DME_RESET.cnf_L()	LA_ResetCnf	No	DME_Reset
DME_ENDPOINTRESET.req()	LA_EndpointResetReq	No	DME_EndpointReset
DME_ENDPOINTRESET.cnf_L()	LA_EndpointResetCnf	No	DME_EndpointReset
DME_ENDPOINTRESET.ind()	LA_EndpointResetInd	Yes	DME_EndpointResetInd
DME_LINKSTARTUP.req()	LA_LinkStartupReq	No	DME_LinkStartup
DME_LINKSTARTUP.cnf_L()	LA_LinkStartupCnf	No	DME_LinkStartup
DME_LINKSTARTUP.ind()	LA_LinkStartupInd	Yes	DME_LinkStartupInd
DME_LINKLOST.ind()	LA_LinkLostInd	Yes	DME_LinkLostInd
DME_HIBERNATE_ENTER.req()	LA_HibernateEnterReq	No	DME_HibernateEnter
DME_HIBERNATE_ENTER.cnf_L()	LA_HibernateEnterCnf	No	DME_HibernateEnter
DME_HIBERNATE_ENTER.ind()	LA_HibernateEnterInd	Yes	DME_HibernateEnter
DME_HIBERNATE_EXIT.req()	LA_HibernateExitReq	No	DME_HibernateExit
DME_HIBERNATE_EXIT.cnf_L()	LA_HibernateExitCnf	No	DME_HibernateExit
DME_HIBERNATE_EXIT.ind()	LA_HibernateExitInd	Yes	DME_HibernateExitInd
DME_POWERMODE.req()	none -> attribute access	No	Several DME_SET.req (for more information, see “Layer 1.5 Configuration” on page 540).
DME_POWERMODE.cnf_L()	Return the value of PWRMODE attribute access	No	Return the value of PWRMODE attribute access
DME_POWERMODE.ind()	LA_PowerModelInd	Yes	DME_PowerModelInd
DME_TEST_MODE.req()	LA_TestModeReq	No	DME_TestMode
DME_TEST_MODE.cnf_L()	LA_TestModeCnf	No	DME_TestMode
DME_TEST_MODE.ind()	LA_TestModeInd	Yes	DME_TestModeInd
DME_ERROR.ind()	■ LA_ErrorPHYInd ■ LA_ErrorPAInd ■ LA_ErrorDInd ■ LA_ErrorNInd ■ LA_ErrorTInd	Yes	■ DME_ErrorPHYInd ■ DME_ErrorPAInd ■ DME_ErrorDInd ■ DME_ErrorNInd ■ DME_ErrorTInd

Most local confirmation primitives are not explicitly implemented as signals. If a system needs the confirmation, an attribute access can be performed.

The power mode related primitives can only be handled by attribute accesses because several attributes are involved. For more information about power mode changes, see “[Layer 1.5 Configuration](#)” on page 540.

2.8.1 DME Control Interface Signals

The DME control interface signals can be used by hardware based applications to control the module and react on external events. Additionally, these signals can be used in systems having a separate interrupt controller. If the interrupt controller is limited in the number of interrupt sources, the separate interrupt signal (LA_Intr) can ease the integration.

The DME Control interface signals are described in “[DME Control Interface Signals](#)” on page 122. The properties are as follows:

- This interface is synchronous to the system clock used for all accesses between DWC_mipi_unipro_20 and the application.
- All input signals of this interface must be active for a single system clock cycle only.
- Output signals are also valid for a single system clock cycle unless specified differently.
- If request signals are not used, they must be tied to 0.

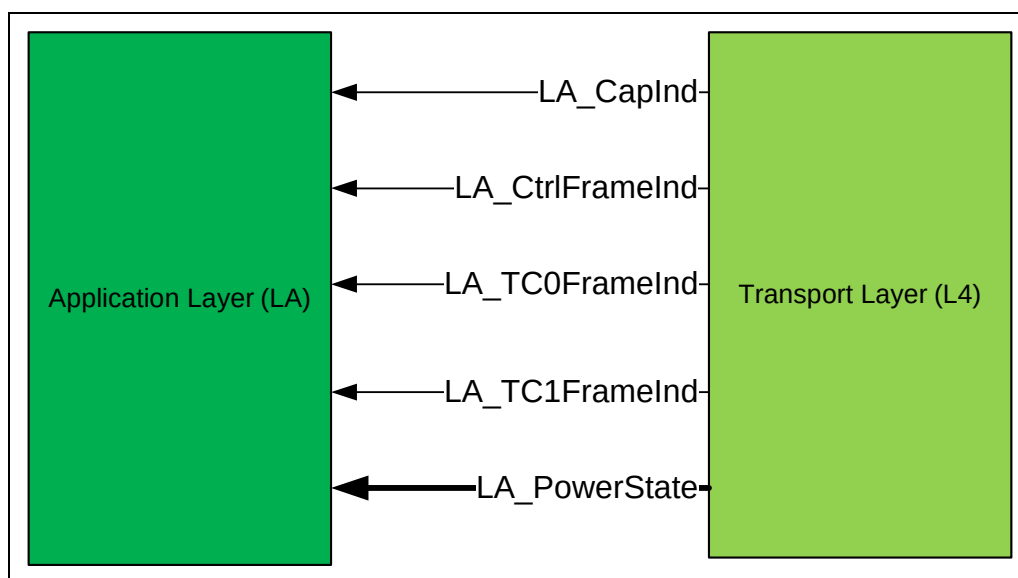
Examples for using these interfaces are at “[Layer 4 Test Feature](#)” on page 62.

2.9 Additional Indication Interface

While all layer 2 frame indications are level sensitive (that is, a high level indicates the availability of the respective frame inside layer 2), the capability indication is a pulse, which is only activated for a single CG_Sysclk cycle each time a capability PACP frame is received.

[Figure 2-10](#) shows all signals of the Additional Indication interface.

Figure 2-10 Additional Indication Interface



The Additional Indication interface signals are described in “[Additional Indication Signals](#)” on page 132.

2.10 Interrupt Interface

DWC_mipi_unipro_20 includes a small interrupt controller to generate a single interrupt signal using several internal sources. The Interrupt interface signals are described in “[Interrupt Interface Signals](#)” on page 134.

Generation of the interrupt can be programmed by configuring the interrupt enable attribute (VS_InterruptEnable). Setting the individual bits to 1 enables the corresponding interrupt source to set the LA_Intr signal to active state (high). This signal stays active until the corresponding indication is read out and cleared. The application can read the source of the interrupt using the VS_InterruptStatus attribute. The individual bits of the status attribute show the status of the corresponding indication independent of the enable bit. This allows any application to check the status of DWC_mipi_unipro_20 with a single attribute access.

All sources for the interrupt signal are shown in [Table 2-5](#).

All the interrupt sources except Debug Counter Overflow and PA_INIT_ERROR are also available as output signals as part of the DME Control interface as shown in [Table 2-5](#). These signals are not level sensitive but are normally asserted for only one clock period.

Table 2-5 Interrupt Sources

Interrupt Source	Interrupt Number	Description
LA_EndpointResetInd	0	Endpoint reset indication
LA_LinkStartupInd	1	Link startup indication
LA_LinkLostInd	2	Link lost indication
LA_HibernateEnterInd	3	Hibernate enter indication
LA_HibernateExitInd	4	Hibernate exit indication
LA_PowerModeInd	5	Power mode indication
LA_TestModeInd	6	Test mode indication
LA_ErrorPHYInd	7	PHY layer error indication
LA_ErrorPAInd	8	PHY adapter layer (L1.5) error indication
LA_ErrorDInd	9	Data link layer (L2) error indication
LA_ErrorNInd	10	Network layer (L3) error indication
LA_ErrorTInd	11	Transport layer (L4) error indication
PA_INIT_ERROR (L2) (EDI_PIE)	12	Error in PA initialization sequence (fatal error)
Debug Counter Overflow (VS_DebugCounterOverflow)	13	Overflow of debug counter detected
LA_LinkStartupCnf	14	Link Startup Confirmation
LA_QosInd	15	PHY Link degradation indication

The generated interrupt (LA_Intr) is level sensitive because it stays active until all interrupt sources are cleared.

Bit 12 of the configurable interrupt sources is an individual error condition and is included in the LA_ErrorInd bus. This special condition is added, because all other interrupt indications of L2 are self-correcting and do not require application interference. The PA_INIT_ERROR however marks a fatal systematic error, which requires a reset of the module and a complete re-initialization.

2.11 Configuration Accesses

DWC_mipi_unipro_20 provides two interfaces to the application to perform the configuration through attribute access. You can select one of these options at the time of controller configuration:

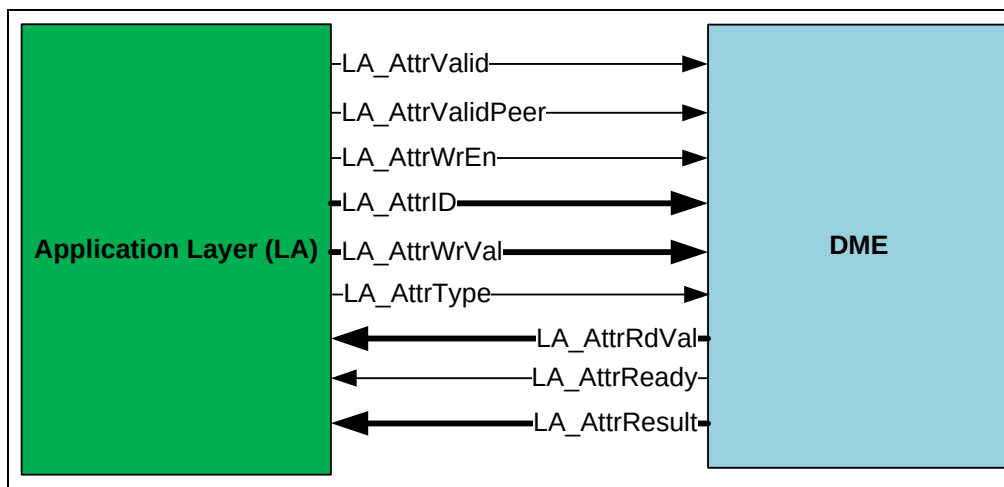
- Native Slave interface
- APB3 Slave interface

These attributes are defined in detail by the MIPI UniPro and M-PHY Specifications as well as described in “APB_Command” on page 166. Each layer of UniPro has a set of attributes depending on the functions of the layer. The configuration port is connected directly (in native interface) or indirectly (in APB3 interface) to the DME, which decodes the addresses and routes the attribute accesses to the respective layers.

2.11.1 Configuration Access Using Native Slave Interface

The native interface provides direct access to the DME module in DWC_mipi_unipro_20. Figure 2-11 shows the list of signals used for the native configuration interface. These signals are described in “Native Slave Interface Signals” on page 116.

Figure 2-11 Native Configuration Interface



The attribute accesses are either write or read accesses to one of the internally stored configuration values. The DME processes these accesses and forwards the data write or read request to the target layer. In general, write accesses target only a single attribute inside DWC_mipi_unipro_20. Exceptions are the M-PHY layer attributes, which are passed through the PHY adapter layer to the individual M-PHY modules. By definition, all attributes in each M-PHY can be accessed individually.

Additionally, the peer device attributes can be accessed by asserting the LA_AttrValidPeer signal (only after link startup). Because these accesses may have a high latency, all attribute write and read accesses are

acknowledged by `DWC_mipi_unipro_20` by flagging its status on the `LA_AttrReady` signal. Depending on the time it takes to return the requested data from the peer device, a peer attribute access has a major influence on the total access time, which must be considered when designing a system using `DWC_mipi_unipro_20`.

When the ready signal is asserted, the `LA_AttrResult` signal shows the result code of the last access to the application.

A subsequent access must be performed only after the previous access is acknowledged.

All signals are expected to be synchronous to the `CG_Sysclk` provided to `DWC_mipi_unipro_20`. If the application layer uses a different clock domain, a separate module must synchronize all the signals between the two clock domains.

The `LA_AttrType` signal defines if the access is performed to the current value of the attribute or to its static value. Because `DWC_mipi_unipro_20` does not use any static attributes, all accesses to local attributes using the `LA_AttrType` signal, which is set to 1, are denied with `INVALID_MIB_ATTRIBUTE` error code. Static attributes on the peer side can be accessed using this special mode.

In contrast to the MIPI UniPro Specification, `DWC_mipi_unipro_20` allows configuration accesses in all power modes. Although all accesses are processed, accesses to some attributes (defined as non-retained in the MIPI UniPro Specification) may have no impact. This is the case for all layer 4 related attributes while the device is in hibernation mode. During hibernation, these attributes are reset and their configured values are no longer valid. Read accesses to these attributes always return 0.

Allowing attribute accesses is required due to the fact that several important control functions (request primitives) are accessible using the configuration attributes.

All result codes are valid for `DWC_mipi_unipro_20` attributes only. The M-PHY attributes (accesses to layer 1) only have the following result codes:

- 0: Success
- 3: Read-only

The read-only result code is generated when writing into blocked attributes like gear or active lane settings.

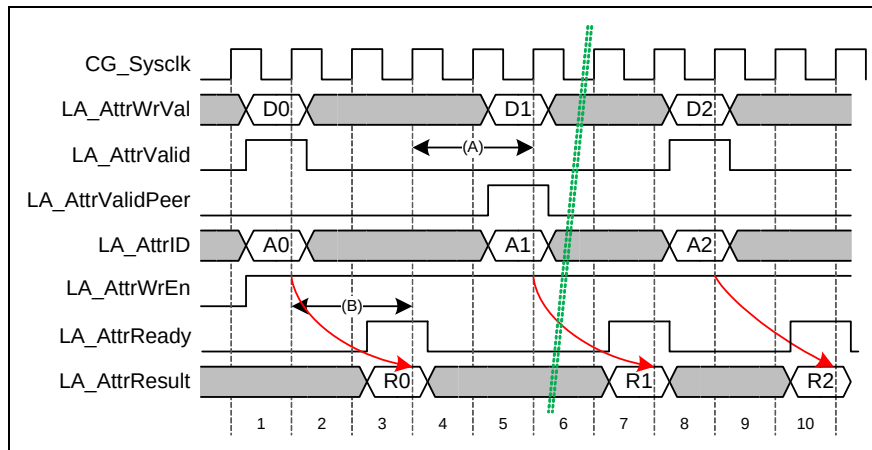


Attribute accesses to PHY attributes do not stop or suspend the ongoing data and control information transfer. It is not possible for the application to predict the current M-PHY state.

2.11.1.1 Attribute Access Timing

Accessing the attributes of the `DWC_mipi_unipro_20` is done in a very straightforward way. The application layer indicates a new access by the activation of the `LA_AttrValid` signal. The `LA_AttrWrEn` signal defines if the access is a write (1) or a read (0) access. The `LA_AttrID` bus carries the address of the attribute.

While the `LA_AttrWrVal` bus delivers the write value, the `LA_AttrRdVal` bus returns the read value of the requested attribute.

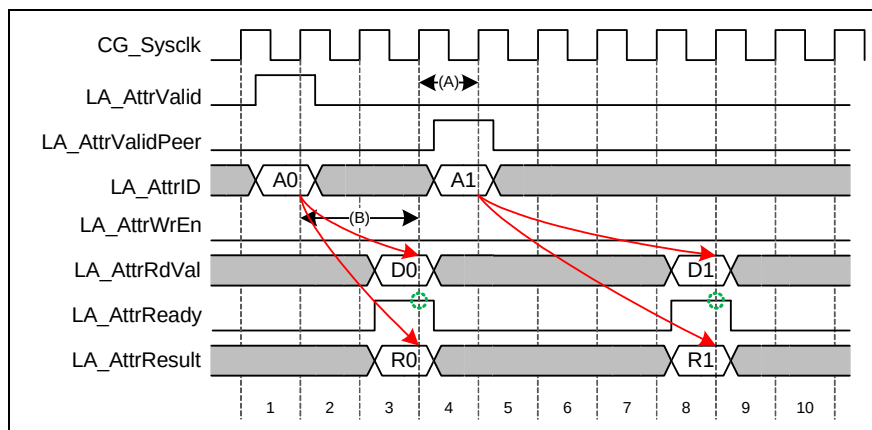
Figure 2-12 Attribute Write Accesses

Data on the LA_AttrWrVal bus is sampled by the DME when LA_AttrValid and LA_AttrWrEn signals are active. At the same time, the value on the LA_AttrID bus is sampled and used as address for the attribute write access. The application uses the LA_AttrReady flag provided by DWC_mipi_unipro_20 to fetch the result code of the access from the LA_AttrResult bus. After the reception of the LA_AttrRdy flag, the application can start the next attribute access request. The application cannot start a new access, until the LA_AttrRdy flag is asserted for the previous access.

To access a peer attribute, the application layer asserts LA_AttrValidPeer instead of LA_AttrValid. DWC_mipi_unipro_20 translates this request to an access over the link. The result code informs the application about the success of the remote access, after the return message is received.

Attribute read accesses use a similar transfer mechanism. The application layer requests an attribute value by activating the LA_AttrValid signal and keeping the LA_AttrWrEn signal low. The DME requests the attribute value from the layer targeted by the LA_AttrID value and keeps its LA_AttrReady signal low as long as the data is not available. When the LA_AttrReady signal is activated by the DME, the application layer can sample the return value from the LA_AttrRdVal bus as well as the result code of the last access.

Peer read accesses are performed in the same way except for activating the LA_AttrValidPeer signal. As these accesses are forwarded to the link, the result may take some time to return to the application.

Figure 2-13 Attribute Read Accesses

**Note**

The application layer must not start a new access while a read or write access is in progress.

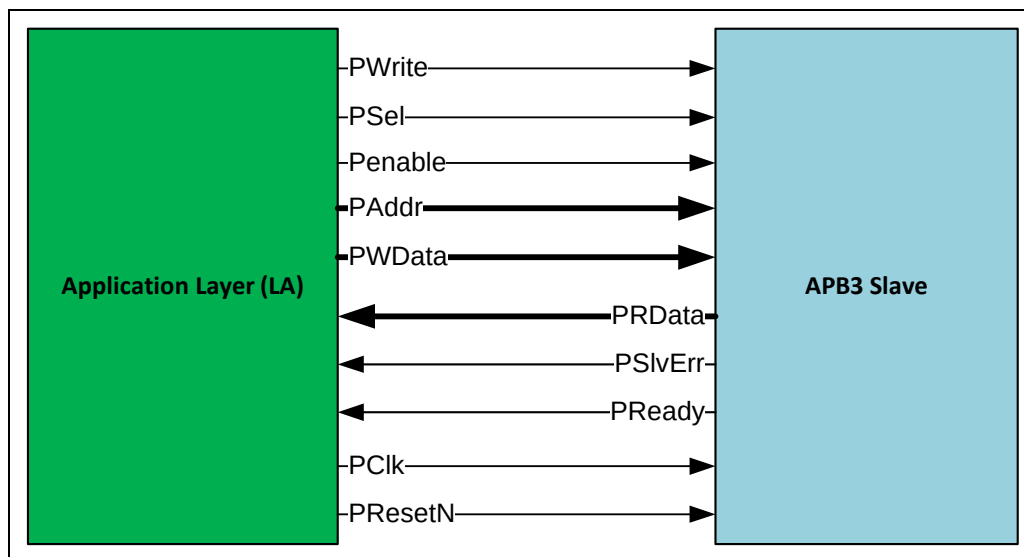
The cycle following a valid LA_AttrRdy signal is the first valid cycle for generating the next attribute read or write access (shown as A in [Figure 2-12](#) and [Figure 2-13](#)). As attribute accesses to local and peer device may need several cycles, the delay of the LA_AttrRdy signal (shown as B) is at least one cycle. The maximum delay of the LA_AttrReady signal is unknown because it depends on the link mode and the reaction time of the peer device. Peer accesses are only possible after link startup. They are not allowed when the link is in hibernate mode.

2.11.2 Configuration Access Using APB3 Slave Interface

This option provides the capability to access the attributes with the AMBA standard APB3 compliant interface. In addition, it reduces the address map of DWC_mipi_unipro_20 APB slave port. All accesses to the DME attributes are indirect.

[Figure 2-14](#) shows the APB3 interface. The APB3 signals are described in “[APB3 Slave Interface Signals](#)” on page 119. In [Figure 2-14](#), the PClk and PResetN signals exist only when ABPB3 Slave interface is configured (by selecting APB3_CLOCK_EN) to contain these signals during configuring the controller. Otherwise, by default, CG_Sysclk and RG_ResetN are used.

Figure 2-14 Access Using APB3



The DME attributes are accessed in a two-step process as the direct address map consists of only five registers as described in “[APB_Command](#)” on page 166. The application must perform the following for a read access to any attribute:

1. Write into the **APB_Command** register with required values for the fields such as attribute ID, attribute type, and so on. Write 0 into the AAT bit (bit 0). This initiates a read access to the required DME attribute within the controller.

2. Poll the **APB_Results** register[0] until it is cleared. A value of 1 indicates that the read command is still in progress.
3. The results of the access is given in the CMDR field (Bits[11:8]) of this register. An **all-zero** value means a successful operation and the application can read the value of the attribute from **APB_Data** register.
4. If the access is unsuccessful, the reason is indicated by the non-zero value of CMDR field.

For writing or setting an attribute, the application must perform these steps:

1. Write the required attribute value in the **APB_Data** register.
2. Write the required values for attribute ID, type, and so on in the **APB_Command** register and set the AAT field. This initiates a write access to the corresponding DME attribute within the controller.
3. Poll the **APB_Results** register[0] until it is cleared. A value of 1 indicates that the write command is still in progress.
4. The results of the access is indicated in the CMDR field (Bits[11:8]) of this register. An all-zero value means a successful completion of the write operation. Otherwise, the reason for the unsuccessful operation is given by the CMDR field.

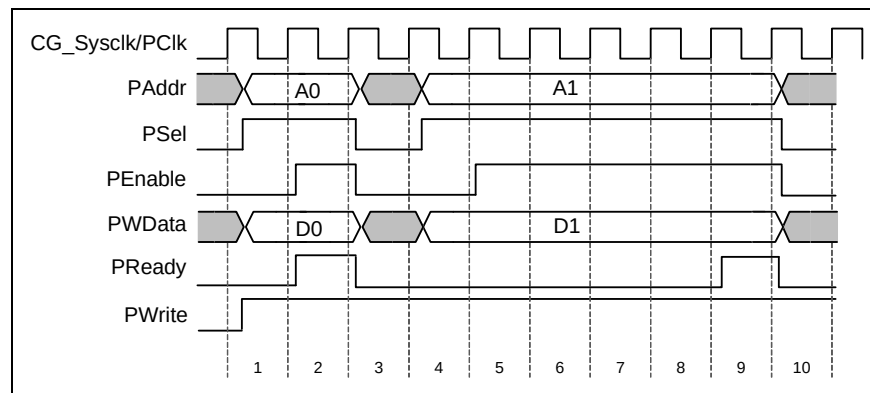
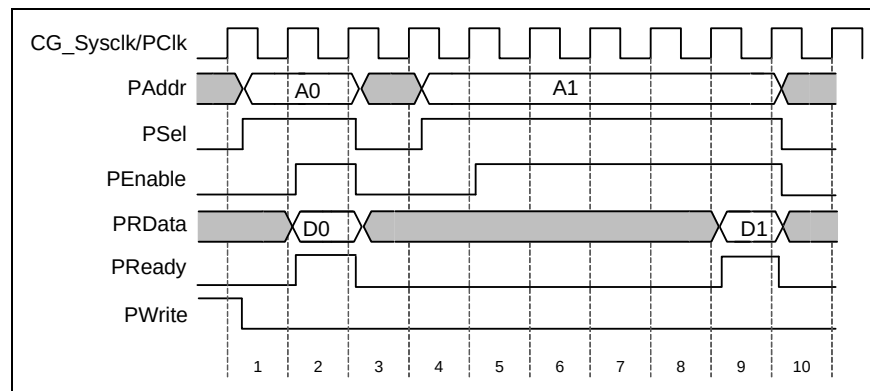
When the application initiates a read or a write operation, the controller internally generates the configuration access to the DME attributes. For more information about detailed operation and latencies of the attribute accesses, see [“Configuration Access Using Native Slave Interface”](#) on page 37.

As explained in [“Configuration Access Using Native Slave Interface”](#) on page 37, the latencies involved in completing a peer device attribute is not predictable and can be large. In such cases, the application can avoid the need for polling the **APB_Results** register by enabling the interrupt mode. It can set the ICE bit (Bit 31) of the **APB_InterruptEnable** register before giving the command in the **APB_Command** register. When the required operation is complete, **DWC_mipi_unipro_20** sets the IC bit (bit 31) of the **APB_InterruptStatus** register and also asserts the **LA_Intr** signal. The application can obtain the results of the operation by reading the **APB_Results** and **APB_Data** registers.

2.11.2.1 APB3 Access Timing

The access to the **APB_Command**, **APB_Results**, and **APB_Data** registers or any of the reserved addresses is completed in two clock cycles. The **PReady** signal is asserted one clock cycle after the **PSel** signal is asserted.

Write and read accesses are shown in [Figure 2-15](#) and [Figure 2-16](#).

Figure 2-15 Write Accesses Using APB**Figure 2-16 Read Accesses Using APB3**

The **APB_InterruptEnable** and **APB_InterruptStatus** registers of the APB3 Slave interface are mapped to the **VS_InterruptEnable** and **VS_InterruptStatus** registers and hence the controller performs an internal access to these attributes before indicating the completion of the APB3 transfer cycle. Accesses to these registers get completed in five clock cycles.

2.12 Layer 2 Memory Interface

DWC_mipi_unipro_20 stores the L2 data frames in external physical memories through this interface. Instantiate the following physical memories external to DWC_mipi_unipro_20 in your design:

- L2 TX Buffer: This is a single dedicated physical memory to store the data for TC0 frames received from the upper layers (layer 3 and layer 4) as well as the data address information. For more information about L2 TX Buffer Size calculations, see [“L2 TX Buffer Size Calculation”](#) on page 566.
- L2 RX Buffer: This is a single dedicated physical memory to store the data for TC0 frames received from the PA layer. For more information about L2 RX Buffer Size calculations, see [“L2 RX Buffer Size Calculation”](#) on page 568.

Figure 2-17 shows the relative placement of these memories.

Figure 2-17 Layer 2 Memory Interface



These memories should be modeled using a two port RAM with registered outputs. The output of these memories do not change when respective enable signals are not asserted.

See the following sections for signal description:

- [“Layer 2 RX Memory Signals \(TC0\)”](#) on page 151
- [“Layer 2 TX Memory Signals”](#) on page 155

Figure 2-18 and Figure 2-19 show the typical write access waveforms.

Figure 2-18 L2 Memory Write Access

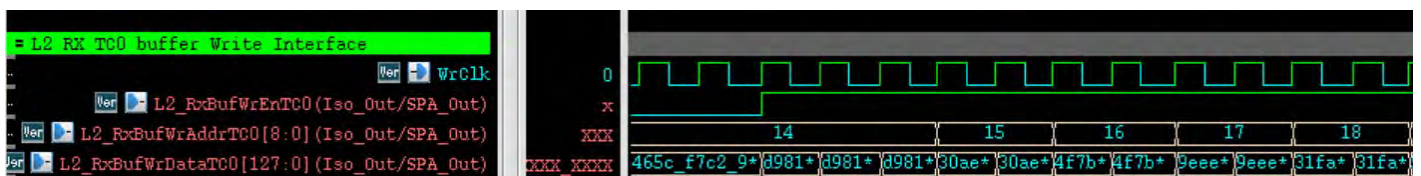


Figure 2-19 L2 Memory Read Access



Make the clock connections as shown in [Table 2-6](#):

Table 2-6 L2 Memory Interface Clock Connections

Type of Clock	Name of the Clock Used
TX Buffer Read Clock	CG_TX_SymbolClk
TX Buffer Write Clock	CG_Sysclk
RX TC0 Buffer Read Clock	CG_Sysclk
RX TC0 Buffer Write Clock	CG_RxSymbolClkSelected
RX TC1 Buffer Read Clock	CG_Sysclk
RX TC1 Buffer Write Clock	CG_RxSymbolClkSelected

2.13 Clock and Reset Interfaces

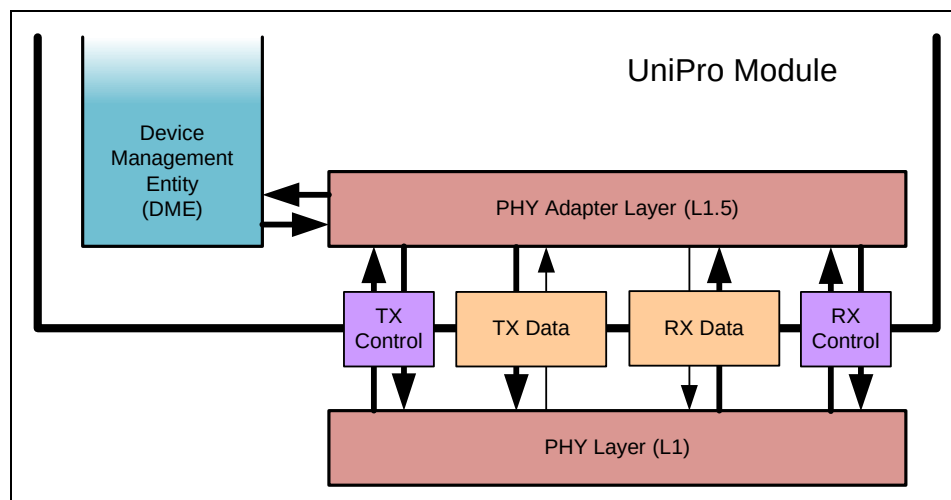
For information about clock and reset interfaces, see *DWC_mipi_unipro_20 Clock Interface* and *DWC_mipi_unipro_20 Reset Handling* sections of the *DesignWare Cores MIPI UniPro 2.0 User Guide*.

2.14 M-PHY Interface

The M-PHY interface has the following parts:

- [TX Data Interface](#)
- [TX Control Interface](#)
- [RX Data Interface](#)
- [RX Control Interface](#)
- [CB Control Interface](#)

The interface between *DWC_mipi_unipro_20* and the M-PHY modules is defined in the MIPI M-PHY Specification. [Figure 2-20](#) shows the different M-PHY interface parts.

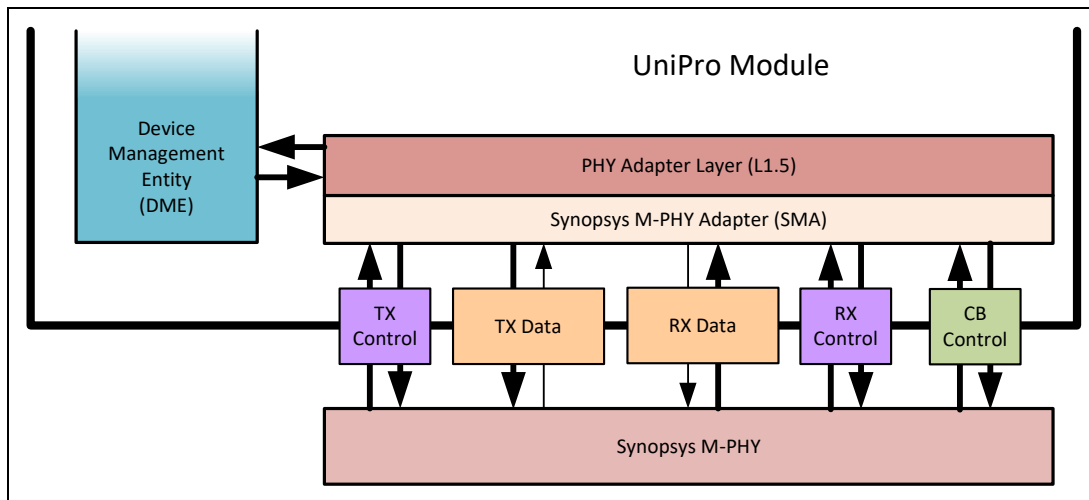
Figure 2-20 M-PHY Interfaces

The PHY adapter layer (L1.5) performs attribute accesses to PHY layer attributes over two control interfaces, while the data interfaces are used to transport the frame data from and towards the UniPro link.

When you choose to use the Synopsys M-PHY Verilog behavioral model by enabling the **SNPS_MPHY** option, the Synopsys M-PHY adapter module is instantiated and interfaces with Synopsys M-PHY.

Figure 2-21 shows the different interface parts in this configuration.

Figure 2-21 Synopsys M-PHY Interfaces

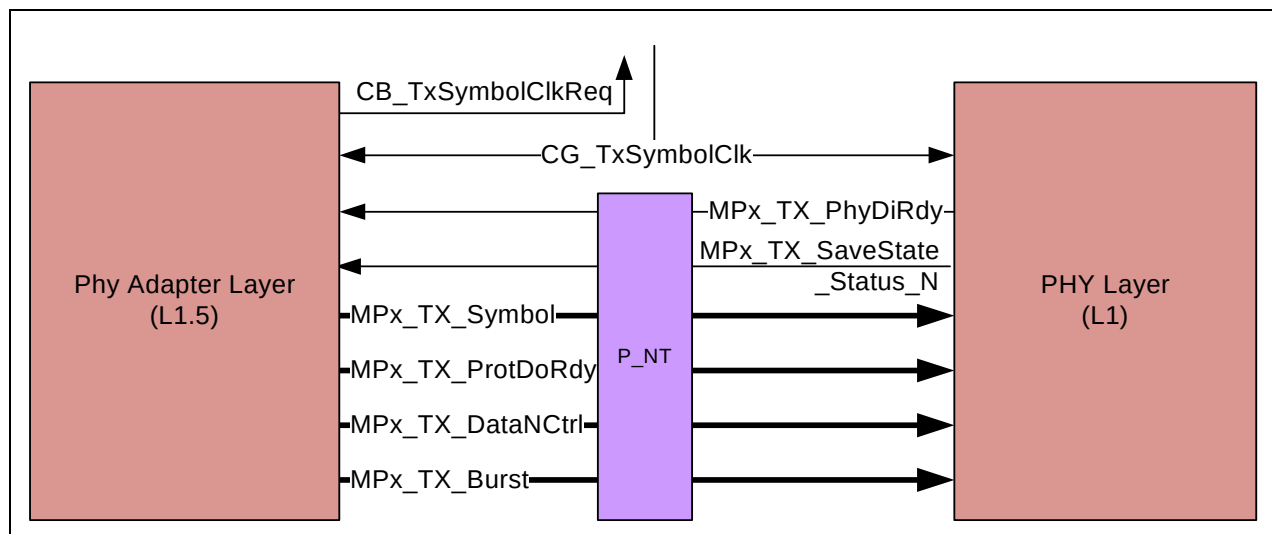


Common Block (CB) control interface is specific to Synopsys M-PHY.

2.14.1 TX Data Interface

The M-PHY TX data interface transports data from the `DWC_mipi_unipro_20` to the M-PHY modules to be transferred to a remote device. The number of signals on this interface depends on the number of TX M-PHY modules available in the system. The `P_NT` parameter defines the number of UniPro symbols processed in parallel with the memory structures of layer 2 to the PHY layer.

Figure 2-22 M-PHY TX Data Interface



The box marked P_NT symbolizes that the width of all the signals running through this box is determined by the parameter P_NT.

The M-PHY TX Data interface signals are described in “M-PHY TX Control Interface Signals” on page 135.

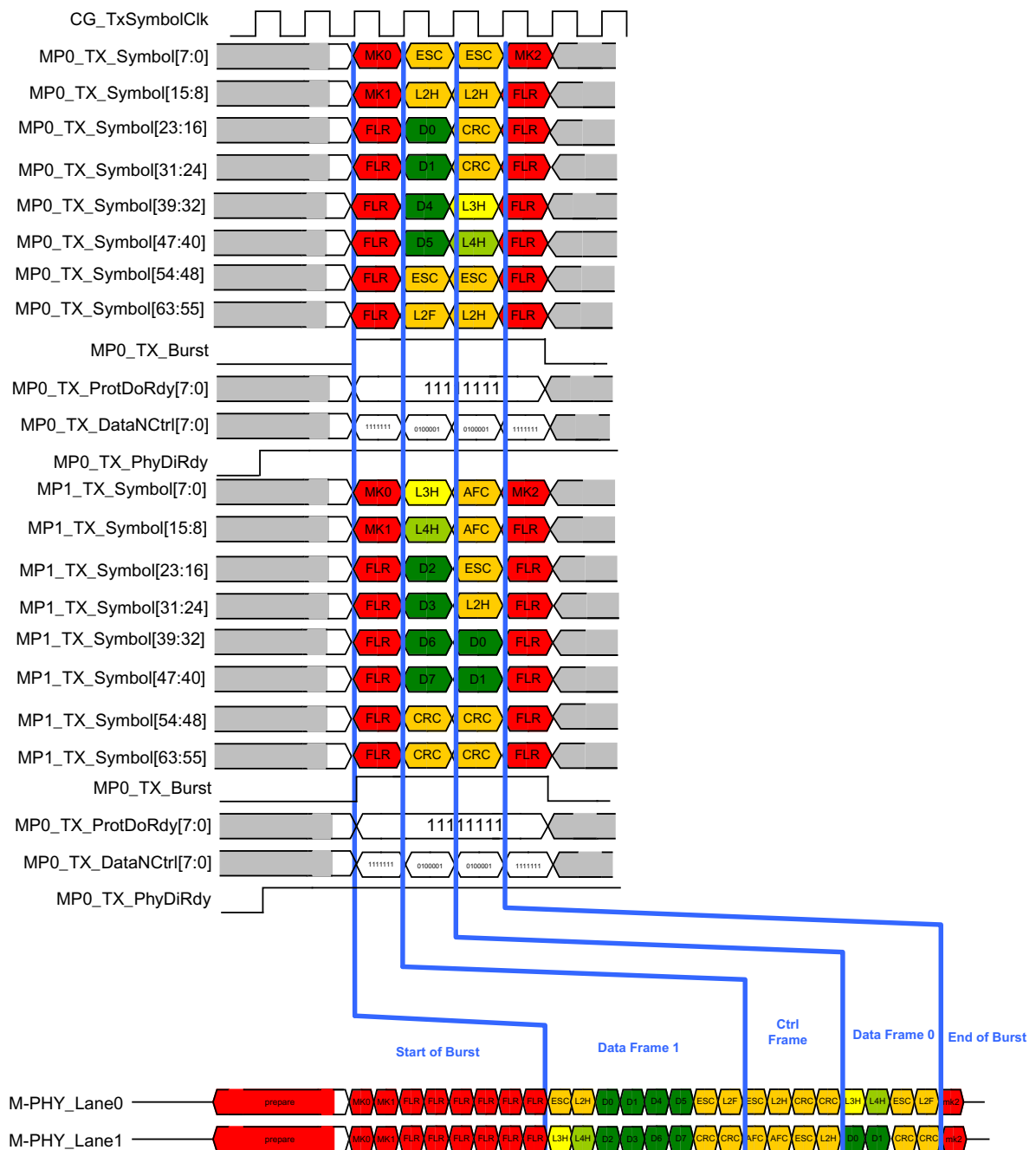
All the signals on the M-PHY TX Data interface are synchronous to **CG_TxSymbolClk**. The frequency of this clock is determined by the gear, series, and mode settings of the TX M-PHY modules.

The PHY adapter layer indicates the start of a new access to the PHY layer by asserting the **MPx_TX_Burst** signal and providing the first marker symbols (MK0) to be sent over the link. All active M-PHY transmitters receive the burst start of an access simultaneously and are expected to react with the same latency. When all the requested M-PHY modules assert the **MPx_TX_PhyDiRdy** signal, **DWC_mipi_unipro_20** starts its transmission.

Figure 2-23 shows a sample sequence of one data frame, one AFC control frame, and another data frame in a single burst. Figure 2-23 also shows the waveform for 80-bit interfaces. In this example, P_NT is equal to two.

Figure 2-23 M-PHY TX Data Access

16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	ESC_DL								SOF			TC0		Reserved		
0	L3s=1	DestDeviceID_Enc (L3H)							L4s=1	DestCPortID_Enc (L4H)				FCT		EOM = 0
0	D0								D1							
0	D2								D3							
0	D4								D5							
0	D6								D7							
1	ESC_DL								EOF_EVEN			Frame Seq. Number = 0				
0	CCITT CRC-16															
1	ESC_DL								AFC			TC		CReq	Reserved	
0	Frame Seq. Number						Reserved			Credit Value						
0	CCITT CRC-16															
1	ESC_DL								SOF			TC0		Reserved		
0	L3s=1	DestDeviceID_Enc (L3H)							L4s=1	DestCPortID_Enc (L4H)				FCT		EOM = 0
0	D0								D1							
1	ESC_DL								EOF_EVEN			Frame Seq. Number = 1				
0	CCITT CRC-16															



With every clock cycle, a new UniPro symbol (four M-PHY symbols) per active lane is provided by layer 1.5. As the complete TX output data path runs on the same clock, there is no need to insert filler (FLR) symbols inside frames. After a frame is sent, some FLR symbols may be inserted to fill the gap to the next frame, which is always started on the lower PHY symbol of the first M-PHY lane (logical lane 0). Layer 1.5 marks these gaps by setting the relevant **MP_x_TX_ProtDoRdy** signal to 0. Using this information, the target M-PHY inserts FLR symbols automatically.

In addition to the filler generation of the M-PHY (through **MPx_TX_ProtDoRdy**), the **DWC_mipi_unipro_20** also generates filler symbols, for example, when issuing the end of burst marker (**MK2**, **FILLER**).

All **MPx_TX_PhyDiRdy** signals of active lanes in a multi-lane environment must be cycle accurate, that is, they must be signaled without any skew from all M-PHY modules of the transmission link.

Control symbols are marked by setting the related bit of **MPx_TX_DataNCtrl** to 1.

[Figure 2-23](#) shows transmissions over lanes 0 and 1, which are identified in this order. UniPro allows any connectivity variant between the two devices and therefore the initial startup sequence of the system must define physical lane 1 as logical lane 0.

Only the lower 64 bits of each symbol of the **TX_Symbol** interface are used by **DWC_mipi_unipro_20** to deliver data to the M-PHY modules. The upper sixteen bits of each symbol are kept at zero. For example, if two lanes are used and actively transmitting data, only bits [63:0] and [143:80] are used for data. The M-PHY symbols at position [15:0] and [95:80] are transmitted first over the M-PHY link, followed by the M-PHY symbols at position [63:16] and [143:95] respectively.

To reduce power consumption, **DWC_mipi_unipro_20** requests the activation of the **CG_TxSymbolClk** signal through the **CB_TX_SymbolClkReq** signal. When this signal is active (high), the external clock switch or PLL controller must activate the **CG_TxSymbolClk** clock input to **DWC_mipi_unipro_20**. The **CG_TxSymbolClk** generation for the M-PHY module may have different requirements. If the PLL needs to be switched on, an additional delay for data output must be taken into account during system design (latency or bandwidth calculations).

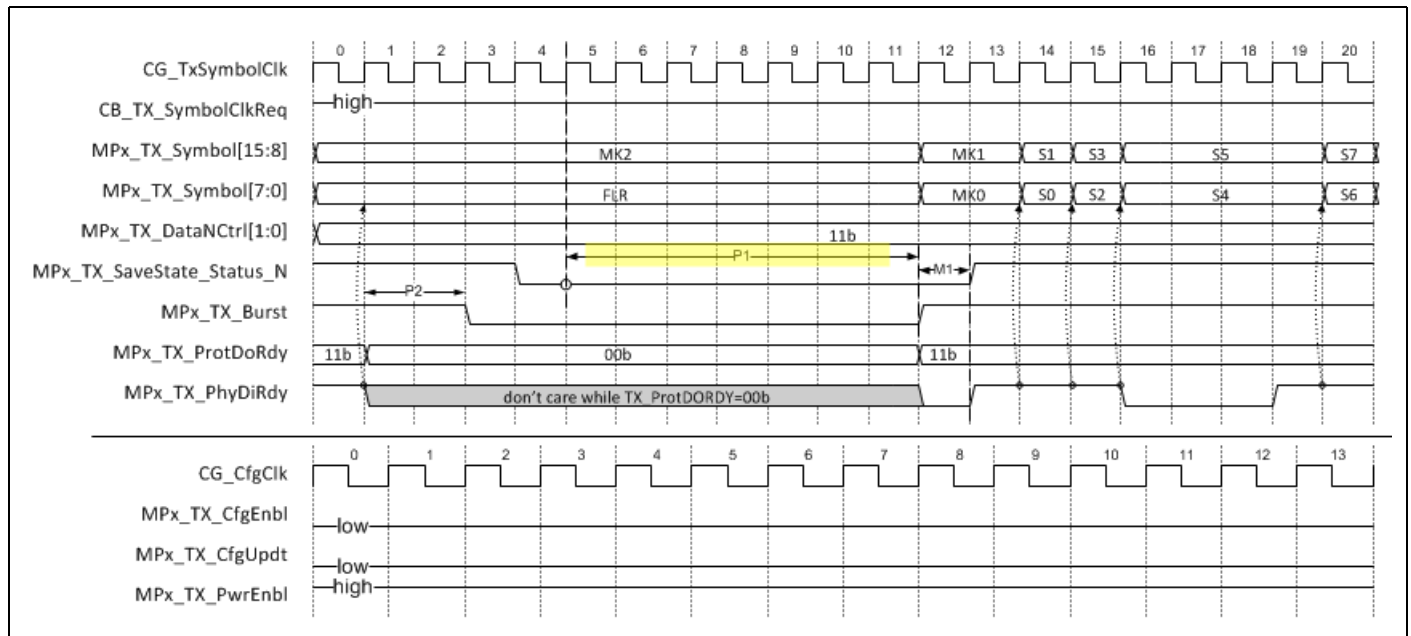


Note

If **DWC_mipi_unipro_20** is configured to use more physical lanes than available in the system, all input signals must be tied to logical 0, except for **MPx_TX_PhyDiRdy**, which must be tied to logical 1.

[Figure 2-24](#) and [Figure 2-25](#) show individual timings that are automatically handled by **DWC_mipi_unipro_20**. [Figure 2-24](#) shows the handling of **SleepNoConfigTime** and **StallNoConfigTime** intervals with two consecutive bursts. When **DWC_mipi_unipro_20** detects a Save state (**MPx_TX_SaveState_Status_N** = low), it waits for the **PA_SleepNoConfigTime** or the **PA_StallNoConfigTime** attribute value before starting a new burst. The **MPx_TX_SaveState_Status_N** signal is expected to signal a high (inactive) value while M-PHY is in the **LineCfg** state.

Timing T2 is defined by the **PA_TxTrailingClocks** attribute, while timing M1 defines the minimum delay for the inactive state of **MPx_TX_SaveState_Status_N** after an active edge of the **MPx_TX_Burst** signal. When the burst is activated, **DWC_mipi_unipro_20** enables its **MPx_TX_ProtDoRdy** signal along with the sync (**MK0** and **MK1**).

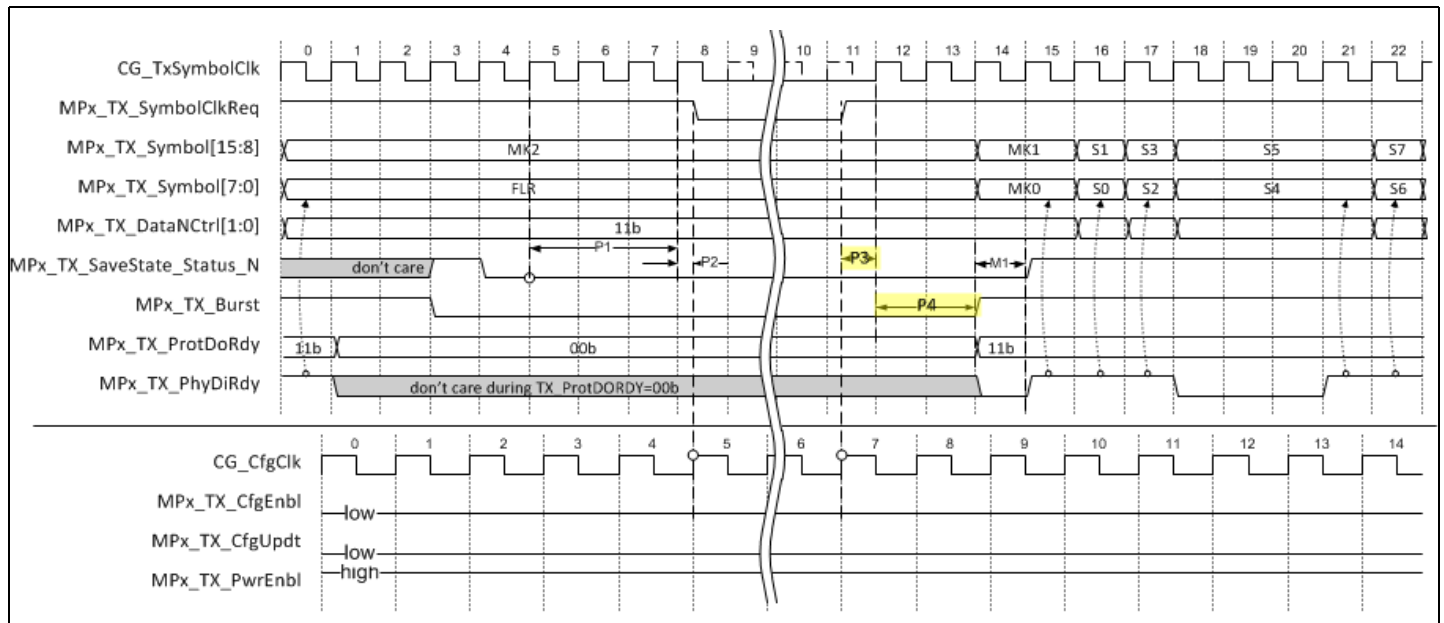
Figure 2-24 SleepNoConfigTime/StallNoConfigTime Handling**Note**

This diagram only represents how SleepNoConfigTime/StallNoConfigTime is arrived at. But the period of P1 varies based on RMMI Data Width selection.

P1 indicates the length of the StallNoConfigTime or SleepNoConfigTime interval. It is counted in **CG_TxSymbolClk** cycles (**PA_StallNoConfigTime** +1/2 or **PA_SleepNoConfigTime** +1/2) ignoring the fractional part.

If DWC_mipi_unipro_20 has no more data, it deactivates the CB_TX_SymbolClkReq signal allowing the application to switch off the CG_TxSymbolClk input to the module. This signal is generated on the CG_Sysclk, and it requires some synchronization logic and some additional cycles (P3 and P4 in [Figure 2-25](#)) before the next burst starts.

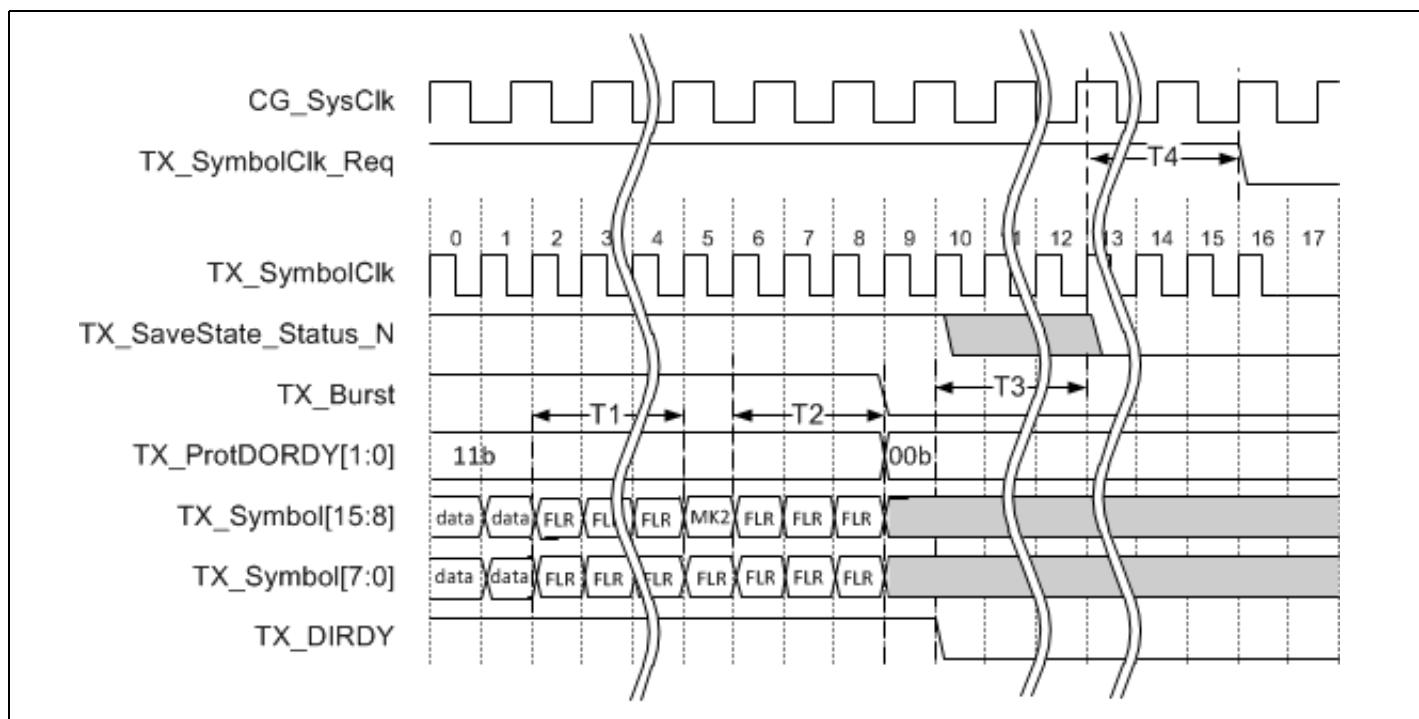
Timing P1 resembles the SleepNoConfigTime, M1 is defined as in the [Figure 2-24](#), and P2 defines the metastability clock sampling of MPx_TX_SaveState_Status_N to the CG_Sysclk domain.

Figure 2-25 TxSymbolClk Gating

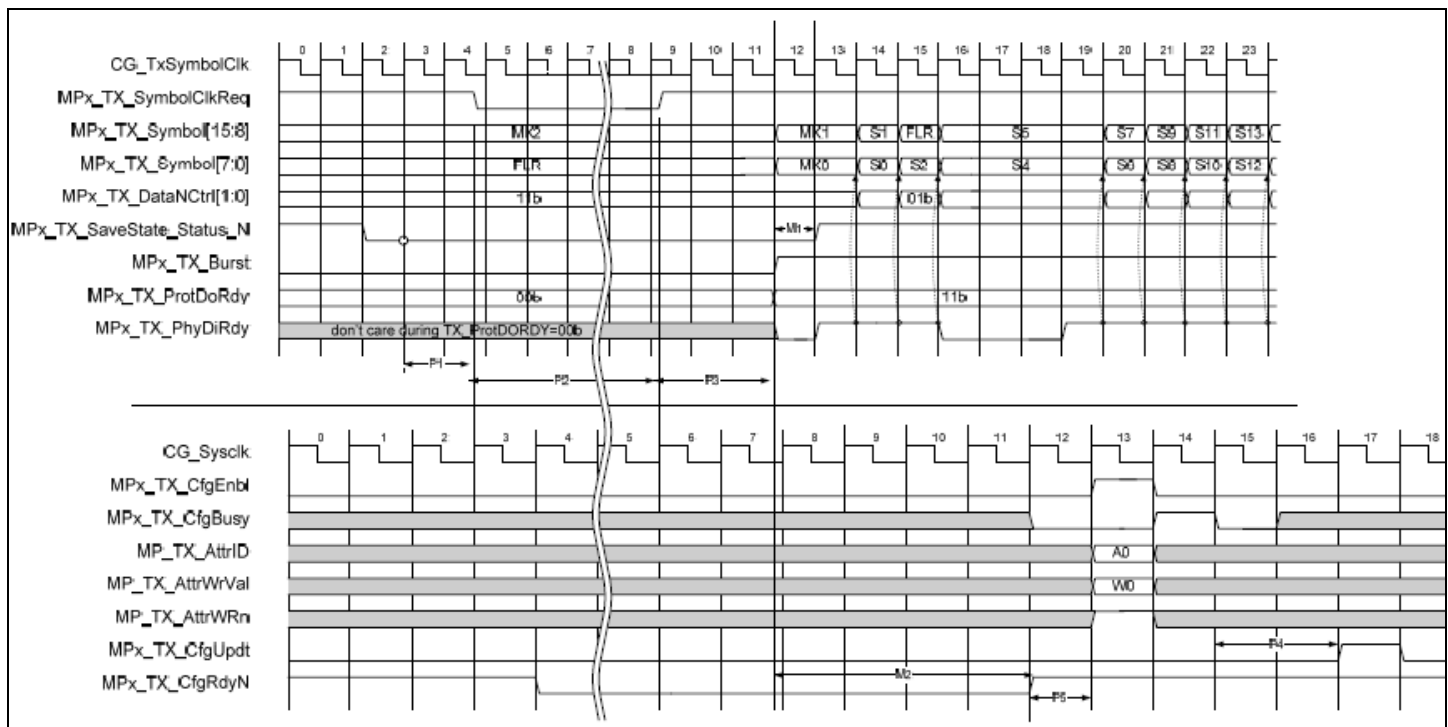
At the end of every burst, the transmitter generates several timings to allow proper data transmission and burst closure. These timings are:

- VS_TxBurstClosureDelay (T1)
- PA_TxTrailingClocks (at least peer PA_MinRxTrailingClocks, T2)
- M-PHY TX induced delay, TX_PWM_BurstClosureExtension (M-PHY, T3)

Figure 2-26 shows these timings. Timing T3 may include the LCC sequence as well. Timing T4 shows the delay of TX_SymbolClkReq de-assertion when a Save state is reached.

Figure 2-26 M-PHY Burst Closure Timing

Programming of M-PHY attributes is usually done while a burst is active. [Figure 2-27](#) shows the timing of a configuration after the burst start when the **MPx_TX_CfgRdyN** M-PHY signal is high followed by the configuration update request through the **MPx_TX_CfgUpdt** signal. In the configuration access shown, the M-PHY uses the **MPx_TX_CfgBusy** signal to extend the write access for one cycle.

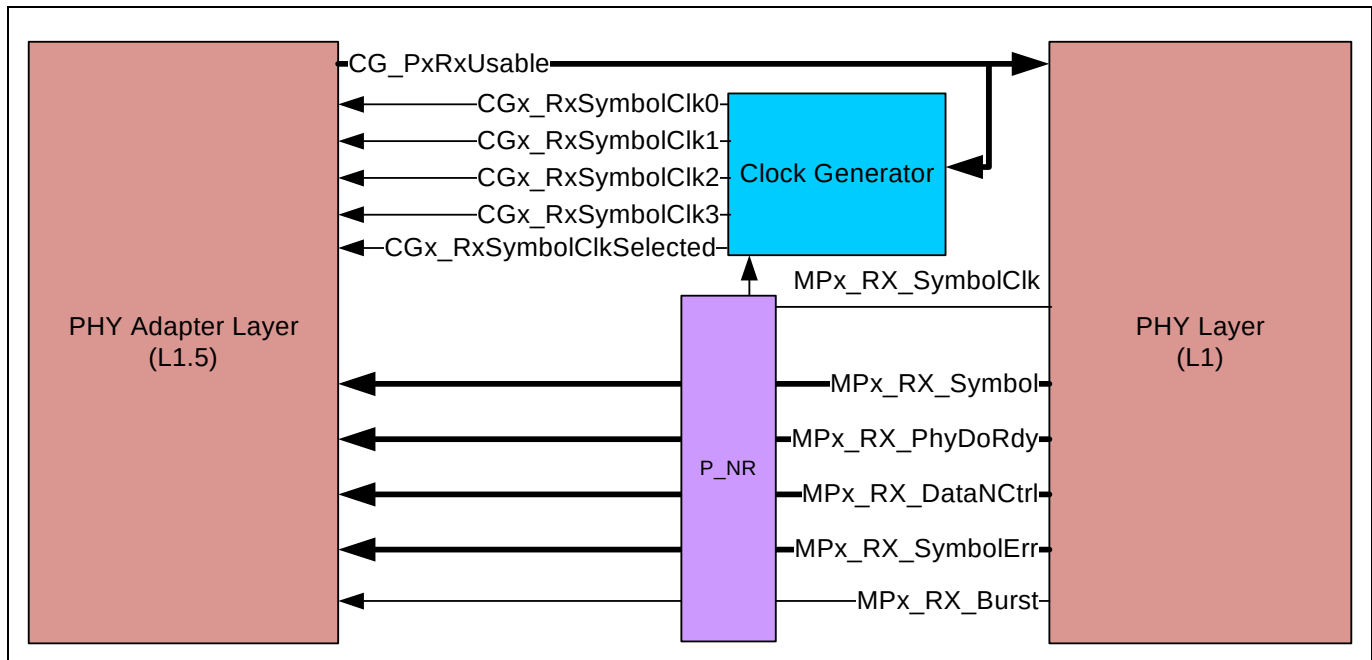
Figure 2-27 Configuration Update Timing

DWC_mipi_unipro_20 does not perform M-PHY configuration accesses in consecutive cycles. There is at least one inactive cycle between two accesses (configuration data or update).

In CG_TxSymbolClk cycle 15, the protocol inserts an FLR symbol in the upper M-PHY symbol by setting the corresponding bit of **MPx_RX_DataNCtrl** to 0 and providing the FLR M-PHY symbol.

2.14.2 RX Data Interface

The RX Data interface transports the data received by the M-PHY modules over the UniPro link to the DWC_mipi_unipro_20. [Figure 2-28](#) shows the connection between the layer 1 and layer 1.5.

Figure 2-28 M-PHY RX Data Interface

The P_NR box indicates that the width of all signals running through this box is determined by the P_NR parameter.

The M-PHY RX Data interface signals are described in “[M-PHY RX Data Interface Signals](#)” on page 149.

The M-PHY module de-serializes and forwards the received data to layer 1.5 over the MPx_RX_Symbol buses. All the signals of this interface scale with the number of available RX M-PHY modules. The P_NR parameter defines the maximum number of UniPro symbols processed in parallel up to the layer 2 memories. Each individual RX M-PHY generates an RxSymbolClk on its own and all the signals of that M-PHY are synchronous to this specific clock. Layer 1.5 uses these (maximum four) clocks, which may vary in phase but not in frequency, to align the data of all receiver lanes.

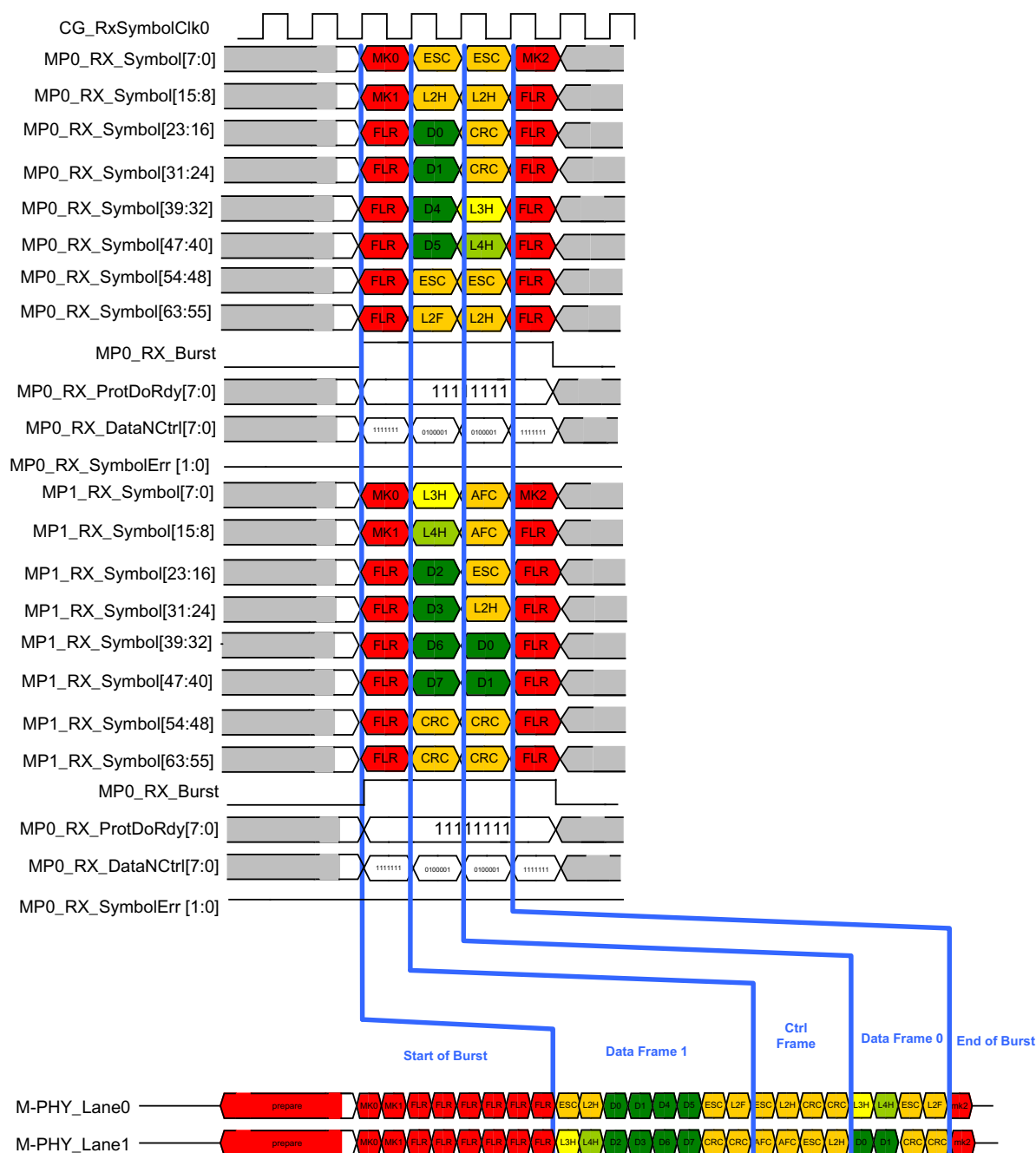
Similar to the transmitter, only the lower 16 bits of each symbol of the RX_Symbol interface are supposed to deliver the data to DWC_mipi_unipro_20. The upper four bits of each symbol are ignored.

Additionally, one of the active RxSymbolClk signals is provided to the module through the **CG_RxSymbolClkSelected** signal. For more information about receive clock handling, see *DesignWare Cores MIPI UniPro 2.0 User Guide*.

[Figure 2-29](#) shows some sample frames received over a 2-lane UniPro interface.

Figure 2-29 M-PHY RX Data Access

16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	ESC_DL								SOF			TC0		Reserved		
0	L3s=1	DestDeviceID_Enc (L3H)							L4s=1	DestCPortID_Enc (L4H)				FCT	EOM = 0	
0	D0								D1							
0	D2								D3							
0	D4								D5							
0	D6								D7							
1	ESC_DL								EOF_EVEN			Frame Seq. Number = 0				
0	CCITT CRC-16															
1	ESC_DL								AFC			TC		CReq	Reserved	
0	Frame Seq. Number						Reserved		Credit Value							
0	CCITT CRC-16															
1	ESC_DL								SOF			TC0		Reserved		
0	L3s=1	DestDeviceID_Enc (L3H)							L4s=1	DestCPortID_Enc (L4H)				FCT	EOM = 0	
0	D0								D1							
1	ESC_DL								EOF_EVEN			Frame Seq. Number = 1				
0	CCITT CRC-16															



The M-PHY asserts the **MPx_RX_Burst** signal to indicate the beginning of data reception to `DWC_mipi_unipro_20` when it detects the start of the PREPARE period. MK0 and MK1 define the beginning of the data stream. To improve readability, [Figure 2-29](#) shows MK0 on the lower byte of the symbol bus. The M-PHY may also output MK0 on the upper byte and MK1 on the lower byte in the following cycle. The deskew buffer in layer 1.5 takes care of realigning the symbols.

In contrast to the TX Data interface, frames on the RX Data interface may start on any lane.

MPx_RX_SymbolErr informs DWC_mipi_unipro that the receiver detected a transmission error (for example, a 10b8b decoding error). This signal is checked only when the burst is active and the lane is synchronized (MK0 or MK1 is detected). The error information is collected inside the DME for statistical purposes, flagged as ErrorPhyInd and forwarded to layer 2 for scheduling a retransmission of failed data.

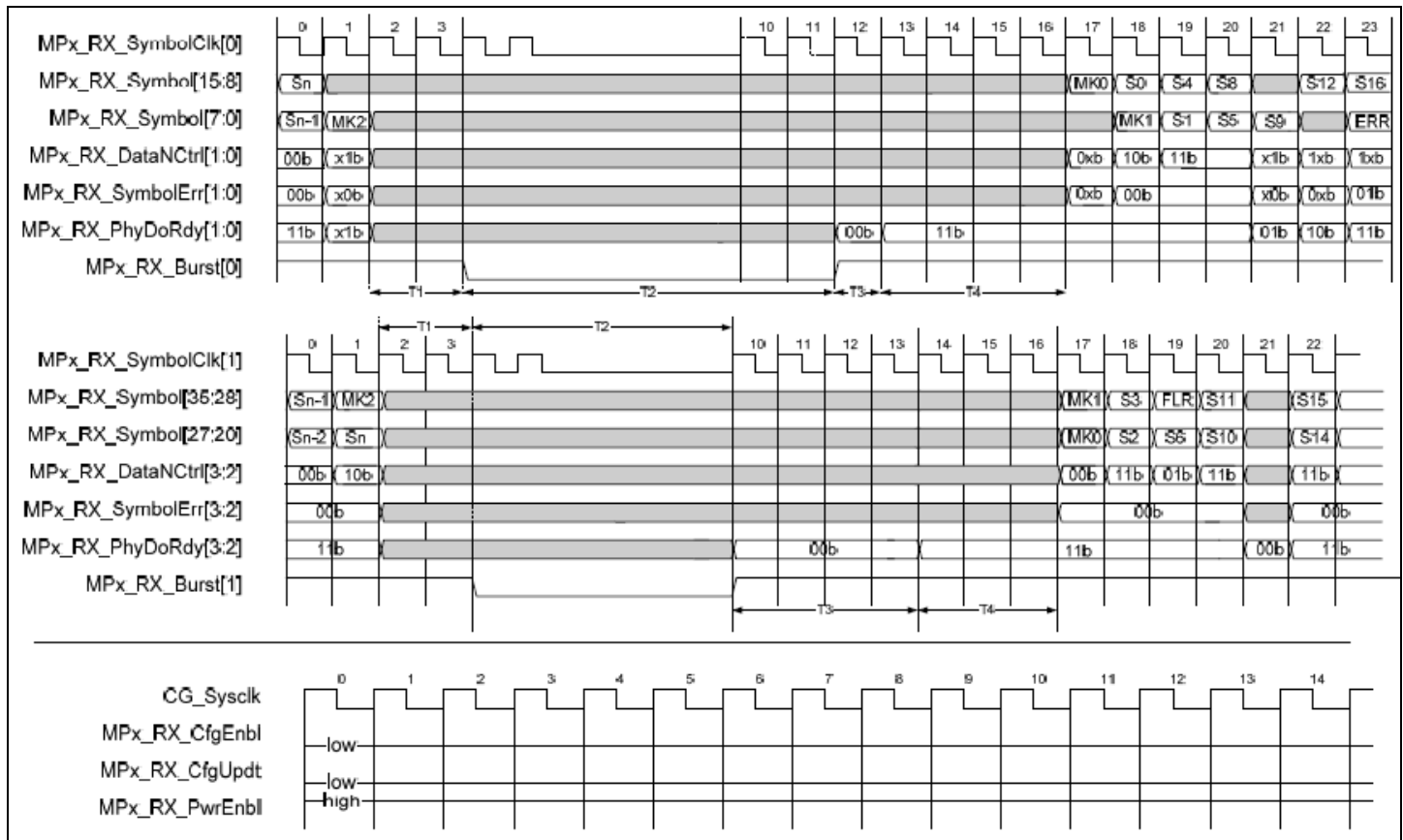
**Note**

If DWC_mipi_unipro_20 is configured to use more physical lanes than available in the system, all input signals of the unused physical lanes must be tied to logical 0.

Figure 2-30 shows several timing and signaling properties between DWC_mipi_unipro_20 and the M-PHY.

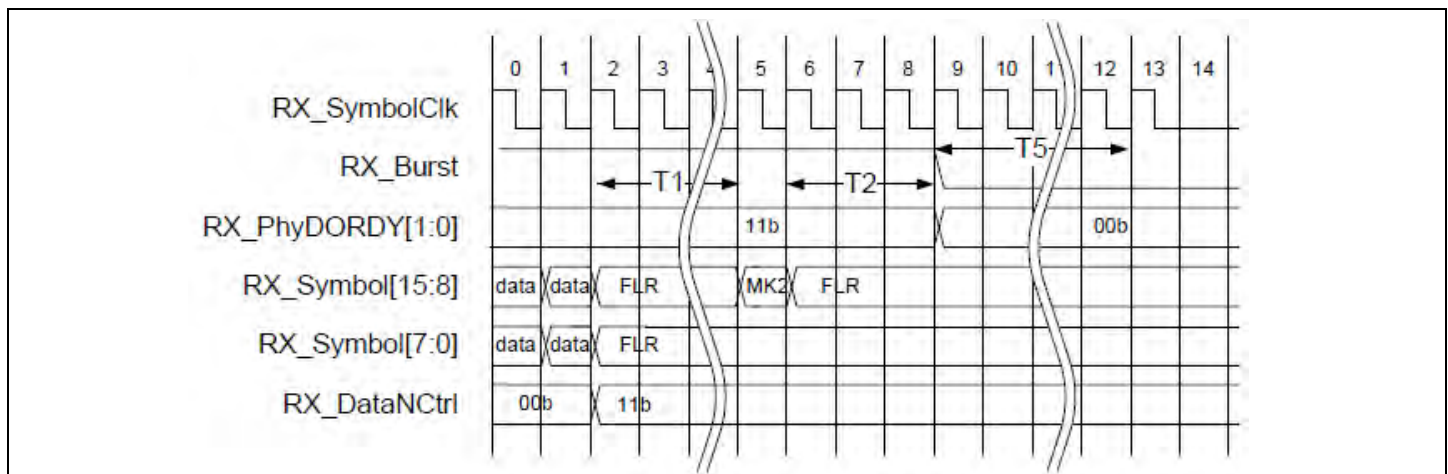
Note the following points in Figure 2-30:

- After receiving the MK2 symbol, all following data are irrelevant and ignored.
- MK0 can be unaligned on different data lanes (cycle 17).
- The MPx_RX_Burst signal must be active at least three cycles before the data is received (timing T4).
- Timing T1 shows the M-PHY burst trailing cycles.
- Timing T3 resembles the SYNC timing of the M-PHY (no limitation imposed by DWC_mipi_unipro_20).
- If the MPx_RX_SymbolErr signal is active, the additional error information of the M-PHY (through **MPx_RX_DataNCtrl**) is not evaluated.

Figure 2-30 M-PHY Receive Timing

When a burst is closed, several timings that the transmitter introduced are visible by the receiver.

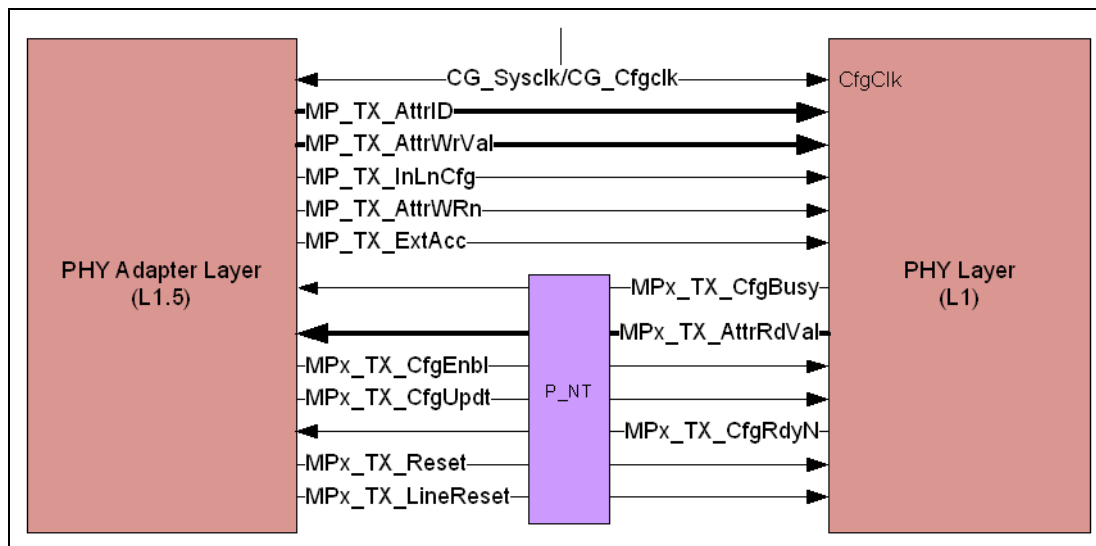
Figure 2-31 shows the BurstClosureDelay (UniPro protocol, T1), the TrailingClocks (M-PHY, T2), and the number of additional clock cycles after the deactivation of the burst signal (minimum of two according to the M-PHY Specification, T5).

Figure 2-31 Receiver Burst End Timings

2.14.3 TX Control Interface

The TX control interface is used by layer 1.5 to configure and read out the attributes of all available TX M-PHY modules.

Figure 2-32 M-PHY TX Control Interface



The P_NT box indicates that the width of all signals running through this box is determined by the P_NT parameter.

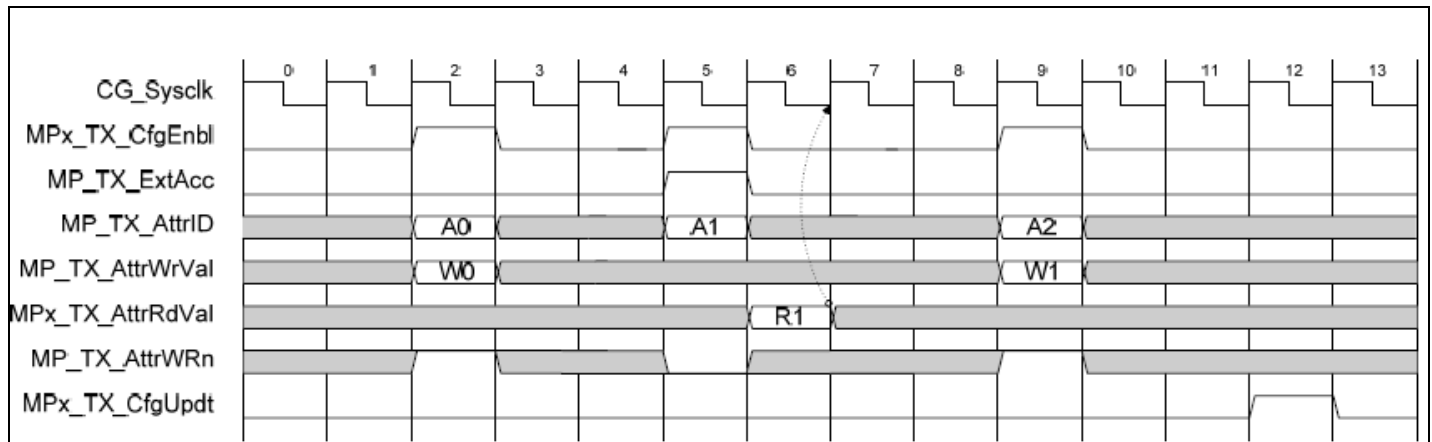
The M-PHY TX Control interface signals are described in “[M-PHY TX Data Interface Signals](#)” on page 147.

All signals of the TX Control interface are synchronous to **CG_Sysclk** provided to DWC_mipi_unipro_20 as shown in [Figure 2-32](#). You can also choose to have these signals synchronous to **CG_Cfgclk** which is independent and asynchronous to CG_Sysclk. Necessary clock-domain crossing synchronizers are added to the interface to transfer the signals between these two clock domains.

The M-PHY attribute configuration accesses coming from the DME are forwarded by the PHY adapter layer to target the M-PHY modules. The PA layer selects the attribute address (**MP_TX_AttrID**) and the type of access (read or write) for all M-PHY modules in common. Target M-PHY modules are selected by asserting the corresponding **MPx_TX_CfgEnbl** signal. While writing attributes, several M-PHY modules can be accessed in parallel by asserting more than one **MPx_TX_CfgEnbl** signal. Read accesses target a single M-PHY only.

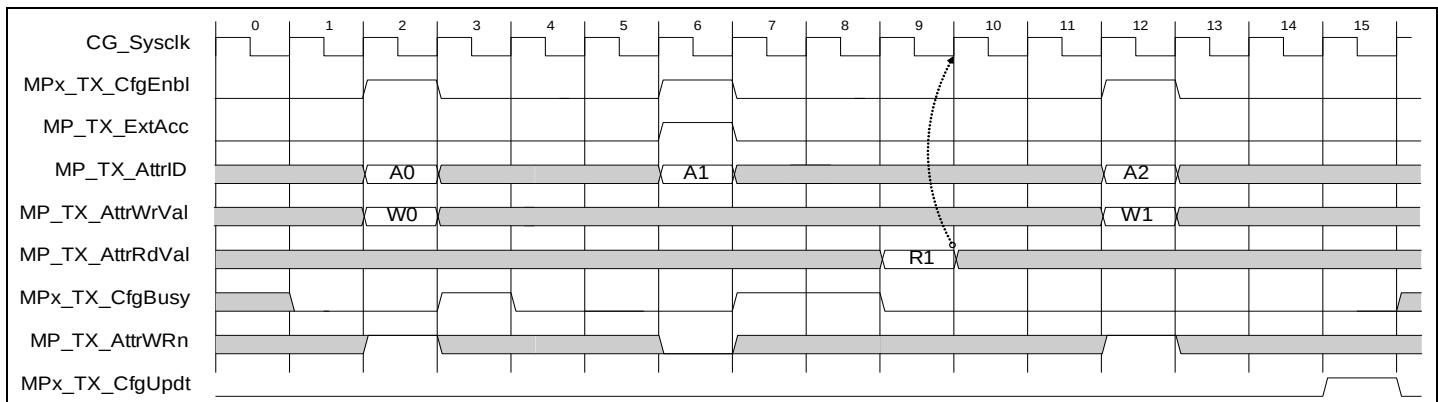
Read data is supposed to be available in the cycle following the read request and should be kept until the next read access is performed.

[Figure 2-33](#) shows sample read and write accesses to TX M-PHY attributes.

Figure 2-33 M-PHY TX Control Access

A read access always targets the effective attribute. The previously configured values are transferred to effective configuration attributes, when the **MPx_TX_CfgUpdt** signal is detected and the burst signal of the individual M-PHY lane is inactive (reconfiguration trigger - RCT).

The M-PHY must return the requested read value in the following cycle. If for example, a non-compliant M-PHY or M-PHY substitute is used, the **MPx_TX_CfgBusy** signal can be used to delay the completion of read or write access and all following accesses. This is shown in [Figure 2-34](#).

Figure 2-34 M-PHY TX Control Access with Busy

In addition, the **MPx_TX_CfgRdyN** signal is used by `DWC_mipi_unipro_20` to check the completion of a line reset request. As long as the **MPx_TX_CfgRdyN** signal is high after issuing the line reset request, `DWC_mipi_unipro_20` waits for the signal to drop low and continues.

The **MP_TX_ExtAcc** signal is used to access attributes in the extended user space of the M-PHY module. These attributes contain implementation specific functions. To activate the **MP_TX_ExtAcc** signal, the application must access the M-PHY attributes using the 0x8000 address offset as described in “[M-PHY Interface](#)” on page 44.

The **MPx_TX_Reset** signal is always asserted asynchronously and de-asserted synchronized to **CG_Sysclk**. The **MPx_TX_LineReset** signal is asserted and de-asserted synchronized to **CG_Cfgclk** if present, otherwise to **CG_Sysclk**.

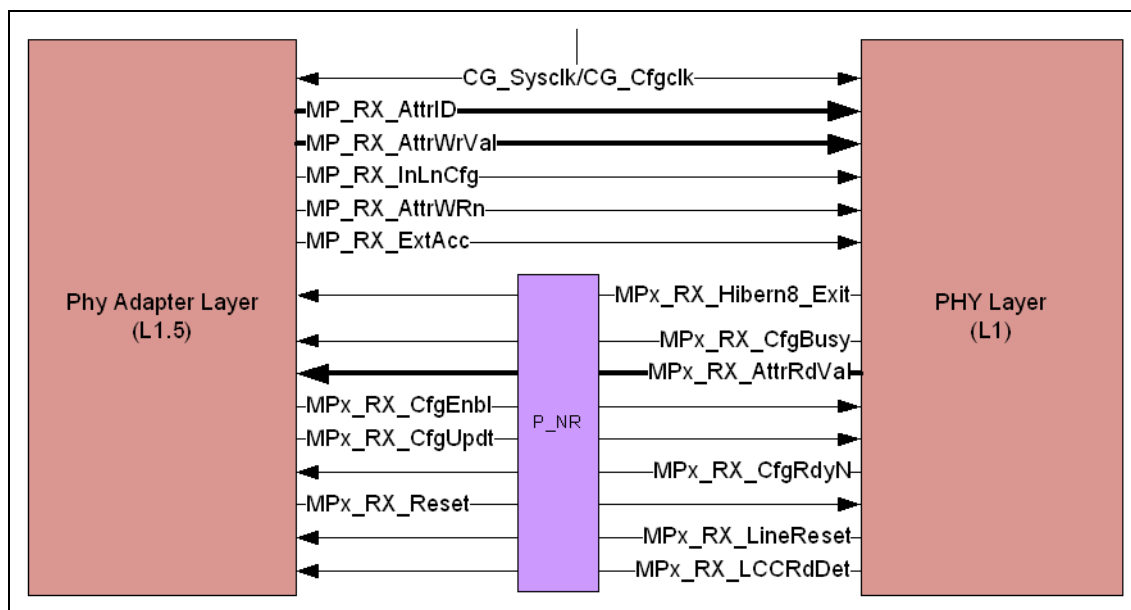
**Note**

- If no physical M-PHY module is connected to a UniPro lane, all input signals of the UniPro lane must be tied to logical 0.
- If the **MPx_TX_PwrEnbl** signal is low, all inputs (data and control) from the corresponding M-PHY module are expected to be static 0.
- The **MP_TX_AttrID**, **MP_TX_ExtAcc**, **MP_RX_AttrWrVal**, and **MP_RX_AttrWRn** output signals are stable until a new access (read or write) is performed.

2.14.4 RX Control Interface

The RX Control interface is used by layer 1.5 to configure and read out the attributes of all available RX M-PHY modules.

Figure 2-35 M-PHY RX Control Interface



The P_NR box indicates that the width of all signals running through this box is determined by the P_NR parameter.

The RX Control interface signals are described in “[M-PHY TX Control Interface Signals](#)” on page 135.

All signals of the RX Control interface are synchronous to CG_Sysclk provided to the DWC_mipi_unipro_20 (except for **MPx_RX_Hibern8_Exit**) as shown in [Figure 2-35](#). This clock must be used as a configuration clock for all M-PHY modules to avoid clock domain crossing. You can also choose to have these signals synchronous to CG_Cfgclk which is independent and asynchronous to CG_Sysclk. Necessary clock-domain crossing synchronizers are added to the interface so as to transfer the signals between these two clock domains.

The RX Control interface is very similar to the TX Control interface. Accessing the attributes is performed in the same way as shown in [Figure 2-33](#).

MPx_RX_Reset is always asserted asynchronously and de-asserted synchronously to **CG_Sysclk**. The **MPx_RX_LineReset** and **MPx_RX_LCCRdDet** signals must be asserted for at least two **CG_Sysclk** cycles. The **RX_LCC_RdDet** signal should have additional delay of one **CG_Cfgclk** cycle, if this clock is present.

2.14.5 CB Control Interface

The Synopsys DWC GEN2 M-PHY Common Block (CB) Control interface exists only when **SNPS_MPHY** is selected. The CB entity is not mentioned in the MIPI M-PHY Specification. To simplify access to the CB and allow similar degree of flexibility, Synopsys M-PHY implements a control signal interface similar to TX-Control and RX-Control interfaces. Access to the PLL and Synopsys M-PHY implementation-specific attributes (CB attributes) is done through this interface.

The CB interface is accessed for

- Application read/write of the CB attributes. The application's DME access is translated to CB control interface access by `DWC_mipi_unipro_20` module.

There is no provision to access a CB directly through `DME_SAP` in the MIPI UniPro Specification. Therefore, the CB attributes are mapped to the UniPro `DME_SAP` memory space reserved for implementation specific GEN2 M-PHY attributes, starting at address `0x8000`, with Target Layer equal to `0x80`. This corresponds to the `DME_SAP` address space from `0x0008000` to `0x00080FF`.

The CB attribute write from the application should be done only before the link startup and must stay stable after the link startup.

- When `MODE`, `GEAR`, `HIBERN8`, and `RATE` attributes in the M-PHY TX or RX lanes change, Synopsys M-PHY Adapter (SMA) updates the corresponding CB attributes through the CB control interface.

During these accesses, `DWC_mipi_unipro_20` write attributes always use `MP_CB_InLnCfg = 0`, addressing the attribute shadow registers. At the end of the configuration, `DWC_mipi_unipro_20` applies `MP_CB_CfgUpdt`.

Figure 2-36 Synopsys M-PHY CB Control Interface

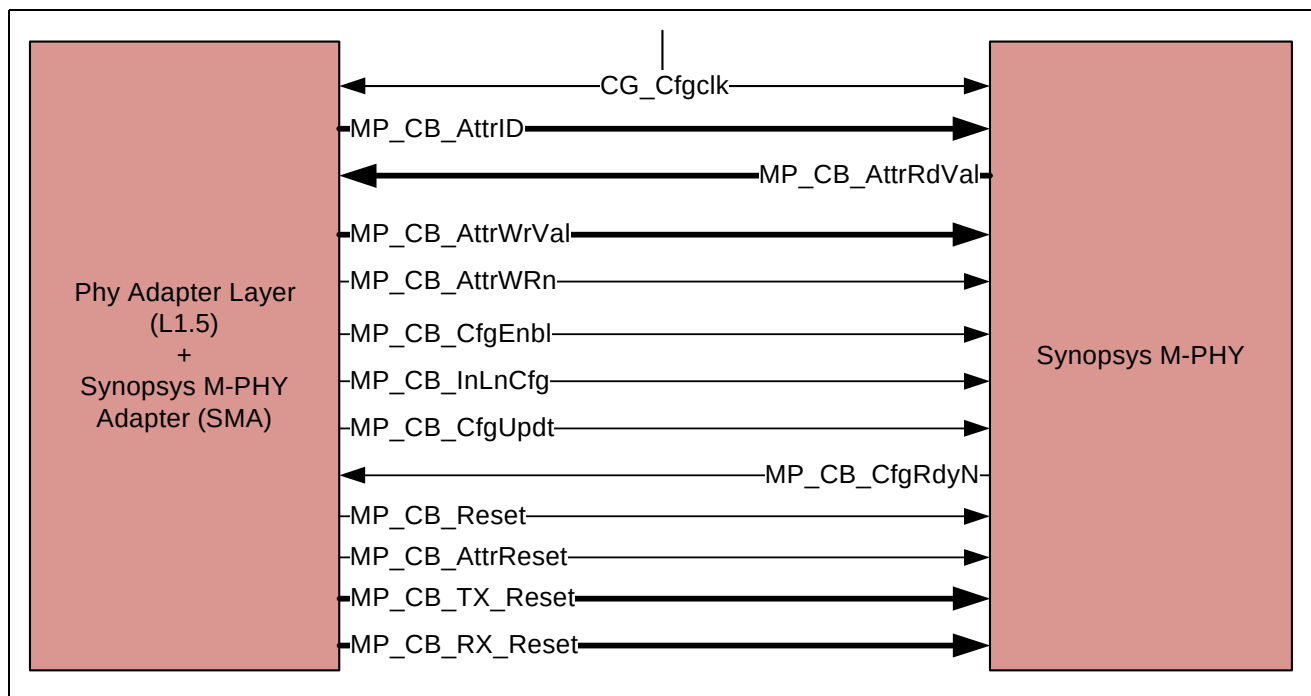
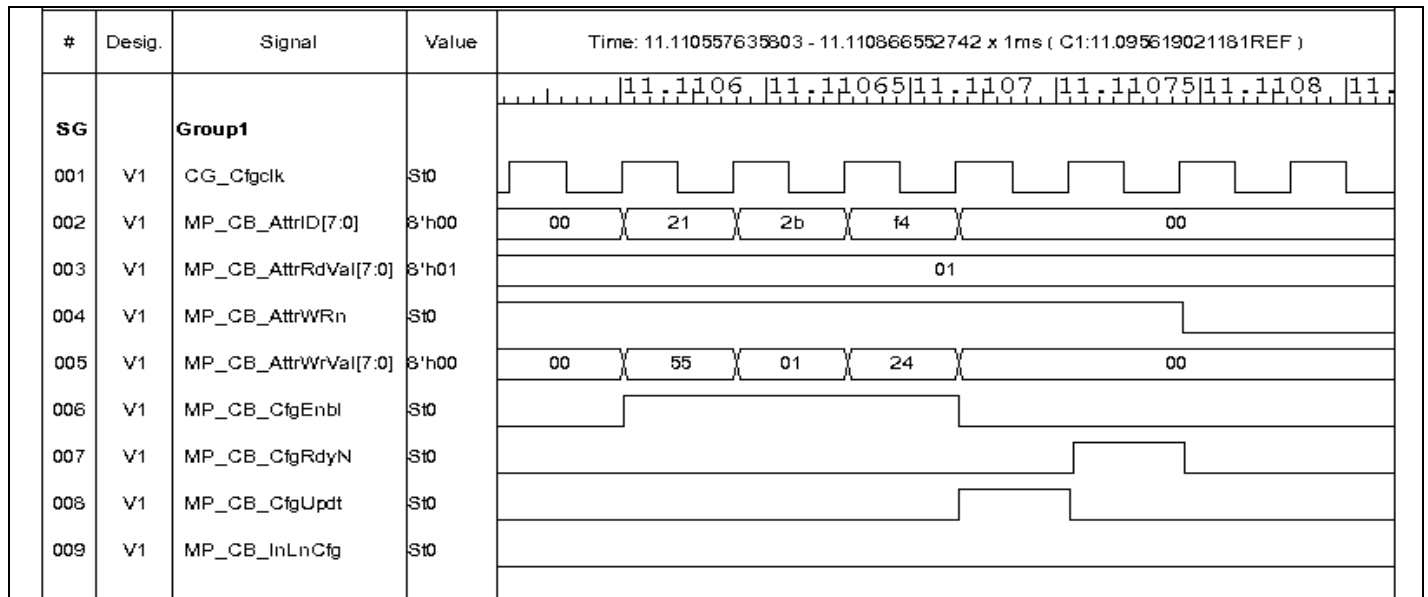


Figure 2-37 shows sample read and write accesses to M-PHY CB attributes.

Figure 2-37 M-PHY CB Control Access



2.15 Layer 4 Test Feature

The test feature specified for layer 4 includes a pattern generator (TstSrc) and a pattern analyzer (TstDst). If enabled, the pattern generator sends a defined data sequence to a target test feature analyzer on the peer device. The analyzer processes the incoming data and reports the errors. The test feature is a simple way of generating layer 4 traffic without data generation by an application and ideally suited for checking initial communication between two devices and link integrity.

Starting and stopping the test feature pattern generation and analysis is supported through layer 4 attributes. This set of attributes also allows a variety of settings to control, such as the number of messages, message sizes, and the gaps between messages.

The test feature even supports the exchange of flow control tokens and credit information.

The number of test features of the `DWC_mipi_unipro_20` can be selected at the compile time using the `P_TF` parameter. A device using only one CPort is required to have a single test feature, while a device using two or more CPorts must have at least two test features. The maximum number of test features is limited by the maximum number of CPorts of that device.



Note

Because the test feature is configurable, the number of test features influences the total size of the `DWC_mipi_unipro_20` to some extent and therefore must be chosen carefully (using the `P_TF` parameter).

To use the test feature, complete the following steps:

1. Start up the link.
2. Configure the local and peer CPorts.

3. Configure the test feature data generators and data checkers.
4. Start the data checkers.
5. Start the data generators.

All data generators are capable of sending either a defined or an infinite number of messages. The message sizes and inter-message gap can be configured individually. Additionally, the flow control interface is updated regularly in parallel to data generation. The update interval and the initial credit amount can be configured.

“T_PeerCPortID” shows the attribute that is common for the data generator and data checker. This attribute must be programmed before activation. Program the attributes in “T_TrafficClass” to “T_LocalBufferSpace” to configure and start the data generator. Program the attributes listed in “T_PeerBufferSpace” to “T_TstSrcMessageSize” to configure and start the data checker. The “T_TstCPortID”, “T_TstSrcOn”, and “T_TstDstOn” attributes are mandatory. If the rest of the attributes are not changed, the default settings transfer 256 messages of 256 bytes each back to back.

If multiple test features are present, the Select bits of LA_AttrID address the corresponding test feature attribute set. The CPorts connected to a test feature cannot be used for regular data traffic.

The P_FC configuration parameter limits the number of credits to be granted by the test feature as well. For example, if P_FC is equal to seven, the test feature can only grant up to 255 bytes at a time (for the initial and regular credit update through the T_TstDstInitialFCCredits and T_TstDstFCCredits signals).

2.16 Layer 1.5 Test Mode

The test mode of layer 1.5 is used to check the M-PHY transmission or reception at a very low (PACP) protocol level. This mode can only be activated by a tester device connected on the peer side of the M-PHY link or by the defined methods of DWC_mipi_unipro_20 to start the test mode (**DME_TestMode**) itself.

In this test mode, predefined M-PHY symbols are exchanged inside PACP frames between both devices. The receiving side checks the PACP packet and flags errors in dedicated error attributes (**PA_PACPErrorCount**). Additionally, the number of received PACP frames is counted in the PA_PACPFrameCount attribute.

The layer 1.5 test mode can only be activated before or during the link startup sequence. After a device enters this test mode, a reset is required to return the device to normal operation.

The DWC_mipi_unipro_20 is capable of handling all local and peer PACP get or set commands sent from the tester or to the peer device while in test mode. These commands can be used to configure the M-PHY attributes or the DWC_mipi_unipro_20 attributes. The test mode results can be checked over the link.

To be able to use the link correctly, the initiator of the test mode must program the following attributes, before transferring test mode data:

- PA_ConnectedTxDataLanes
- PA_ConnectedRxDataLanes
- PA_LogicalLaneMap
- PA_SleepNoConfigTime
- PA_StallNoConfigTime

■ PA_SaveConfigTime

The layer 1.5 test mode implemented in DWC_mipi_unipro_20 supports CJTPAT and CRPAT patterns as defined in UniPro Specification.

**Note**

DWC_mipi_unipro_20 does not support peer initiated PACP_SET_REQ with cnf flag = 1 in PHY Test Mode.

2.17 Advance Low-Power Flow with UPF Methodology Support

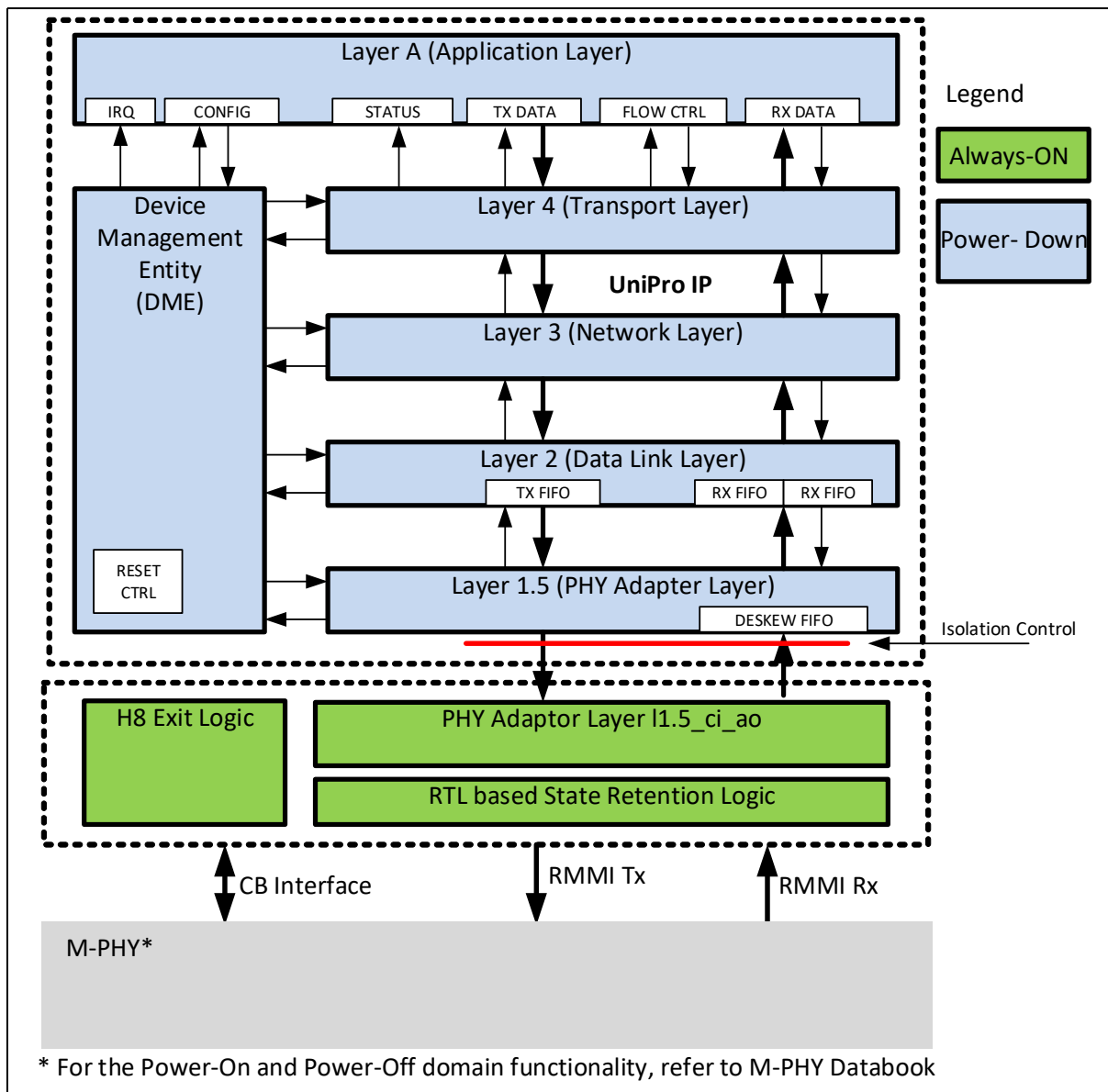
The advance low-power mode supports power gated UniPro Hibernate Enter/Exit low-power states.

Unified Power Format (UPF) provides the ability for electronic systems to be designed with power as a key consideration early in the process. UPF supports the entire design flow. It provides a consistent format to specify power-aware design information. Based on the identification of the power-on and power-off blocks, DWC_mipi_unipro_20 groups these blocks into two separate hierarchies, one for the power-on modules and the other modules in the power-off domain as shown in [Figure 2-38](#). In this methodology, the RTL-based state retention is used there by avoiding the need for special retention register library cells.

Isolation of signals from the power-off domain to power-on domain during low-power states is to avoid the corruption of signals in the power-on domain. This is because removing power results in signals getting tri-stated. Isolation cells are introduced to achieve this objective by the UPF scripts.

Low-power finite state machine built into DWC_mipi_unipro_20 handles the low-power mode save and restoration of states during a power gated hibernate low-power mode.

Enable the advance low-power features in DWC_mipi_unipro_20 using the UPRO_LPM_EN configuration option. For more information about location of UPF files, see the *Performing ASIC Synthesis with UPF* section of the *DesignWare Cores MIPI UniPro 2.0 User Guide*.

Figure 2-38 DWC_mipi_unipro_20 Power-On and Power-Off Blocks

2.17.1 Controlling Power Rails during Low-Power Modes

The different power rails can be controlled using power switches. The power-gated block receives its power through a power-switching network. This network switches either VDD or VSS to the power gated block. The switching fabric typically consists of a large number of CMOS switches distributed around or within the power gated block. The power gating controller controls the CMOS switches that provide power to the gated block.

2.17.2 Hierarchy Definition for Power-Off and Power-On Blocks

Based on the identification of the power-on and power-off blocks, group these blocks into two separate hierarchies, one for the power-on modules and the other modules in the power-off domain. This ensures easy tool flow creation.

2.17.3 Power-Gated Hibernate Enter

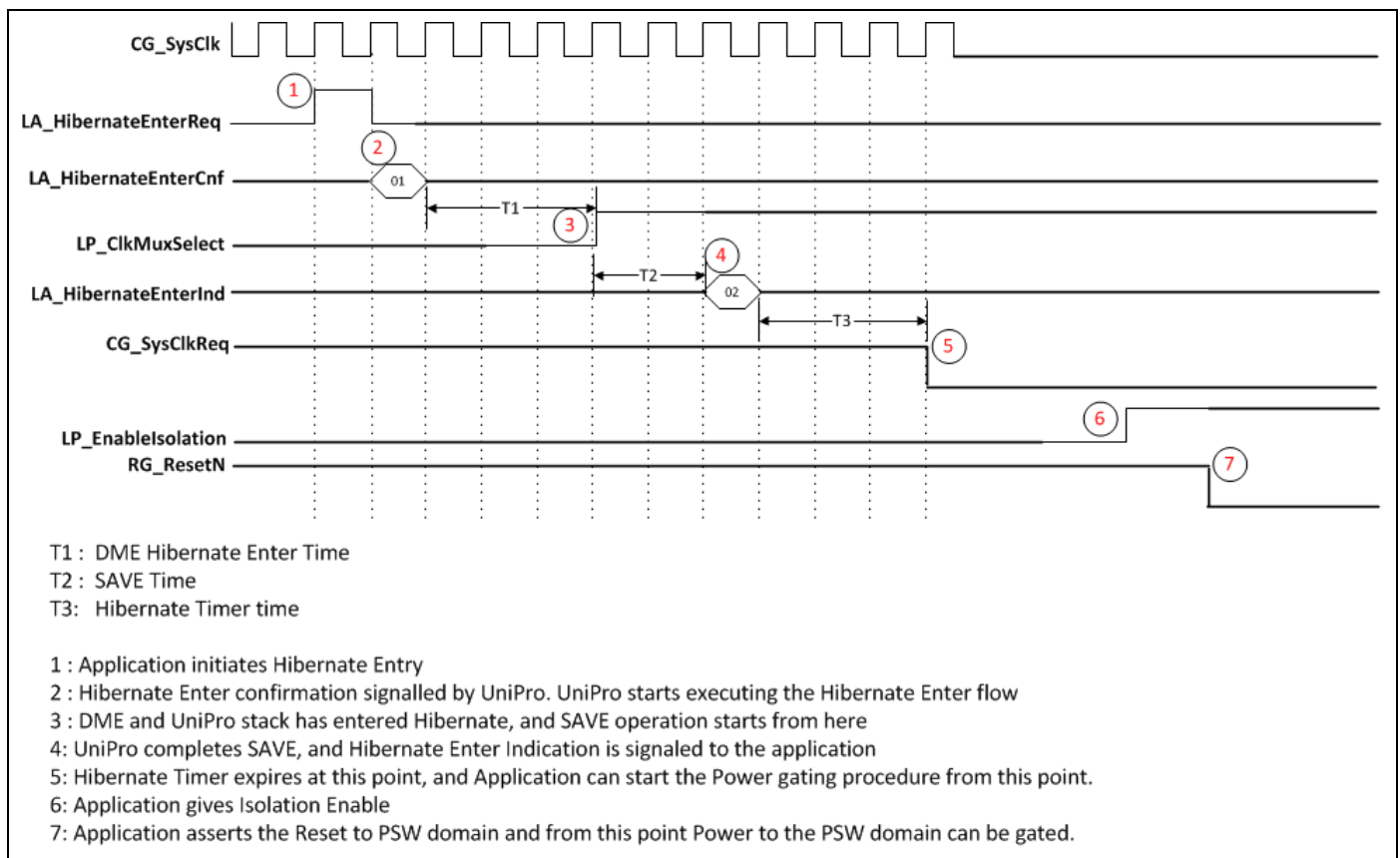
2.17.3.1 Local Hibernate Enter

The application layer can issue the hibernate enter request either through the LA_HibernateEnterReq signal or using the DME_HibernateEnter attribute. The link partner accepts the request by sending the power mode change confirmation PACP_PWR.cnf with the status POWER_OK. This completes the hibernate entry procedure with the link partner.

After this, the low-power FSM in DWC_mipi_unipro_20 stores the required attributes that need to be retained in the RTL-based retention flops. This RTL-based state retention logic is present in the Always-On domain.

Once the process of storing the attributes is complete, DWC_mipi_unipro_20 indicates to the application layer that the hibernate entry is successful through the LA_HibernateEnterInd signal or the DME_HibernateEnterInd attribute by setting the value PWR_LOCAL.

Figure 2-39 Low-Power Hibernate Enter



2.17.3.2 Peer Hibernate Enter

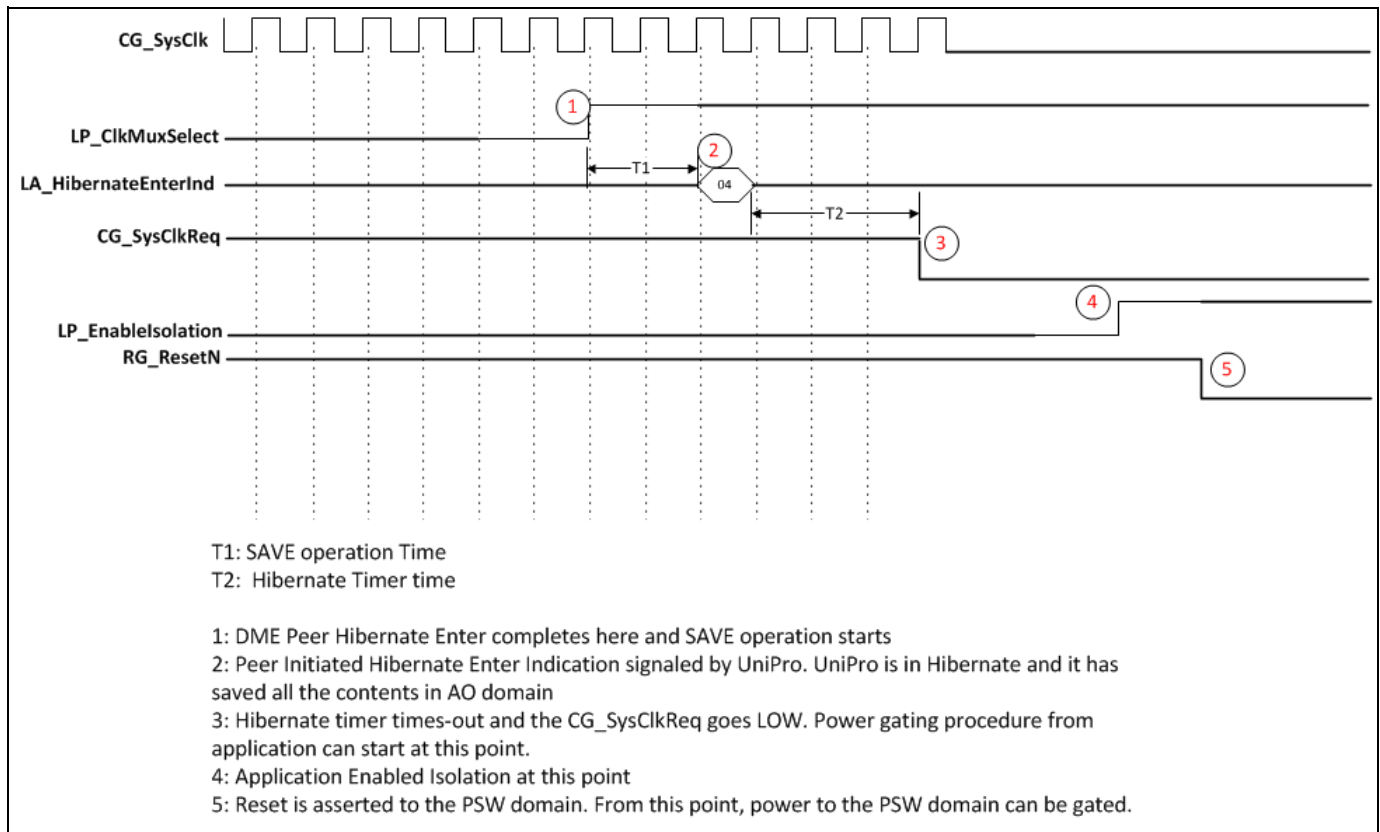
The peer initiates the hibernate entry by sending the power mode change request (PACP_PWR_req) for hibernate. The local DWC_mipi_unipro_20 responds with acceptance by sending the power mode change confirmation PACP_PWR_cnf with the status set to PWR_OK. Once the local DWC_mipi_unipro_20 closes the TX burst, low-power controller FSM of DWC_mipi_unipro_20 stores the required attributes that need to be retained in RTL-based retention flops. After this, DWC_mipi_unipro_20 issues the hibernate enter indication (LA_HibernateEnter.Ind) to the application layer which proceeds with switching off the power to the DWC_mipi_unipro_20 power switchable domain.



Note

The isolation enable LP_EnableIsolation need to be asserted before power to power switchable domain is removed.

Figure 2-40 Peer Hibernate Enter



2.17.4 Power-Gated Hibernate Exit

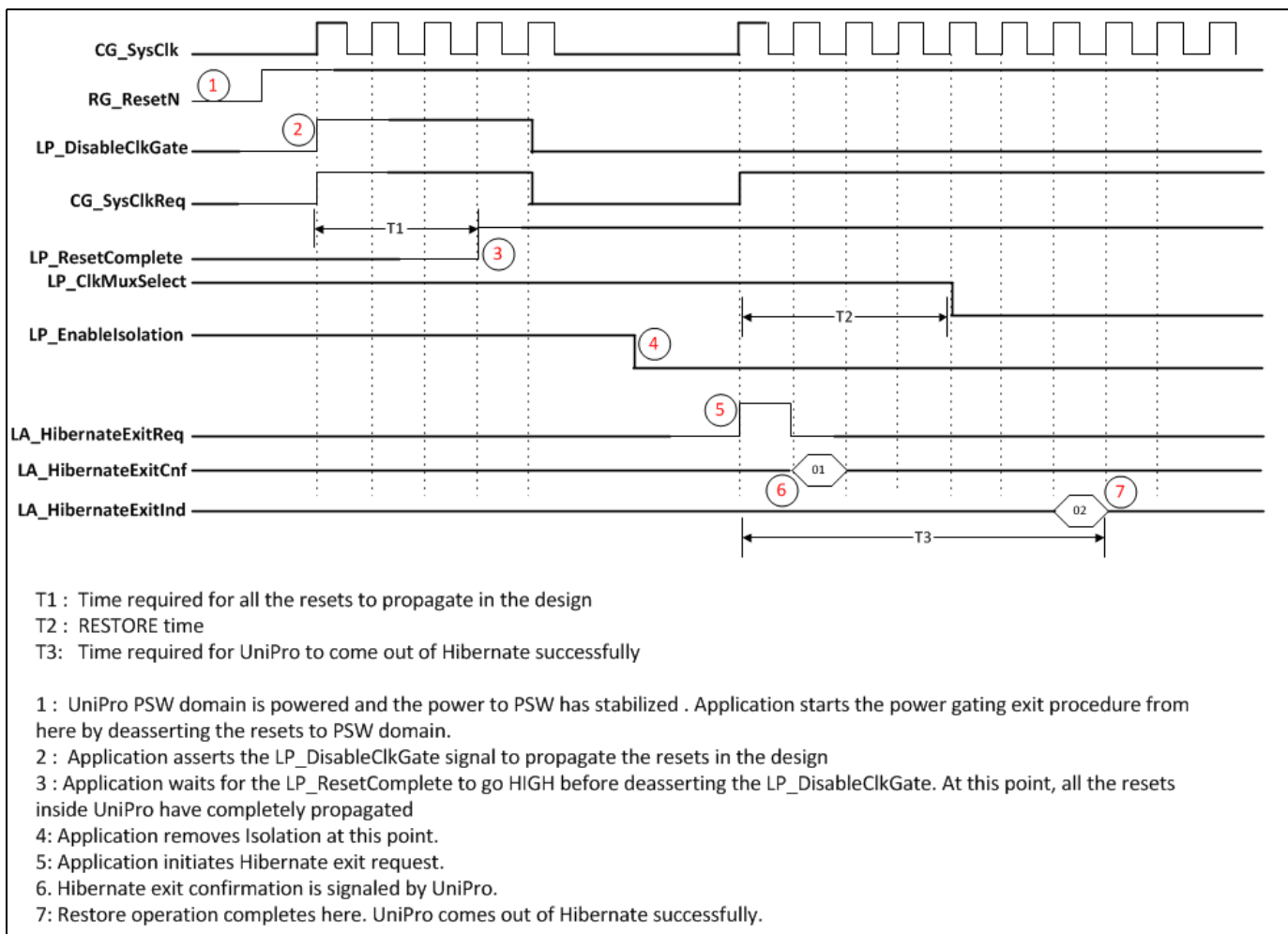
2.17.4.1 Local Hibernate Exit

The application layer is aware of when to exit the hibernate state. When it decides to do so, it must follow this sequence:

1. Restore the power to the power switchable domain.
2. De-assert the RG_ResetN signal.
3. Assert the LP_DisableClkGate signal to propagate the reset to all clock domains.
4. De-assert LP_EnableIsolation.
5. Issue LA_HibernateExitReq.

See [Figure 2-41](#). Once DWC_mipi_unipro_20 receives LA_HibernateExitReq, it starts restoring the previously stored attributes from the Always-On to power switchable domain. Rest of the hibernate exit process follows as described in the specification. For more information, see [“Hibernation Mode”](#) on page 561.

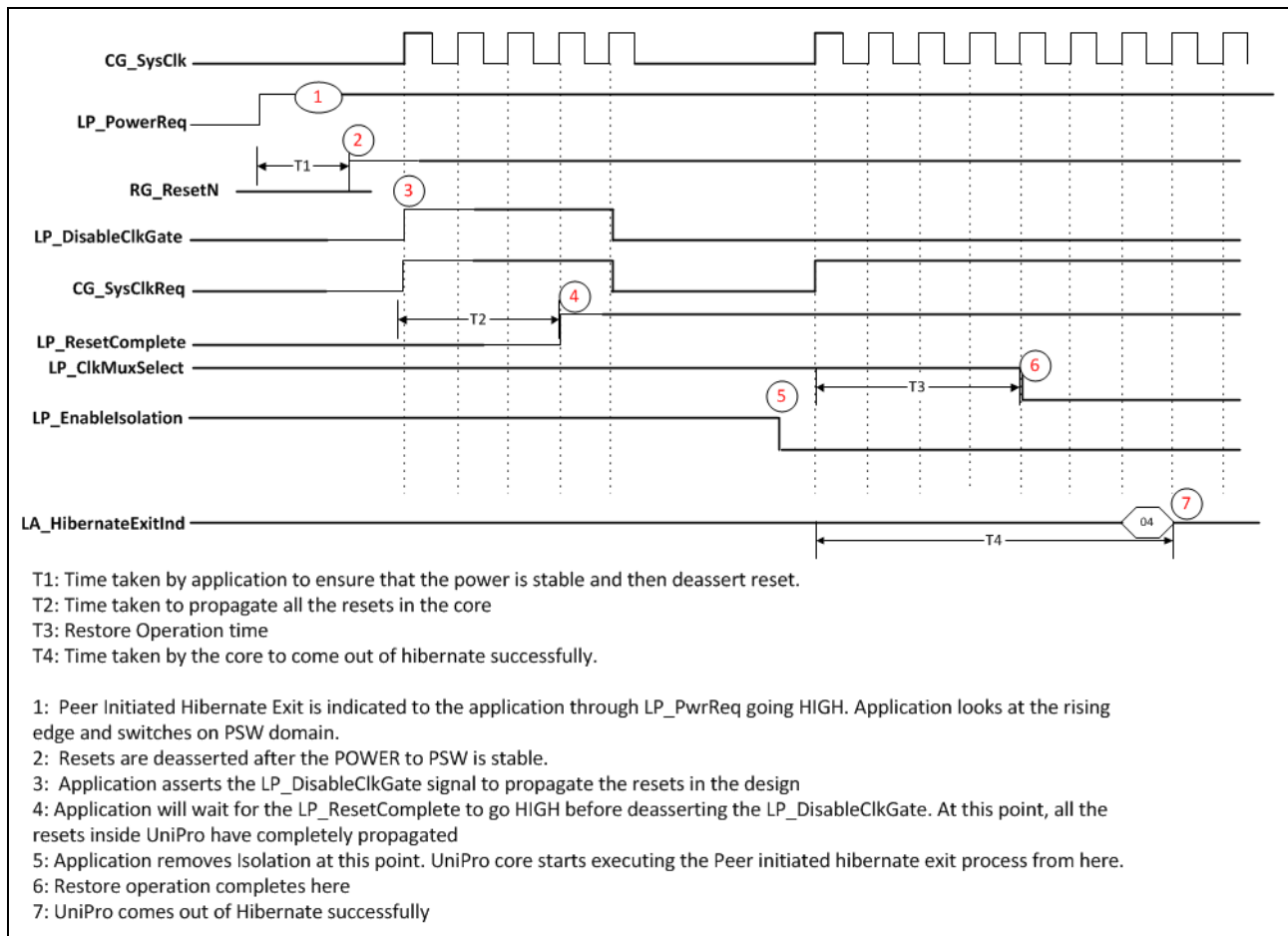
Figure 2-41 Low-Power Hibernate Exit



2.17.4.2 Peer Hibernate Exit

When M-PHY detects the hibernate exit (DIF-Z to DIF-N transition), it asserts `MPx_RX_Hibern8_Exit`. This port is used as input in the Always-On power domain of `DWC_mipi_unipro_20`. Once this port is sampled high, `DWC_mipi_unipro_20` low-power controller asserts `LP_PowerReq` with high. It is expected that once this port is set to high, the application restores the power to the OFF domain of `DWC_mipi_unipro_20`. The application completes the power-up procedure followed by the de-assertion of reset and removing the isolation control. Once the isolation is removed, `DWC_mipi_unipro_20` restores all the saved attributes and performs a peer-initiated hibernate exit. Then, `DWC_mipi_unipro_20` comes out of hibernate state by indicating `LA_HibernateExitInd`.

Figure 2-42 Peer Hibernate Exit



2.17.5 Retained State Information

While in low-power state, the following information gathered during the link start-up sequence are retained in `DWC_mipi_unipro_20`'s Always-On domain:

- ❑ `PA_ConnectedTxDataLanes`
- ❑ `PA_ConnectedRxDataLanes`
- ❑ `PA_LogicalLaneMap`
- ❑ `PA_MaxRxPWMGear`

- ❑ PA_MaxRxHSGear
- ❑ PA_RemoteVerInfo
- ❑ PA_SleepNoConfigTime
- ❑ PA_StallNoConfigTime
- ❑ PA_SaveConfigTime
- ❑ PA_RxHSUntermCap
- ❑ PA_RxPWMTermCap
- ❑ PA_TActivate
- ❑ PA_Granularity
- ❑ PA_TxTrailingClocks
- ❑ PA_PACPRReqTimeout
- ❑ PA_PACPRReqEoBTimeout
- ❑ PA_PeerScrambling
- ❑ PA_TxSkip
- ❑ PA_Local_TX_LCC_Enable
- ❑ PA_Peer_TX_LCC_Enable
- ❑ PA_TxMk2Extension
- ❑ PA_Hibern8Time
- ❑ PA_TxHsG1SyncLength
- ❑ PA_TxHsG1PrepareLength
- ❑ PA_TxHsG2SyncLength
- ❑ PA_TxHsG2PrepareLength
- ❑ PA_TxHsG3SyncLength
- ❑ PA_TxHsG3PrepareLength
- ❑ PA_TxHsG4SyncLength
- ❑ PA_TxHsG4PrepareLength
- ❑ PA_PeerRxHsG4AdaptRefresh
- ❑ PA_PeerRxHsG4AdaptInitial
- ❑ PA_TxHsG5SyncLength
- ❑ PA_TxHsG5PrepareLength
- ❑ PA_PeerRxHsG5AdaptRefresh
- ❑ PA_PeerRxHsG5AdaptInitial
- ❑ PA_MK2ExtensionGuardBand
- ❑ PA_PWRMode
- ❑ PA_AvailTxDataLanes

- ❑ PA_AvailRxDataLanes
- ❑ PA_TxGear
- ❑ PA_RxGear
- ❑ PA_HSSeries
- ❑ PA_TxTermination
- ❑ PA_RxTermination
- ❑ PA_Scrambling

While in low-power state, the following information from Layer 2, Layer 3, Layer 4 and DME_QOS are retained in DWC_mipi_unipro_20's Always-On domain:

- DL_TC0TxMaxSDUSize
- DL_TC0RxInitCreditVal
- DL_TC1TxMaxSDUSize
- DL_TC1RxInitCreditVal
- DL_TC0TxBufferSize
- DL_TC1TxBufferSize
- DL_MTU
- DL_SYMBOL_MTU
- DL_CreditUnitSize
- DL_TC0TXFCThreshold
- DL_FC0ProtectionTimeOutVal
- DL_TC0ReplayTimeOutVal
- DL_AFC0ReqTimeOutVal
- DL_AFC0CreditThreshold
- DL_TC0OutAckThreshold
- DL_TC1TXFCThreshold
- DL_FC1ProtectionTimeOutVal
- DL_TC1ReplayTimeOutVal
- DL_AFC1ReqTimeOutVal
- DL_AFC1CreditThreshold
- DL_TC1OutAckThreshold
- N_DeviceID
- N_DeviceID_valid
- N_TC0TxMaxSDUSize

- N_TC1TxMaxSDUSize
- T_TC0TxMaxSDUSize
- T_TC1TxMaxSDUSize
- DME_TX_DATA_OFL
- DME_TX_NAC_RECEIVED
- DME_TX_QoS_COUNT
- DME_TX_DL_LM_ERROR
- DME_RX_DATA_OFL
- DME_RX_CRC_ERROR
- DME_RX_QoS_COUNT
- DME_RX_DL_LM_ERROR
- DME_TXRX_DATA_OFL
- DME_TXRX_PA_INIT_REQUEST
- DME_TXRX_QoS_COUNT
- DME_TXRX_DL_LM_ERROR
- DME_QoS_ENABLE
- DME_QoS_STATUS

2.17.6 Isolating Signals Crossing the Power Domain

It is important to isolate signals from the power-off domain to power-on domain during low-power states. This is to avoid corruption of signals in the power-on domain because removing the power results in signals getting tri-stated. Isolation cells are introduced to achieve this objective.

2.17.7 Low-Power Control Finite State Machine

Low-power control Finite State Machine (FSM) controls DWC_mipi_unipro_20's power-gated hibernate enter and exit. The application requests hibernate through **LA_HibernateEnterReq**. On receiving the hibernate request, the FSM sends the hibernate primitive request to the PA layer. If no previous power mode change requests are in pipeline, the PA layer acknowledges hibernate confirmation with SUCCESS. Otherwise, it responds with the parameter set to FAILURE. When the procedure has completed successfully, it indicates that the hibernate enter request is accepted by DWC_mipi_unipro_20 and the FSMs current state changes to Save state. In this state, all required information is saved and waits for Save.complete. Upon assertion of Save.complete, the FSM state moves to hibernate state. The application is indicated about this state by driving **LA_HibernateEnterInd**. In this hibernate state, the input **LP_EnableIsolation** signal can be asserted and followed by the assertion of the active low reset **RG_ResetN**. It is now safe to remove the power. The whole stack including M-PHY is in the hibernate state.

When the application layer decides to wake from power-gated hibernation, it must follow this sequence:

- Restore power to the off domain

- De-assert RG_ResetN
- Assert LP_DisableClkGate to propagate reset to all clock domains
- De-assert LP_EnableIsolation
- Issue LA_HibernateExitReq

On receiving a **LA_HibernateExitReq**, low-power control state machine moves from hibernate state to restore state and saved information is restored in this state. On restore.complete, the FSM state changes to **Check Hibernate Exit Confirmation** state. With hibernate exit confirmation and SUCCESS indication, the FSM drives **LA_HibernateExitInd** and the state machine moves to Run state.

Figure 2-43 Low-Power Control FSM

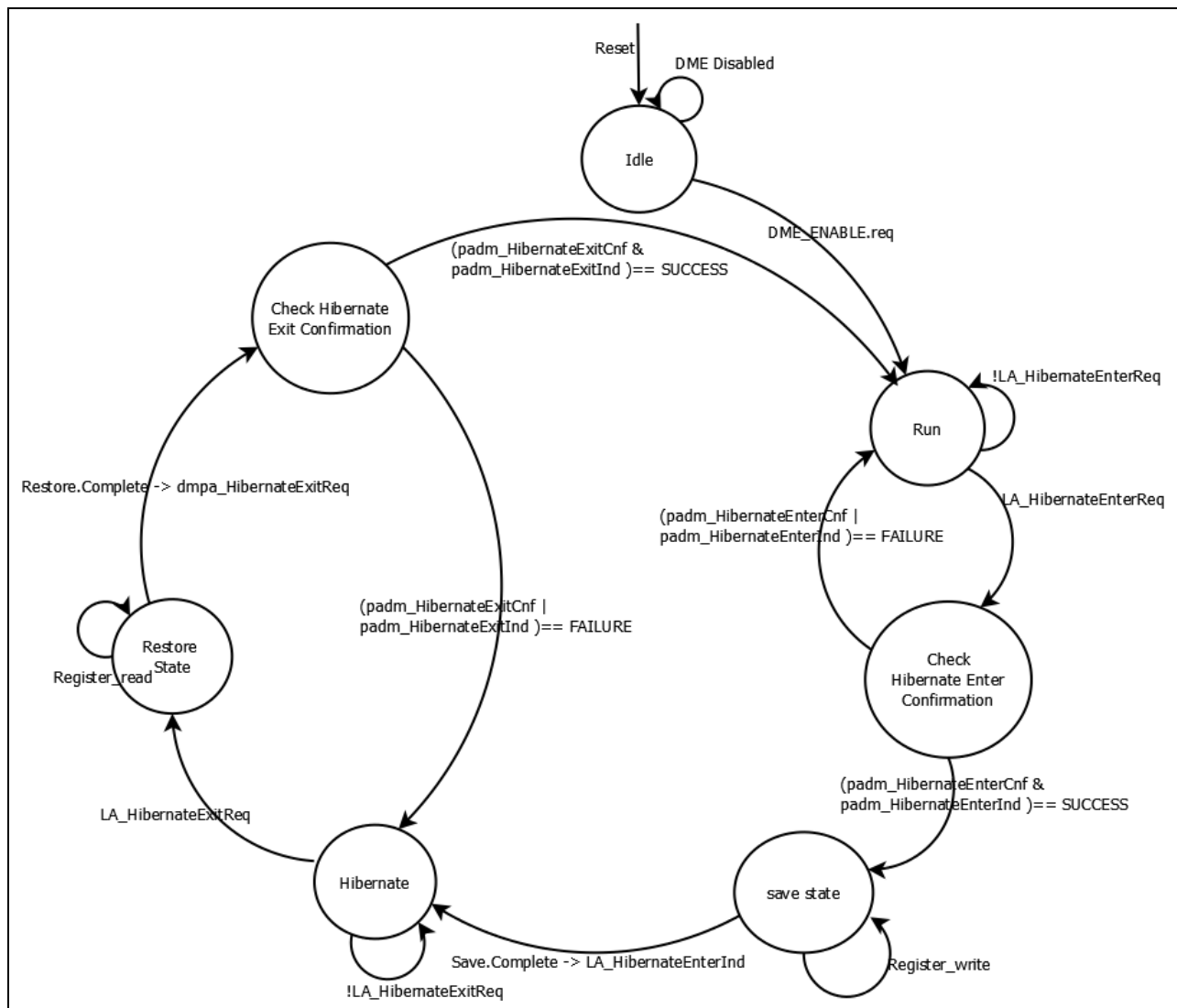


Table 2-7 FSM Encoding Table for the DSLPFSM Field of the VS_DebugStates Register

FSM State	Encoding (3 bits)
Idle	3'b000
Run	3'b001
Check Hibernate Enter Confirmation	3'b010
Save	3'b011
Hibernate	3'b100
Restore	3'b110
Check Hibernate Exit Confirmation	3'b101

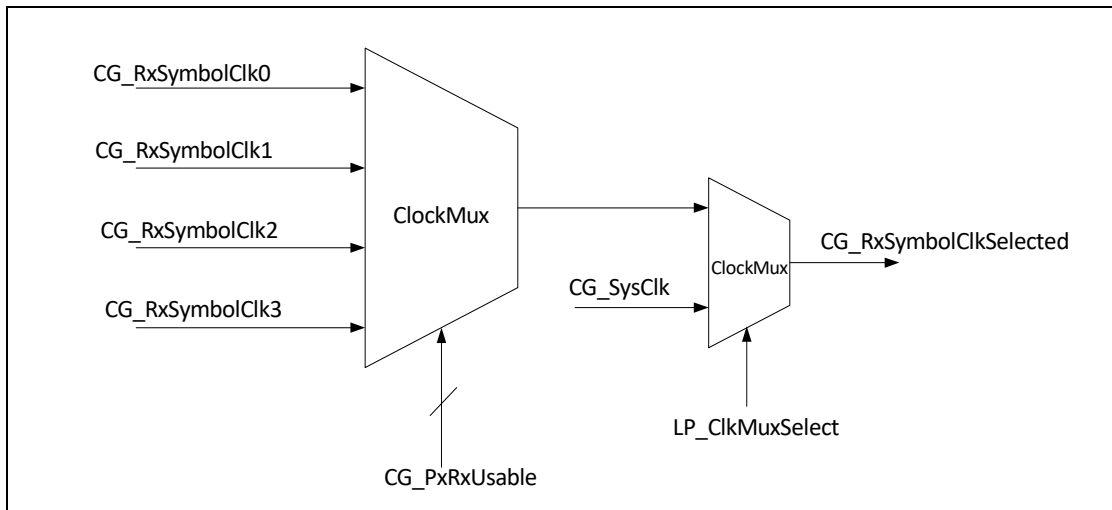
2.17.8 Hibernate Enter Interrupts

The interrupt is generated on hibernate enter if the vendor-specific attribute **VS_InterruptEnable** [IE_HENI] is enabled and the interrupt status is logged in **VS_InterruptStatus** [IS_HENI]. This interrupt can be taken as a final indication for application layer or processor for safe removal of power to the OFF domain.

2.17.9 Low-Power Mode Requirements

2.17.9.1 RX Clock Multiplexer

The **LP_ClkMuxSelect** signal indicates which signal is used as the clock input to the controller. In power-gated low-power hibernate state, if **CG_RxSymbolClkSelected** is not present, there is a need of multiplexer between **CG_RxSymbolClkSelected** and the system clock **CG_Sysclk** as shown in [Figure 2-44](#). This multiplexer should be instantiated outside **DWC_mipi_unipro_20**. The **LP_ClkMuxSelect** signal must be used as the select pin for this multiplexer. In low-power mode, **LP_ClkMuxSelect** remains high during reset propagation and **Save and Restore** FSM state, so the system clock is selected as the input clock to **DWC_mipi_unipro_20**. In normal mode, **LP_ClkMuxSelect** is set to low and therefore **CG_RxSymbolClkSelected** is used as the input clock.

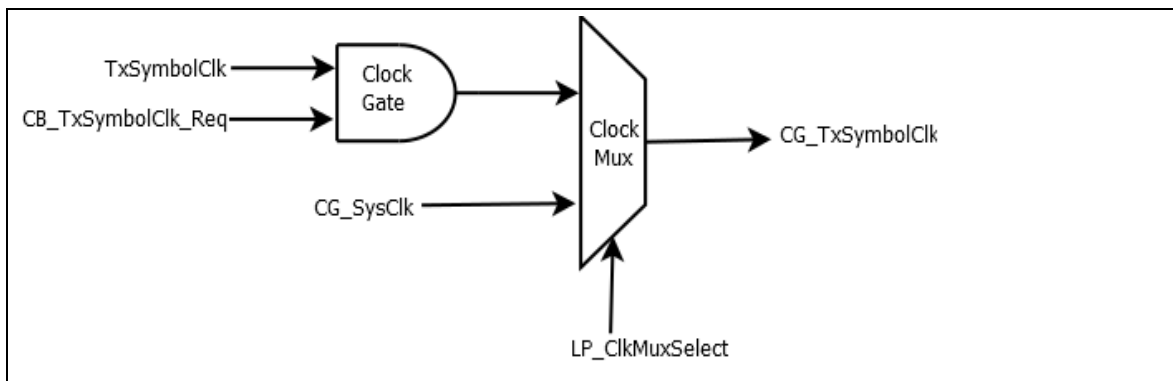
Figure 2-44 Low-Power Mode RX Clock Multiplexer**Note**

The multiplexer between CG_RxSymbolClkSelected and the system clock CG_SysClk, as shown in Figure 2-44, is an additional clock multiplexer which is required only in low-power implementation, if CG_RxSymbolClk0/1/2/3 clocks are gated in the hibernate state. The design of this clock multiplexer ensures glitch free output at the time of clock switching. The design is also capable of switching to the other clock even when the currently selected clock is not toggling.

2.17.9.2 TX Clock Multiplexer

As shown in Figure 2-45, TxSymbolClk is the symbol clock given by M-PHY for driving the data on the RMMI TX lanes. The CG_TxSymbolClk clock is the symbol clock input to DWC_mipi_unipro_20.

If TxSymbolClk is gated then it is not available to the controller because the CB_TX_SymbolClkReq signal is set to low in the hibernate state. However, if the clock gating is not done to TxSymbolClk and if it is available, then the multiplexer is not required. In power-gated low-power hibernate state, if TxSymbolClk is not available then there is a need of a multiplexer between TxSymbolClk and the system clock CG_SysClk as shown in Figure 2-45. This multiplexer should be instantiated outside DWC_mipi_unipro_20. The LP_ClkMuxSelect signal must be used as the select pin for this multiplexer. In low-power mode, LP_ClkMuxSelect remains high during reset propagation, **Save and Restore FSM** state, so the system clock is selected as the input clock to DWC_mipi_unipro_20. In normal mode, LP_ClkMuxSelect is set to low and therefore, TxSymbolClk is used as the input clock.

Figure 2-45 Low-Power Mode TX Clock Multiplexer**Note**

The Clock Gate in [Figure 2-45](#) is not an AND gate. The AND gate symbol is shown to imply that Clock Gating or other appropriate cell can be used.

**Note**

This is an additional clock multiplexer which is required only in low-power implementation, if the CG_TxSymbolClk clock is not present in the hibernate state. The design of this clock multiplexer ensures glitch free output at the time of clock switching. When SNPS_MPHY_EXT_EN=0, the clock multiplexer design should be such that the clock switching should happen even when the current clock is not toggling.

2.17.9.3 Programming Requirements

In the event of power-gated hibernate, if the system requires, the software must save and write back the attributes shown in [Table 2-8](#). This is because, these attributes get reset to their reset values after power-gated hibernate.

These attributes are updated to L2 and retained in the Low-Power flow. Therefore, if the software does not save and write back these attributes, there is no functional impact.

However, if the software wants to know current link configuration and therefore reads values corresponding to these DME/PA attributes after power-gated hibernate, it gets the default reset values and not the actual values of power mode configuration before entering hibernate.

If the system does not require to read the link configuration after power-gated hibernate, these steps can be skipped.

Table 2-8 Attributes that are Reset After Power-Gated Hibernate

Address Offset	Attribute Name
0xd041	DME_FC0ProtectionTimeOutVal
0xd042	DME_TC0ReplayTimeOutVal
0xd043	DME_AFC0ReqTimeOutVal
0xd044	DME_FC1ProtectionTimeOutVal
0xd045	DME_TC1ReplayTimeOutVal

Table 2-8 Attributes that are Reset After Power-Gated Hibernate (Continued)

Address Offset	Attribute Name
0xd046	DME_AFC1ReqTimeOutVal
0x15B0 - 0x15BB	PA_PWRModeUserData0/1/2/3/4/5/6/7/8/9/10/11
0x15C0	PA_PACPFramCount

2.18 General Purpose Input and Output Ports

DWC_mipi_unipro_20 provides the option to drive up to 96 bits of output ports and to read the status of up to 96 input ports. This feature can be used to set or get ports of a UniPro device through UniPro host DME software interface.

2.18.1 Enabling General Purpose Input and Output

To enable this feature while configuring the controller in the coreConsultant GUI, specify a value for the *P_GPIO* parameter. The supported values for this parameter are 0 to 96. When a non-zero value is set for this parameter, the corresponding number of input (LA_GPI) and output (LA_GPO) ports gets added into DWC_mipi_unipro_20 controller.

2.18.2 General Purpose IO Registers

The following registers are available to support the General Purpose Input and Output:

- VS_GPO_Ctrl_0
- VS_GPO_Ctrl_1
- VS_GPO_Ctrl_2
- VS_GPI_Status_0
- VS_GPI_Status_1
- VS_GPI_Status_2

For more information about these registers, see [“Register Descriptions”](#) on page 161.

2.18.3 General Purpose IO Signals

The following signals are available to support the General Purpose Input and Output:

- LA_GPO
- LA_GPI

For more information about these signals, see [“Signal Descriptions”](#) on page 93.

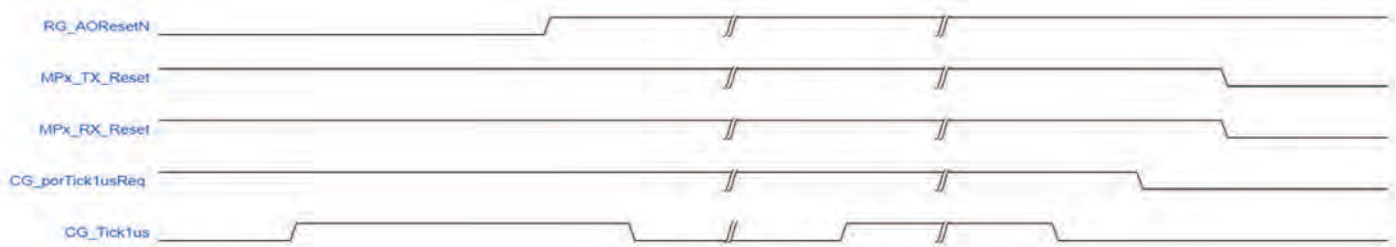
2.19 Ports Added for Link Startup in High-Speed Mode

2.19.1 CG_porTick1usReq

Tick 1us Request Signal after Power on Reset. CG_porTick1usReq is an active level high signal which requests application layer to provide CG_Tick1us signal after power on reset. This port is available only when DWC_mipi_unipro_20 is configured with SNPS MPHY V5.0 and P_PHYRESET value is greater than 63. The reset value of CG_porTick1usReq port is 1. This port goes low after P_PHYRESET timer expires.

The functionality of CG_porTick1usReq signal is captured in [Figure 2-46](#).

Figure 2-46 CG_porTick1usReq Signal Behavior



2.19.2 Reset Mode Selection

2.19.2.1 LA_ResetModeReq

When P_UPRO_LSS_HS is set to 1, LA_ResetModeReq defines the link startup Mode. LA_ResetModeReq controls the reset value of following PA layer attributes as defined in UniPro V2.0 Specification:

- PA_TxTrailingClocks (0x1564)
- PA_TActivate (0x15A8)
- PA_PWRMode (0x1571)
- PA_TxTermination (0x1569)
- PA_RxTermination (0x1584)

A value of 0 indicates LS_MODE and value of 1 indicates HS_MODE.

LA_ResetModeReq is selected before Power on Reset and should stay with same value until next Power on Reset is applied. This port should have a valid value before RG_ResetN and RG_AOResetN is released. The value that is assigned to this port is driven on MPx_TX_ResetMode and MPx_RX_ResetMode ports.

2.19.2.2 MPx_TX_ResetMode

MPx_TX_ResetMode[P_NT-1:0] defines the ResetMode for each MPHY TX lane. A value of 0 indicates LS_MODE and value of 1 indicates HS_MODE. The value assigned to LA_ResetModeReq is reflected on this port.

2.19.2.3 MPx_RX_ResetMode

MPx_RX_ResetMode[P_Nr-1:0] defines the ResetMode for each MPHY RX lane. A value of 0 indicates LS_MODE and value of 1 indicates HS_MODE. The value assigned to LA_ResetModeReq is reflected on this port.

The functionality of ResetMode selection signal is captured in [Figure 2-47](#).

Figure 2-47 ResetMode Selection Signal Behavior



2.19.3 PMC Intervention

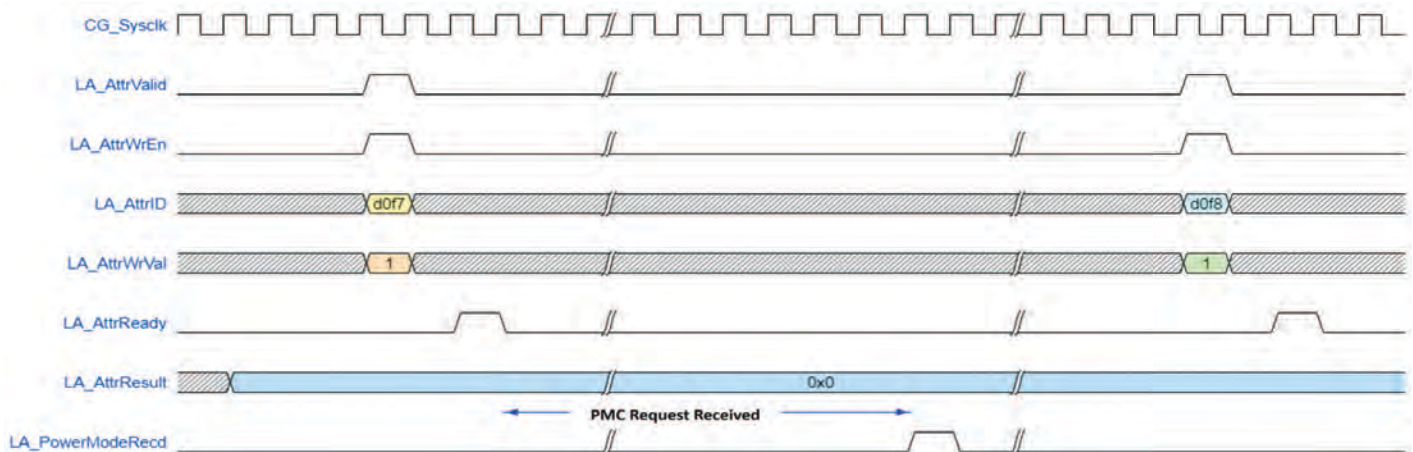
2.19.3.1 LA_PowerModeRecd

LA_PowerModeRecd is a pulse signal generated in CG_Sysclk. When PMC intervention feature is enabled, the UniPro controller generates a pulse on LA_PowerModeRecd port when it receives PMC request from remote side, and waits for conformation about the PMC Intervention completion from the application.

PMC intervention feature can be enabled by setting PMCINTEN programmable bit to 1 in VS_PmcInterventionEn (0xd0f7) attribute. PMC intervention completion is indicated by setting PMCINTDONE bit to 1 in VS_PmcInterventionDone (0xd0f8) attribute.

The functionality of LA_PowerModeRecd signal is captured in [Figure 2-48](#).

Figure 2-48 LA_PowerModeRecd Signal Behavior



2.19.3.2 LA_PowerModeAdptRecd

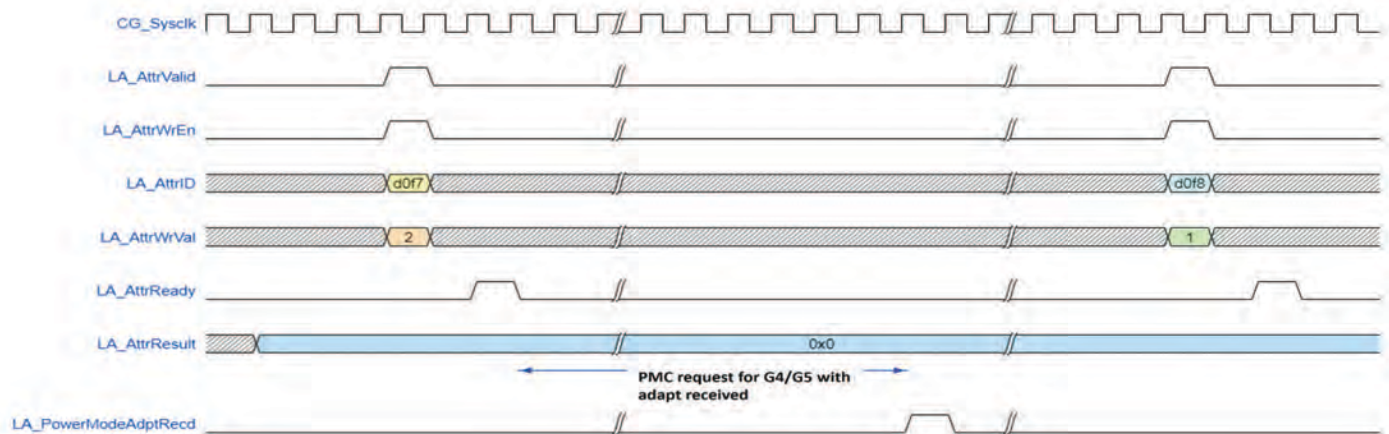
LA_PowerModeAdptRecd is a pulse signal generated in CG_Sysclk. When PMC intervention with adapt feature is enabled, the UniPro controller generates a pulse on LA_PowerModeAdptRecd port when it

receives PMC request for G4/G5 with adapt enabled from remote side, and waits for conformation about the PMC Intervention completion from the application.

PMC intervention feature with adapt can be enabled by setting PMCINTADPTEN programmable bit to 1 in VS_PmcInterventionEn (0xd0f7) attribute. PMC intervention completion is indicated by setting PMCINTDONE bit to 1 in VS_PmcinterventionDone (0xd0f8) attribute.

The functionality of LA_PowerModeAdptRecd signal is captured in [Figure 2-49](#).

Figure 2-49 LA_PowerModeAdptRecd Signal Behavior



3

Parameter Descriptions

This chapter details all the configuration parameters. **You can use the coreConsultant GUI configuration reports to determine the actual configured state of the controller.** Some expressions might refer to TCL functions or procedures (sometimes identified as **<functionof>**) that coreConsultant uses to make calculations. The exact formula used by these TCL functions is not provided in this chapter. However, when you configure the controller in coreConsultant, all TCL functions and parameters are evaluated completely; and the resulting values are displayed where appropriate in the coreConsultant GUI reports.

The parameter descriptions in this chapter include the **Enabled:** attribute which indicates the values required to be set on other parameters before you can change the value of this parameter.

These tables define all of the configuration options for this component.

- Main on [page 82](#)
- Misc on [page 90](#)

3.1 Main Parameters

Table 3-1 Main Parameters

Label	Description
UniPro Features	
Select UniPro Version	Select the desired UniPro specification version to be supported by the DWC_mipi_unipro controller. 0: UniPro V 1.61 1: UniPro V 1.8 2: UniPro V 2.0 By default, UniPro V1.8 is enabled. Values: - v1_61 (0) - v1_8 (1) - v2_0 (2) Default Value: UPRO_V18_EN ? 1 : 0 Enabled: Always Parameter Name: P_UPRO_VER
Select RMMI Interface Data Width	Select the desired RMMI Data interface width per lane. 20: 20-bit RMMI Interface 40: 40-bit RMMI Interface 80: 80-bit RMMI Interface NOTE: If P_UPRO_VER is selected as v2.0 then allowed configurable values is only 80-bit RMMI interface. If P_UPRO_VER is selected as v1.8 then allowed configurable values is only 40-bit RMMI interface. If P_UPRO_VER is selected as v1.61 then allowed configurable values is only 20-bit RMMI interface. By default, 40-bit RMMI Interface is enabled. Values: - RMMI_20bit (20) - RMMI_40bit (40) - RMMI_80bit (80) Default Value: (P_UPRO_VER==2) ? 80 : (((P_UPRO_VER==1) (P_RMMI_40B_EN==1)) ? 40 : 20) Enabled: Always Parameter Name: P_RMMI_DATAWIDTH

Table 3-1 Main Parameters (Continued)

Label	Description
Select L2 CreditUnitMode	This Parameter enables supporting CreditUnitMode-128' in Unipro controller. By enabling the CreditUnitMode-128, the Maximum payload length is modified to 1144 bytes from 272 bytes. If the parameter is set to 1, the controller can operate in CreditUnitMode-128 or CreditUnitMode-32 based on programming of the register VS_PACapabilityUserData. If the parameter is set to 0, the controller will operate only in CreditUnitMode-32 NOTE: Please ensure L2 Tx and Rx buffer sizes configured are sufficient to meet the functionality and throughput based on the Maximum Payload Length of chosen CreditUnitMode. Values: ■ CreditUnitMode-32 (0) ■ CreditUnitMode-128 (1) Default Value: (P_UPRO_VER==2) ? 1 : 0 Enabled: P_UPRO_VER==2 Parameter Name: P_L2_BUF_CREDUNIT_MODE
UniPro C-Port Interface	
Data width of TX interface	Select the data width of the CPort TX interface from the application. Values: 16, 32, 64, 128 Default Value: (IS_UFS) ? (UFS_P_DWTX) : (IS_CSI3) ? 32 : 16 Enabled: (!(CC_DEVICE_TYPE == CC_UFS)) Parameter Name: P_DWTX
Data width of RX interface	Select the data width of the CPort RX interface from the application. Values: 16, 32, 64, 128 Default Value: (IS_UFS) ? (UFS_P_DWRX) : (IS_CSI3) ? 32 : 16 Enabled: (!(CC_DEVICE_TYPE == CC_UFS)) Parameter Name: P_DWRX
Number of CPort	Configure the maximum number of logical CPorts. Values: (NMUX_CPORT_SEL) ? 2 : 1, ..., (NMUX_CPORT_SEL) ? 128: 2048 Default Value: NMUX_CPORT_SEL) ? 128 : ((IS_CSI3) ? 4 : (IS_UFS_UME) ? (UFS_P_CP) : 1 Enabled: Always Parameter Name: P_CP
Width of L4 E2E FC interface bus	Configure the width of the L4 End-2-End flow control interface bus. This value influences the maximum amount of credits to be granted at a time. Values: 0, ..., 31 Default Value: (IS_UFS) ? 0 : (IS_CSI3) ? 8 : 16 Enabled: Always Parameter Name: P_FC

Table 3-1 Main Parameters (Continued)

Label	Description
Number of L4 test features	Configure the number of L4 test features. This value should be less than or equal to the number of CPorts. Values: 1, ..., P_CP Default Value: 1 Enabled: Always Parameter Name: P_TF
Enable non-multiplexed CPORTS	Enable non-multiplexed CPORT for configuration. By default, non-multiplexed CPORT is disabled. Values: 0, 1 Default Value: 0 Enabled: Always Parameter Name: NMUX_CPORT_SEL
Enable RX Data Byte Packing	To enable byte packing logic on RXDATA interface. By default, byte packing logic is disabled. Values: 0, 1 Default Value: 0 Enabled: P_UPRO_VER != 2 Parameter Name: VS_RD_DATABYTE_PACK
UniPro M-PHY Interface	
Number of M-PHY TX lanes	Select the number of M-PHY TX lanes, each carrying a 16-bit symbol. In 20-bit RMMI data interface mode (P_RMMI_DATAWIDTH = 20), M-PHY TX lanes are configurable with the values 1, 2, 3, or 4. In 40-bit RMMI data interface mode (P_RMMI_DATAWIDTH = 40), M-PHY TX lanes are configurable with the values 1 or 2. In 80-bit RMMI data interface mode (P_RMMI_DATAWIDTH = 80), M-PHY TX lanes are configurable with the values 1 or 2. Values: ■ 1 (1) ■ 2 (2) ■ 3 (3) ■ 4 (4) Default Value: 1 Enabled: Always Parameter Name: P_NT

Table 3-1 Main Parameters (Continued)

Label	Description
Number of M-PHY RX lanes	Select the number of M-PHY RX lanes, each carrying a 16-bit symbol. In 20-bit RMMI data interface mode (P_RMMI_DATAWIDTH = 20), M-PHY RX lanes are configurable with the values 1, 2, 3, or 4. In 40-bit RMMI data interface mode (P_RMMI_DATAWIDTH = 40), M-PHY RX lanes are configurable with the values 1 or 2. In 80-bit RMMI data interface mode (P_RMMI_DATAWIDTH = 80), M-PHY RX lanes are configurable with the values 1 or 2. Values: ■ 1 (1) ■ 2 (2) ■ 3 (3) ■ 4 (4) Default Value: 1 Enabled: Always Parameter Name: P_NR
Select Synopsys M-PHY interface	Select the desired version of Synopsys M-PHY Values: ■ NO_SNPS_MPHY (0) ■ SNPS_MPHY_GEN2 (1) ■ SNPS_MPHY_GEN3 (2) ■ SNPS_MPHY_GEN3_V41 (3) ■ SNPS_MPHY_V5 (4) Default Value: IS_UFS ? UFS_SNPS_MPHY : 0 Enabled: !(CC_DEVICE_TYPE == CC_UFS) Parameter Name: SNPS_MPHY
Remove the M-PHY CFG clock port	Enable the RMMI control interface to operate with CG_Sysclk when this option is selected. By default, the RMMI control interface operates with CG_Cfgclk. ■ 0: RMMI control interface is synchronous to CG_Cfgclk ■ 1: RMMI control interface is synchronous to CG_Sysclk Values: 0, 1 Default Value: 0 Enabled: !(SNPS_MPHY_EXT_EN SNPS_GEN3_MPHY_EXT_EN) Parameter Name: DISBL_MCFG_CLK
Enable common RX Symbol Clocks for lanes.	Enables the common RX Symbol Clock for all lanes. By default, the RX Symbol clocks are separate for all lanes. Values: 0, 1 Default Value: 0 Enabled: Always Parameter Name: COMMON_RX_SYM_CLK

Table 3-1 Main Parameters (Continued)

Label	Description
Minimum trailing clocks for RxBurst	Configure the initial value of the PA_MinRxTrailingClocks attribute. For more information about the calculation of minimum requirement of PA_MinRxTrailingClocks, see the <i>Minimum RX Trailing Clocks</i> section in the <i>DWC MIPI UniPro User Guide</i>. Values: 17, ..., 255 Default Value: (P_RMMI_DATAWIDTH > 20) ? 132 : 64 Enabled: Always Parameter Name: P_RXTRAIL
Length of M-PHY reset	Configures the duration of the following MPHY reset port signals after POR exit. MPx_TX_reset MPx_RX_reset (x indicates the lane number, x = 0,1) P_PHYRESET[6] indicates the duration steps in one microsecond when set to 1 else one CG_Sysclk period when set to 0 P_PHYRESET[5:0] indicates the number of steps Values: 1, ..., 127 Default Value: 13 Enabled: Always Parameter Name: P_PHYRESET
Skip Request for Rx Interface	Enable the skip requirement for the RX interface in EXT_1_Cap packet. By default, skip request is disabled. ■ 0: RX interface does not need skip symbols. ■ 1: RX interface needs skip symbols. Values: 0x0, 0x1 Default Value: 0x0 Enabled: Always Parameter Name: RX_SKIP_REQUEST
MK2 Extension Support	To enable/disable the support for MK2 Extension during EXT1 capability exchange. If set to '1', MK2 extension support field in the EXT1 capability packet is set to '1' during LSS. Values: 0x0, 0x1 Default Value: P_UPRO_VER != 2 Enabled: P_UPRO_VER != 2 Parameter Name: MK2_EXTN_SUPPORT

Table 3-1 Main Parameters (Continued)

Label	Description
Configuration Interface	
Enable APB3 configuration interface	Enable the APB3 interface for configuration, otherwise, native DME interface is used. The default (0) is the native DME interface. 0: APB3 interface is disabled and native DME interface is used. 1: APB3 interface is enabled. Values: 0, 1 Default Value: (IS_CSI3) ? 1 : (IS_UFS) ? UFS_P_UPRO_APB3_EN : 0 Enabled: Always Parameter Name: APB3_EN
Enable Independent clock and reset for APB	Enable the independent clock (PClk) and the reset (PResetN) for the APB interface. The default APB clock is same as the CG_Sysclk system clock and the reset is same as the RG_ResetN system reset. 0: PClk and PResetN for the APB interface are disabled. 1: PClk and PResetN for the APB interface are enabled. Values: 0, 1 Default Value: 0 Enabled: APB3_EN Parameter Name: APB3_CLOCK_EN
Enable Low Power Support	Enable the advance low-power features: 0: Advance low-power features are disabled. 1: Advance low-power features are enabled. Values: 0, 1 Default Value: 1 Enabled: Always Parameter Name: UPRO_LPM_EN
GPIO	Configure General Purpose Input Output Signals. When configured to a non-zero value, this adds the corresponding number of sideband general purpose IO signals. Values: 0, ..., 96 Default Value: 0 Enabled: Always Parameter Name: P_GPIO

Table 3-1 Main Parameters (Continued)

Label	Description
UniPro L2 Buffers Size Parameters	
Depth of L2 TX buffer (TC0)	Configure the L2 TX TC0 buffer depth. For more information about the calculation of TX buffer depth for your system considerations, see the <i>L2 TX Buffer Size Calculation</i> section in the <i>DWC MIPI UniPro Databook</i>. Values: 28, ..., 2256 Default Value: (IS_UFS) ? 218 : ((P_L2_BUF_CREDUNIT_MODE==1) ? 1160 : 480) Enabled: Always Parameter Name: P_TBD0
Depth of L2 RX buffer (TC0)	Configure the L2 RX TC0 buffer depth. For more information about the calculation of RX buffer depth for your system considerations, see the <i>L2 RX Buffer Size Calculation</i> section in the <i>DWC MIPI UniPro Databook</i>. Values: 34, ..., 2272 Default Value: (IS_UFS) ? 226 : ((P_L2_BUF_CREDUNIT_MODE==1) ? 1168 : 512) Enabled: Always Parameter Name: P_RBD0
Disable support for TC1	Disable the support for TC1 in DWC_mipi_unipro by selecting this option: ■ 0: Support for TC1 is enabled. ■ 1: Support for TC1 is disabled. Values: 0, 1 Default Value: (IS_UFS) ? (!IS_UFS_UME) : 1 Enabled: Always Parameter Name: DISBL_TC1
Depth of L2 TX buffer (TC1)	Configure the L2 TX TC1 buffer depth. When DISBL_TC1 = 1, the TC1 buffer is disabled and the depth is zero. When enabled, the value must be between 28 and 2256. For more information about the calculation of TX buffer depth for your system considerations, see the <i>L2 TX Buffer Size Calculation</i> section in the <i>DWC MIPI UniPro Databook</i>. ■ 0 (When support for TC1 is disabled) ■ 28 to 2256 (When support for TC1 is enabled) Values: 0, ..., 2256 Default Value: (DISBL_TC1) ? 0 : 256 Enabled: !DISBL_TC1 Parameter Name: P_TBD1

Table 3-1 Main Parameters (Continued)

Label	Description
Depth of L2 RX buffer (TC1)	Configure the L2 RX TC1 buffer depth. When DISBL_TC1 = 1, the TC1 buffer is disabled and the depth is zero. When enabled, the value must be between 34 and 2272. For more information about the calculation of RX buffer depth for your system considerations, see the <i>L2 RX Buffer Size Calculation</i> section in the <i>DWC MIPI UniPro Databook</i>. ■ 0 (When support for TC1 is disabled) ■ 34 to 2272 (When support for TC1 is enabled) Values: 0, ..., 2272 Default Value: (DISBL_TC1) ? 0 : 256 Enabled: !DISBL_TC1 Parameter Name: P_RBD1

3.2 Misc Parameters

Table 3-2 Misc Parameters

Label	Description
UniPro DDB Parameters	
DDB revision ID	Configure the DDB L1 revision ID. Configure it as Major-Rev.Minor Rev. ■ The default is "1.0" = 8'h10 (16). ■ If a non-zero value is set during the configuration, all other DDB attributes become read-only. ■ If 0x00 value is set during the configuration, the host modifies the DDB attributes through programming. Values: 0x0, ..., 0xff Default Value: ((CC_DEVICE_TYPE == CC_UFS)) ? 1 : 0 Enabled: Always Parameter Name: P_DDB_REV
DDB level	Configure the DDB level support: ■ 1 - DDB level 1 support ■ 3 - DDB level 2 get support ■ 7 - DDB level 3 set support Values: 0x1, 0x3, 0x7 Default Value: 0x1 Enabled: Always Parameter Name: P_DDB_LVL
Device class	Specify the UniPro device class. For example, UFS Device = 0x02 and UFS HC = 0x03 Values: 0x0, ..., 0xff Default Value: ((CC_DEVICE_TYPE == CC_UFS)) ? 1 : 0 Enabled: Always Parameter Name: P_DDB_DCL
Manufacturer ID	Specify the UniPro manufacturer ID. Values: 0x0, ..., 0xffff Default Value: ((CC_DEVICE_TYPE == CC_UFS)) ? 0x0148 : 0 Enabled: Always Parameter Name: P_DDB_MID
Product ID	Specify the user-defined product ID. Values: 0x0, ..., 0xffff Default Value: 0x0 Enabled: Always Parameter Name: P_DDB_PID

Table 3-2 Misc Parameters (Continued)

Label	Description
Length of DDB level 2	Configure the length of DDB level 2. For DDB L1, the value is equal to 0. Values: 0x0, ..., 0xff Default Value: ((CC_DEVICE_TYPE == CC_UFS)) ? 1 : 0 Enabled: (P_DDB_LVL == 1 && !IS_UFS) ? 0 : 1 Parameter Name: P_DDB_LGT

4

Signal Descriptions

This chapter details all possible I/O signals in the controller. For configurable IP titles, your actual configuration might not contain all of these signals.

Inputs are on the left of the signal diagrams; outputs are on the right.

Attention: For configurable IP titles, do not use this document to determine the exact I/O footprint of the controller. It is for reference purposes only.

When you configure the controller in coreConsultant, you must access the I/O signals for your actual configuration at workspace/report/IO.html or workspace/report/IO.xml after you have completed the report creation activity. That report comes from the exact same source as this chapter but removes all the I/O signals that are not in your actual configuration. This does not apply to non-configurable IP titles. In addition, all parameter expressions are evaluated to actual values. Therefore, the widths might change depending on your actual configuration.

Some expressions might refer to TCL functions or procedures (sometimes identified as **<functionof>**) that coreConsultant uses to make calculations. The exact formula used by these TCL functions is not provided in this chapter. However, when you configure the controller in coreConsultant, all TCL functions and parameters are evaluated completely; and the resulting values are displayed where appropriate in the coreConsultant GUI reports.

In addition to describing the function of each signal, the signal descriptions in this chapter include the following information:

Active State: Indicates whether the signal is active high or active low. When a signal is not intended to be used in a particular application, then this signal needs to be tied or driven to the inactive state (opposite of the active state).

Registered: Indicates whether or not the signal is registered directly inside the IP boundary without intervening logic (excluding simple buffers). A value of No does not imply that the signal is not synchronous, only that there is some combinatorial logic between the signal's origin or destination register and the boundary of the controller. A value of N/A indicates that this information is not provided for this IP title.

Synchronous to: Indicates which clocks in the IP sample this input (drive for an output) when considering all possible configurations. A particular configuration might not have all of the clocks listed. This clock might not be the same as the clock that your application logic should use to clock (sample/drive) this pin. For more details, consult the clock section in the databook. The presence of the postfix SuperList indicates a list of all possible clocks over all possible configs. Consult coreConsultant report for which clock applies to your specific configuration.

Exists: Names of configuration parameters that populate this signal in your configuration.

Validated by: Assertion or de-assertion of signals that validates the signal being described.

The I/O signals are grouped as follows:

- Clock Interface Signals on [page 95](#)
- Reset Interface Signals on [page 99](#)
- Low-Power Mode Signals on [page 100](#)
- Scan Mode Signals on [page 103](#)
- Data Write Interface Signals on [page 104](#)
- Data Read Interface Signals on [page 108](#)
- Flow Control Interface Signals on [page 112](#)
- Status Interface Signals on [page 115](#)
- Native Slave Interface Signals on [page 116](#)
- APB3 Slave Interface Signals on [page 119](#)
- DME Control Interface Signals on [page 122](#)
- Additional Indication Signals on [page 132](#)
- Interrupt Interface Signals on [page 134](#)
- M-PHY TX Control Interface Signals on [page 135](#)
- Synopsys V5.0 MPHY on [page 140](#)
- M-PHY RX Control Interface Signals on [page 142](#)
- M-PHY TX Data Interface Signals on [page 147](#)
- M-PHY RX Data Interface Signals on [page 149](#)
- Layer 2 RX Memory Signals (TC0) on [page 151](#)
- Layer 2 RX Memory Signals (TC1) on [page 153](#)
- Layer 2 TX Memory Signals on [page 155](#)
- GPIO Signals on [page 157](#)
- Debug Interface Signals on [page 158](#)

4.1 Clock Interface Signals

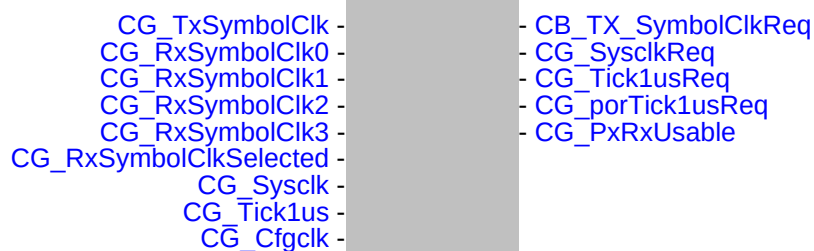


Table 4-1 Clock Interface Signals

Port Name	I/O	Description
CG_TxSymbolClk	I	Transmit Symbol Clock. This is the symbol clock generated inside the M-PHY PLL for data transmission. The clock should carry the scan clock in SCANTEST mode. For more information, see the <i>Gating CG_TxSymbolClk</i> section of the <i>DWC MIPI UniPro User Guide</i>. Exists: Always Synchronous To: None Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: N/A
CB_TX_SymbolClkReq	O	TX Symbol Clock Request. This signal indicates to an external clock multiplexer or control logic that the CG_TxSymbolClk clock is required, and it should be activated. Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: High
CG_RxSymbolClk0	I	Receive Symbol Clock of Lane 0. This is the M-PHY output clock of the received symbols of lane 0. For more information, see the <i>Gating CG_RxSymbolClk</i> section of the <i>DWC MIPI UniPro User Guide</i>. Exists: !COMMON_RX_SYM_CLK Synchronous To: None Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: N/A

Table 4-1 Clock Interface Signals (Continued)

Port Name	I/O	Description
CG_RxSymbolClk1	I	Receive Symbol Clock of Lane 1. This is the M-PHY output clock of the received symbols of lane 1. For more information, see the <i>Gating CG_RxSymbolClk</i> section of the <i>DWC MIPI UniPro User Guide</i>. Note: Tie this signal to 0 when lane 1 is not present. Exists: P_NR>1 && !COMMON_RX_SYM_CLK Synchronous To: None Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: N/A
CG_RxSymbolClk2	I	Receive Symbol Clock of Lane 2. This is the M-PHY output clock of the received symbols of lane 2. For more information, see the <i>Gating CG_RxSymbolClk</i> section of the <i>DWC MIPI UniPro User Guide</i>. Note: Tie this signal to 0 when lane 2 is not present. Exists: P_NR>2 && !COMMON_RX_SYM_CLK Synchronous To: None Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: N/A
CG_RxSymbolClk3	I	Receive Symbol Clock of Lane 3. This is the M-PHY output clock of the received symbols of lane 3. For more information, see the <i>Gating CG_RxSymbolClk</i> section of the <i>DWC MIPI UniPro User Guide</i>. Note: Tie this signal to 0 when lane 3 is not present. Exists: P_NR>3 && !COMMON_RX_SYM_CLK Synchronous To: None Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: N/A
CG_RxSymbolClkSelected	I	Selected RX Symbol Clock. This clock is used for data processing. Exists: Always Synchronous To: None Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: N/A

Table 4-1 Clock Interface Signals (Continued)

Port Name	I/O	Description
CG_Sysclk	I	System Clock. This is the main clock of the UniPro module. For more information, see the <i>Gating CG_Sysclk</i> section of the <i>DWC MIPI UniPro User Guide</i>. Exists: Always Synchronous To: None Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: N/A
CG_SysclkReq	O	CG_SysClk Request signal. An active (high) level requires the application to activate the CG_Sysclk input. When low, the application can gate-off the CG_Sysclk input to the DWC_mipi_unipro. This signal goes low only when the DWC_mipi_unipro is in the hibernate mode. For more information, see the <i>Gating CG_Sysclk</i> section of the <i>DWC MIPI UniPro User Guide</i>. Exists: Always Synchronous To: CG_Sysclk at the time of transitioning from high to low; asynchronous when transitioning from low to high. Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: High
CG_Tick1us	I	1 us Tick Input. This signal is used for microsecond timer implementation. This signal toggles once every microsecond. Note: Connect a 1 MHz clock (Synopsys M-PHY signal CB_CFGCLK_DIV) to this port when Synopsys GEN 2 M-PHY is selected. Exists: Always Synchronous To: Asynchronous Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
CG_Tick1usReq	O	1 us Tick Request Signal. An active level requires the application to activate the CG_Tick1us signal. Exists: !GEN2_FPGA_CONF_EN && !SNPS_FPGA_CONF_EN && !SNPS_MPHY_EXT_EN Synchronous To: CG_Sysclk Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

Table 4-1 Clock Interface Signals (Continued)

Port Name	I/O	Description
CG_porTick1usReq	O	1 us Tick Request Signal after Power on Reset. An active level high indicates application to provide CG_Tick1us signal. This Port is available only for SNPS MPHY V5.0 when P_PHYRESET value is greater than 63. When CG_porTick1usReq is set to HIGH then 1 us Tick signal shall be supplied failing which the system will not progress. Exists: SNPS_MPHY_V5_TICKUS_EN Synchronous To: CG_Sysclk Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: High
CG_Cfgclk	I	M-PHY Configuration Clock. M-PHY Configuration clock. Exists: !DISBL_MCFG_CLK Synchronous To: None Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: N/A
CG_PxRxUsable[(P_NR-1):0]	O	RX Clock Selection Indicator. This signal indicates logical lane 0 for selecting the RX clock input. Only one bit is active at any time. Each bit of this signal is synchronous to negative edge of the corresponding CG_RxSymbolClk clock. ■ CG_PxRxUsable[0] is synchronous to negative edge of CG_RxSymbolClk0 ■ CG_PxRxUsable[1] is synchronous to negative edge of CG_RxSymbolClk1 ■ CG_PxRxUsable[2] is synchronous to negative edge of CG_RxSymbolClk2 ■ CG_PxRxUsable[3] is synchronous to negative edge of CG_RxSymbolClk3 Exists: P_NR>1 && !COMMON_RX_SYM_CLK Synchronous To: Depending on the lane number, this signal is synchronous to the negative edge of CG_RxSymbolClkn, where n = 0, 1, 2, or 3. Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

4.2 Reset Interface Signals

RG_ResetN -



Table 4-2 Reset Interface Signals

Port Name	I/O	Description
RG_ResetN	I	System Reset. Reset of the module from the system. Exists: Always Synchronous To: CG_Sysclk at the time of transitioning from low to high; asynchronous when transitioning from high to low Registered: N/A Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: Low

4.3 Low-Power Mode Signals

RG_AOResetN	-	LP_PowerReq
LP_PowerDown	-	LP_ClkMuxSelect
LP_DisableClkGate	-	LP_ResetComplete
LP_EnableIsolation	-	
LP_RetentionEnable	-	
LP_TopIsolationEnable	-	

Table 4-3 Low-Power Mode Signals

Port Name	I/O	Description
RG_AOResetN	I	System reset to the Always-On power domain. Reset of the module from the system to the Always-On Power domain required only in low-power mode. Exists: UPRO_LPM_EN Synchronous To: CG_Sysclk at the time of transitioning from low to high; asynchronous when transitioning from high to low Registered: N/A Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: Low
LP_PowerDown	I	UPF-based low-power switch simulation control. This signal is used only in UniPro standalone low-power simulation where the power switch inferred though the UPF script is controlled though this signal in UPF simulation. Exists: UPRO_LPM_EN Synchronous To: Asynchronous Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: High (Power switch is off)
LP_DisableClkGate	I	Low-power Mode Disable CG_Sysclk Clock Gating. This signal must be driven high if the application layer needs to perform a hibernate exit. Note: The application layer must drive this signal low on receiving LP_ResetComplete as HIGH. Exists: UPRO_LPM_EN Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

Table 4-3 Low-Power Mode Signals (Continued)

Port Name	I/O	Description
LP_PowerReq	O	Low-power Mode Power Restore Request. This signal indicates to an external power management block to restore the power when a hibernate exit request is received from the peer or device. Exists: UPRO_LPM_EN Synchronous To: Asynchronous Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: High
LP_EnableIsolation	I	Low-power Mode Isolation Cell Enable Control. This signal must be driven high, if the application layer needs to enable the isolation cell in a UPF methodology-based power gated hibernation. Exists: UPRO_LPM_EN Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: High
LP_ClkMuxSelect	O	Low-power Mode TX and RX Clock Select. This signal indicates to an external clock multiplexer or control logic that the CG_RxSymbolClkSelected and CG_TxSymbolClk signals must be driven with a clock which is used in hibernate mode. Exists: UPRO_LPM_EN Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: High
LP_ResetComplete	O	Low-power Mode Reset Complete Handshake Signal. This signal indicates to an external power management block that all the resets have completely propagated to all the clock domains in the design during Low Power mode. Seeing this Handshake signal power management block can go ahead and de-assert the LP_EnableIsolation input. Exists: UPRO_LPM_EN Synchronous To: CG_Sysclk Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: High

Table 4-3 Low-Power Mode Signals (Continued)

Port Name	I/O	Description
LP_RetentionEnable	I	Low-power Mode Retention Enable Control. This signal is used by the UPF script to save the contents of AO domain in retention flops. This acts as a control for single pin based SRPG cells. Exists: UPRO_LPM_EN && P_RETENTION_EN Synchronous To: Asynchronous Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: High
LP_TopIsolationEnable	I	Low-power Mode Top Port Isolation Cell Enable Control. This signal is used by the UPF script to isolate the top ports of the UniPro controller when used in single power domain retention based low power flow. Exists: UPRO_LPM_EN && P_RETENTION_EN Synchronous To: Asynchronous Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: High

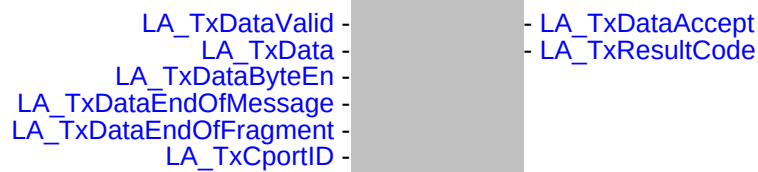
4.4 Scan Mode Signals



Table 4-4 Scan Mode Signals

Port Name	I/O	Description
TC_ScanMode	I	Scan Mode Control. This input enables bypassing of the internally generated resets and directly connecting these resets to the external reset input RG_ResetN. Tie this input to zero and set it only during scan testing. Exists: Always Synchronous To: None Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: High

4.5 Data Write Interface Signals



LA_TxDataValid -
 LA_TxData -
 LA_TxDataByteEn -
 LA_TxDataEndOfMessage -
 LA_TxDataEndOfFragment -
 LA_TxCportID -

LA_TxDataAccept
 LA_TxResultCode

Table 4-5 Data Write Interface Signals

Port Name	I/O	Description
LA_TxDataValid[(P_NUMCPORTS-1):0]	I	TX Data Valid. The application layer uses this signal to indicate the availability of new data. If LA_TxDataValid and LA_TxDataAccept are active in the same cycle, the transport layer captures new data and CPortID. When Non-Multiplexed CPort is enabled (NMUX_CPORT_SEL = 1), bit width parameter P_NUMCPORTS = P_CP, that is the number of Non-Multiplexed CPorts. When Non-Multiplexed CPort is disabled (NMUX_CPORT_SEL = 0), bit width parameter P_NUMCPORTS = 1 Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
LA_TxDataAccept[(P_NUMCPORTS-1):0]	O	TX Data Accept. The transport layer uses this signal to indicate the data processing capability to the Application layer. When Non-Multiplexed CPort is enabled (NMUX_CPORT_SEL = 1), bit width parameter P_NUMCPORTS = P_CP, that is the number of Non-Multiplexed CPorts. When Non-Multiplexed CPort is disabled (NMUX_CPORT_SEL = 0), bit width parameter P_NUMCPORTS = 1 Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

Table 4-5 Data Write Interface Signals (Continued)

Port Name	I/O	Description
LA_TxData[(((P_NUMCPORTS*P_ST)*16)-1):0]	I	TX Data. The bus width is a multiple of 16, depending on the number of symbols to be processed in parallel. The first byte to be sent is located in the highest byte of this bus and all valid bytes in any access need to be adjacent to this or another valid byte. When Non-Multiplexed CPort is enabled (NMUX_CPORT_SEL = 1), bit width parameter P_NUMCPORTS = P_CP, that is the number of Non-Multiplexed CPorts. When Non-Multiplexed CPort is disabled (NMUX_CPORT_SEL = 0), bit width parameter P_NUMCPORTS = 1 Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
LA_TxDataByteEn[(((P_NUMCPORTS*P_ST)*2)-1):0]	I	TX Byte Enable. This bus indicates validation of single bytes of the TX data bus. A value of 1 for a single bit of this bus indicates presence of new data for the respective byte of the data bus. All bytes are located adjacent to the MSB of the data bus. Therefore, only some combinations of the byte enable bus are valid. For example, for P_ST = 2 (2 symbols in parallel, 32-bit data interface, and 4-bit byte enables), only the following values are allowed: ■ b'1111 ■ b'1110 ■ b'1100 ■ b'1000 When Non-Multiplexed CPort is enabled (NMUX_CPORT_SEL = 1), bit width parameter P_NUMCPORTS = P_CP, that is the number of Non-Multiplexed CPorts. When Non-Multiplexed CPort is disabled (NMUX_CPORT_SEL = 0), bit width parameter P_NUMCPORTS = 1 Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

Table 4-5 Data Write Interface Signals (Continued)

Port Name	I/O	Description
LA_TxDataEndOfMessage[(P_NUMCPORTS-1):0]	I	TX End of Message Indicator. When high, this signal indicates to the transport layer that the LSB of this access is the final byte of the current message. This information is used to set the EOM flag inside the Frame header. When Non-Multiplexed CPort is enabled (NMUX_CPORT_SEL = 1), bit width parameter P_NUMCPORTS = P_CP, that is the number of Non-Multiplexed CPorts. When Non-Multiplexed CPort is disabled (NMUX_CPORT_SEL = 0), bit width parameter P_NUMCPORTS = 1 Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
LA_TxDataEndOfFragment[(P_NUMCPORTS-1):0]	I	TX End of Fragment Indicator. When high, this signal indicates to the transport layer that the LSB of this access is the final byte of the message fragment. The current fragment is finalized and any new data to the same CPort starts a new fragment. This signal can be used to control the size of the frames link. If this functionality is not required, this signal must be tied to 0. When Non-Multiplexed CPort is enabled (NMUX_CPORT_SEL = 1), bit width parameter P_NUMCPORTS = P_CP, that is the number of Non-Multiplexed CPorts. When Non-Multiplexed CPort is disabled (NMUX_CPORT_SEL = 0), bit width parameter P_NUMCPORTS = 1 Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

Table 4-5 Data Write Interface Signals (Continued)

Port Name	I/O	Description
LA_TxResultCode[((P_NUMCPORTS*3)-1):0]	O	TX Result Code. This signal provides the status of the current write access to the application. 0: Access successful 1: Local CPort is unconnected 2: Reserved, not used for data transmission 3: Local CPort is not available 4: Reserved 5: Reserved 6: Peer device does not support selected TC 7: Message of 0 size rejected (EOM false) This result is valid one cycle after the LA_TxDataValid and LA_TxDataAccept are asserted. When Non-Multiplexed CPort is enabled (NMUX_CPORT_SEL = 1), bit width parameter P_NUMCPORTS = P_CP, that is the number of Non-Multiplexed CPorts. When Non-Multiplexed CPort is disabled (NMUX_CPORT_SEL = 0), bit width parameter P_NUMCPORTS = 1 Exists: Always Synchronous To: CG_Sysclk Registered: (NMUX_CPORT_SEL == 1) ? "No" : "Yes" Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
LA_TxCportID[(P_CA-1):0]	I	TX CPort Identification. This is the CPort ID (address) of the current access. The width of this bus depends on the maximum number of CPorts defined for the system. Exists: !NMUX_CPORT_SEL Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A

4.6 Data Read Interface Signals

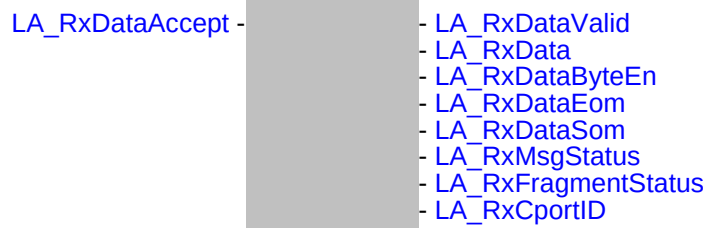


Table 4-6 Data Read Interface Signals

Port Name	I/O	Description
LA_RxDataValid[(P_NUMCPORTS-1):0]	O	RX Data Valid. The transport layer uses this signal to indicate the availability of new data. If LA_RxDataValid and LA_RxDataAccept are active in the same cycle, the application layer captures new data and address. When Non-Multiplexed CPort is enabled (NMUX_CPORT_SEL = 1), bit width parameter P_NUMCPORTS = P_CP, that is the number of Non-Multiplexed CPorts. When Non-Multiplexed CPort is disabled (NMUX_CPORT_SEL = 0), bit width parameter P_NUMCPORTS = 1 Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
LA_RxDataAccept[(P_NUMCPORTS-1):0]	I	RX Data Accept. The application layer uses this signal to indicate the data processing capability to the transport layer. When Non-Multiplexed CPort is enabled (NMUX_CPORT_SEL = 1), bit width parameter P_NUMCPORTS = P_CP, that is the number of Non-Multiplexed CPorts. When Non-Multiplexed CPort is disabled (NMUX_CPORT_SEL = 0), bit width parameter P_NUMCPORTS = 1 Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

Table 4-6 Data Read Interface Signals (Continued)

Port Name	I/O	Description
LA_RxData[(((P_NUMCPORTS*P_SR)*16)-1):0]	O	RX Data. The bus width is a multiple of 16, depending on the number of symbols to be processed in parallel. The first received byte is located the highest byte of this bus and all valid bytes in any access are adjacent to this or another valid byte. All bytes of a single frame are delivered in one part, that is, there are no spaces between the bytes. When Non-Multiplexed CPort is enabled (NMUX_CPORT_SEL = 1), bit width parameter P_NUMCPORTS = P_CP, that is the number of Non-Multiplexed CPorts. When Non-Multiplexed CPort is disabled (NMUX_CPORT_SEL = 0), bit width parameter P_NUMCPORTS = 1 Exists: Always Synchronous To: CG_Sysclk Registered: (NMUX_CPORT_SEL == 1) ? "No" : "Yes" Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
LA_RxDataByteEn[(((P_NUMCPORTS*P_SR)*2)-1):0]	O	RX Byte Enable. This bus indicates validation of single bytes of the RX data bus. A value of 1 for a single bit of this bus indicates presence of new data for the respective byte of the data bus. All bytes are located adjacent to the MSB of the data bus. Therefore, only some combinations of the byte enable bus are valid. For example, for P_SR = 2 (2 symbols in parallel, 32-bit data interface, and 4-bit byte enables), only the following values are allowed: ■ b'1111 ■ b'1110 ■ b'1100 ■ b'1000 When Non-Multiplexed CPort is enabled (NMUX_CPORT_SEL = 1), bit width parameter P_NUMCPORTS = P_CP, that is the number of Non-Multiplexed CPorts. When Non-Multiplexed CPort is disabled (NMUX_CPORT_SEL = 0), bit width parameter P_NUMCPORTS = 1 Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

Table 4-6 Data Read Interface Signals (Continued)

Port Name	I/O	Description
LA_RxDataEom[(P_NUMCPORTS-1):0]	O	RX End of Message Indicator. When high, this signal indicates to the application layer that the LSB of this access is the final byte of the current message. When Non-Multiplexed CPort is enabled (NMUX_CPORT_SEL = 1), bit width parameter P_NUMCPORTS = P_CP, that is the number of Non-Multiplexed CPorts. When Non-Multiplexed CPort is disabled (NMUX_CPORT_SEL = 0), bit width parameter P_NUMCPORTS = 1 Exists: Always Synchronous To: CG_Sysclk Registered: (NMUX_CPORT_SEL == 1) ? "No" : "Yes" Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
LA_RxDataSom[(P_NUMCPORTS-1):0]	O	RX Start of Message Indicator. When high, this signal indicates to the application layer that the MSB of this access is the first byte of the current message. When Non-Multiplexed CPort is enabled (NMUX_CPORT_SEL = 1), bit width parameter P_NUMCPORTS = P_CP, that is the number of Non-Multiplexed CPorts. When Non-Multiplexed CPort is disabled (NMUX_CPORT_SEL = 0), bit width parameter P_NUMCPORTS = 1 Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
LA_RxMsgStatus[(P_NUMCPORTS-1):0]	O	RX Message Status. This signal indicates detection of corrupted data until the next EoM is found. When Non-Multiplexed CPort is enabled (NMUX_CPORT_SEL = 1), bit width parameter P_NUMCPORTS = P_CP, that is the number of Non-Multiplexed CPorts. When Non-Multiplexed CPort is disabled (NMUX_CPORT_SEL = 0), bit width parameter P_NUMCPORTS = 1 Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

Table 4-6 Data Read Interface Signals (Continued)

Port Name	I/O	Description
LA_RxFragmentStatus[((P_NUMCPORTS*2)-1):0]	O	RX Fragment Status. This signal provides the status of the current write access. 0: Message fragment is correctly received 1: Current message fragment is incomplete 2: Data is dropped because of CSV/CSD in the preceding cycle 3: Current fragment is incomplete and the data is dropped in the preceding cycle When Non-Multiplexed CPort is enabled (NMUX_CPORT_SEL = 1), bit width parameter P_NUMCPORTS = P_CP, that is the number of Non-Multiplexed CPorts. When Non-Multiplexed CPort is disabled (NMUX_CPORT_SEL = 0), bit width parameter P_NUMCPORTS = 1 Exists: Always Synchronous To: CG_Sysclk Registered: (NMUX_CPORT_SEL == 1) ? "Yes" : "No" Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
LA_RxCportID[(P_CA-1):0]	O	RX CPort ID. This is the CPort ID (address) of the current access. The width of this bus depends on the maximum number of CPorts defined for the system. Exists: !NMUX_CPORT_SEL Synchronous To: CG_Sysclk Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A

4.7 Flow Control Interface Signals

LA_FcValid -
 LA_FcCredits -
 LA_FcCportID -



LA_FcAccept
 LA_FcResultCode
 LA_FcCreditsAccepted

Table 4-7 Flow Control Interface Signals

Port Name	I/O	Description
LA_FcValid[(P_NUMCPORTS-1):0]	I	Flow Control Valid. This signal indicates to the transport layer that new credit information is available. When Non-Multiplexed CPort is enabled (NMUX_CPORT_SEL = 1), bit width parameter P_NUMCPORTS = P_CP, that is the number of Non-Multiplexed CPorts. When Non-Multiplexed CPort is disabled (NMUX_CPORT_SEL = 0), bit width parameter P_NUMCPORTS = 1 Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
LA_FcAccept[(P_NUMCPORTS-1):0]	O	Flow Control Accept. The transport layer uses this signal to indicate that it is capable of processing new credit information. When Non-Multiplexed CPort is enabled (NMUX_CPORT_SEL = 1), bit width parameter P_NUMCPORTS = P_CP, that is the number of Non-Multiplexed CPorts. When Non-Multiplexed CPort is disabled (NMUX_CPORT_SEL = 0), bit width parameter P_NUMCPORTS = 1 Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

Table 4-7 Flow Control Interface Signals (Continued)

Port Name	I/O	Description
LA_FcCredits[(((P_NUMCPORTS*(P_FC+1))-1):0]	I	Flow Control Credits. This signal indicates the number of credits for current flow control access. When Non-Multiplexed CPort is enabled (NMUX_CPORT_SEL = 1), bit width parameter P_NUMCPORTS = P_CP, that is the number of Non-Multiplexed CPorts. When Non-Multiplexed CPort is disabled (NMUX_CPORT_SEL = 0), bit width parameter P_NUMCPORTS = 1 Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
LA_FcResultCode[(((P_NUMCPORTS*3)-1):0]	O	Flow Control Result Code. This signal provides the status of the current flow control access back to the application. When Non-Multiplexed CPort is enabled (NMUX_CPORT_SEL = 1), bit width parameter P_NUMCPORTS = P_CP, that is the number of Non-Multiplexed CPorts. When Non-Multiplexed CPort is disabled (NMUX_CPORT_SEL = 0), bit width parameter P_NUMCPORTS = 1 ■ 0: FC access successful ■ 1: Local CPort is not connected ■ 2: Number of credits exceeds the maximum counter value ■ 3: Local CPort is not available ■ 4'7: Reserved, not used for flow control. This result code is valid one cycle after the LA_FcValid and LA_FcAccept are asserted. Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
LA_FcCreditsAccepted[(((P_NUMCPORTS*(P_FC+1))-1):0]	O	Flow Control Accepted Credits. This signal indicates the number of credits accepted for the last flow control access. When Non-Multiplexed CPort is enabled (NMUX_CPORT_SEL = 1), bit width parameter P_NUMCPORTS = P_CP, that is the number of Non-Multiplexed CPorts. When Non-Multiplexed CPort is disabled (NMUX_CPORT_SEL = 0), bit width parameter P_NUMCPORTS = 1 Exists: Always Synchronous To: CG_Sysclk Registered: (NMUX_CPORT_SEL == 1) ? "No" : "Yes" Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

Table 4-7 Flow Control Interface Signals (Continued)

Port Name	I/O	Description
LA_FcCportID[(P_CA-1):0]	I	Flow Control CPort ID. This is the CPort ID of the current flow control access. Exists: !NMUX_CPORT_SEL Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A

4.8 Status Interface Signals

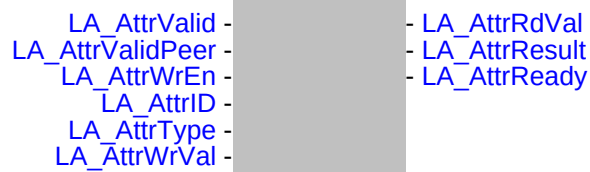


- LA_TxSaCportStatus
- LA_TxSaFcMaxSegment
- LA_TxSaEndSegment

Table 4-8 Status Interface Signals

Port Name	I/O	Description
LA_TxSaCportStatus[((P_CP*2)-1):0]	O	CPort Status. This signal indicates the CPort Status. 0: CPort is configured and usable 1: CPort is not connected 2: Reserved 3: CPort is connected to a TC which is not present in the peer UniPro node Exists: !NMUX_CPORT_SEL Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
LA_TxSaFcMaxSegment[(P_CP-1):0]	O	Credits for Maximum Segment. This signal indicates the availability of credits for a maximum sized segment per CPort. The bit is useful only if end-to-end flow control is enabled. If end-to-end flow control is disabled, it outputs static high state. Exists: !NMUX_CPORT_SEL Synchronous To: CG_Sysclk Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
LA_TxSaEndSegment[(P_CP-1):0]	O	End of Segment. This signal indicates the completion of a segment for sending to the peer device. Exists: !NMUX_CPORT_SEL Synchronous To: CG_Sysclk Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

4.9 Native Slave Interface Signals



LA_AttrValid -
 LA_AttrValidPeer -
 LA_AttrWrEn -
 LA_AttrID -
 LA_AttrType -
 LA_AttrWrVal -

- LA_AttrRdVal
 - LA_AttrResult
 - LA_AttrReady

Table 4-9 Native Slave Interface Signals

Port Name	I/O	Description
LA_AttrValid	I	Attribute Access Valid. When this signal is high, the DME captures and processes the current attribute access the application layer interface. This must be a pulse of 1 clock cycle. The host must perform the next attribute access only after the current access is completed which is indicated by the active LA_AttrReady. Exists: !APB3_EN Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: High
LA_AttrValidPeer	I	Peer Attribute Access Valid. This signal indicates that the attribute access is forwarded to the peer device. This must be a pulse of 1 clock cycle. The host must perform the next attribute access only after the current access is completed which is indicated by the active LA_AttrReady. Exists: !APB3_EN Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: High
LA_AttrWrEn	I	Attribute Write Enable. An active value of this signal identifies the current access as write access; an inactive value indicates a read access. Exists: !APB3_EN Synchronous To: CG_Sysclk Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

Table 4-9 Native Slave Interface Signals (Continued)

Port Name	I/O	Description
LA_AttrID[26:0]	I	Attribute Identifier (address). The DME captures the address of this bus and translates it to access the target layer. Exists: !APB3_EN Synchronous To: CG_Sysclk Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
LA_AttrType	I	Attribute Type. This signal indicates the type of the attribute. ■ 0: Normal ■ 1: Static Exists: !APB3_EN Synchronous To: CG_Sysclk Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
LA_AttrWrVal[31:0]	I	Attribute Write Value. For a write access, this bus holds the value to be written to an attribute. Exists: !APB3_EN Synchronous To: CG_Sysclk Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
LA_AttrRdVal[31:0]	O	Attribute Read Value. For a read access, this bus holds the value to be read from an attribute. Exists: !APB3_EN Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A

Table 4-9 Native Slave Interface Signals (Continued)

Port Name	I/O	Description
LA_AttrResult[3:0]	O	Attribute Access Result Code. This signal returns the status of the current access to the application. ■ 0: Access is successful ■ 1: Attribute ID is invalid ■ 2: Attribute value is invalid ■ 3: Write access to a read-only attribute ■ 4: Read access to a write-only attribute ■ 5: Selected CPort/M-PHY is out of range ■ 6: Write access to a connected CPort is rejected or accessed M-PHY lane is busy ■ 7: Selected test feature is out of range ■ 8: PACP remote configuration error ■ 9: Attribute access is rejected as layer is busy with previous operation ■ 10: Addressed layer is in the wrong power state Exists: !APB3_EN Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
LA_AttrReady	O	Attribute Access Ready. The DME signals to the application layer that the last access is processed and the result code is available. If the last access is a read access, the read data should be sampled from LA_AttrRdVal. Exists: !APB3_EN Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

4.10 APB3 Slave Interface Signals

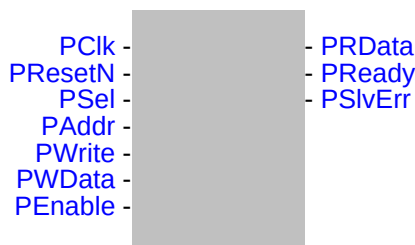


Table 4-10 APB3 Slave Interface Signals

Port Name	I/O	Description
PClk	I	APB Clock. This is the clock used for the APB3 slave interface if APB3_CLOCK_EN is selected during controller configuration. Otherwise, CG_Sysclk is used by default. The APB interface signals are synchronous to this clock. Exists: APB3_EN && APB3_CLOCK_EN Synchronous To: None Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
PRresetN	I	APB Reset. This is the reset signal used for the APB interface if APB3_CLOCK_EN is selected during controller configuration. Otherwise, RG_ResetN is used by default. Exists: APB3_EN && APB3_CLOCK_EN Synchronous To: PClk Registered: N/A Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: Low
PSel	I	APB Slave Select. This indicates that the master has selected this slave port for read or write operation. All APB3 signals are taken as valid only when this signal is asserted. Exists: APB3_EN Synchronous To: PClk/CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

Table 4-10 APB3 Slave Interface Signals (Continued)

Port Name	I/O	Description
PAddr[7:0]	I	APB Address. This is decoded to select the register accessed during the read or write operation. Bits [7:4] are currently unused and can be tied to 0. Exists: APB3_EN Synchronous To: (APB3_CLOCK_EN == 1) ? "PClk" : "CG_Sysclk" Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
PWrite	I	APB Write Enable. When asserted, it indicates that a write operation is initiated by the host. When de-asserted, a read operation is performed. Exists: APB3_EN Synchronous To: (APB3_CLOCK_EN == 1) ? "PClk" : "CG_Sysclk" Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
PWData[31:0]	I	APB Write Data. This contains the data to be written into the register indicated by the PAddr signal. Exists: APB3_EN Synchronous To: (APB3_CLOCK_EN == 1) ? "PClk" : "CG_Sysclk" Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
PEnable	I	APB Enable. When high, it is used to indicate the data transfer cycles. It should be set to high 1 clock after the PSEL signal is high. It should be low for 1 clock after PReady signal is asserted. Exists: APB3_EN Synchronous To: (APB3_CLOCK_EN == 1) ? "PClk" : "CG_Sysclk" Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

Table 4-10 APB3 Slave Interface Signals (Continued)

Port Name	I/O	Description
PRData[31:0]	O	APB Read Data. This contains the data read from the register indicated by the PAddr signal. Exists: APB3_EN Synchronous To: (APB3_CLOCK_EN == 1) ? "PClk" : "CG_Sysclk" Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
PReady	O	APB Ready. The slave uses this signal to extend an APB transfer. Exists: APB3_EN Synchronous To: (APB3_CLOCK_EN == 1) ? "PClk" : "CG_Sysclk" Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
PSlvErr	O	APB Slave Error. Flag for the slave error response from APB. This signal is always driven 0. Exists: APB3_EN Synchronous To: (APB3_CLOCK_EN == 1) ? "PClk" : "CG_Sysclk" Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

4.11 DME Control Interface Signals

LA_ResetReqWarm	-	LA_ResetCnf
LA_ResetReqCold	-	LA_EndpointResetCnf
LA_ResetModeReq	-	LA_EndpointResetInd
LA_EndpointResetReq	-	LA_LinkStartupInd
LA_LinkStartupReq	-	LA_LinkStartupCnf
LA_EnableReq	-	LA_EnableCnf
LA_TestModeReq	-	LA_PowerModeInd
LA_HibernateEnterReq	-	LA_PowerModeRecd
LA_HibernateExitReq	-	LA_PowerModeAdptRecd
		LA_TestModeInd
		LA_TestModeCnf
		LA_LinkLostInd
		LA_HibernateEnterCnf
		LA_HibernateEnterInd
		LA_HibernateExitCnf
		LA_HibernateExitInd
		LA_ErrorTInd
		LA_ErrorNInd
		LA_ErrorDInd
		LA_ErrorPAInd
		LA_ErrorPHYInd

Table 4-11 DME Control Interface Signals

Port Name	I/O	Description
LA_ResetReqWarm	I	Reset Request (warm). This signal initiates a warm reset of the module. It resets all attributes except the statistics attributes. The signal must be activated synchronously to CG_Sysclk for at least one cycle. Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
LA_ResetReqCold	I	Reset Request (cold). This signal initiates a cold reset of the module. It resets all attributes including the statistics attributes. The signal must be activated synchronously to CG_Sysclk for at least one cycle. Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

Table 4-11 DME Control Interface Signals (Continued)

Port Name	I/O	Description
LA_ResetModeReq	I	Reset Mode Request. This signal defines the link startup Mode and it controls the reset value of certain PA layer attributes as defined in V2.0 Specification. This signal shall be set before RG_ResetN or RG_AOResetN port is de-asserted. Value 0 - LS_MODE Value 1 - HS_MODE If set to 1, this signal is not expected to be set to 0. Exists: UPRO_V2_EN Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
LA_ResetCnf	O	Reset Confirmation. This signal indicates that the (cold or warm) reset procedure is complete. Exists: Always Synchronous To: CG_Sysclk Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: High
LA_EndpointResetReq	I	Endpoint Reset Request. This signal triggers the transmission of an endpoint reset to the peer device. Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: High
LA_EndpointResetCnf[1:0]	O	Endpoint Reset Confirmation. This signal indicates that the endpoint reset procedure is complete. - Bit 0: Endpoint reset procedure successfully ended - Bit 1: Endpoint reset procedure ended with error Exists: Always Synchronous To: CG_Sysclk Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

Table 4-11 DME Control Interface Signals (Continued)

Port Name	I/O	Description
LA_EndpointResetInd	O	Endpoint Reset Indication. This signal indicates that the an endpoint reset is received from the peer device. Exists: Always Synchronous To: CG_Sysclk Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
LA_LinkStartupInd	O	Link Startup Indication. This signal indicates that the initial link startup request is received from the peer device. Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
LA_LinkStartupCnf[1:0]	O	Link Startup Confirmation. This signal indicates completion of the link startup procedure. - Bit 0: Link startup procedure successfully ended - Bit 1: Link startup procedure ended with error Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
LA_LinkStartupReq	I	Link Startup Request. This signal triggers the link startup procedure in the local module. Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: High
LA_EnableReq	I	Enable Request. This signal starts the layer enable procedure. Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: High

Table 4-11 DME Control Interface Signals (Continued)

Port Name	I/O	Description
LA_EnableCnf[1:0]	O	Enable Confirmation. This signal indicates that the layer enable procedure is complete. - Bit 0: Layer enable procedure ended successfully - Bit 1: Layer enable procedure ended with error Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
LA_PowerModeInd[5:0]	O	Power Mode Change Indication. This signal indicates the change in the power mode. - Bit 0: Reserved - Bit 1: PWR_LOCAL (The local request is successfully applied.) - Bit 2: PWR_REMOTE (The remote request is successfully applied.) - Bit 3: PWR_BUSY (The request is aborted because of concurrent requests.) - Bit 4: PWR_ERROR_CAP (The request is rejected because of an invalid configuration.) - Bit 5: PWR_FATAL_ERROR (The request is aborted because of a communication problem. The link may be inoperable.) This signal is asserted for one cycle in case of a successful or an aborted power mode change. Exists: Always Synchronous To: CG_Sysclk Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
LA_PowerModeRecd	O	Power Mode Received Pulse. This signal indicates that a power mode change is received when PMC Intervention feature is enabled. Exists: Always Synchronous To: CG_Sysclk Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

Table 4-11 DME Control Interface Signals (Continued)

Port Name	I/O	Description
LA_PowerModeAdptRecd	O	Power Mode Received Pulse with Adapt enabled. This signal sends a pulse of one clock period of Cg_sysclk indicating that PMC request is received for HS G4/G5 with adapt value as 00/01 from peer when PMC intervention with adapt feature is enabled. Exists: Always Synchronous To: CG_Sysclk Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
LA_TestModeInd	O	Test Mode Indication. This signal indicates that a test mode request is received from the peer device. The DWC_mipi_unipro is now in the test mode. Exists: Always Synchronous To: CG_Sysclk Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
LA_TestModeReq	I	Test Mode Request. This signal starts the test mode procedure for layer 1.5. Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: High
LA_TestModeCnf[1:0]	O	Test Mode Confirmation. This signal indicates that the test mode request is sent to the peer device. - Bit 0: Request is sent successfully - Bit 1: An error occurs in the request Exists: Always Synchronous To: CG_Sysclk Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

Table 4-11 DME Control Interface Signals (Continued)

Port Name	I/O	Description
LA_LinkLostInd	O	Link Lost Indication. This signal indicates that the link is lost. The module should be reset using the LA_ResetReqWarm or LA_ResetReqCold signal. After disabling the DWC_mipi_unipro, the application should again trigger the boot sequence. Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
LA_HibernateEnterReq	I	Hibernation Entry Request. This signal indicates the request for hibernate mode. This signal should be valid only for one cycle. Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: High
LA_HibernateEnterCnf[1:0]	O	Hibernation Entry Confirmation. This signal indicates the status of the hibernation request. - Bit 0: Successful hibernate entry - Bit 1: Error in hibernate entry Exists: Always Synchronous To: CG_Sysclk Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
LA_HibernateEnterInd[5:0]	O	Hibernate Entry Indication. This signal indicates completion of the hibernate entry sequence. Bit mapping is similar to the mapping for the LA_PowerModeInd signal. Exists: Always Synchronous To: CG_Sysclk Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: High

Table 4-11 DME Control Interface Signals (Continued)

Port Name	I/O	Description
LA_HibernateExitReq	I	Hibernate Exit Request. This signal indicates the request to exit the hibernate mode. This signal should be valid only for one cycle. Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: High
LA_HibernateExitCnf[1:0]	O	Hibernate Exit Confirmation. This signal indicates the status of the hibernation request. - Bit 0: Successful hibernate exit - Bit 1: Error in hibernate exit Exists: Always Synchronous To: CG_Sysclk Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
LA_HibernateExitInd[5:0]	O	Hibernate Exit Indication. This signal indicates completion of the hibernate exit sequence. Bit mapping is similar to the mapping for the LA_PowerModeInd signal. Exists: Always Synchronous To: CG_Sysclk Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

Table 4-11 DME Control Interface Signals (Continued)

Port Name	I/O	Description
LA_ErrorTInd[6:0]	O	Layer 4 Error. This signal indicates that a TL layer error is detected. The discard T_PDU feature has been performed and the reason for the error is provided. - Bit 0: UNSUPPORTED_HEADER_TYPE - Bit 1: UNKNOWN_CPORTID - Bit 2: NO_CONNECTION_RX - Bit 3: CONTROLLED_SEGMENT_DROPPING - Bit 4: BAD_TC - Bit 5: E2E_CREDIT_OVERFLOW - Bit 6: SAFETY_VALVE_DROPPING Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
LA_ErrorNInd[2:0]	O	Layer 3 Error. This signal indicates that a NL layer error is detected. The discard N_PDU feature has been performed and the reason for the error is provided. - Bit 0: UNSUPPORTED_HEADER_TYPE - Bit 1: BAD_DEVICEID_ENC - Bit 2: LHDR_TRAP_PACKET_DROPPING Exists: Always Synchronous To: CG_Sysclk Registered: N/A Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

Table 4-11 DME Control Interface Signals (Continued)

Port Name	I/O	Description
LA_ErrorDInd[(ERRDIND_WD-1):0]	O	Layer 2 Error. This signal indicates that a DL layer error is detected. - Bit 0: NAC_RECEIVED - Bit 1: TCx_REPLAY_TIMER_EXPIRED - Bit 2: AFCx_REQUEST_TIMER_EXPIRED - Bit 3: FCx_PROTECTION_TIMER_EXPIRED - Bit 4: CRC_ERROR - Bit 5: RX_BUFFER_OVERFLOW - Bit 6: MAX_FRAME_LENGTH_EXCEEDED - Bit 7: WRONG_SEQUENCE_NUMBER - Bit 8: AFC_FRAME_SYNTAX_ERROR - Bit 9: NAC_FRAME_SYNTAX_ERROR - Bit 10: EOF_SYNTAX_ERROR - Bit 11: FRAME_SYNTAX_ERROR - Bit 12: BAD_CTRL_SYMBOL_TYPE - Bit 13: PA_INIT_ERROR - Bit 14: PA_ERROR_IND_RECEIVED - Bit 15: PA_INIT Exists: Always Synchronous To: CG_Sysclk Registered: N/A Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
LA_ErrorPAInd[1:0]	O	Layer 1.5 Error. This signal indicates that a PA layer line reset is detected. This error is flagged after the reception of a line reset. - Bit 0: Line reset initiated by the local device is detected - Bit 1: Line reset initiated by the peer device is detected Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

Table 4-11 DME Control Interface Signals (Continued)

Port Name	I/O	Description
LA_ErrorPHYInd[(P_NR-1):0]	O	M-PHY Layer Error. This signal indicates detection of a symbol error received separately each lane. The symbol error can have multiple conditions. Exists: Always Synchronous To: CG_Sysclk Registered: (P_RMMI_DATAWIDTH == 20) ? "Yes" : "No" Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

4.12 Additional Indication Signals

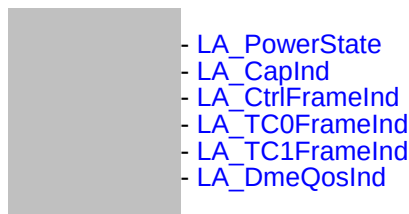


Table 4-12 Additional Indication Signals

Port Name	I/O	Description
LA_PowerState[2:0]	O	Power State Indication. This signal indicates the power state of the DWC_mipi_unipro generated CG_Sysclk. ■ 0: Disabled state ■ 1: Link Down state ■ 2: Link Up state ■ 3: Hibernate state ■ 4: Link Lost state ■ 5: Link Configuration state ■ 6-7: Reserved Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: N/A
LA_CapInd	O	Received Capability (L1.5). This signal indicates that the DWC_mipi_unipro received a capability PACP frame. It is used in the last state of the Link Startup Sequence to notify the remote PA of the local M-TX capabilities. Exists: Always Synchronous To: CG_Sysclk Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

Table 4-12 Additional Indication Signals (Continued)

Port Name	I/O	Description
LA_CtrlFrameInd	O	Control Frame Indication (L2). This signal indicates that one or more control frames are queued for transmission in layer 2. Exists: Always Synchronous To: CG_Sysclk Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
LA_TC0FrameInd	O	TC0 Frame Indication (L2). This signal indicates that one or more TC0 data frames are queued in the layer 2 TX buffer. Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
LA_TC1FrameInd	O	TC1 Frame Indication (L2). This signal indicates that one or more TC1 data frames are queued in the layer 2 TX buffer. Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
LA_DmeQosInd[2:0]	O	QoS Indication from DME. This signal indicates the DME user, that a PHY Link degradation has been monitored. Exists: UPRO_V18_EN Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

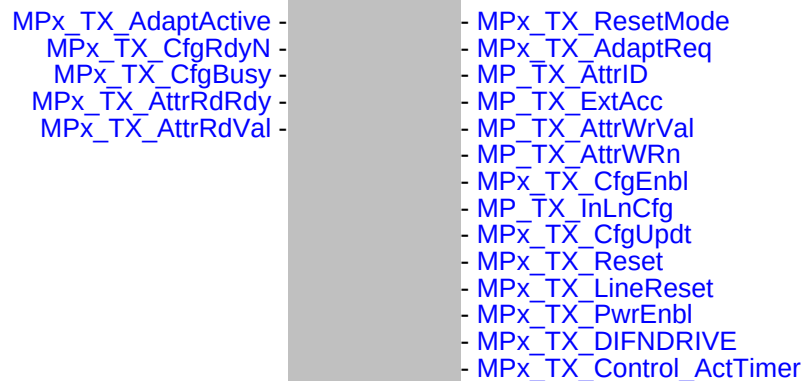
4.13 Interrupt Interface Signals



Table 4-13 Interrupt Interface Signals

Port Name	I/O	Description
LA_Intr	O	Interrupt Output. This signal indicates occurrence of one of the selected internal interrupt sources. Exists: Always Synchronous To: CG_Sysclk Registered: (APB3_EN == 1) ? "No" : "Yes" Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

4.14 M-PHY TX Control Interface Signals



MPx_TX_AdaptActive - MPx_TX_ResetMode
 MPx_TX_CfgRdyN - MPx_TX_AdaptReq
 MPx_TX_CfgBusy - MP_TX_AttrID
 MPx_TX_AttrRdRdy - MP_TX_ExtAcc
 MPx_TX_AttrRdVal - MP_TX_AttrWrVal
 MP_TX_AttrWRn
 MPx_TX_CfgEnbl
 MP_TX_InLnCfg
 MPx_TX_CfgUpdt
 MPx_TX_Reset
 MPx_TX_LineReset
 MPx_TX_PwrEnbl
 MPx_TX_DIFNDRIVE
 MPx_TX_Control_ActTimer

Table 4-14 M-PHY TX Control Interface Signals

Port Name	I/O	Description
MPx_TX_ResetMode[(P_NT-1):0]	O	TX Reset Mode. This signal defines the ResetMode for each TX lane. Value 0 - LS_MODE Value 1 - HS_MODE The value assigned to LA_ResetModeReq is reflected on this port. Exists: UPRO_LSS_HS_EN Synchronous To: None Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
MPx_TX_AdaptReq[(P_NT-1):0]	O	TX adapt request. This is adapt request signal to transmit adapt pattern on the TX interface. Exists: UPRO_V18_EN Synchronous To: CG_TxSymbolClk Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
MPx_TX_AdaptActive[(P_NT-1):0]	I	TX adapt active indication. This port indicates the adapt transmission on the TX interface. Exists: UPRO_V18_EN Synchronous To: CG_TxSymbolClk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A

Table 4-14 M-PHY TX Control Interface Signals (Continued)

Port Name	I/O	Description
MP_TX_AttrID[7:0]	O	TX Attribute ID. This is the address of the target M-PHY TX attribute for read and write operations. Exists: !SNPS_FPGA_CONF_EN && (!GEN2_FPGA_CONF_EN && !GEN3_FPGA_CONF_EN) Synchronous To: DISBL_MCFG_CLK ? "CG_Sysclk" : "CG_Cfgclk" Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
MP_TX_ExtAcc	O	TX Extended Access Flag. An active level of this signal marks the access to implementation-specific attributes of the target M-PHY TX module. Exists: !SNPS_FPGA_CONF_EN && !GEN2_FPGA_CONF_EN && !SNPS_MPHY_EXT_EN Synchronous To: DISBL_MCFG_CLK ? "CG_Sysclk" : "CG_Cfgclk" Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
MP_TX_AttrWrVal[7:0]	O	TX Attribute Write Value. This is the attribute write value sent to all M-PHY TX modules. Exists: !SNPS_FPGA_CONF_EN && (!GEN2_FPGA_CONF_EN && !GEN3_FPGA_CONF_EN) Synchronous To: DISBL_MCFG_CLK ? "CG_Sysclk" : "CG_Cfgclk" Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
MP_TX_AttrWRn	O	Attribute Write Enable. A high (1) value signals a write access, whereas a low (0) value signals a read access to all M-PHY TX modules. Exists: !SNPS_FPGA_CONF_EN && (!GEN2_FPGA_CONF_EN && !GEN3_FPGA_CONF_EN) Synchronous To: DISBL_MCFG_CLK ? "CG_Sysclk" : "CG_Cfgclk" Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A

Table 4-14 M-PHY TX Control Interface Signals (Continued)

Port Name	I/O	Description
MPx_TX_CfgEnbl[(P_NT-1):0]	O	TX Configuration Enable. This indicates to the M-PHY TX lane x to process the attribute access. Exists: !SNPS_FPGA_CONF_EN && (!GEN2_FPGA_CONF_EN && !GEN3_FPGA_CONF_EN) Synchronous To: DISBL_MCFG_CLK ? "CG_Sysclk" : "CG_Cfgclk" Registered: (SNPS_MPHY == 1) ? "No" : "Yes" Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
MP_TX_InLnCfg	O	TX In Lane Configuration. This signal indicates if a write access is performed to the shadow attribute or the effective attribute bank. This signal is not used by DWC_mipi_unipro and is always set to 0. Exists: !SNPS_FPGA_CONF_EN && (!GEN2_FPGA_CONF_EN && !GEN3_FPGA_CONF_EN) Synchronous To: None Registered: DISBL_MCFG_CLK ? "No" : "Yes" Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
MPx_TX_CfgRdyN[(P_NT-1):0]	I	TX Configuration Ready. By setting this signal to low (0), the M-PHY TX lane x shows that all configuration accesses are directly transferred to the effective attribute bank. A high (1) value shows that the shadow attribute bank is used. Exists: Always Synchronous To: DISBL_MCFG_CLK ? "CG_Sysclk" : "CG_Cfgclk" Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: Low
MPx_TX_CfgBusy[(P_NT-1):0]	I	TX Configuration Access Busy. When set, it indicates that the last configuration access to M-PHY TX lane x is not yet complete and more than one cycle is required to process the request. If this signal is not used by the configuration interface, it needs to be tied to 0. Exists: !SNPS_FPGA_CONF_EN && !GEN2_FPGA_CONF_EN && !SNPS_MPHY_EXT_EN Synchronous To: DISBL_MCFG_CLK ? "CG_Sysclk" : "CG_Cfgclk" Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

Table 4-14 M-PHY TX Control Interface Signals (Continued)

Port Name	I/O	Description
MPx_TX_AttrRdRdy[(P_NT-1):0]	I	TX Attribute Read Value Ready. Exists: VS_ATTR_RD_RDY Synchronous To: DISBL_MCFG_CLK ? "CG_Sysclk" : "CG_Cfgclk" Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
MPx_TX_AttrRdVal[(P_NT*8)-1:0]	I	TX Attribute Read Value. This is the attribute read value returned by each M-PHY TX. Exists: !SNPS_FPGA_CONF_EN && (!GEN2_FPGA_CONF_EN && !GEN3_FPGA_CONF_EN) Synchronous To: DISBL_MCFG_CLK ? "CG_Sysclk" : "CG_Cfgclk" Registered: (SNPS_MPHY >= 3) ? "Yes" : "No" Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
MPx_TX_CfgUpdt[(P_NT-1):0]	O	TX Configuration Update. This signal is used by target M-PHY TX lane x to initiate the transfer of attributes of the shadow attribute bank to the effective attribute bank in the next Save state. DWC_mipi_unipro activates this signal only during an active burst (except for a direct request of the application using VS_MphyCfgUpdt) and when the MPx_TX_CfgRdyN signal is de-asserted (high). The only exception is the hibernate exit request to the M-PHY lane. Exists: !SNPS_FPGA_CONF_EN && (!GEN2_FPGA_CONF_EN && !GEN3_FPGA_CONF_EN) Synchronous To: DISBL_MCFG_CLK ? "CG_Sysclk" : "CG_Cfgclk" Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
MPx_TX_Reset[(P_NT-1):0]	O	TX Reset. This is the asynchronous reset for M-PHY TX lane x to reset all state machines and attributes to their default value. Exists: Always Synchronous To: DISBL_MCFG_CLK ? "CG_Sysclk" : "CG_Cfgclk" Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: High

Table 4-14 M-PHY TX Control Interface Signals (Continued)

Port Name	I/O	Description
MPx_TX_LineReset[(P_NT-1):0]	O	TX Line Reset Request. This signal is asserted to send a line reset sequence M-PHY TX lane x. Exists: Always Synchronous To: DISBL_MCFG_CLK ? "CG_Sysclk" : "CG_Cfgclk" Registered: DISBL_MCFG_CLK ? "No" : "Yes" Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
MPx_TX_PwrEnbl[(P_NT-1):0]	O	TX PHY Power Enable. A value of 0 shows that the corresponding M-PHY TX lane is disconnected and can be powered-down. A value of 1 indicates that the M-PHY is active. Reset value: 1 Exists: Always Synchronous To: DISBL_MCFG_CLK ? "CG_Sysclk" : "CG_Cfgclk" Registered: (UPRO_LPM_EN == 0) ? "Yes" : "No" Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: High
MPx_TX_DIFNDRIVE[(P_NT-1):0]	O	Transmit DIFN driver. This port controls the DIFN level of the serial transmit lines when high TX output impedance is used in Sleep/Stall state. Exists: Always Synchronous To: DISBL_MCFG_CLK ? "CG_Sysclk" : "CG_Cfgclk" Registered: DISBL_MCFG_CLK ? "No" : "Yes" Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
MPx_TX_Control_ActTimer[(P_NT-1):0]	O	Transmit controller act timer. This port selects the Tactivate timer control between the MPHY and protocol. Exists: Always Synchronous To: None Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: High

4.15 Synopsys V5.0 MPHY Signals

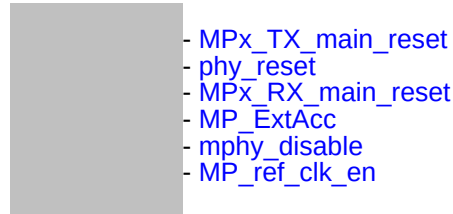


Table 4-15 Synopsys V5.0 MPHY Signals

Port Name	I/O	Description
MPx_TX_main_reset[(P_NT-1):0]	O	TX Lane Main Reset Signal. This is the asynchronous reset for the logic inside the Synopsys M-PHY TX lane x. Exists: SNPS_MPHY_EXT_EN SNPS_GEN3_MPHY_EXT_EN GEN3_FPGA_CONF_EN Synchronous To: None Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: High
phy_reset	O	This port signal is intended for Synopsys MPHY. This signal is driven with the value of the register VS_MphyRst.PRST. The default value of this signal after POR is high (asserted). Application need to de-assert this port through VS_MphyRst.PRST attribute after RMMI registers are programmed in the Synopsys MPHY. Exists: SNPS_GEN3_MPHY_EXT_EN Synchronous To: DISBL_MCFG_CLK ? "CG_Sysclk" : "CG_Cfgclk" Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: High
MPx_RX_main_reset[(P_NR-1):0]	O	RX Lane Main Reset Signal. This is the asynchronous reset for logic inside the Synopsys M-PHY RX lane x. Exists: SNPS_MPHY_EXT_EN SNPS_GEN3_MPHY_EXT_EN Synchronous To: None Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: High

Table 4-15 Synopsys V5.0 MPHY Signals (Continued)

Port Name	I/O	Description
MP_ExtAcc	O	MPHY Common Attribute Access Signal. This is the port to access the common attribute space of the MPHY with help of the TX_LANE 0 config interface. Exists: !SNPS_FPGA_CONF_EN && !GEN2_FPGA_CONF_EN && !SNPS_MPHY_EXT_EN Synchronous To: (DISBL_MCFG_CLK == 1) ? "CG_Sysclk" : "CG_Cfgclk" Registered: DISBL_MCFG_CLK ? "No" : "Yes" Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
mphy_disable	O	MPHY disable state control signal. This is the port to keep the GEN3 MPHY into disable state after reset release. Exists: SNPS_GEN3_MPHY_EXT_EN GEN3_FPGA_CONF_EN Synchronous To: CG_Sysclk Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: High
MP_ref_clk_en	O	Ref Clock Enable This signal indicates to the M-phy that ref clock is available and asserts, when the power mode is set to HS(Fast Mode/FastAuto Mode) and de-asserts in non-HS(Slow Mode/Slow Auto Mode). This signal needs to be connected to Mphy input port (ref_clk_en_unipro in synopsys Mphy), when unipro along with Mphy is used in device Mode It is left unconnected in host mode Exists: (P_UPRO_VER == 1) (P_UPRO_VER == 2) Synchronous To: CG_Sysclk Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

4.16 M-PHY RX Control Interface Signals

MPx_RX_CfgRdyN	-	MPx_RX_ResetMode
MPx_RX_CfgBusy	-	MP_RX_AttrID
MPx_RX_AttrRdRdy	-	MP_RX_ExtAcc
MPx_RX_AttrRdVal	-	MP_RX_AttrWrVal
MPx_RX_LineReset	-	MP_RX_AttrWRn
MPx_RX_LCCRdDet	-	MPx_RX_CfgEnbl
MPx_RX_Hibern8_Exit	-	MP_RX_InLnCfg
		MPx_RX_CfgUpdt
		MPx_RX_Reset
		MPx_RX_PwrEnbl

Table 4-16 M-PHY RX Control Interface Signals

Port Name	I/O	Description
MPx_RX_ResetMode[(P_NR-1):0]	O	RX Reset Mode. This signal defines the ResetMode for each RX lane. Value 0 - LS_MODE Value 1 - HS_MODE The value assigned to LA_ResetModeReq is reflected on this port. Exists: UPRO_LSS_HS_EN Synchronous To: None Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
MP_RX_AttrID[7:0]	O	RX Attribute ID. This is the address of the target attribute for read and write operations to all M-PHY RX lanes. Exists: !SNPS_FPGA_CONF_EN && (!GEN2_FPGA_CONF_EN && !GEN3_FPGA_CONF_EN) Synchronous To: DISBL_MCFG_CLK ? "CG_Sysclk" : "CG_Cfgclk" Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
MP_RX_ExtAcc	O	RX Extended Access Flag. An active level of this signal marks the access to implementation-specific attributes of the target M-PHY RX module. Exists: !SNPS_FPGA_CONF_EN && !GEN2_FPGA_CONF_EN && !SNPS_MPHY_EXT_EN Synchronous To: DISBL_MCFG_CLK ? "CG_Sysclk" : "CG_Cfgclk" Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

Table 4-16 M-PHY RX Control Interface Signals (Continued)

Port Name	I/O	Description
MP_RX_AttrWrVal[7:0]	O	RX Attribute Write Value. This is the attribute write value sent to all M-PHY RX modules. Exists: !SNPS_FPGA_CONF_EN && (!GEN2_FPGA_CONF_EN && !GEN3_FPGA_CONF_EN) Synchronous To: DISBL_MCFG_CLK ? "CG_Sysclk" : "CG_Cfgclk" Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
MP_RX_AttrWRn	O	RX Attribute Write Enable. A high (1) value signals a write access whereas a low (0) value signals a read access to all M-PHY modules. Exists: !SNPS_FPGA_CONF_EN && (!GEN2_FPGA_CONF_EN && !GEN3_FPGA_CONF_EN) Synchronous To: DISBL_MCFG_CLK ? "CG_Sysclk" : "CG_Cfgclk" Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
MPx_RX_CfgEnbl[(P_NR-1):0]	O	RX Configuration Enable. This signal indicates to the M-PHY RX lane x to process the current attribute access. Exists: !SNPS_FPGA_CONF_EN && (!GEN2_FPGA_CONF_EN && !GEN3_FPGA_CONF_EN) Synchronous To: DISBL_MCFG_CLK ? "CG_Sysclk" : "CG_Cfgclk" Registered: (SNPS_MPHY == 1) ? "No" : "Yes" Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
MP_RX_InLnCfg	O	RX In Lane Configuration. This signal indicates if a write access is performed to the shadow attribute or the effective attribute bank. This signal is unused by DWC_mipi_unipro and is always set to 0. Exists: !SNPS_FPGA_CONF_EN && (!GEN2_FPGA_CONF_EN && !GEN3_FPGA_CONF_EN) Synchronous To: None Registered: DISBL_MCFG_CLK ? "No" : "Yes" Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

Table 4-16 M-PHY RX Control Interface Signals (Continued)

Port Name	I/O	Description
MPx_RX_CfgRdyN[(P_NR-1):0]	I	RX Configuration Ready. By setting this signal to low (0), the M-PHY RX lane x shows that all write accesses are directly transferred to the effective attribute bank. A high (1) value shows that the shadow attribute bank is used. When a local reset (MPx_RX_Reset) is issued, the corresponding M-PHY asynchronously sets the MPx_RX_CfgRdyN signal to 1. Exists: Always Synchronous To: DISBL_MCFG_CLK ? "CG_Sysclk" : "CG_Cfgclk" Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: Low
MPx_RX_CfgBusy[(P_NR-1):0]	I	RX Configuration Access Busy. An active level signals that the last configuration access to M-PHY lane x is not yet complete and more than one cycle is required to process the request. If this signal is not used by the configuration interface, it needs to be tied to 0. Exists: !SNPS_FPGA_CONF_EN && !GEN2_FPGA_CONF_EN && !SNPS_MPHY_EXT_EN Synchronous To: DISBL_MCFG_CLK ? "CG_Sysclk" : "CG_Cfgclk" Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
MPx_RX_AttrRdRdy[(P_NR-1):0]	I	RX Attribute Read Value Ready. Exists: VS_ATTR_RD_RDY Synchronous To: DISBL_MCFG_CLK ? "CG_Sysclk" : "CG_Cfgclk" Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
MPx_RX_AttrRdVal[(P_NR*8)-1:0]	I	RX Attribute Read Value. This is the attribute read value returned from each M-PHY RX lane. Exists: !SNPS_FPGA_CONF_EN && (!GEN2_FPGA_CONF_EN && !GEN3_FPGA_CONF_EN) Synchronous To: DISBL_MCFG_CLK ? "CG_Sysclk" : "CG_Cfgclk" Registered: (SNPS_MPHY >= 3) ? "Yes" : "No" Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A

Table 4-16 M-PHY RX Control Interface Signals (Continued)

Port Name	I/O	Description
MPx_RX_CfgUpdt[(P_NR-1):0]	O	RX Configuration Update. This signal is used by target M-PHY RX lane x to initiate the transfer of attributes of the shadow attribute bank to the effective attribute bank in the next Save state. DWC_mipi_unipro activates this signal only during an active burst (except for a direct request of the application using VS_MphyCfgUpdt) and when the MPx_RX_CfgRdyN signal is de-asserted (high). The only exception is the hibernate exit request to the M-PHY lane. Exists: !SNPS_FPGA_CONF_EN && (!GEN2_FPGA_CONF_EN && !GEN3_FPGA_CONF_EN) Synchronous To: DISBL_MCFG_CLK ? "CG_Sysclk" : "CG_Cfgclk" Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
MPx_RX_Reset[(P_NR-1):0]	O	RX Reset. This is the asynchronous reset for M-PHY RX lane x to reset all state machines and attributes to their default value. Exists: Always Synchronous To: DISBL_MCFG_CLK ? "CG_Sysclk" : "CG_Cfgclk" Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: High
MPx_RX_LineReset[(P_NR-1):0]	I	RX Line Reset Request. This signal notifies the PA layer 1.5 that a line reset sequence is detected in the M-PHY RX lane x. Exists: Always Synchronous To: DISBL_MCFG_CLK ? "CG_Sysclk" : "CG_Cfgclk" Registered: (COMMON_RX_SYM_CLK == 1) ? "No" : ((SNPS_MPHY == 1) (SNPS_FPGA == 1)) ? "No" : "Yes" Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
MPx_RX_LCCRdDet[(P_NR-1):0]	I	RX LCC Read Detected. This signal informs layer 1.5 that a LCC read sequence is detected in the M-PHY lane x. Only the bit corresponding to logical lane 0 is processed internally. OMC capabilities from this lane are read out afterwards. Exists: P_LCC_LGC_EN Synchronous To: DISBL_MCFG_CLK ? "CG_Sysclk" : "CG_Cfgclk" Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

Table 4-16 M-PHY RX Control Interface Signals (Continued)

Port Name	I/O	Description
MPx_RX_Hibern8_Exit[(P_NR-1):0]	I	RX Hibernate State. This signal indicates if the M-PHY lane x is in Hibernate state (0) or outside (1). This input can be asynchronous because it is synchronized internally to the required clock signal as per the selected configuration. Exists: Always Synchronous To: None Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: High
MPx_RX_PwrEnbl[(P_NR-1):0]	O	RX PHY Power Enable. A value of 0 shows that the corresponding RX M-PHY lane is disconnected and can be powered down. A value of 1 indicates that the M-PHY is active. Reset value: 1 Exists: Always Synchronous To: DISBL_MCFG_CLK ? "CG_Sysclk" : "CG_Cfgclk" Registered: (UPRO_LPM_EN == 0) ? "Yes" : "No" Power Domain: {P_RETENTION_EN == 1 ? "TOP" : "ALWAYS_ON"} Active State: High

4.17 M-PHY TX Data Interface Signals

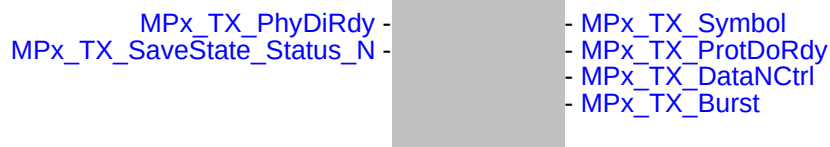


Table 4-17 M-PHY TX Data Interface Signals

Port Name	I/O	Description
MPx_TX_Symbol[((P_NT_SYM*20)-1):0]	O	TX Symbol Interface. Represents the RMMI/M-PHY Tx Interface over which one or more PA Layer symbols are transferred from UniPro to M-PHY. The number of such PA Layer symbols concurrently transferred is based on Number of Tx lanes (P_NT) and width of RMMI Interface (P_RMMI_DATAWIDTH) For every 16 bits of PA Layer Symbol, upper 4 bits are reserved. For example, For P_NT=1, MPx_TX_Symbol can be depicted as follows for various values of P_RMMI_DATAWIDTH: 20 bit RMMI : [19:16] = Reserved, [15:0]=PA[15:0] 40 bit RMMI : [39:32] = Reserved, [31:0]=PA[31:0] 80 bit RMMI : [79:64] = Reserved, [63:0]=PA[63:0] Exists: Always Synchronous To: CG_TxSymbolClk Registered: (P_NT == 2) ? "Yes" : (P_RMMI_DATAWIDTH == 40) ? "No" : "Yes" Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
MPx_TX_ProtDoRdy[((P_NT_SYM*2)-1):0]	O	Protocol Data Output Ready. This signal represents whether data is available on the TX Symbol bus for each byte (10-bit symbol) The number of such PA Layer data valid signals is based on Number of Tx lanes (P_NT) and width of RMMI Interface (P_RMMI_DATAWIDTH) Exists: Always Synchronous To: CG_TxSymbolClk Registered: ((SNPS_FPGA == 1) (SNPS_FPGA == 2)) ? "Yes" : "No" Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

Table 4-17 M-PHY TX Data Interface Signals (Continued)

Port Name	I/O	Description
MPx_TX_DataNCtrl[(P_NT_SYM*2)-1]:0]	O	Data/Control Select. Represents whether Tx Symbol on TX Symbol bus carries either data or control symbol for each byte (10-bit symbol). Value of 0 represents data byte and 1 represents control byte. The number of such PA Layer Data/control select signals is based on Number of Tx lanes (P_NT) and width of RMMI Interface (P_RMMI_DATAWIDTH) Exists: Always Synchronous To: CG_TxSymbolClk Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
MPx_TX_PhyDiRdy[(P_NT-1):0]	I	M-PHY Data Input Ready. This signal indicates that the M-PHY TX lane x is ready to receive new data. Exists: Always Synchronous To: (P_UPRO_VER == 2) ? "CG_TxSymbolClk" : "{CG_TxSymbolClk}, CG_Sysclk" Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
MPx_TX_Burst[(P_NT-1):0]	O	Burst Indicator. This signal indicates if the link should be activated and kept in the burst mode. Exists: Always Synchronous To: CG_TxSymbolClk Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
MPx_TX_SaveState_Status_N[(P_NT-1):0]	I	Save state status signal. This signal indicates if the M-PHY is in a Save state (Sleep/Stall). Exists: Always Synchronous To: CG_TxSymbolClk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: Low

4.18 M-PHY RX Data Interface Signals

MPx_RX_Symbol
MPx_RX_PhyDoRdy
MPx_RX_DataNCtrl
MPx_RX_SymbolErr
MPx_RX_Burst



Table 4-18 M-PHY RX Data Interface Signals

Port Name	I/O	Description
MPx_RX_Symbol[(((P_NR_SYM*20)-1):0)]	I	RX Symbol. Represents the RMMI/M-PHY Rx Interface over which one or more PA Layer symbols are received from M-PHY to UniPro. The number of such PA Layer symbols concurrently received is based on Number of Rx lanes (P_NR) and width of RMMI Interface (P_RMMI_DATAWIDTH). For every 16 bits of PA Layer Symbol, upper 4 bits are reserved. For example, For P_NR=1, MPx_RX_Symbol can be depicted as follows for various values of P_RMMI_DATAWIDTH : 20 bit RMMI : [19:16] = Reserved, [15:0]=PA[15:0] 40 bit RMMI : [39:32] = Reserved, [31:0]=PA[31:0] 80 bit RMMI : [79:64] = Reserved, [63:0]=PA[63:0] Exists: Always Synchronous To: Depending on the lane number, the corresponding 20-bit symbol is synchronous to CG_RxSymbolClkn, where n = 0, 1, 2, or 3. Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
MPx_RX_PhyDoRdy[(((P_NR_SYM*2)-1):0)]	I	M-PHY RX Data Output Ready. This signal represents whether the data is available on the RX Symbol bus for each byte (10-bit symbol). The number of such PA Layer data valid signals is based on Number of Rx lanes (P_NR) and width of RMMI Interface (P_RMMI_DATAWIDTH) Exists: Always Synchronous To: Depending on the lane number, the corresponding 2-bit signal is synchronous to CG_RxSymbolClkn, where n = 0, 1, 2, or 3. Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

Table 4-18 M-PHY RX Data Interface Signals (Continued)

Port Name	I/O	Description
MPx_RX_DataNCtrl[((P_NR_SYM*2)-1):0]	I	Data/Control Select. This signal represents whether RX Symbol bus carries either data or control symbol for each byte (10-bit symbol). Value of 0 represents data byte and 1 represents control byte. The number of such PA Layer Data/control select signals is based on Number of Rx lanes (P_NR) and width of RMMI Interface (P_RMMI_DATAWIDTH) Exists: Always Synchronous To: Depending on the lane number, the corresponding 2-bit signal is synchronous to CG_RxSymbolClkn, where n = 0, 1, 2, or 3. Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
MPx_RX_SymbolErr[((P_NR_SYM*2)-1):0]	I	M-PHY RX Error Indicator. Each bit indicates if an error is detected for the corresponding 10-bit M-PHY symbol. This signal is only processed if the burst is active, the lane synchronization is detected, if the MPx_RX_PhyDoRdy signal is active, and the end of burst (MK2) is not yet detected. Exists: Always Synchronous To: Depending on the lane number, the corresponding 2-bit signal is synchronous to CG_RxSymbolClkn, where n = 0, 1, 2, or 3. Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
MPx_RX_Burst[(P_NR-1):0]	I	M-PHY RX Lane x Burst Indicator. This signal indicates if the link is actively receiving. During link startup an active level of this signal defines that the lane is checked for occurrence of the trigger sequence. Exists: Always Synchronous To: Depending on the lane number, the corresponding 2-bit signal is synchronous to CG_RxSymbolClkn, where n = 0, 1, 2, or 3. Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High

4.19 Layer 2 RX Memory Signals (TC0)



Table 4-19 Layer 2 RX Memory Signals (TC0)

Port Name	I/O	Description
L2_RxBufWrAddrTC0[x-1:0]	O	Write Port Address. This is the address bus for TC0 RxFIFO write port. The width (x) of this bus depends on the depth of L2 TC0 RX Buffer selected during controller configuration. Exists: Always Synchronous To: CG_RxSymbolClkSelected Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
L2_RxBufWrEnTC0	O	Write Port Data Enable. When high, this signal indicates that L2_RxBufWrDataTC0 is written into the memory at the address specified by L2_RxBufWrAddrTC0. Exists: Always Synchronous To: CG_RxSymbolClkSelected Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
L2_RxBufWrDataTC0[((16*P_L2_RXBUF_WIDTH)-1):0]	O	Write Port Data. This bus carries the data to be written in the TC0 RxFIFO. It is valid when L2_RxBufWrEnTC0 is asserted high. The width of this data bus depends on the width of the RX CPort and the maximum number of M-PHY RX lanes selected during RTL configuration. The width is the maximum of the RX CPort width and 16 times the number of RX lanes. Exists: Always Synchronous To: CG_RxSymbolClkSelected Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A

Table 4-19 Layer 2 RX Memory Signals (TC0) (Continued)

Port Name	I/O	Description
L2_RxBufRdAddrTC0[x-1:0]	O	Read Port Address. This is the address bus for TC0 RxFIFO read port. The width (x) of this bus is the same as of L2_RxBufWrAddrTC0. Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
L2_RxBufRdEnTC0	O	Read Port Data Enable. When high, this signal indicates a read access from the memory location addressed by L2_RxBufRdAddrTC0. Exists: Always Synchronous To: CG_Sysclk Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
L2_RxBufRdDataTC0[((16*P_L2_RXBUF_WIDTH)-1):0]	I	Read Port Data Bus. This bus carries the data read from the TC0 RxFIFO corresponding to the L2_RxBufRdAddrTC0 given in the previous clock cycle. The width of this bus is the same as L2_RxBufWrDataTC0. Exists: Always Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A

4.20 Layer 2 RX Memory Signals (TC1)



Table 4-20 Layer 2 RX Memory Signals (TC1)

Port Name	I/O	Description
L2_RxBufWrAddrTC1[x-1:0]	O	Write Port Address. This is the address bus for TC1 RxFIFO write port. The width (x) of this bus depends on the depth of L2 TC1 RX Buffer selected during controller configuration. Exists: UPRO_HAS_TC1 Synchronous To: CG_RxSymbolClkSelected Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
L2_RxBufWrEnTC1	O	Write Port Data Enable. When high, this signal indicates that L2_RxBufWrDataTC1 is written into the memory at the address specified by L2_RxBufWrAddrTC1. Exists: UPRO_HAS_TC1 Synchronous To: CG_RxSymbolClkSelected Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
L2_RxBufWrDataTC1[((16*P_L2_RXBUF_WIDTH)-1):0]	O	Write Port Data. This bus carries the data to be written in the TC1 RxFIFO. It is valid when L2_RxBufWrEnTC1 is asserted high. The width of this data bus depends on the width of the RX CPort and the maximum number of M-PHY RX lanes selected during RTL configuration. The width is the maximum of the RX CPort width and 16 times the number of RX lanes. Exists: UPRO_HAS_TC1 Synchronous To: CG_RxSymbolClkSelected Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A

Table 4-20 Layer 2 RX Memory Signals (TC1) (Continued)

Port Name	I/O	Description
L2_RxBufRdAddrTC1[x-1:0]	O	Read Port Address. This is the address bus for TC1 RxFIFO read port. The width (x) of this bus is the same as that of L2_RxBufWrAddrTC1. Exists: UPRO_HAS_TC1 Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
L2_RxBufRdEnTC1	O	Read Port Data Enable. When high, this signal indicates a read access from the memory location addressed by L2_RxBufRdAddrTC1. Exists: UPRO_HAS_TC1 Synchronous To: CG_Sysclk Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
L2_RxBufRdDataTC1[((16*P_L2_RXBUF_WIDTH)-1):0]	I	Read Port Data Bus. This bus carries the data read from the TC1 RxFIFO corresponding to the L2_RxBufRdAddrTC1 given in the previous clock cycle. The width of this bus is the same as L2_RxBufWrDataTC1. Exists: UPRO_HAS_TC1 Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A

4.21 Layer 2 TX Memory Signals



Table 4-21 Layer 2 TX Memory Signals

Port Name	I/O	Description
<code>L2_TxBufWrAddr[x-1:0]</code>	O	Write Port Address. This is the address bus for common TC0-TC1 TxFIFO write port. The width (x) of this bus depends on the depth of TC0 and TC1 TX Buffer selected during controller configuration. Exists: Always Synchronous To: <code>CG_Sysclk</code> Registered: Yes Power Domain: {<code>P_RETENTION_EN</code> == 1 ? "TOP" : (<code>UPRO_TOP_SW</code> == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
<code>L2_TxBufWrEn</code>	O	Write Port Data Enable. When high, this signal indicates that <code>L2_TxBufWrData</code> is written into the memory at the address specified by <code>L2_TxBufWrAddr</code>. Exists: Always Synchronous To: <code>CG_Sysclk</code> Registered: Yes Power Domain: {<code>P_RETENTION_EN</code> == 1 ? "TOP" : (<code>UPRO_TOP_SW</code> == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
<code>L2_TxBufWrData[((16*P_L2_TXBUF_WIDTH)-1):0]</code>	O	Write Port Data. This bus carries the data to be written in the TxFIFO. It is valid when <code>L2_TxBufWrEn</code> is asserted high. The width of this data bus depends on the width of the TX CPort and the maximum number of M-PHY TX lanes selected during controller configuration. The width is the maximum of the TX CPort width and 16 times the number of TX lanes. Exists: Always Synchronous To: <code>CG_Sysclk</code> Registered: No Power Domain: {<code>P_RETENTION_EN</code> == 1 ? "TOP" : (<code>UPRO_TOP_SW</code> == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A

Table 4-21 Layer 2 TX Memory Signals (Continued)

Port Name	I/O	Description
L2_TxBufRdAddr[x-1:0]	O	Read Port Address. This is address bus for Tx FIFO read port. The width (x) of this bus is same as of L2_TxBufWrAddr. Exists: Always Synchronous To: CG_TxSymbolClk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
L2_TxBufRdEn	O	Read Port Data Enable. When high, this signal indicates a read access from the memory location addressed by L2_TxBufRdAddr. Exists: Always Synchronous To: CG_TxSymbolClk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
L2_TxBufRdData[((16*P_L2_TXBUF_WIDTH)-1):0]	I	Read Port Data Bus. This bus carries the data read from the Tx FIFO corresponding to the L2_TxBufRdAddr given in the previous clock cycle. The width of this bus is the same as L2_TxBufWrData. Exists: Always Synchronous To: CG_TxSymbolClk Registered: (P_DPP_EN == 1) ? "No" : "Yes" Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A

4.22 GPIO Signals

LA_Gpi -  - LA_Gpo

Table 4-22 GPIO Signals

Port Name	I/O	Description
LA_Gpo[(P_GPIO-1):0]	O	General Purpose Output. This signal reflects GPO field values in the VS_GPO_Ctrl_0/1/2 register. Exists: P_GPIO>0 Synchronous To: CG_Sysclk Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
LA_Gpi[(P_GPIO-1):0]	I	General Purpose Input. Value driven on this input is stored in GPI fields of VS_GPI_Ctrl_0/1/2 register. Exists: P_GPIO>0 Synchronous To: None Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A

4.23 Debug Interface Signals

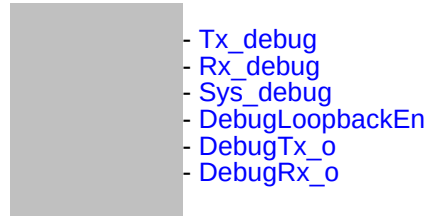


Table 4-23 Debug Interface Signals

Port Name	I/O	Description
Tx_debug[(P_TX_DBG_WID-1):0]	O	TX clock Debug. This signal is for monitoring purposes of Tx clock domain. Exists: P_DEBUG_PORTS_ENABLE Synchronous To: CG_TxSymbolClk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
Rx_debug[(P_RX_DBG_WID-1):0]	O	RX clock Debug. This signal is for monitoring purposes of Rx clock domain. Exists: P_DEBUG_PORTS_ENABLE Synchronous To: CG_RxSymbolClkSelected Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A
Sys_debug[(P_SYS_DBG_WID-1):0]	O	Sys clock Debug. This signal is for monitoring purposes of SYS clock domain. Exists: P_DEBUG_PORTS_ENABLE Synchronous To: CG_Sysclk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A

Table 4-23 Debug Interface Signals (Continued)

Port Name	I/O	Description
DebugLoopbackEn	O	Loopback Enable. This signal specifies that the controller is configured for the RMMI interface level loopback operation. The phase and frequency of the CG_TxSymbolClk and CG_RxSymbolClk signals must be same in the loopback mode. Power-on reset is required to exit from the loopback mode. Exists: Always Synchronous To: CG_Sysclk Registered: Yes Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: High
DebugTx_o[15:0]	O	TX Debug. This signal provides additional visibility into the PA lane distribution FSM in the TX side for monitoring purposes. ■ Bits [7:0]: Lane distribution FSM ■ Bit[8] : TXDataAccept from PA layer to DL layer ■ Bit[9] : TxsymbolEnable from DL layer to PA layer ■ Bit[10] : TXDataValid from DL to PA layer ■ Bit[11] : Data layer resume request ■ Bit[12] : Data layer pause indication ■ Bit[13] : Data layer pause request ■ Bit[14] : PA layer idle indication ■ Bit[15] : TX lane align indication w.r.t PhyDiRdy on all lanes Exists: Always Synchronous To: CG_TxSymbolClk Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A

Table 4-23 Debug Interface Signals (Continued)

Port Name	I/O	Description
DebugRx_o[(P_NR+16):0]	O	RX Debug. This signal provides additional visibility into the RX PA layer. ■ Bits[7:0] : Lane merge FSM ■ Bits[P_NR+7:8]: Sync error detection at the start of RX burst in each lane ■ Bit [P_NR+8] : Error during RxBurst de-assertion ■ Bit [P_NR+9] : RX SymbolEnable from PA layer to DL layer ■ Bit [P_NR+10] : RX DataValid from PA layer to DL layer ■ Bit [P_NR+11] : Power mode change request received on RX ■ Bit [P_NR+12] : Peer Get request received on RX ■ Bit [P_NR+13] : Peer Set request received on RX Exists: Always Synchronous To: CG_RxSymbolClkSelected Registered: No Power Domain: {P_RETENTION_EN == 1 ? "TOP" : (UPRO_TOP_SW == 0 ? "ALWAYS_ON" : "PWR_SW")} Active State: N/A

5

Register Descriptions

This chapter details all possible registers in the IP. They are arranged hierarchically into maps and blocks (banks). Your actual configuration might not contain all of these registers.

Attention: For configurable IP titles, do not use this document to determine the exact attributes of your register map. It is for reference purposes only.

When you configure the controller in coreConsultant, you must access the register attributes for your actual configuration at `workspace/report/ComponentRegisters.html` or `workspace/report/ComponentRegisters.xml` after you have completed the report creation activity. That report comes from the exact same source as this chapter but removes all the registers that are not in your actual configuration. This does not apply to non-configurable IP titles. In addition, all parameter expressions are evaluated to actual values. Therefore, the Offset and Memory Access values might change depending on your actual configuration.

Some expressions might refer to TCL functions or procedures (sometimes identified as **<functionof>**) that coreConsultant uses to make calculations. The exact formula used by these TCL functions is not provided in this chapter. However, when you configure the controller in coreConsultant, all TCL functions and parameters are evaluated completely; and the resulting values are displayed where appropriate in the coreConsultant GUI reports.

Exists Expressions

These expressions indicate the combination of configuration parameters required for a register, field, or block to exist in the memory map. The expression is only valid in the local context and does not indicate the conditions for existence of the parent. For example, the expression for a bit field in a register assumes that the register exists and does not include the conditions for existence of the register.

Offset

The term *Offset* is synonymous with *Address*.

Memory Access Attributes

The Memory Access attribute is defined as **<ReadBehavior>/<WriteBehavior>** which are defined in the following table.

Table 5-1 Possible Read and Write Behaviors

Read (or Write) Behavior	Description
RC	A read clears this register field.
RS	A read sets this register field.
RM	A read modifies the contents of this register field.
Wo	You can only write once to this register field.
W1C	A write of 1 clears this register field.
W1S	A write of 1 sets this register field.
W1T	A write of 1 toggles this register field.
W0C	A write of 0 clears this register field.
W0S	A write of 0 sets this register field.
W0T	A write of 0 toggles this register field.
WC	Any write clears this register field.
WS	Any write sets this register field.
WM	Any write toggles this register field.
no Read Behavior attribute	You cannot read this register. It is Write-Only.
no Write Behavior attribute	You cannot write to this register. It is Read-Only.

Table 5-2 Memory Access Examples

Memory Access	Description
R	Read-only register field.
W	Write-only register field.
R/W	Read/write register field.
R/W1C	You can read this register field. Writing 1 clears it.
RC/W1C	Reading this register field clears it. Writing 1 clears it.
R/Wo	You can read this register field. You can only write to it once.

Special Optional Attributes

Some register fields might use the following optional attributes.

Table 5-3 Optional Attributes

Attribute	Description
Volatile	As defined by the IP-XACT specification. If true, indicates in the case of a write followed by read, or in the case of two consecutive reads, there is no guarantee as to what is returned by the read on the second transaction or that this return value is consistent with the write or read of the first transaction. The element implies there is some additional mechanism by which this field can acquire new values other than by reads/writes/resets and other access methods known to IP-XACT. For example, when the core updates the register field contents.
Testable	As defined by the IP-XACT specification. Possible values are unconstrained, untestable, readOnly, writeAsRead, restore. Untestable means that this field is untestable by a simple automated register test. For example, the read-write access of the register is controlled by a pin or another register. readOnly means that you should not write to this register; only read from it. This might apply for a register that modifies the contents of another register.
Reset Mask	As defined by the IP-XACT specification. Indicates that this register field has an unknown reset value. For example, the reset value is set by another register or an input pin; or the register is implemented using RAM.
* Varies	Indicates that the memory access (or reset) attribute (read, write behavior) is not fixed. For example, the read-write access of the register is controlled by a pin or another register. Or when the access depends on some configuration parameter; in this case the post-configuration report in coreConsultant gives the actual access value.

Component Banks/Blocks

The following table shows the address blocks for each memory map. Follow the link for an address block to see a table of its registers.

Table 5-4 Address Banks/Blocks for Memory Map: DWC_mipi_unipro_map

Address Block	Description
DWC_mipi_unipro_apb_block on page 165	APB Indirect Access registers Exists: Always
DWC_mipi_unipro_l15_block on page 178	Layer 1.5 Attributes Exists: Always
DWC_mipi_unipro_l2_block on page 257	Layer 2 attributes Exists: Always

Table 5-4 Address Banks/Blocks for Memory Map: DWC_mipi_unipro_map (Continued)

Address Block	Description
DWC_mipi_unipro_l3_block on page 286	Layer 3 Attributes Exists: Always
DWC_mipi_unipro_l4_block on page 293	Layer 4 Attributes Exists: Always
DWC_mipi_unipro_ddb_block on page 332	DDB attributes Exists: Always
DWC_mipi_unipro_dme_qos_block on page 339	Quality of Service Attributes Exists: Always
DWC_mipi_unipro_vsl4_block on page 354	Vendor Specific Layer 4 Attributes Exists: Always
DWC_mipi_unipro_Dme_n_VS_block on page 357	Dme and Vendor Specific attributes Exists: Always

5.1 DWC_mipi_unipro_map/DWC_mipi_unipro_apb_block Registers

APB Indirect Access registers. Follow the link for the register to see a detailed description of the register.

Table 5-5 Registers for Address Block: DWC_mipi_unipro_map/DWC_mipi_unipro_apb_block

Register	Offset	Description
APB_Command on page 166	0x0	APB_Command
APB_Results on page 168	0x4	APB_Results
APB_Data on page 171	0x8	APB_Results
APB_InterruptEnable on page 172	0x10	APB_InterruptEnable
APB_InterruptStatus on page 175	0x14	APB_InterruptStatus

5.1.1 APB_Command

- **Name:** APB_Command
- **Description:** This register controls the access to the DME attributes when using the APB interface. It contains the Attribute ID and the type of attribute access to be performed.
- **Size:** 32 bits
- **Offset:** 0x0
- **Exists:** APB3_EN

31	30:20	19:4	3	2	1	0
Rsvd	SEL	AID	Rsvd	AT	DT	AAT

Table 5-6 Fields for Register: APB_Command

Bits	Name	Memory Access	Description
31			Reserved Field: Yes
30:20	SEL	R/W	Gen selector: This selects the target M-PHY port or C-Port. Value ranges from 0-7 for M-PHY port and 0-2047 for C-Port. Value After Reset: 0x0 Exists: Always
19:4	AID	R/W	Attribute ID: Indicates the attribute to be accessed. The upper eight bits (AID[19:12]) indicate the UniPro layer to be accessed and the lower eight bits indicate the attribute within the layer. Value After Reset: 0x0 Exists: Always
3			Reserved Field: Yes
2	AT	R/W	Attribute type: This controls the type of attribute access. Values: ■ 0x1 (ENABLED): Static(currently not supported for the local device) ■ 0x0 (DISABLED): Normal Value After Reset: 0x0 Exists: Always

Table 5-6 Fields for Register: APB_Command (Continued)

Bits	Name	Memory Access	Description
1	DT	R/W	Device type: This controls whether the attribute access is done on the local. UniPro device or forwarded to the peer device. Values: ■ 0x1 (ENABLED): Peer device attribute access ■ 0x0 (DISABLED): Local device attribute access Value After Reset: 0x0 Exists: Always
0	AAT	R/W	Attribute Access type: Indicates whether a read (get attribute) or write (set attribute) is performed in this access. Values: ■ 0x1 (ENABLED): Set or Write ■ 0x0 (DISABLED): Get or Read Value After Reset: 0x0 Exists: Always

5.1.2 APB_Results

- **Name:** APB_Results
- **Description:** This register indicates the result of the command given in the APB_Command register.
- **Size:** 32 bits
- **Offset:** 0x4
- **Exists:** APB3_EN

31:12	11:8	7:1	0
Rsvd	CMDR	Rsvd	CMDB

Table 5-7 Fields for Register: APB_Results

Bits	Name	Memory Access	Description
31:12			Reserved Field: Yes

Table 5-7 Fields for Register: APB_Results (Continued)

Bits	Name	Memory Access	Description
11:8	CMDR	R	This field outputs the result of the command. # - 0: Access is successful # - 1: Attribute ID is invalid # - 2: Attribute value is invalid # - 3: Write access to a read-only attribute # - 4: Read access to a write-only attribute # - 5: Selected CPort or PHY is out of range # - 6: Write access to a connected CPort is rejected or accessed PHY lane is busy # - 7: Selected test feature is out of range or PHY Error is flagged # - 8: PACP remote configuration error # - 9: Attribute access is rejected as layer is busy with previous operation # - 10: Addressed layer is in the wrong power state - Others: Reserved Values: 0x0 (SUCCESS): Access is successful 0x1 (ATTR_ID_INVALID): Attribute ID is invalid 0x2 (ATTR_VAL_INVALID): Attribute value is invalid 0x3 (WR_ACCESS): Write access to a read-only attribute 0x4 (RD_ACCESS): Read access to a write-only attribute 0x5 (OUT_OF_RANGE): Selected CPort or PHY is out of range 0x6 (BUSY): Write access to a connected CPort is rejected or accessed PHY lane is busy 0x7 (TST_FUTURE): Selected test feature is out of range or PHY Error is flagged 0x8 (PACP_ERROR): PACP remote configuration error 0x9 (ACCESS_REJECT): Attribute access is rejected as layer is busy with previous operation 0xa (WRONG_PWR_STATE): Addressed layer is in the wrong power state Value After Reset: 0x0 Exists: Always
7:1			Reserved Field: Yes

Table 5-7 Fields for Register: APB_Results (Continued)

Bits	Name	Memory Access	Description
0	CMDB	R	Host must write the next attribute access command in APB_Command register only when this bit is low: Values: ■ 0x0 (ABSENT): Command is executed and the result is indicated in the CMDR field ■ 0x1 (PRESENT): Command execution is in progress. This bit is set whenever a write operation is performed in the APB_Command register Value After Reset: 0x0 Exists: Always

5.1.3 APB_Data

- **Name:** APB_Results
- **Description:** This register contains the data read during GET_attribute access or the data to be written during SET_attribute access.
- **Size:** 32 bits
- **Offset:** 0x8
- **Exists:** APB3_EN

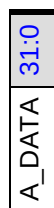


Table 5-8 Fields for Register: APB_Data

Bits	Name	Memory Access	Description
31:0	A_DATA	R/W	Data is written with the DME_SET command and data is read with the DME_GET command Value After Reset: 0x0 Exists: Always

5.1.4 APB_InterruptEnable

- **Name:** APB_InterruptEnable
- **Description:** Individual interrupt sources can be activated by writing 1 into the corresponding bit location. Initial value of each bit is 0 = disabled.
- **Size:** 32 bits
- **Offset:** 0x10
- **Exists:** APB3_EN

31	ICE
30:16	Rsvd
15	IE_QI
14	IE_LSC
13	IE_DC
12	IE_IE
11	IE_ETIE
10	IE_EN
9	IE_ED
8	IE_EPA
7	IE_EP
6	IE_TMI
5	IE_PMI
4	IE_HEXI
3	IE_HENI
2	IE_LLI
1	IE_LSI
0	IE_ERI

Table 5-9 Fields for Register: APB_InterruptEnable

Bits	Name	Memory Access	Description
31	ICE	R/W	When set to 1, DWC_mipi_unipro gives an interrupt when the command given in APB_Command register gets executed and the results are present in APB_Results/Data register. Value After Reset: 0x0 Exists: Always
30:16			Reserved Field: Yes
15	IE_QI	R/W	When set to 1, this bit enables the PHY link degradation indication interrupt (for more information, see QosInd). Value After Reset: 0x0 Exists: Always
14	IE_LSC	R/W	When set to 1, this bit enables the link startup confirmation interrupt (for more information, see LinkStartupCnf). Value After Reset: 0x0 Exists: Always
13	IE_DC	R/W	When set to 1, this bit enables the debug counter overflow interrupt (for more information, see VS_DebugCounterOverflow). Value After Reset: 0x0 Exists: Always

Table 5-9 Fields for Register: APB_InterruptEnable (Continued)

Bits	Name	Memory Access	Description
12	IE_IE	R/W	When set to 1, this bit enables the PA initialization error (L2) interrupt (for more information, see PA Init Error) Value After Reset: 0x0 Exists: Always
11	IE_ETIE	R/W	When set to 1, this bit enables the transport layer error indication PA interrupt (for more information, see ErrorTInd) Value After Reset: 0x0 Exists: Always
10	IE_EN	R/W	When set to 1, this bit enables the network layer error indication interrupt (for more information, see ErrorNInd) Value After Reset: 0x0 Exists: Always
9	IE_ED	R/W	When set to 1, this bit enables the data link layer error indication interrupt (for more information, see ErrorDInd) Value After Reset: 0x0 Exists: Always
8	IE_EPA	R/W	When set to 1, this bit enables the PA layer line reset error indication interrupt (for more information, see ErrorPAInd) Value After Reset: 0x0 Exists: Always
7	IE_EP	R/W	When set to 1, this bit enables the PHY error indication interrupt (for more information, see ErrorPHYInd) Value After Reset: 0x0 Exists: Always
6	IE_TMI	R/W	When set to 1, this bit enables the test mode indication interrupt (for more information, see TestModeInd) Value After Reset: 0x0 Exists: Always
5	IE_PMI	R/W	When set to 1, this bit enables the power mode indication interrupt (for more information, see PowerModeInd) Value After Reset: 0x0 Exists: Always
4	IE_HEXI	R/W	When set to 1, this bit enables the hibernate exit indication interrupt (for more information, see HibernateExitInd) Value After Reset: 0x0 Exists: Always

Table 5-9 Fields for Register: APB_InterruptEnable (Continued)

Bits	Name	Memory Access	Description
3	IE_HENI	R/W	When set to 1, this bit enables the hibernate enter indication interrupt (for more information, see HibernateEnterInd) Value After Reset: 0x0 Exists: Always
2	IE_LLI	R/W	When set to 1, this bit enables the link loss indication interrupt (for more information, see LinkLostInd) Value After Reset: 0x0 Exists: Always
1	IE_LSI	R/W	When set to 1, this bit enables the link startup indication interrupt (for more information, see LinkStartupInd) Value After Reset: 0x0 Exists: Always
0	IE_ERI	R/W	When set to 1, this bit enables the endpoint reset indication interrupt (for more information, see EndpointResetInd) Value After Reset: 0x0 Exists: Always

5.1.5 APB_InterruptStatus

- **Name:** APB_InterruptStatus
- **Description:** This register shows the occurrence of any of the defined interrupt sources independent of the interrupt enable setting.
- **Size:** 32 bits
- **Offset:** 0x14
- **Exists:** APB3_EN

31	IC
30:16	Rsvd
15	IS_QI
14	IS_LSC
13	IS_DC
12	IS_IE
11	IS_ETIE
10	IS_EN
9	IS_ED
8	IS_EPA
7	IS_EP
6	IS_TMI
5	IS_PMI
4	IS_HEXI
3	IS_HENI
2	IS_LLI
1	IS_LSI
0	IS_ERI

Table 5-10 Fields for Register: APB_InterruptStatus

Bits	Name	Memory Access	Description
31	IC	R	This is set when ICE is set and the command initiated in CR gets executed (In Prog bit gets cleared). This bit is cleared when the host reads this Interrupt Status Register. Value After Reset: 0x0 Exists: Always
30:16			Reserved Field: Yes
15	IS_QI	R	When this bit reads 1, it indicates PHY link degradation (for more information, see QosInd) Value After Reset: 0x0 Exists: Always
14	IS_LSC	R	When this bit reads 1, it indicates link startup confirmation (for more information, see LinkStartupCnf) Value After Reset: 0x0 Exists: Always
13	IS_DC	R	When this bit reads 1, it indicates debug counter overflow interrupt (for more information, see VS_DebugCounterOverflow) Value After Reset: 0x0 Exists: Always

Table 5-10 Fields for Register: APB_InterruptStatus (Continued)

Bits	Name	Memory Access	Description
12	IS_IE	R	When this bit reads 1, it indicates PA initialization error (L2) interrupt (for more information, see PA Init Error) Value After Reset: 0x0 Exists: Always
11	IS_ETIE	R	When this bit reads 1, it indicates transport layer error indication PA interrupt (for more information, see ErrorTInd) Value After Reset: 0x0 Exists: Always
10	IS_EN	R	When this bit reads 1, it indicates network layer error indication interrupt (for more information, see ErrorNInd) Value After Reset: 0x0 Exists: Always
9	IS_ED	R	When this bit reads 1, it indicates data link layer error indication interrupt (for more information, see ErrorDInd) Value After Reset: 0x0 Exists: Always
8	IS_EPA	R	When this bit reads 1, it indicates PA layer line reset error indication interrupt (for more information, see ErrorPAInd) Value After Reset: 0x0 Exists: Always
7	IS_EP	R	When this bit reads 1, it indicates PHY error indication interrupt (for more information, see ErrorPHYInd) Value After Reset: 0x0 Exists: Always
6	IS_TMI	R	When this bit reads 1, it indicates test mode indication interrupt (for more information, see TestModeInd) Value After Reset: 0x0 Exists: Always
5	IS_PMI	R	When this bit reads 1, it indicates power mode indication interrupt (for more information, see PowerModeInd) Value After Reset: 0x0 Exists: Always
4	IS_HEXI	R	When this bit reads 1, it indicates hibernate exit indication interrupt (for more information, see HibernateExitInd) Value After Reset: 0x0 Exists: Always

Table 5-10 Fields for Register: APB_InterruptStatus (Continued)

Bits	Name	Memory Access	Description
3	IS_HENI	R	When this bit reads 1, it indicates hibernate enter indication interrupt (for more information, see HibernateEnterInd) Value After Reset: 0x0 Exists: Always
2	IS_LLI	R	When this bit reads 1, it indicates link loss indication interrupt (for more information, see LinkLostInd) Value After Reset: 0x0 Exists: Always
1	IS_LSI	R	When this bit reads 1, it indicates link startup indication interrupt (for more information, see LinkStartupInd) Value After Reset: 0x0 Exists: Always
0	IS_ERI	R	When this bit reads 1, it indicates endpoint reset indication interrupt (for more information, see EndpointResetInd) Value After Reset: 0x0 Exists: Always

5.2 DWC_mipi_unipro_map/DWC_mipi_unipro_l15_block Registers

Layer 1.5 Attributes. Follow the link for the register to see a detailed description of the register.

Table 5-11 Registers for Address Block: DWC_mipi_unipro_map/DWC_mipi_unipro_l15_block

Register	Offset	Description
PA_PhyType on page 181	0x1500	PA_PhyType Register
PA_AvailTxDataLanes on page 182	0x1520	PA_AvailTxDataLanes
PA_AvailRxDataLanes on page 183	0x1540	PA_AvailRxDataLanes
PA_MinRxTrailingClocks on page 184	0x1543	PA_MinRxTrailingClocks
PA_TxHsG1SyncLength on page 185	0x1552	PA_TxHsG1SyncLength
PA_TxHsG1PrepareLength on page 186	0x1553	PA_TxHsG1PrepareLength
PA_TxHsG2SyncLength on page 187	0x1554	PA_TxHsG2SyncLength
PA_TxHsG2PrepareLength on page 188	0x1555	PA_TxHsG2PrepareLength
PA_TxHsG3SyncLength on page 189	0x1556	PA_TxHsG3SyncLength
PA_TxHsG3PrepareLength on page 190	0x1557	PA_TxHsG3PrepareLength
PA_TxMk2Extension on page 191	0x155a	PA_TxMk2Extension
PA_PeerScrambling on page 192	0x155b	PA_PeerScrambling
PA_TxSkip on page 193	0x155c	PA_TxSkip
PA_TxSkipPeriod on page 194	0x155d	PA_TxSkipPeriod
PA_Local_TX_LCC_Enable on page 195	0x155e	PA_Local_TX_LCC_Enable
PA_peer_TX_LCC_Enable on page 196	0x155f	PA_peer_TX_LCC_Enable
PA_ActiveTxDataLanes on page 197	0x1560	PA_ActiveTxDataLanes
PA_ConnectedTxDataLanes on page 198	0x1561	PA_ConnectedTxDataLanes
PA_TxTrailingClocks on page 199	0x1564	PA_TxTrailingClocks
PA_EoBDelay on page 200	0x1565	PA_EoBDelay
PA_TxPWRStatus on page 201	0x1567	PA_TxPWRStatus
PA_TxGear on page 202	0x1568	PA_TxGear
PA_TxTermination on page 203	0x1569	PA_TxTermination
PA_HSSeries on page 204	0x156a	PA_HSSeries
PA_PWRMode on page 205	0x1571	PA_PWRMode

Table 5-11 Registers for Address Block: DWC_mipi_unipro_map/DWC_mipi_unipro_l15_block (Continued)

Register	Offset	Description
PA_ActiveRxDataLanes on page 207	0x1580	PA_ActiveRxDataLanes
PA_ConnectedRxDataLanes on page 208	0x1581	PA_ConnectedRxDataLanes
PA_RxPWRStatus on page 209	0x1582	PA_RxPWRStatus
PA_RxGear on page 210	0x1583	PA_RxGear
PA_RxTermination on page 211	0x1584	PA_RxTermination
PA_Scrambling on page 212	0x1585	PA_Scrambling
PA_MaxRxPWMGear on page 213	0x1586	PA_MaxRxPWMGear
PA_MaxRxHSGear on page 214	0x1587	PA_MaxRxHSGear
PA_PACPRReqTimeout on page 215	0x1590	PA_PACPRReqTimeout
PA_PACPRReqEoBTimeout on page 216	0x1591	PA_PACPRReqEoBTimeout
PA_RemoteVerInfo on page 217	0x15a0	PA_RemoteVerInfo
PA_LogicalLaneMap on page 218	0x15a1	PA_LogicalLaneMap
PA_SleepNoConfigTime on page 220	0x15a2	PA_SleepNoConfigTime
PA_StallNoConfigTime on page 221	0x15a3	PA_StallNoConfigTime
PA_SaveConfigTime on page 222	0x15a4	PA_SaveConfigTime
PA_RxHSUntermCap on page 223	0x15a5	PA_RxHSUntermCap
PA_RxLSTerminationCapability on page 224	0x15a6	PA_RxLSTerminationCapability
PA_Hibern8Time on page 225	0x15a7	PA_Hibern8Time
PA_TActivate on page 226	0x15a8	PA_TActivate
PA_LocalVerInfo on page 227	0x15a9	PA_LocalVerInfo
PA_Granularity on page 228	0x15aa	PA_Granularity
PA_Mk2ExtensionGuardBand on page 229	0x15ab	PA_Mk2ExtensionGuardBand
PA_PWRModeUserData0 on page 230	0x15b0	PA_PWRModeUserData0
PA_PWRModeUserData1 on page 231	0x15b1	PA_PWRModeUserData1
PA_PWRModeUserData2 on page 232	0x15b2	PA_PWRModeUserData2
PA_PWRModeUserData3 on page 233	0x15b3	PA_PWRModeUserData3
PA_PWRModeUserData4 on page 234	0x15b4	PA_PWRModeUserData4

Table 5-11 Registers for Address Block: DWC_mipi_unipro_map/DWC_mipi_unipro_l15_block (Continued)

Register	Offset	Description
PA_PWRRModeUserData5 on page 235	0x15b5	PA_PWRRModeUserData5
PA_PWRRModeUserData6 on page 236	0x15b6	PA_PWRRModeUserData6
PA_PWRRModeUserData7 on page 237	0x15b7	PA_PWRRModeUserData7
PA_PWRRModeUserData8 on page 238	0x15b8	PA_PWRRModeUserData8
PA_PWRRModeUserData9 on page 239	0x15b9	PA_PWRRModeUserData9
PA_PWRRModeUserData10 on page 240	0x15ba	PA_PWRRModeUserData10
PA_PWRRModeUserData11 on page 241	0x15bb	PA_PWRRModeUserData11
PA_PACPFrameCount on page 242	0x15c0	PA_PACPFrameCount
PA_PACPErrorCount on page 243	0x15c1	PA_PACPErrorCount
PA_PhyTestControl on page 244	0x15c2	PA_PhyTestControl
VS_MphyRst on page 246	0x15c3	VS_MphyRst
PA_TxHsG4SyncLength on page 247	0x15d0	PA_TxHsG4SyncLength
PA_TxHsG4PrepareLength on page 248	0x15d1	PA_TxHsG4PrepareLength
PA_PeerRxHsAdaptRefresh on page 249	0x15d2	PA_PeerRxHsAdaptRefresh
PA_PeerRxHsAdaptInitial on page 250	0x15d3	PA_PeerRxHsAdaptInitial
PA_TxHsAdaptType on page 251	0x15d4	PA_TxHsAdaptType
PA_AdaptAfterLRSTInPA_INIT on page 252	0x15d5	PA_AdaptTypeInPa_Init
PA_TxHsG5SyncLength on page 253	0x15d6	PA_TxHsG5SyncLength
PA_TxHsG5PrepareLength on page 254	0x15d7	PA_TxHsG5PrepareLength
PA_PeerRxHsG5AdaptRefresh on page 255	0x15d8	PA_PeerRxHsG5AdaptRefresh
PA_PeerRxHsG5AdaptInitial on page 256	0x15d9	PA_PeerRxHsG5AdaptInitial

5.2.1 PA_PhyType

- **Name:** PA_PhyType Register
- **Description:** This attribute indicates the type of M-PHY.
- **Size:** 32 bits
- **Offset:** 0x1500
- **Exists:** Always

31:1	0
Rsvd	PT

Table 5-12 Fields for Register: PA_PhyType

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	PT	R	M-PHY type Values: ■ 0x0 (VALUE_OF_0): D-PHY (not implemented) ■ 0x1 (VALUE_OF_1): M-PHY Value After Reset: 0x1 Exists: Always

5.2.2 PA_AvailTxDataLanes

- **Name:** PA_AvailTxDataLanes
- **Description:** This attribute specifies the number of available TX data lanes.
- **Size:** 32 bits
- **Offset:** 0x1520
- **Exists:** Always

31:3	2:0
Rsvd	TXDL

Table 5-13 Fields for Register: PA_AvailTxDataLanes

Bits	Name	Memory Access	Description
31:3			Reserved Field: Yes
2:0	TXDL	R	This field returns the number of available TX data lanes. Values: ■ 0x0 (VALUE_OF_0): 1 lane ■ 0x1 (VALUE_OF_1): 2 lanes ■ 0x2 (VALUE_OF_2): 3 lanes ■ 0x3 (VALUE_OF_3): 4 lanes Value After Reset: P_NT Exists: Always

5.2.3 PA_AvailRxDataLanes

- **Name:** PA_AvailRxDataLanes
- **Description:** This attribute specifies the number of available RX data lanes.
- **Size:** 32 bits
- **Offset:** 0x1540
- **Exists:** Always

31:3	2:0
Rsvd	RXDL

Table 5-14 Fields for Register: PA_AvailRxDataLanes

Bits	Name	Memory Access	Description
31:3			Reserved Field: Yes
2:0	RXDL	R	This field returns the number of available RX data lanes. Values: ■ 0x0 (VALUE_OF_0): 1 lane ■ 0x1 (VALUE_OF_1): 2 lanes ■ 0x2 (VALUE_OF_2): 3 lanes ■ 0x3 (VALUE_OF_3): 4 lanes Value After Reset: P_NR Exists: Always

5.2.4 PA_MinRxTrailingClocks

- **Name:** PA_MinRxTrailingClocks
- **Description:** This attribute shows the minimum number of byte clock cycles without data after end of the burst.
- **Size:** 32 bits
- **Offset:** 0x1543
- **Exists:** Always



Table 5-15 Fields for Register: PA_MinRxTrailingClocks

Bits	Name	Memory Access	Description
x:0	MRTC	R	Minimum number of M-PHY byte clock cycles without the data to receive before a mode change from FAST_STATE or SLOW_STATE to SLEEP_STATE, or before a pause in data transmission while in FAST_STATE or SLOW_STATE. NOTE: The Width is increased to 10 bits from 8 bits to support the value up to 1023 for Gear-5 and future Gear speeds. In the Current version, it is limited to 255 for backward compatibility purposes (Interacting with lower version v1.8). Value After Reset: P_RXTRAIL Exists: Always Range Variable[x]: "UPRO_V2_EN ? \"10\" : \"8\" - 1

5.2.5 PA_TxHsG1SyncLength

- **Name:** PA_TxHsG1SyncLength
- **Description:** This attribute gives value of Sync length in high-speed gear 1 mode.
- **Size:** 8 bits
- **Offset:** 0x1552
- **Exists:** Always

7:6	5:0
SYNRANGE	SYNLEN

Table 5-16 Fields for Register: PA_TxHsG1SyncLength

Bits	Name	Memory Access	Description
7:6	SYNRANGE	R/W	This bit gives the fine or course range of the synch for high-speed gear 1: Values: ■ 0x1 (ENABLED): Coarse ■ 0x0 (DISABLED): Fine Value After Reset: 0x1 Exists: Always
5:0	SYNLEN	R/W	This value gives the value of synch length with fine or course value selection for high-speed gear 1. Value After Reset: 0xf Exists: Always

5.2.6 PA_TxHsG1PrepareLength

- **Name:** PA_TxHsG1PrepareLength
- **Description:** This attribute gives value of prepare length in high-speed gear 1 mode.
- **Size:** 8 bits
- **Offset:** 0x1553
- **Exists:** Always

7:4	3:0
Rsvd	PRPLEN

Table 5-17 Fields for Register: PA_TxHsG1PrepareLength

Bits	Name	Memory Access	Description
7:4			Reserved Field: Yes
3:0	PRPLEN	R/W	This value gives the prepare length value for high-speed gear 1 mode. Value After Reset: 0xf Exists: Always

5.2.7 PA_TxHsG2SyncLength

- **Name:** PA_TxHsG2SyncLength
- **Description:** This attribute gives value of Sync length in high-speed gear 2 mode.
- **Size:** 8 bits
- **Offset:** 0x1554
- **Exists:** Always

7:6	5:0
SYNRANGE	SYNLEN

Table 5-18 Fields for Register: PA_TxHsG2SyncLength

Bits	Name	Memory Access	Description
7:6	SYNRANGE	R/W	This bit gives the fine or course range of the synch for high-speed gear 2: Values: ■ 0x1 (ENABLED): Coarse ■ 0x0 (DISABLED): Fine Value After Reset: 0x1 Exists: Always
5:0	SYNLEN	R/W	This value gives the value of synch length with fine or course value selection for high-speed gear 2. Value After Reset: 0xf Exists: Always

5.2.8 PA_TxHsG2PrepareLength

- **Name:** PA_TxHsG2PrepareLength
- **Description:** This attribute gives value of prepare length in high-speed gear 2 mode.
- **Size:** 8 bits
- **Offset:** 0x1555
- **Exists:** Always

7:4	3:0
Rsvd	PRPLEN

Table 5-19 Fields for Register: PA_TxHsG2PrepareLength

Bits	Name	Memory Access	Description
7:4			Reserved Field: Yes
3:0	PRPLEN	R/W	This value gives the prepare length value for high-speed gear 2 mode. Value After Reset: 0xf Exists: Always

5.2.9 PA_TxHsG3SyncLength

- **Name:** PA_TxHsG3SyncLength
- **Description:** This attribute gives value of Sync length in high-speed gear 3 mode.
- **Size:** 8 bits
- **Offset:** 0x1556
- **Exists:** Always

7:6	5:0
SYNRANGE	SYNLEN

Table 5-20 Fields for Register: PA_TxHsG3SyncLength

Bits	Name	Memory Access	Description
7:6	SYNRANGE	R/W	This bit gives the fine or course range of the synch for high-speed gear 3: Values: ■ 0x1 (ENABLED): Coarse ■ 0x0 (DISABLED): Fine Value After Reset: 0x1 Exists: Always
5:0	SYNLEN	R/W	This value gives the value of synch length with fine or course value selection for high-speed gear 3. Value After Reset: 0xf Exists: Always

5.2.10 PA_TxHsG3PrepareLength

- **Name:** PA_TxHsG3PrepareLength
- **Description:** This attribute gives value of prepare length in high-speed gear 3 mode.
- **Size:** 8 bits
- **Offset:** 0x1557
- **Exists:** Always

7:4	3:0
Rsvd	PRPLEN

Table 5-21 Fields for Register: PA_TxHsG3PrepareLength

Bits	Name	Memory Access	Description
7:4			Reserved Field: Yes
3:0	PRPLEN	R/W	This value gives the prepare length value for high-speed gear 3 mode. Value After Reset: 0xf Exists: Always

5.2.11 PA_TxMk2Extension

- **Name:** PA_TxMk2Extension
- **Description:** This attribute specifies whether peer supports MK2 extension or not.
- **Size:** 32 bits
- **Offset:** 0x155a
- **Exists:** Always

31:1	
Rsvd	MK2EXT
	0

Table 5-22 Fields for Register: PA_TxMk2Extension

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	MK2EXT	R/W	This bit specifies that peer supports MK2 extension or not: Values: ■ 0x0 (VALUE_OF_0): Peer does not support MK2 extension. ■ 0x1 (VALUE_OF_1): Peer supports MK2 extension. Value After Reset: 0x0 Exists: Always

5.2.12 PA_PeerScrambling

- **Name:** PA_PeerScrambling
- **Description:** This attribute specifies whether peer supports scrambling or not.
- **Size:** 32 bits
- **Offset:** 0x155b
- **Exists:** Always

31:1	
Rsvd	PEERSCR
	0

Table 5-23 Fields for Register: PA_PeerScrambling

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	PEERSCR	R/W	This bit specifies that peer supports scrambling or not Values: ■ 0x0 (VALUE_OF_0): Peer does not support scrambling ■ 0x1 (VALUE_OF_1): Peer supports scrambling Value After Reset: 0x0 Exists: Always

5.2.13 PA_TxSkip

- **Name:** PA_TxSkip
- **Description:** This attribute specifies whether peer supports skip symbols or not.
- **Size:** 32 bits
- **Offset:** 0x155c
- **Exists:** Always

31:1	
Rsvd	PEERSKIP
	0

Table 5-24 Fields for Register: PA_TxSkip

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	PEERSKIP	R/W	This bit specifies that peer requires skip symbols or not Values: ■ 0x0 (VALUE_OF_0): Peer does not require skip symbols. ■ 0x1 (VALUE_OF_1): Peer require skip symbols. Value After Reset: 0x0 Exists: Always

5.2.14 PA_TxSkipPeriod

- **Name:** PA_TxSkipPeriod
- **Description:** This attribute specifies the number of data symbols after which the skip symbol should be added on all lanes.
- **Size:** 32 bits
- **Offset:** 0x155d
- **Exists:** Always

31:14	Rsvd
13:0	SKPTIME

Table 5-25 Fields for Register: PA_TxSkipPeriod

Bits	Name	Memory Access	Description
31:14			Reserved Field: Yes
13:0	SKPTIME	R/W	This value specifies the number of data symbols after which skip symbol should be added on lane if skip insertion is enabled. Value After Reset: 0xfa Exists: Always

5.2.15 PA_Local_TX_LCC_Enable

- **Name:** PA_Local_TX_LCC_Enable
- **Description:** This attribute specifies that local M-PHY line configuration is enabled or not.
- **Size:** 32 bits
- **Offset:** 0x155e
- **Exists:** Always

31:1	
Rsvd	LTXLCCEN
	0

Table 5-26 Fields for Register: PA_Local_TX_LCC_Enable

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	LTXLCCEN	R/W	This bit specifies that local M-PHY line configuration is enabled or not Values: ■ 0x0 (DISABLED): Local M-PHY line configuration is disabled ■ 0x1 (ENABLED): Local M-PHY line configuration is enabled Value After Reset: 0x0 Exists: Always

5.2.16 PA_peer_TX_LCC_Enable

- **Name:** PA_peer_TX_LCC_Enable
- **Description:** This attribute specifies that peer M-PHY line configuration is enabled or not.
- **Size:** 32 bits
- **Offset:** 0x155f
- **Exists:** Always

31:1	
Rsvd	PTXLCCEN
	0

Table 5-27 Fields for Register: PA_peer_TX_LCC_Enable

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	PTXLCCEN	R/W	This bit specifies that peer M-PHY line configuration is enabled or not Values: ■ 0x1 (ENABLED): Peer M-PHY line configuration is enabled ■ 0x0 (DISABLED): Peer M-PHY line configuration is disabled Value After Reset: 0x0 Exists: Always

5.2.17 PA_ActiveTxDataLanes

- **Name:** PA_ActiveTxDataLanes
- **Description:** This attribute specifies the number of TX data lanes in use. Write happens only to the shadow registers. The values are visible on read only after the successful power mode change. Follow the power-mode change sequence as described in *Power Mode Change* in the *Databook*.
- **Size:** 32 bits
- **Offset:** 0x1560
- **Exists:** Always

31:3	Rsvd
2:0	UTXDL

Table 5-28 Fields for Register: PA_ActiveTxDataLanes

Bits	Name	Memory Access	Description
31:3			Reserved Field: Yes
2:0	UTXDL	R/W	This field describes the number of TX data lanes in use Value After Reset: 0x1 Exists: Always

5.2.18 PA_ConnectedTxDataLanes

- **Name:** PA_ConnectedTxDataLanes
- **Description:** This attribute specifies the number of connected TX data lanes.
- **Size:** 32 bits
- **Offset:** 0x1561
- **Exists:** Always

31:2	1:0
Rsvd	CTXDL

Table 5-29 Fields for Register: PA_ConnectedTxDataLanes

Bits	Name	Memory Access	Description
31:2			Reserved Field: Yes
1:0	CTXDL	R	This field specifies the number of connected TX data lanes. Value After Reset: 0x0 Exists: Always

5.2.19 PA_TxTrailingClocks

- **Name:** PA_TxTrailingClocks
- **Description:** This attribute specifies the number of M-PHY byte clock cycles. Register Reset value is based on RESET_MODE field. If RESET_MODE is set as HS_MODE then Reset value will be 1023 otherwise 255
- **Size:** 32 bits
- **Offset:** 0x1564
- **Exists:** Always

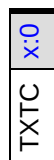


Table 5-30 Fields for Register: PA_TxTrailingClocks

Bits	Name	Memory Access	Description
x:0	TXTC	R/W	Number of M-PHY byte clock cycles after the EoB marker without the data before a mode change from FAST_STATE or SLOW_STATE to SLEEP_STATE, or before a data transmission pause while in FAST_STATE or SLOW_STATE. NOTE: The Width is increased to 10 bits from 8 bits to support the value up to 1023 for Gear-5 support and future Gear speeds. In current release, MSB two bits (TXTC[9:8]) are currently ignored and reserved. Value After Reset: "(UPRO_V2_EN==1) ? 1023 : 255" Exists: Always Range Variable[x]: "(UPRO_V2_EN) ? \"10\" : \"8\" - 1"

5.2.20 PA_EoBDelay

- **Name:** PA_EoBDelay
- **Description:** This attribute specifies Minimum number of PHY Symbols per lane to delay EoB transmission in FastAuto_Mode
- **Size:** 32 bits
- **Offset:** 0x1565
- **Exists:** Always

31:16	Rsvd
15:0	EOBDLY

Table 5-31 Fields for Register: PA_EoBDelay

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	EOBDLY	R/W	Specifies the minimum number of PHY symbols(FILLER symbols) per lane between the end of a valid frame and before the End of Burst in Fast Auto Mode. Value After Reset: 0x0 Exists: Always

5.2.21 PA_TxPWRStatus

- **Name:** PA_TxPWRStatus
- **Description:** This attribute specifies the TX power state.
- **Size:** 32 bits
- **Offset:** 0x1567
- **Exists:** Always

31:3	2:0
Rsvd	TXPS

Table 5-32 Fields for Register: PA_TxPWRStatus

Bits	Name	Memory Access	Description
31:3			Reserved Field: Yes
2:0	TXPS	R	This field indicates TX power state status: Values: ■ 0x0 (VALUE_OF_0): OFF_STATE ■ 0x1 (VALUE_OF_1): FAST_STATE ■ 0x2 (VALUE_OF_2): SLOW_STATE ■ 0x3 (VALUE_OF_3): HIBERNATE_STATE ■ 0x4 (VALUE_OF_4): SLEEP_STATE/STALL_STATE Value After Reset: 0x3 Exists: Always

5.2.22 PA_TxGear

- **Name:** PA_TxGear
- **Description:** This attribute specifies the TX gear in PWM or HS mode. Write happens only to shadow registers. The values are visible on read only after the successful power mode change. Follow the power-mode change sequence as described in *Power Mode Change* in the *Databook*.
- **Size:** 32 bits
- **Offset:** 0x1568
- **Exists:** Always

31:3	2:0
Rsvd	TXG

Table 5-33 Fields for Register: PA_TxGear

Bits	Name	Memory Access	Description
31:3			Reserved Field: Yes
2:0	TXG	R/W	TX gear in PWM mode: ■ 1: PWM_G1 ■ 2: PWM_G2 ■ 3: PWM_G3 ■ 4: PWM_G4 ■ 5: PWM_G5 ■ 6: PWM_G6 ■ 7: PWM_G7 TX gear in HS mode: ■ 1: HS_G1 ■ 2: HS_G2 ■ 3: HS_G3 ■ 4: HS_G4 ■ 5: HS_G5 Value After Reset: 0x1 Exists: Always

5.2.23 PA_TxTermination

- **Name:** PA_TxTermination
- **Description:** This attribute enables the TX termination. Write happens only to shadow registers. The values are visible on read only after the successful power mode change. Follow the power-mode change sequence as described in *Power Mode Change* in the *Databook*. Register Reset value is based on RESET_MODE field. If ResetMode is set as HS_MODE, then reset value will be 0x1(Enabled) otherwise 0x0(Disabled).
- **Size:** 32 bits
- **Offset:** 0x1569
- **Exists:** Always

31:1	0
Rsvd	TXT

Table 5-34 Fields for Register: PA_TxTermination

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	TXT	R/W	When set to 1, this bit enables the TX termination. Value After Reset: 0x0 Exists: Always

5.2.24 PA_HSSeries

- **Name:** PA_HSSeries
- **Description:** This attribute specifies the frequency series in high-speed mode. Write happens only to shadow registers. The values are visible on read only after the successful power mode change. Follow the power-mode change sequence as described in *Power Mode Change* in the *Databook*.
- **Size:** 32 bits
- **Offset:** 0x156a
- **Exists:** Always

31:2	1:0
Rsvd	HSS

Table 5-35 Fields for Register: PA_HSSeries

Bits	Name	Memory Access	Description
31:2			Reserved Field: Yes
1:0	HSS	R/W	This field specifies the frequency series in high-speed mode. Values: ■ 0x1 (VALUE_OF_1): Series A ■ 0x2 (VALUE_OF_2): Series B Value After Reset: 0x1 Exists: Always

5.2.25 PA_PWRMode

- **Name:** PA_PWRMode
- **Description:** This attribute specifies the TX/RX power mode. Write happens only to shadow registers. The values are visible on read only after the successful power mode change. Follow the power-mode change sequence as described in *Power Mode Change* in the *Databook*. Register Reset value is based on RESET_MODE field. If ResetMode is set as HS_MODE, then reset value will be 0x44(FastAuto_Mode) otherwise 0x55(SlowAuto_Mode).
- **Size:** 32 bits
- **Offset:** 0x1571
- **Exists:** Always

	31:8	7:4	3:0
Rsvd	PMRX	PMTX	

Table 5-36 Fields for Register: PA_PWRMode

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:4	PMRX	R/W	RX Power Mode: Writing this attribute triggers a power mode change which includes exchanging PACP packets between local and peer. Values: ■ 0x1 (VALUE_OF_1): Fast_Mode ■ 0x2 (VALUE_OF_2): Slow_Mode(Invalid in HS_MODE) ■ 0x4 (VALUE_OF_4): FastAuto_Mode ■ 0x5 (VALUE_OF_5): SlowAuto_Mode(Invalid in HS_MODE) ■ 0x7 (VALUE_OF_7): UNCHANGED Value After Reset: 0x5 Exists: Always

Table 5-36 Fields for Register: PA_PWRMode (Continued)

Bits	Name	Memory Access	Description
3:0	PMTX	R/W	TX Power Mode: Writing this attribute triggers a power mode change which includes exchanging PACP packets between local and peer. Values: ■ 0x1 (VALUE_OF_1): Fast_Mode ■ 0x2 (VALUE_OF_2): Slow_Mode(Invalid in HS_MODE) ■ 0x4 (VALUE_OF_4): FastAuto_Mode ■ 0x5 (VALUE_OF_5): SlowAuto_Mode(Invalid in HS_MODE) ■ 0x7 (VALUE_OF_7): UNCHANGED Value After Reset: 0x5 Exists: Always

5.2.26 PA_ActiveRxDataLanes

- **Name:** PA_ActiveRxDataLanes
- **Description:** This attribute specifies the number of RX data lanes. Write happens only to shadow registers. The values are visible on read only after the successful power mode change. Follow the power-mode change sequence as described in *Power Mode Change* in the *Databook*.
- **Size:** 32 bits
- **Offset:** 0x1580
- **Exists:** Always

31:3	URXDL
Rsvd	2:0

Table 5-37 Fields for Register: PA_ActiveRxDataLanes

Bits	Name	Memory Access	Description
31:3			Reserved Field: Yes
2:0	URXDL	R/W	This field specifies the number of RX data lanes in use. Value After Reset: 0x1 Exists: Always

5.2.27 PA_ConnectedRxDataLanes

- **Name:** PA_ConnectedRxDataLanes
- **Description:** This attribute specifies the number of connected RX data lanes.
- **Size:** 32 bits
- **Offset:** 0x1581
- **Exists:** Always

31:2	Rsvd
1:0	CRXDL

Table 5-38 Fields for Register: PA_ConnectedRxDataLanes

Bits	Name	Memory Access	Description
31:2			Reserved Field: Yes
1:0	CRXDL	R	This field specifies the number of connected RX data lanes Value After Reset: 0x0 Exists: Always

5.2.28 PA_RxPWRStatus

- **Name:** PA_RxPWRStatus
- **Description:** This attribute specifies the RX power state.
- **Size:** 32 bits
- **Offset:** 0x1582
- **Exists:** Always

31:3	2:0
Rsvd	RXPS

Table 5-39 Fields for Register: PA_RxPWRStatus

Bits	Name	Memory Access	Description
31:3			Reserved Field: Yes
2:0	RXPS	R	RX Power State Status: Values: ■ 0x0 (VALUE_OF_0): OFF_STATE ■ 0x1 (VALUE_OF_1): FAST_STATE ■ 0x2 (VALUE_OF_2): SLOW_STATE ■ 0x3 (VALUE_OF_3): HIBERNATE_STATE ■ 0x4 (VALUE_OF_4): SLEEP_STATE/STALL_STATE Value After Reset: 0x3 Exists: Always

5.2.29 PA_RxGear

- **Name:** PA_RxGear
- **Description:** This attribute specifies the RX gear in PWM or HS mode. Write happens only to shadow registers. The values are visible on read only after the successful power mode change. Follow the power-mode change sequence as described in *Power Mode Change* in the *Databook*.
- **Size:** 32 bits
- **Offset:** 0x1583
- **Exists:** Always

31:3	2:0
Rsvd	RXG

Table 5-40 Fields for Register: PA_RxGear

Bits	Name	Memory Access	Description
31:3			Reserved Field: Yes
2:0	RXG	R/W	RX Gear in PWM Mode: ■ 1: PWM_G1 ■ 2: PWM_G2 ■ 3: PWM_G3 ■ 4: PWM_G4 ■ 5: PWM_G5 ■ 6: PWM_G6 ■ 7: PWM_G7 RX Gear in HS Mode: ■ 1: HS_G1 ■ 2: HS_G2 ■ 3: HS_G3 ■ 4: HS_G4 ■ 5: HS_G5 Value After Reset: 0x1 Exists: Always

5.2.30 PA_RxTermination

- **Name:** PA_RxTermination
- **Description:** This attribute enables the RX termination. Write happens only to shadow registers. The values are visible on read only after the successful power mode change. Follow the power-mode change sequence as described in *Power Mode Change* in the *Databook*. Register Reset value is based on RESET_MODE field. If ResetMode is set as HS_MODE, then reset value will be 0x1(Enabled) otherwise 0x0(Disabled).
- **Size:** 32 bits
- **Offset:** 0x1584
- **Exists:** Always

31:1	0
Rsvd	RXT

Table 5-41 Fields for Register: PA_RxTermination

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	RXT	R/W	When set to 1, this bit enables the RX termination. Value After Reset: 0x0 Exists: Always

5.2.31 PA_Scrambling

- **Name:** PA_Scrambling
- **Description:** This attribute controls the scrambling operation.
- **Size:** 32 bits
- **Offset:** 0x1585
- **Exists:** Always

31:1	
Rsvd	SCREN

Table 5-42 Fields for Register: PA_Scrambling

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	SCREN	R/W	This bit is used to request the scrambling operation after it is successful power mode change if PA_PeerScrambling is enable #- 0 : scrambling request off #- 1 : scrambling request on Note: Write happens only to the shadow register. Read returns the effective value. If the value read is 1, it means the link is currently using the scrambler. If the value read is 0, it means the scrambler is not used by link currently. Values: ■ 0x1 (ENABLED): scrambling request on ■ 0x0 (DISABLED): scrambling request off Value After Reset: 0x0 Exists: Always

5.2.32 PA_MaxRxPWMGear

- **Name:** PA_MaxRxPWMGear
- **Description:** This attribute specifies the maximum RX low-speed gears (PWM).
- **Size:** 32 bits
- **Offset:** 0x1586
- **Exists:** Always

31:3	Rsvd
2:0	MRXPG

Table 5-43 Fields for Register: PA_MaxRxPWMGear

Bits	Name	Memory Access	Description
31:3			Reserved Field: Yes
2:0	MRXPG	R/W	Maximum RX Low-Speed Gears (PWM) Values: ■ 0x1 (VALUE_OF_1): PWM_G1 ■ 0x2 (VALUE_OF_2): PWM_G2 ■ 0x3 (VALUE_OF_3): PWM_G3 ■ 0x4 (VALUE_OF_4): PWM_G4 ■ 0x5 (VALUE_OF_5): PWM_G5 ■ 0x6 (VALUE_OF_6): PWM_G6 ■ 0x7 (VALUE_OF_7): PWM_G7 Value After Reset: 0x1 Exists: Always

5.2.33 PA_MaxRxHSGear

- **Name:** PA_MaxRxHSGear
- **Description:** This attribute specifies the maximum RX high-speed gears (HS).
- **Size:** 32 bits
- **Offset:** 0x1587
- **Exists:** Always

31:3	2:0
Rsvd	MRXHG

Table 5-44 Fields for Register: PA_MaxRxHSGear

Bits	Name	Memory Access	Description
31:3			Reserved Field: Yes
2:0	MRXHG	R/W	Maximum RX High-Speed Gears (HS). A value of 0 indicates that high speed is not available Values: ■ 0x0 (VALUE_OF_0): NO_HS ■ 0x1 (VALUE_OF_1): HS_G1 ■ 0x2 (VALUE_OF_2): HS_G2 ■ 0x3 (VALUE_OF_3): HS_G3 ■ 0x4 (VALUE_OF_4): HS_G4 ■ 0x5 (VALUE_OF_5): HS_G5 Value After Reset: 0x0 Exists: Always

5.2.34 PA_PACPReqTimeout

- **Name:** PA_PACPReqTimeout
- **Description:** This attribute specifies the expiration value of the PACP_REQUEST_TIMER.
- **Size:** 32 bits
- **Offset:** 0x1590
- **Exists:** Always

31:8	7:0
Rsvd	PRT

Table 5-45 Fields for Register: PA_PACPReqTimeout

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	PRT	R/W	Expiration value of the PACP_REQUEST_TIMER. The actual value is the value defined by this attribute +/- 10%. Value After Reset: 0x3f Exists: Always

5.2.35 PA_PACPReqEoBTimeout

- **Name:** PA_PACPReqEoBTimeout
- **Description:** This attribute specifies the expiry value of the PACP_REQUEST_EoB_TIMER.
- **Size:** 32 bits
- **Offset:** 0x1591
- **Exists:** Always

31:4	3:0
Rsvd	PRET

Table 5-46 Fields for Register: PA_PACPReqEoBTimeout

Bits	Name	Memory Access	Description
31:4			Reserved Field: Yes
3:0	PRET	R/W	Expiry value of the PACP_REQUEST_EoB_TIMER. The actual value is the value defined by the attribute +/- 10%. Value After Reset: 0xf Exists: Always

5.2.36 PA_RemoteVerInfo

- **Name:** PA_RemoteVerInfo
- **Description:** This attribute specifies the peer device version.
- **Size:** 32 bits
- **Offset:** 0x15a0
- **Exists:** Always

31:16	15:0
Rsvd	RVI

Table 5-47 Fields for Register: PA_RemoteVerInfo

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	RVI	R	This field specifies the version information of the peer device. Value After Reset: 0x0 Exists: Always

5.2.37 PA_LogicalLaneMap

- **Name:** PA_LogicalLaneMap
- **Description:** This attribute provides the logical lane map as calculated during link startup phase. For example, in case of 4 TX and 4 RX, all lanes connected in parallel, the lane map reads 0xe4e4. This means, TX lane 0 is logical TX lane 0, lane 1 is logical lane 1, and so on. If physical lane 2 TX and physical lane 1 RX are connected, the lane map is 0x0102.
- **Size:** 32 bits
- **Offset:** 0x15a1
- **Exists:** Always

	31:16
	15:14
	13:12
	11:10
	9:8
	7:6
	5:4
	3:2
	1:0
Rsvd	
LLMRX3	
LLMRX2	
LLMRX1	
LLMRX0	
LLMTX3	
LLMTX2	
LLMTX1	
LLMTX0	

Table 5-48 Fields for Register: PA_LogicalLaneMap

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:14	LLMRX3	R	Physical lane of logical RX lane 3 Value After Reset: 0x0 Exists: Always
13:12	LLMRX2	R	Physical lane of logical RX lane 2 Value After Reset: 0x0 Exists: Always
11:10	LLMRX1	R	Physical lane of logical RX lane 1 Value After Reset: 0x0 Exists: Always
9:8	LLMRX0	R	Physical lane of logical RX lane 0 Value After Reset: 0x0 Exists: Always
7:6	LLMTX3	R	Physical lane of logical TX lane 3 Value After Reset: 0x0 Exists: Always

Table 5-48 Fields for Register: PA_LogicalLaneMap (Continued)

Bits	Name	Memory Access	Description
5:4	LLMTX2	R	Physical lane of logical TX lane 2 Value After Reset: 0x0 Exists: Always
3:2	LLMTX1	R	Physical lane of logical TX lane 1 Value After Reset: 0x0 Exists: Always
1:0	LLMTX0	R	Physical lane of logical TX lane 0 Value After Reset: 0x0 Exists: Always

5.2.38 PA_SleepNoConfigTime

- **Name:** PA_SleepNoConfigTime
- **Description:** This attribute specifies the minimum time (in SI) needed in the SLEEP state when inline configuration is not performed, that is, M-PHY is not reprogrammed during burst.
- **Size:** 32 bits
- **Offset:** 0x15a2
- **Exists:** Always

31:16	15:0
Rsvd	SLNCT

Table 5-49 Fields for Register: PA_SleepNoConfigTime

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	SLNCT	R/W	This field specifies the minimum time (in SI) needed in the SLEEP state when the inline configuration is not performed, that is, M-PHY is not reprogrammed during burst. Value After Reset: 0xf Exists: Always

5.2.39 PA_StallNoConfigTime

- **Name:** PA_StallNoConfigTime
- **Description:** This attribute configures the minimum time (in SI) needed in the STALL state when inline configuration is not performed.
- **Size:** 32 bits
- **Offset:** 0x15a3
- **Exists:** Always

31:16	Rsvd
15:0	STNCT

Table 5-50 Fields for Register: PA_StallNoConfigTime

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	STNCT	R/W	This field specifies the minimum time (in SI) needed in the STALL state when the inline configuration is not performed. Value After Reset: 0xff Exists: Always

5.2.40 PA_SaveConfigTime

- **Name:** PA_SaveConfigTime
- **Description:** This attribute configures the minimum reconfiguration time. Reset value is 100us during LSS in HS_MODE and 10us during LSS in LS_MODE
- **Size:** 32 bits
- **Offset:** 0x15a4
- **Exists:** Always

31:16	15:0
Rsvd	SACT

Table 5-51 Fields for Register: PA_SaveConfigTime

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	SACT	R/W	This field specifies the minimum reconfiguration time in 1us steps if Unipro operates in V2.0 mode, else reconfiguration time is in 40ns steps. This applies only to SLEEP and STALL states Value After Reset: 0xa Exists: Always

5.2.41 PA_RxHSUntermCap

- **Name:** PA_RxHSUntermCap
- **Description:** This attribute specifies the HS untermination capability.
- **Size:** 32 bits
- **Offset:** 0x15a5
- **Exists:** Always

31:1	
Rsvd	RXHSUC
	0

Table 5-52 Fields for Register: PA_RxHSUntermCap

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	RXHSUC	R/W	When read 1, this bit indicates that HS untermination capability is received with PACP_CAP packet. Value After Reset: 0x0 Exists: Always

5.2.42 PA_RxLSTerminationCapability

- **Name:** PA_RxLSTerminationCapability
- **Description:** This attribute specifies the PWM termination capability.
- **Size:** 32 bits
- **Offset:** 0x15a6
- **Exists:** Always

31:1	
Rsvd	RXPTC

Table 5-53 Fields for Register: PA_RxLSTerminationCapability

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	RXPTC	R	When read 1, this bit indicates that PWM termination capability is received with PACP_CAP packet. Value After Reset: 0x0 Exists: Always

5.2.43 PA_Hibern8Time

- **Name:** PA_Hibern8Time
- **Description:** The PA layer delays any hibernate exit operations until the timer has expired. A value of zero disables the waiting of HIBERN8 time.
- **Size:** 32 bits
- **Offset:** 0x15a7
- **Exists:** Always

31:14	13:0
Rsvd	H8T

Table 5-54 Fields for Register: PA_Hibern8Time

Bits	Name	Memory Access	Description
31:14			Reserved Field: Yes
13:0	H8T	R/W	The PA layer starts the timer when a TX lane enters the HIBERN8 state. The PA layer delays any hibernate exit operations until the timer has expired. A value of zero disables the waiting of HIBERN8 time. Note: Internally UniPro selects the max of the downgraded value and written value of this attribute by application after lss process. Value After Reset: 0x80 Exists: Always

5.2.44 PA_TActivate

- **Name:** PA_TActivate
- **Description:** This attribute specifies the time to wait in the Save state before activating a burst. Register Reset value is based on RESET_MODE. If RESET_MODE is set as HS_MODE then reset value will be 4 otherwise 16
- **Size:** 32 bits
- **Offset:** 0x15a8
- **Exists:** Always

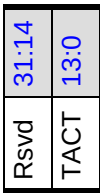


Table 5-55 Fields for Register: PA_TActivate

Bits	Name	Memory Access	Description
31:14			Reserved Field: Yes
13:0	TACT	R/W	This field specifies the time to wait in the Save state before activating a burst to wakeup. OMC and remote M-RX. Note: Internally UniPro selects the max of the downgraded value and written value of this attribute by application after lss process. Value After Reset: 0x10 Exists: Always

5.2.45 PA_LocalVerInfo

- **Name:** PA_LocalVerInfo
- **Description:** This attribute specifies the local version.
- **Size:** 32 bits
- **Offset:** 0x15a9
- **Exists:** Always

31:16	15:0
Rsvd	LVI

Table 5-56 Fields for Register: PA_LocalVerInfo

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	LVI	R/W	This field specifies the local version information. Value After Reset: "(UPRO_V2_EN==1) ? \"0x6\" : \"0x5\"" Exists: Always

5.2.46 PA_Granularity

- **Name:** PA_Granularity
- **Description:** This attribute specifies the granularity for TActivate and Hibern8Time attributes.
- **Size:** 32 bits
- **Offset:** 0x15aa
- **Exists:** Always

31:3	2:0
Rsvd	GRNL

Table 5-57 Fields for Register: PA_Granularity

Bits	Name	Memory Access	Description
31:3			Reserved Field: Yes
2:0	GRNL	R/W	This value specifies the different granularity for TActivate and Hibern8Time: # - 1 : 1 us # - 2 : 4 us # - 3 : 8 us # - 4 : 16 us # - 5 : 32 us # - 6 : 100 us Note: Values 0 and 7 are reserved. By default, 100 us is set for these reserved values. Values: ■ 0x1 (VALUE_OF_1): 1 us ■ 0x2 (VALUE_OF_2): 4 us ■ 0x3 (VALUE_OF_3): 8 us ■ 0x4 (VALUE_OF_4): 16 us ■ 0x5 (VALUE_OF_5): 32 us ■ 0x6 (VALUE_OF_6): 100 us Value After Reset: 0x6 Exists: Always

5.2.47 PA_Mk2ExtensionGuardBand

- **Name:** PA_Mk2ExtensionGuardBand
- **Description:** This attribute specifies the MK2 extension time required between start of M-LANE-PREPARE.confirm and the M-LANE-SYMBOL.request for the <MK0>, Head-of-Burst.

Note: When this attribute is set to '0'(default), the additional guard time follows the attribute PA_TActivate.

- **Size:** 32 bits
- **Offset:** 0x15ab
- **Exists:** Always

31:14	Rsvd
13:0	MK2EXTGUARD

Table 5-58 Fields for Register: PA_Mk2ExtensionGuardBand

Bits	Name	Memory Access	Description
31:14			Reserved Field: Yes
13:0	MK2EXTGUARD	R/W	This field specifies the required time extension between M-LANEPREPARE.confirm and the M-LANE-SYMBOL request for transmission of the head of the burst. When it is set to 0, the additional guard time follows the attribute PA_TActivate Note: The unit of this attribute depends on PA_granularity Note: This additional guard time value is used to run a MicroSecond timer inside DWC_mipi_unipro. The timer has a tolerance value of +/-10% or +/- 1 microsecond whichever is more Value After Reset: 0x0 Exists: Always

5.2.48 PA_PWRModeUserData0

- **Name:** PA_PWRModeUserData0
- **Description:** This attribute specifies the user mode data 0 to transmit.
- **Size:** 32 bits
- **Offset:** 0x15b0
- **Exists:** Always

31:16	15:0
Rsvd	UD0

Table 5-59 Fields for Register: PA_PWRModeUserData0

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	UD0	R/W	This field specifies the user mode data to transmit with PACP_PWR.req and PACP_PWR.cnf frames. Value After Reset: 0x0 Exists: Always

5.2.49 PA_PWRModeUserData1

- **Name:** PA_PWRModeUserData1
- **Description:** This attribute specifies the user mode data 1 to transmit.
- **Size:** 32 bits
- **Offset:** 0x15b1
- **Exists:** Always

31:16	15:0
Rsvd	UD1

Table 5-60 Fields for Register: PA_PWRModeUserData1

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	UD1	R/W	This field specifies user mode data to transmit with PACP_PWR.req and PACP_PWR.cnf frames. Value After Reset: 0x0 Exists: Always

5.2.50 PA_PWRModeUserData2

- **Name:** PA_PWRModeUserData2
- **Description:** This attribute specifies the user mode data 2 to transmit.
- **Size:** 32 bits
- **Offset:** 0x15b2
- **Exists:** Always

31:16	15:0
Rsvd	UD2

Table 5-61 Fields for Register: PA_PWRModeUserData2

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	UD2	R/W	This field specifies user mode data to transmit with PACP_PWR.req and PACP_PWR.cnf frames. Value After Reset: 0x0 Exists: Always

5.2.51 PA_PWRModeUserData3

- **Name:** PA_PWRModeUserData3
- **Description:** This attribute specifies the user mode data 3 to transmit.
- **Size:** 32 bits
- **Offset:** 0x15b3
- **Exists:** Always

31:16	15:0
Rsvd	UD3

Table 5-62 Fields for Register: PA_PWRModeUserData3

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	UD3	R/W	This field specifies user mode data to transmit with PACP_PWR.req and PACP_PWR.cnf frames. Value After Reset: 0x0 Exists: Always

5.2.52 PA_PWRModeUserData4

- **Name:** PA_PWRModeUserData4
- **Description:** This attribute specifies the user mode data 4 to transmit.
- **Size:** 32 bits
- **Offset:** 0x15b4
- **Exists:** Always

31:16	15:0
Rsvd	UD4

Table 5-63 Fields for Register: PA_PWRModeUserData4

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	UD4	R/W	This field specifies user mode data to transmit with PACP_PWR.req and PACP_PWR.cnf frames. Value After Reset: 0x0 Exists: Always

5.2.53 PA_PWRModeUserData5

- **Name:** PA_PWRModeUserData5
- **Description:** This attribute specifies the user mode data 5 to transmit.
- **Size:** 32 bits
- **Offset:** 0x15b5
- **Exists:** Always

31:16	15:0
Rsvd	UD5

Table 5-64 Fields for Register: PA_PWRModeUserData5

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	UD5	R/W	This field specifies user mode data to transmit with PACP_PWR.req and PACP_PWR.cnf frames. Value After Reset: 0x0 Exists: Always

5.2.54 PA_PWRModeUserData6

- **Name:** PA_PWRModeUserData6
- **Description:** This attribute specifies the user mode data 6 to transmit.
- **Size:** 32 bits
- **Offset:** 0x15b6
- **Exists:** Always

31:16	15:0
Rsvd	UD6

Table 5-65 Fields for Register: PA_PWRModeUserData6

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	UD6	R/W	This field specifies user mode data to transmit with PACP_PWR.req and PACP_PWR.cnf frames. Value After Reset: 0x0 Exists: Always

5.2.55 PA_PWRModeUserData7

- **Name:** PA_PWRModeUserData7
- **Description:** This attribute specifies the user mode data 7 to transmit.
- **Size:** 32 bits
- **Offset:** 0x15b7
- **Exists:** Always

31:16	15:0
Rsvd	UD7

Table 5-66 Fields for Register: PA_PWRModeUserData7

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	UD7	R/W	This field specifies user mode data to transmit with PACP_PWR.req and PACP_PWR.cnf frames. Value After Reset: 0x0 Exists: Always

5.2.56 PA_PWRModeUserData8

- **Name:** PA_PWRModeUserData8
- **Description:** This attribute specifies the user mode data 8 to transmit.
- **Size:** 32 bits
- **Offset:** 0x15b8
- **Exists:** Always

31:16	15:0
Rsvd	UD8

Table 5-67 Fields for Register: PA_PWRModeUserData8

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	UD8	R/W	This field specifies user mode data to transmit with PACP_PWR.req and PACP_PWR.cnf frames. Value After Reset: 0x0 Exists: Always

5.2.57 PA_PWRModeUserData9

- **Name:** PA_PWRModeUserData9
- **Description:** This attribute specifies the user mode 9 data to transmit.
- **Size:** 32 bits
- **Offset:** 0x15b9
- **Exists:** Always

31:16	15:0
Rsvd	UD9

Table 5-68 Fields for Register: PA_PWRModeUserData9

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	UD9	R/W	This field specifies user mode data to transmit with PACP_PWR.req and PACP_PWR.cnf frames. Value After Reset: 0x0 Exists: Always

5.2.58 PA_PWRModeUserData10

- **Name:** PA_PWRModeUserData10
- **Description:** This attribute specifies the user mode data 10 to transmit.
- **Size:** 32 bits
- **Offset:** 0x15ba
- **Exists:** Always

31:16	15:0
Rsvd	UD10

Table 5-69 Fields for Register: PA_PWRModeUserData10

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	UD10	R/W	This field specifies user mode data to transmit with PACP_PWR.req and PACP_PWR.cnf frames. Value After Reset: 0x0 Exists: Always

5.2.59 PA_PWRModeUserData11

- **Name:** PA_PWRModeUserData11
- **Description:** This attribute specifies the user mode data 11 to transmit.
- **Size:** 32 bits
- **Offset:** 0x15bb
- **Exists:** Always

31:16	15:0
Rsvd	UD11

Table 5-70 Fields for Register: PA_PWRModeUserData11

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	UD11	R/W	This field specifies user mode data to transmit with PACP_PWR.req and PACP_PWR.cnf frames. Value After Reset: 0x0 Exists: Always

5.2.60 PA_PACPFrameCount

- **Name:** PA_PACPFrameCount
- **Description:** This attribute specifies the number of valid PACP frames received. Perform a dummy write to clear the counter.
- **Size:** 32 bits
- **Offset:** 0x15c0
- **Exists:** Always

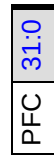


Table 5-71 Fields for Register: PA_PACPFrameCount

Bits	Name	Memory Access	Description
31:0	PFC	R/W	This field specifies the number of valid PACP frames received. Value After Reset: 0x0 Exists: Always

5.2.61 PA_PACPErrrorCount

- **Name:** PA_PACPErrrorCount
- **Description:** This attribute specifies the number of erroneous PACP frames received Perform a dummy write to clear the counter.
- **Size:** 32 bits
- **Offset:** 0x15c1
- **Exists:** Always

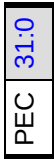


Table 5-72 Fields for Register: PA_PACPErrrorCount

Bits	Name	Memory Access	Description
31:0	PEC	R/W	This field specifies the number of erroneous PACP frames received Value After Reset: 0x0 Exists: Always

5.2.62 PA_PhyTestControl

- **Name:** PA_PhyTestControl
- **Description:** This attribute controls the M-PHY test feature.
- **Size:** 32 bits
- **Offset:** 0x15c2
- **Exists:** Always

31:6	5	4	3	2	1	0
Rsvd	TADP	TPS	TPT	LRST	CRD	CTBT

Table 5-73 Fields for Register: PA_PhyTestControl

Bits	Name	Memory Access	Description
31:6			Reserved Field: Yes
5	TADP	R/W	ADAPT The PA layer generates the adapt request for all active tx data lanes. Value After Reset: 0x0 Exists: (UPRO_V18_EN)
4	TPS	R/W	TestPatternSelect The PA layer generates the CJTPAT sequence when TestPatternSelect is set to 0 and generates the CRPAT sequence when TestPatternSelect is set to 1. Value After Reset: 0x0 Exists: Always
3	TPT	R/W	TestPatternTransmit The PA layer transmits the test patterns when TestPatternTransmit is set. It stops the transmission when it is cleared. Value After Reset: 0x0 Exists: Always
2	LRST	R/W	LineReset When LineReset is set to 1, the PA layer TX issues a MLANELINERESET request on all PA_ConnectedTxDataLanes. This bit is automatically cleared. Value After Reset: 0x0 Exists: Always

Table 5-73 Fields for Register: PA_PhyTestControl (Continued)

Bits	Name	Memory Access	Description
1	CRD	R/W	CfgReady. When CfgReady is set to 1, the PA layer TX issues a MLANECFGREADY request to all M-TXs and M-RXs. This bit is automatically cleared. Value After Reset: 0x0 Exists: Always
0	CTBT	R/W	ContBurst Values: 0x0 (VALUE_OF_0): If ContBurst is set to 0, the PA layer TX creates a new burst for each transmitted PACP frame. When there is no data to be sent, the M-TX is in Save state. 0x1 (VALUE_OF_1): If ContBurst is set to 1, the PA layer TX keeps the M-TX in Burst mode. When there is no data to be sent, the PA layer TX sends fillers. Value After Reset: 0x0 Exists: Always

5.2.63 VS_MphyRst

- **Name:** VS_MphyRst
- **Description:** This attribute controls phy_reset output port
- **Size:** 32 bits
- **Offset:** 0x15c3
- **Exists:** Always

31:1	
Rsvd	PRST

Table 5-74 Fields for Register: VS_MphyRst

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	PRST	R/W	The value of this field controls the phy_reset output port. Clearing this bit will de-assert the phy_reset output signal Value After Reset: 0x1 Exists: Always

5.2.64 PA_TxHsG4SyncLength

- **Name:** PA_TxHsG4SyncLength
- **Description:** This attribute gives value of Sync length in high-speed gear 4 mode.
- **Size:** 8 bits
- **Offset:** 0x15d0
- **Exists:** Always

7:6	5:0
SYNRANGE	SYNLEN

Table 5-75 Fields for Register: PA_TxHsG4SyncLength

Bits	Name	Memory Access	Description
7:6	SYNRANGE	R/W	This bit gives the fine or coarse range of the synch for high-speed gear 4: Values: ■ 0x1 (ENABLED): Coarse ■ 0x0 (DISABLED): Fine Value After Reset: 0x1 Exists: Always
5:0	SYNLEN	R/W	This value gives the value of synch length with fine or course value selection for high-speed gear 4. Value After Reset: 0xf Exists: Always

5.2.65 PA_TxHsG4PrepareLength

- **Name:** PA_TxHsG4PrepareLength
- **Description:** This attribute gives value of prepare length in high-speed gear 4 mode.
- **Size:** 8 bits
- **Offset:** 0x15d1
- **Exists:** Always

7:4	3:0
Rsvd	PRPLEN

Table 5-76 Fields for Register: PA_TxHsG4PrepareLength

Bits	Name	Memory Access	Description
7:4			Reserved Field: Yes
3:0	PRPLEN	R/W	This value gives the prepare length value for high-speed gear 4 mode. Value After Reset: 0xf Exists: Always

5.2.66 PA_PeerRxHsAdaptRefresh

- **Name:** PA_PeerRxHsAdaptRefresh
- **Description:** This attribute gives value of Refresh adaption pattern length.
- **Size:** 8 bits
- **Offset:** 0x15d2
- **Exists:** Always

7	6:0
ADAPTSEL	ADAPTREF

Table 5-77 Fields for Register: PA_PeerRxHsAdaptRefresh

Bits	Name	Memory Access	Description
7	ADAPTSEL	R/W	Fine or coarse select for Refresh adapt pattern. # - 0 : Fine # - 1 : coarse Values: ■ 0x0 (VALUE_OF_0): Fine ■ 0x1 (VALUE_OF_1): Coarse Value After Reset: 0x0 Exists: Always
6:0	ADAPTREF	R/W	Refresh adapt pattern length for selected fine or coarse adapt. Value After Reset: 0x0 Exists: Always

5.2.67 PA_PeerRxHsAdaptInitial

- **Name:** PA_PeerRxHsAdaptInitial
- **Description:** This attribute gives value of initial adaption pattern length.
- **Size:** 8 bits
- **Offset:** 0x15d3
- **Exists:** Always

7	6:0
ADAPTSEL	ADAPTINIT

Table 5-78 Fields for Register: PA_PeerRxHsAdaptInitial

Bits	Name	Memory Access	Description
7	ADAPTSEL	R/W	Fine or coarse select for Initial adapt pattern. # - 0 : Fine # - 1 : coarse Values: ■ 0x0 (VALUE_OF_0): Fine ■ 0x1 (VALUE_OF_1): Coarse Value After Reset: 0x0 Exists: Always
6:0	ADAPTINIT	R/W	Initial adapt pattern length for selected fine or coarse adapt. Value After Reset: 0x0 Exists: Always

5.2.68 PA_TxHsAdaptType

- **Name:** PA_TxHsAdaptType
- **Description:** This attribute gives value to select adapt pattern type.
- **Size:** 8 bits
- **Offset:** 0x15d4
- **Exists:** Always

7:2	1:0
Rsvd	ADAPTTYPE

Table 5-79 Fields for Register: PA_TxHsAdaptType

Bits	Name	Memory Access	Description
7:2			Reserved Field: Yes
1:0	ADAPTTYPE	R/W	Adapt pattern type. # - 00 : Refresh Adapt # - 01 : Initial Adapt # - 10 : Reserved # - 11 : No Adapt Values: ■ 0x0 (VAL_00): Refresh Adapt ■ 0x1 (VAL_01): Initial Adapt ■ 0x2 (VAL_10): Reserved ■ 0x3 (VAL_11): No Adapt Value After Reset: 0x3 Exists: Always

5.2.69 PA_AdaptAfterLRSTInPA_INIT

- **Name:** PA_AdaptTypeInPa_Init
- **Description:** This attribute gives value to select adapt pattern type during PA_Init operation. During Link startup procedure in HS_MODE this attribute stands invalid and write operation has no effect on functionality.
- **Size:** 8 bits
- **Offset:** 0x15d5
- **Exists:** Always

7:2	1:0
Rsvd	ADAPTTYPE

Table 5-80 Fields for Register: PA_AdaptAfterLRSTInPA_INIT

Bits	Name	Memory Access	Description
7:2			Reserved Field: Yes
1:0	ADAPTTYPE	R/W	Adapt pattern type. # - 00 : Refresh Adapt # - 01 : Initial Adapt # - 10 : Adapt Hold # - 11 : No Adapt Values: ■ 0x0 (VAL_00): Refresh Adapt ■ 0x1 (VAL_01): Initial Adapt ■ 0x2 (VAL_10): Adapt Hold ■ 0x3 (VAL_11): No Adapt Value After Reset: 0x2 Exists: Always

5.2.70 PA_TxHsG5SyncLength

- **Name:** PA_TxHsG5SyncLength
- **Description:** This attribute gives value of Sync length in high-speed gear 5 mode.
- **Size:** 8 bits
- **Offset:** 0x15d6
- **Exists:** Always

7:6	5:0
SYNRANGE	SYNLEN

Table 5-81 Fields for Register: PA_TxHsG5SyncLength

Bits	Name	Memory Access	Description
7:6	SYNRANGE	R/W	This bit gives the fine or coarse range of the synch for high-speed gear 5 Values: ■ 0x1 (ENABLED): Coarse ■ 0x0 (DISABLED): Fine Value After Reset: 0x1 Exists: Always
5:0	SYNLEN	R/W	This value gives the value of synch length with fine or course value selection for high-speed gear 5. Value After Reset: 0xf Exists: Always

5.2.71 PA_TxHsG5PrepareLength

- **Name:** PA_TxHsG5PrepareLength
- **Description:** This attribute gives value of prepare length in high-speed gear 5 mode.
- **Size:** 8 bits
- **Offset:** 0x15d7
- **Exists:** Always

7:4	3:0
Rsvd	PRPLEN

Table 5-82 Fields for Register: PA_TxHsG5PrepareLength

Bits	Name	Memory Access	Description
7:4			Reserved Field: Yes
3:0	PRPLEN	R/W	This value gives the prepare length value for high-speed gear 5 mode. Value After Reset: 0xf Exists: Always

5.2.72 PA_PeerRxHsG5AdaptRefresh

- **Name:** PA_PeerRxHsG5AdaptRefresh
- **Description:** This attribute gives value of Refresh adaption pattern length.
- **Size:** 8 bits
- **Offset:** 0x15d8
- **Exists:** Always

7	6:0
ADAPTSEL	ADAPTREF

Table 5-83 Fields for Register: PA_PeerRxHsG5AdaptRefresh

Bits	Name	Memory Access	Description
7	ADAPTSEL	R/W	Fine or coarse select for Refresh adapt pattern. # - 0 : Fine # - 1 : coarse Values: ■ 0x0 (VALUE_OF_0): Fine ■ 0x1 (VALUE_OF_1): Coarse Value After Reset: 0x0 Exists: Always
6:0	ADAPTREF	R/W	Refresh adapt pattern length for selected fine or coarse adapt. Value After Reset: 0x0 Exists: Always

5.2.73 PA_PeerRxHsG5AdaptInitial

- **Name:** PA_PeerRxHsG5AdaptInitial
- **Description:** This attribute gives value of initial adaption pattern length.
- **Size:** 8 bits
- **Offset:** 0x15d9
- **Exists:** Always

7	6:0
ADAPTSEL	ADAPTINIT

Table 5-84 Fields for Register: PA_PeerRxHsG5AdaptInitial

Bits	Name	Memory Access	Description
7	ADAPTSEL	R/W	Fine or coarse select for Initial adapt pattern. # - 0 : Fine # - 1 : coarse Values: ■ 0x0 (VALUE_OF_0): Fine ■ 0x1 (VALUE_OF_1): Coarse Value After Reset: 0x0 Exists: Always
6:0	ADAPTINIT	R/W	Initial adapt pattern length for selected fine or coarse adapt. Value After Reset: 0x0 Exists: Always

5.3 DWC_mipi_unipro_map/DWC_mipi_unipro_l2_block Registers

Layer 2 attributes. Follow the link for the register to see a detailed description of the register.

Table 5-85 Registers for Address Block: DWC_mipi_unipro_map/DWC_mipi_unipro_l2_block

Register	Offset	Description
DL_TxPreemptionCap on page 259	0x2000	DL_TxPreemptionCap
DL_TC0TxMaxSDUSize on page 260	0x2001	DL_TC0TxMaxSDUSize
DL_TC0RxInitCreditVal on page 261	0x2002	DL_TC0RxInitCreditVal
DL_TC1TxMaxSDUSize on page 262	0x2003	DL_TC1TxMaxSDUSize
DL_TC1RxInitCreditVal on page 263	0x2004	DL_TC1RxInitCreditVal
DL_TC0TxBufferSize on page 264	0x2005	DL_TC0TxBufferSize
DL_TC1TxBufferSize on page 265	0x2006	DL_TC1TxBufferSize
DL_MTU on page 266	0x2007	DL_MTU
DL_SYMBOL_MTU on page 267	0x2008	DL_SYMBOL_MTU
DL_CreditUnitSize on page 268	0x2009	DL_CreditUnitSize
DL_TC0TxFCThreshold on page 269	0x2040	DL_TC0TxFCThreshold
DL_FC0ProtectionTimeOutVal on page 270	0x2041	DL_FC0ProtectionTimeOutVal
DL_TC0ReplayTimeOutVal on page 271	0x2042	DL_TC0ReplayTimeOutVal
DL_AFC0ReqTimeOutVal on page 272	0x2043	DL_AFC0ReqTimeOutVal
DL_AFC0CreditThreshold on page 273	0x2044	DL_AFC0CreditThreshold
DL_TC0OutAckThreshold on page 274	0x2045	DL_TC0OutAckThreshold
DL_PeerTC0Present on page 275	0x2046	DL_PeerTC0Present
DL_PeerTC0RxInitCreditVal on page 276	0x2047	DL_PeerTC0RxInitCreditVal
DL_TC1TXFCThreshold on page 277	0x2060	DL_TC1TXFCThreshold
DL_FC1ProtectionTimeOutVal on page 278	0x2061	DL_FC1ProtectionTimeOutVal
DL_TC1ReplayTimeOutVal on page 279	0x2062	DL_TC1ReplayTimeOutVal
DL_AFC1ReqTimeOutVal on page 280	0x2063	DL_AFC1ReqTimeOutVal
DL_AFC1CreditThreshold on page 281	0x2064	DL_AFC1CreditThreshold
DL_TC1OutAckThreshold on page 282	0x2065	DL_TC1OutAckThreshold

Table 5-85 Registers for Address Block: DWC_mipi_unipro_map/DWC_mipi_unipro_l2_block (Continued)

Register	Offset	Description
DL_PeerTC1Present on page 283	0x2066	DL_PeerTC1Present
DL_PeerTC1RxInitCreditVal on page 284	0x2067	DL_PeerTC1RxInitCreditVal
DL_ReInitCount on page 285	0x2100	DL_ReInitCount

5.3.1 DL_TxPreemptionCap

- **Name:** DL_TxPreemptionCap
- **Description:** This attribute specifies the preemption capability on transmit side for both traffic classes. The DWC_mipi_unipro supports preemption on transmit side, thus this attribute is always read back as 1 (TRUE).
- **Size:** 32 bits
- **Offset:** 0x2000
- **Exists:** Always



Table 5-86 Fields for Register: DL_TxPreemptionCap

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	TXPC	R	Preemption capability on transmit side for both traffic classes. The DWC_mipi_unipro supports preemption on transmit side, thus this attribute is always read back as 1 (TRUE). Value After Reset: 0x1 Exists: Always

5.3.2 DL_TC0TxMaxSDUSize

- **Name:** DL_TC0TxMaxSDUSize
- **Description:** This attribute specifies the TC0 transmitter maximum SDU Size.
- **Size:** 32 bits
- **Offset:** 0x2001
- **Exists:** Always



Table 5-87 Fields for Register: DL_TC0TxMaxSDUSize

Bits	Name	Memory Access	Description
x:0	DL_TC0TX_SDU	R	This field specifies the maximum SDU size of the TC0 transmitter Value After Reset: "(P_L2_BUF_CREDUNIT_MODE==1) ? \"580\" : \"144\"" Exists: Always Range Variable[x]: "(P_L2_CREDUNIT_128==1) ? \"10\" : \"8\" - 1"

5.3.3 DL_TC0RxInitCreditVal

- **Name:** DL_TC0RxInitCreditVal
- **Description:** This attribute specifies the initial credit value (initial register A value) of the TC0 receiver.
- **Size:** 32 bits
- **Offset:** 0x2002
- **Exists:** Always

31:8	Rsvd
7:0	TC0_RX_ICV

Table 5-88 Fields for Register: DL_TC0RxInitCreditVal

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	TC0_RX_ICV	R	This field specifies the initial credit value (initial register A value) of the TC0 receiver. Value After Reset: *Varies with the parameter P_RBD0 Exists: Always

5.3.4 DL_TC1TxMaxSDUSize

- **Name:** DL_TC1TxMaxSDUSize
- **Description:** This attribute specifies the maximum SDU size of the TC1 transmitter.
- **Size:** 32 bits
- **Offset:** 0x2003
- **Exists:** Always



Table 5-89 Fields for Register: DL_TC1TxMaxSDUSize

Bits	Name	Memory Access	Description
x:0	DL_TC1TX_SDU	R	This attribute specifies maximum SDU size of the TC1 transmitter. Value After Reset: "(P_L2_BUF_CREDUNIT_MODE==1) ? \"580\" : \"144\"" Exists: Always Range Variable[x]: "(P_L2_CREDUNIT_128==1) ? \"10\" : \"8\" - 1"

5.3.5 DL_TC1RxInitCreditVal

- **Name:** DL_TC1RxInitCreditVal
- **Description:** This attribute specifies the initial credit value (initial register A value) of the TC1 receiver.
- **Size:** 32 bits
- **Offset:** 0x2004
- **Exists:** Always

31:8	Rsvd
7:0	TC1_RX_ICV

Table 5-90 Fields for Register: DL_TC1RxInitCreditVal

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	TC1_RX_ICV	R	This field specifies the initial credit value (initial register A value) of the TC1 receiver. Value After Reset: *Varies with the parameter P_RBD1 Exists: Always

5.3.6 DL_TC0TxBufferSize

- **Name:** DL_TC0TxBufferSize
- **Description:** This attribute specifies the TC0 transmitter buffer size. It is measured in L2 credits (32 bytes).
- **Size:** 32 bits
- **Offset:** 0x2005
- **Exists:** Always

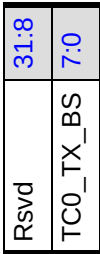


Table 5-91 Fields for Register: DL_TC0TxBufferSize

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	TC0_TX_BS	R	This field specifies the TC0 transmitter buffer size. It is measured in L2 credits (32 bytes) Value After Reset: *Varies with the parameter P_TBD0 Exists: Always

5.3.7 DL_TC1TxBufferSize

- **Name:** DL_TC1TxBufferSize
- **Description:** This attribute specifies the TC1 transmitter buffer size. Measured in L2 credits (32 bytes).
- **Size:** 32 bits
- **Offset:** 0x2006
- **Exists:** Always

31:8	
7:0	TC1_TX_BS

Table 5-92 Fields for Register: DL_TC1TxBufferSize

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	TC1_TX_BS	R	This field specifies the TC1 transmitter buffer size. It is measured in L2 credits (32 bytes) Value After Reset: *Varies with the parameter P_TBD1 Exists: Always

5.3.8 DL_MTU

- **Name:** DL_MTU
- **Description:** This attribute represents Maximum Payload Length of Data Link Layer.
- **Size:** 32 bits
- **Offset:** 0x2007
- **Exists:** Always

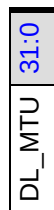


Table 5-93 Fields for Register: DL_MTU

Bits	Name	Memory Access	Description
31:0	DL_MTU	R	This field specifies the Maximum Payload Length of DL layer. In DLCreditUnit32_Mode, DL_MTU equals to 288. In DLCreditUnit128_Mode, DL_MTU equals to 1160. Value After Reset: *Varies with the parameter P_L2_BUF_CREDUNIT_MODE Exists: Always

5.3.9 DL_SYMBOL_MTU

- **Name:** DL_SYMBOL_MTU
- **Description:** This attribute represents Maximum number of symbols in DL_MTU size payload.
- **Size:** 32 bits
- **Offset:** 0x2008
- **Exists:** Always

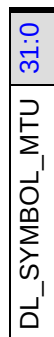


Table 5-94 Fields for Register: DL_SYMBOL_MTU

Bits	Name	Memory Access	Description
31:0	DL_SYMBOL_MTU	R	This field specifies the Maximum number of symbols in DL_MTU size payload. In DLCreditUnit32_Mode, DL_SYMBOL_MTU equals to 144. In DLCreditUnit128_Mode, DL_SYMBOL_MTU equals to 580. Value After Reset: *Varies with the parameter P_L2_BUF_CREDUNIT_MODE Exists: Always

5.3.10 DL_CreditUnitSize

- **Name:** DL_CreditUnitSize
- **Description:** This attribute represents size of one Credit Unit
- **Size:** 32 bits
- **Offset:** 0x2009
- **Exists:** Always



Table 5-95 Fields for Register: DL_CreditUnitSize

Bits	Name	Memory Access	Description
31:0	DL_CreditUnitSize	R	This field specifies the size of one Credit Unit. In DLCreditUnit32_Mode, DL_CreditUnitSize equals to 32. In DLCreditUnit128_Mode, DL_CreditUnitSize equals to 128. Value After Reset: *Varies with the parameter P_L2_BUF_CREDUNIT_MODE Exists: Always

5.3.11 DL_TC0TxFCThreshold

- **Name:** DL_TC0TxFCThreshold
- **Description:** This attribute specifies the threshold for triggering an AFC0 frame with CReq bit set to request flow control update to remote end.
- **Size:** 32 bits
- **Offset:** 0x2040
- **Exists:** Always

31:8	Rsvd
7:0	TC0_TX_FCT

Table 5-96 Fields for Register: DL_TC0TxFCThreshold

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	TC0_TX_FCT	R/W	This field specifies the threshold for triggering an AFC0 frame with CReq bit set to request flow control update to remote end Value After Reset: 0x9 Exists: Always

5.3.12 DL_FC0ProtectionTimeOutVal

- **Name:** DL_FC0ProtectionTimeOutVal
- **Description:** This attribute configures the expiry value of the FC0_PROTECTION_TIMER in microseconds.
- **Size:** 32 bits
- **Offset:** 0x2041
- **Exists:** Always

31:16	
15:0	DL_FC0PTV

Table 5-97 Fields for Register: DL_FC0ProtectionTimeOutVal

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	DL_FC0PTV	R/W	This field specifies the expiry value of the FC0_PROTECTION_TIMER in microseconds.The actual duration of the time out period is the value set in this attribute +/- 10%. Value After Reset: 0x1fff Exists: Always

5.3.13 DL_TC0ReplayTimeOutVal

- **Name:** DL_TC0ReplayTimeOutVal
- **Description:** This attribute configures the expiry value of the TC0_REPLAY_TIMER in microseconds.
- **Size:** 32 bits
- **Offset:** 0x2042
- **Exists:** Always

31:16	DL_TC0_RTV
Rsvd	15:0

Table 5-98 Fields for Register: DL_TC0ReplayTimeOutVal

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	DL_TC0_RTV	R/W	This field specifies the expiry value of the TC0_REPLAY_TIMER in microseconds. The actual duration of the time out period is the value set in this attribute +/- 10%. Value After Reset: 0xffff Exists: Always

5.3.14 DL_AFC0ReqTimeOutVal

- **Name:** DL_AFC0ReqTimeOutVal
- **Description:** This attribute configures the expiry value of the AFC0_REQUEST_TIMER in microseconds.
- **Size:** 32 bits
- **Offset:** 0x2043
- **Exists:** Always

Rsvd	31:16
AFC0_RQTV	15:0

Table 5-99 Fields for Register: DL_AFC0ReqTimeOutVal

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	AFC0_RQTV	R/W	This field specifies the expiry value of the AFC0_REQUEST_TIMER in microseconds. The actual duration of the time out period is the value set in this attribute +/- 10%. Value After Reset: 0x7fff Exists: Always

5.3.15 DL_AFC0CreditThreshold

- **Name:** DL_AFC0CreditThreshold
- **Description:** This attribute configures the Threshold for AFC0 triggering based on available TC0 credits at receiver.
- **Size:** 32 bits
- **Offset:** 0x2044
- **Exists:** Always

31:7	Rsvd
6:0	AFC0_CT

Table 5-100 Fields for Register: DL_AFC0CreditThreshold

Bits	Name	Memory Access	Description
31:7			Reserved Field: Yes
6:0	AFC0_CT	R/W	Threshold for AFC0 triggering based on available TC0 credits at receiver Value After Reset: "(P_L2_CREDUNIT_128==1) ? \"0x8\" : \"0x0\"" Exists: Always

5.3.16 DL_TC0OutAckThreshold

- **Name:** DL_TC0OutAckThreshold
- **Description:** This attribute specifies the number of outstanding acknowledgments for TC0 (number of TC0 data frames received before an AFC0 frame is sent).
- **Size:** 32 bits
- **Offset:** 0x2045
- **Exists:** Always

31:5	Rsvd
4:0	TC0_OAT

Table 5-101 Fields for Register: DL_TC0OutAckThreshold

Bits	Name	Memory Access	Description
31:5			Reserved Field: Yes
4:0	TC0_OAT	R/W	This field specifies the number of outstanding acknowledgments for TC0 (number of TC0 data frames received before an AFC0 frame is sent). Value After Reset: 0x0 Exists: Always

5.3.17 DL_PeerTC0Present

- **Name:** DL_PeerTC0Present
- **Description:** This attribute specifies if the peer node supports the TC0 or not.
- **Size:** 32 bits
- **Offset:** 0x2046
- **Exists:** Always

31:1	
Rsvd	PTC0P

Table 5-102 Fields for Register: DL_PeerTC0Present

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	PTC0P	R	Specifies if the peer supports TC0 ■ 1: Peer node supports TC0 ■ 0: Peer node does not support TC0 Values: ■ 0x1 (ENABLED): Peer node supports TC0 ■ 0x0 (DISABLED): Peer node does not support TC0 Value After Reset: 0x0 Exists: Always

5.3.18 DL_PeerTC0RxInitCreditVal

- **Name:** DL_PeerTC0RxInitCreditVal
- **Description:** This attribute specifies the initial credit value (initial register A value) of the peer TC0 receiver.
- **Size:** 32 bits
- **Offset:** 0x2047
- **Exists:** Always

31:8	Rsvd
7:0	PTC0_RX_ICV

Table 5-103 Fields for Register: DL_PeerTC0RxInitCreditVal

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	PTC0_RX_ICV	R	This field specifies the initial credit value (initial register A value) of the peer TC0 receiver. Value After Reset: 0x0 Exists: Always

5.3.19 DL_TC1TXFCTThreshold

- **Name:** DL_TC1TXFCTThreshold
- **Description:** This attribute specifies the threshold for triggering an AFC1 frame with CReq bit set to request flow control update to remote end.
- **Size:** 32 bits
- **Offset:** 0x2060
- **Exists:** Always

31:8	Rsvd
7:0	TC1_TX_FCT

Table 5-104 Fields for Register: DL_TC1TXFCTThreshold

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	TC1_TX_FCT	R/W	This field specifies the threshold for triggering an AFC1 frame with CReq bit set to request flow control update to remote end. Value After Reset: 0x9 Exists: Always

5.3.20 DL_FC1ProtectionTimeOutVal

- **Name:** DL_FC1ProtectionTimeOutVal
- **Description:** This attribute configures the expiry value of the FC1_PROTECTION_TIMER in microseconds.
- **Size:** 32 bits
- **Offset:** 0x2061
- **Exists:** Always



Table 5-105 Fields for Register: DL_FC1ProtectionTimeOutVal

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	DL_FC1_PTV	R/W	This field specifies the expiry value of the FC1_PROTECTION_TIMER in microseconds. The actual duration of the time out period is the value set in this attribute +/- 10%. Value After Reset: 0x1fff Exists: Always

5.3.21 DL_TC1ReplayTimeOutVal

- **Name:** DL_TC1ReplayTimeOutVal
- **Description:** This attribute configures the expiry value of the TC1_REPLAY_TIMER in microseconds.
- **Size:** 32 bits
- **Offset:** 0x2062
- **Exists:** Always

31:16	
15:0	DL_RC1_RTV

Table 5-106 Fields for Register: DL_TC1ReplayTimeOutVal

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	DL_RC1_RTV	R/W	This field specifies the expiry value of the TC1_REPLAY_TIMER in microseconds. The actual duration of the time out period is the value set in this attribute +/- 10%. Value After Reset: 0xffff Exists: Always

5.3.22 DL_AFC1ReqTimeOutVal

- **Name:** DL_AFC1ReqTimeOutVal
- **Description:** This attribute configures the expiry value of the AFC1_REQUEST_TIMER in microseconds.
- **Size:** 32 bits
- **Offset:** 0x2063
- **Exists:** Always

Rsvd	31:16
AFC1_RQTV	15:0

Table 5-107 Fields for Register: DL_AFC1ReqTimeOutVal

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	AFC1_RQTV	R/W	This field specifies the expiry value of the AFC1_REQUEST_TIMER in microseconds. The actual duration of the time out period is the value set in this attribute +/- 10%. Value After Reset: 0x7fff Exists: Always

5.3.23 DL_AFC1CreditThreshold

- **Name:** DL_AFC1CreditThreshold
- **Description:** This attribute configures the threshold for AFC1 triggering based on available TC1 credits at receiver.
- **Size:** 32 bits
- **Offset:** 0x2064
- **Exists:** Always

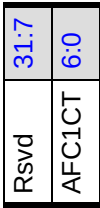


Table 5-108 Fields for Register: DL_AFC1CreditThreshold

Bits	Name	Memory Access	Description
31:7			Reserved Field: Yes
6:0	AFC1CT	R/W	This field specifies the threshold for AFC1 triggering based on available TC1 credits at the receiver. Value After Reset: "(P_L2_CREDUNIT_128==1) ? \"0x8\" : \"0x0\"" Exists: Always

5.3.24 DL_TC1OutAckThreshold

- **Name:** DL_TC1OutAckThreshold
- **Description:** This attribute specifies the number of outstanding acknowledgments for TC1.
- **Size:** 32 bits
- **Offset:** 0x2065
- **Exists:** Always

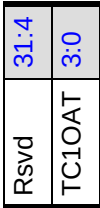


Table 5-109 Fields for Register: DL_TC1OutAckThreshold

Bits	Name	Memory Access	Description
31:4			Reserved Field: Yes
3:0	TC1OAT	R/W	This field specifes the number of outstanding acknowledgments for TC1 (number of TC1 data frames received before an AFC1 frame is sent) Value After Reset: 0x0 Exists: Always

5.3.25 DL_PeerTC1Present

- **Name:** DL_PeerTC1Present
- **Description:** This attribute specifies if the peer node supports the TC1 or not.
- **Size:** 32 bits
- **Offset:** 0x2066
- **Exists:** Always

31:1	
Rsvd	PTC1P

Table 5-110 Fields for Register: DL_PeerTC1Present

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	PTC1P	R	Specifies if the peer node supports the TC1 Values: ■ 0x1 (ENABLED): Peer node supports TC1 ■ 0x0 (DISABLED): Peer node does not support TC1 Value After Reset: 0x0 Exists: Always

5.3.26 DL_PeerTC1RxInitCreditVal

- **Name:** DL_PeerTC1RxInitCreditVal
- **Description:** This attribute specifies the initial credit value (initial register A value) of the peer TC1 receiver.
- **Size:** 32 bits
- **Offset:** 0x2067
- **Exists:** Always

31:8	Rsvd
7:0	PTC1_RXICV

Table 5-111 Fields for Register: DL_PeerTC1RxInitCreditVal

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	PTC1_RXICV	R	This field specifies the initial Credit Value (initial register A value) of the peer TC1 receiver. Value After Reset: 0x0 Exists: Always

5.3.27 DL_ReInitCount

- **Name:** DL_ReInitCount
- **Description:** This attribute specifies the number of times the DL layer has issued a PA_INIT.req to the PA layer.
- **Size:** 32 bits
- **Offset:** 0x2100
- **Exists:** Always

31:16	
15:0	REINITCOUNT
	Rsvd

Table 5-112 Fields for Register: DL_ReInitCount

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	REINITCOUNT	R/W	This field specifies the number of times the DL layer has issued a PA_INIT.req to PA layer due to some errors. Value After Reset: 0x0 Exists: Always

5.4 DWC_mipi_unipro_map/DWC_mipi_unipro_l3_block Registers

Layer 3 Attributes. Follow the link for the register to see a detailed description of the register.

Table 5-113 Registers for Address Block: DWC_mipi_unipro_map/DWC_mipi_unipro_l3_block

Register	Offset	Description
N_DeviceID on page 287	0x3000	N_DeviceID
N_DeviceID_Valid on page 288	0x3001	N_DeviceID_Valid
N_TC0TxMaxSDUSize on page 289	0x3020	N_TC0TxMaxSDUSize
N_TC1TxMaxSDUSize on page 290	0x3021	N_TC1TxMaxSDUSize
N_MaxPacketSize on page 291	0x3022	N_MaxPacketSize
N_MTU on page 292	0x3023	N_MTU

5.4.1 N_DeviceID

- **Name:** N_DeviceID
- **Description:** This attribute specifies the local device ID.
- **Size:** 32 bits
- **Offset:** 0x3000
- **Exists:** Always

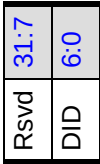


Table 5-114 Fields for Register: N_DeviceID

Bits	Name	Memory Access	Description
31:7			Reserved Field: Yes
6:0	DID	R/W	This field specifies the local device ID. Received packets are checked against this device ID. Packets that do not match the ID are discarded. Value After Reset: {(IS_UFS?0:(!IS_UFS && IS_UFS_DEVICE)?1:(!IS_UFS && !IS_UFS_DEVICE)?0:0)} Exists: Always

5.4.2 N_DeviceID_Valid

- **Name:** N_DeviceID_Valid
- **Description:** This attribute specifies the valid indication for local device ID.
- **Size:** 32 bits
- **Offset:** 0x3001
- **Exists:** Always

31:1	0
Rsvd	DIDV

Table 5-115 Fields for Register: N_DeviceID_Valid

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	DIDV	R/W	This bit specifies the valid indication for local device ID. If the local device ID is not set, that is, the valid bit is de-asserted, the received packets are not checked for device IDs, that is, the device ID in the header is not compared against N_DeviceID. # - 0: Disables the check for device ID in the received packet header against N_DeviceID. # - 1: Enables the check for device ID in the received packet header against N_DeviceID. Values: ■ 0x0 (ENABLED): Disables the check for device ID in the received packet header against N_DeviceID. ■ 0x1 (DISABLED): Enables the check for device ID in the received packet header against N_DeviceID. Value After Reset: {(IS_UFS?1:(!IS_UFS && IS_UFS_DEVICE)?1:(!IS_UFS && !IS_UFS_DEVICE)?0:0)} Exists: Always

5.4.3 N_TC0TxMaxSDUSize

- **Name:** N_TC0TxMaxSDUSize
- **Description:** This attribute defines the maximum SDU size for traffic class TC0, defined at the compile time.
- **Size:** 32 bits
- **Offset:** 0x3020
- **Exists:** Always



Table 5-116 Fields for Register: N_TC0TxMaxSDUSize

Bits	Name	Memory Access	Description
x:0	N_TC0TX_SDU	R	This field defines the maximum SDU size for traffic class TC0, defined at compile time Value After Reset: "(P_L2_BUF_CREDUNIT_MODE==1) ? \"1145\" : \"273\"" Exists: Always Range Variable[x]: "(P_L2_CREDUNIT_128==1) ? \"11\" : \"9\" - 1

5.4.4 N_TC1TxMaxSDUSize

- **Name:** N_TC1TxMaxSDUSize
- **Description:** This attribute defines the maximum SDU size for traffic class TC1, defined at the compile time.
- **Size:** 32 bits
- **Offset:** 0x3021
- **Exists:** Always

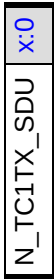


Table 5-117 Fields for Register: N_TC1TxMaxSDUSize

Bits	Name	Memory Access	Description
x:0	N_TC1TX_SDU	R	This attribute defines maximum SDU size for traffic class TC1, defined at compile time Value After Reset: "(P_L2_BUF_CREDUNIT_MODE==1) ? \"1145\" : \"273\"" Exists: Always Range Variable[x]: "(P_L2_CREDUNIT_128==1) ? \"11\" : \"9\" - 1

5.4.5 N_MaxPacketSize

- **Name:** N_MaxPacketSize
- **Description:** This attribute represents the Maximum Packet size of Network layer.
- **Size:** 32 bits
- **Offset:** 0x3022
- **Exists:** Always

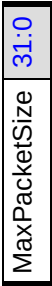


Table 5-118 Fields for Register: N_MaxPacketSize

Bits	Name	Memory Access	Description
31:0	MaxPacketSize	R	This attribute reflects the Maximum Packet size of Network Layer. In DLCreditUnit32_Mode, MaxPacketSize equals to 274. In DLCreditUnit128_Mode, MaxPacketSize equals to 1146. Value After Reset: *Varies with the parameter P_L2_BUF_CREDUNIT_MODE Exists: Always

5.4.6 N_MTU

- **Name:** N_MTU
- **Description:** This attribute represents the Maximum Transfer Unit of Network layer.
- **Size:** 32 bits
- **Offset:** 0x3023
- **Exists:** Always



Table 5-119 Fields for Register: N_MTU

Bits	Name	Memory Access	Description
31:0	MTU	R	This attribute reflects the Maximum Transfer Unit of Network Layer. In DLCreditUnit32_Mode, MTU equals to 273. In DLCreditUnit128_Mode, MaxPacketSize equals to 1145. Value After Reset: *Varies with the parameter P_L2_BUF_CREDUNIT_MODE Exists: Always

5.5 DWC_mipi_unipro_map/DWC_mipi_unipro_l4_block Registers

Layer 4 Attributes. Follow the link for the register to see a detailed description of the register.

Table 5-120 Registers for Address Block: DWC_mipi_unipro_map/DWC_mipi_unipro_l4_block

Register	Offset	Description
T_NumCPorts on page 295	0x4000	T_NumCPorts
T_NumTestFeatures on page 296	0x4001	T_NumTestFeatures
T_ConnectionState on page 297	0x4020	T_ConnectionState
T_PeerDeviceID on page 298	0x4021	T_PeerDeviceID
T_PeerCPortID on page 299	0x4022	T_PeerCPortID
T_TrafficClass on page 300	0x4023	T_TrafficClass
T_ProtocolID on page 301	0x4024	T_ProtocolID
T_CPortFlags on page 302	0x4025	T_CPortFlags
T_TxTokenValue on page 304	0x4026	T_TxTokenValue
T_RxTokenValue on page 305	0x4027	T_RxTokenValue
T_LocalBufferSpace on page 306	0x4028	T_LocalBufferSpace
T_PeerBufferSpace on page 307	0x4029	T_PeerBufferSpace
T_CreditsToSend on page 308	0x402a	T_CreditsToSend
T_CPortMode on page 309	0x402b	T_CPortMode
T_TC0TxMaxSDUSize on page 310	0x4060	T_TC0TxMaxSDUSize
T_TC1TxMaxSDUSize on page 311	0x4061	T_TC1TxMaxSDUSize
T_MaxSegmentSize on page 312	0x4062	T_MaxSegmentSize
T_MTU on page 313	0x4063	T_MTU
T_TstCPortID on page 314	0x4080	T_TstCPortID
T_TstSrcOn on page 315	0x4081	T_TstSrcOn
T_TstSrcPattern on page 316	0x4082	T_TstSrcPattern
T_TstSrcIncrement on page 317	0x4083	T_TstSrcIncrement
T_TstSrcMessageSize on page 318	0x4084	T_TstSrcMessageSize
T_TstSrcMessageCount on page 319	0x4085	T_TstSrcMessageCount
T_TstSrcInterMessageGap on page 320	0x4086	T_TstSrcInterMessageGap

Table 5-120 Registers for Address Block: DWC_mipi_unipro_map/DWC_mipi_unipro_l4_block (Continued)

Register	Offset	Description
T_TstDstOn on page 321	0x40a1	T_TstDstOn
T_TstDstErrorDetectionEnable on page 322	0x40a2	T_TstDstErrorDetectionEnable
T_TstDstPattern on page 323	0x40a3	T_TstDstPattern
T_TstDstIncrement on page 324	0x40a4	T_TstDstIncrement
T_TstDstMessageCount on page 325	0x40a5	T_TstDstMessageCount
T_TstDstMessageOffset on page 326	0x40a6	T_TstDstMessageOffset
T_TstDstMessageSize on page 327	0x40a7	T_TstDstMessageSize
T_TstDstFCCredits on page 328	0x40a8	T_TstDstFCCredits
T_TstDstInterFCTokenGap on page 329	0x40a9	T_TstDstInterFCTokenGap
T_TstDstInitialFCCredits on page 330	0x40aa	T_TstDstInitialFCCredits
T_TstDstErrorCode on page 331	0x40ab	T_TstDstErrorCode

5.5.1 T_NumCPorts

- **Name:** T_NumCPorts
- **Description:** This attribute specifies the number of CPorts to be handled by L4. This attribute is controlled by top-level parameter P_CP.
- **Size:** 32 bits
- **Offset:** 0x4000
- **Exists:** Always

31:12	11:0
Rsvd	NCP

Table 5-121 Fields for Register: T_NumCPorts

Bits	Name	Memory Access	Description
31:12			Reserved Field: Yes
11:0	NCP	R	This field specifies the number of CPorts to be handled by L4. This attribute is controlled by the P_CP top-level parameter. Value After Reset: P_CP Exists: Always

5.5.2 T_NumTestFeatures

- **Name:** T_NumTestFeatures
- **Description:** This attribute specifies the number of CPort test features.
- **Size:** 32 bits
- **Offset:** 0x4001
- **Exists:** Always

31:12	11:0
Rsvd	NTF

Table 5-122 Fields for Register: T_NumTestFeatures

Bits	Name	Memory Access	Description
31:12			Reserved Field: Yes
11:0	NTF	R	This field specifies the number of CPort test features. The number of test features is lower or equal to number of CPorts. This attribute is controlled by the P_TF toplevel parameter. Value After Reset: P_TF Exists: Always

5.5.3 T_ConnectionState

- **Name:** T_ConnectionState
- **Description:** This attribute specifies the CPort connection status.
- **Size:** 32 bits
- **Offset:** 0x4020
- **Exists:** Always

31:1	0
Rsvd	CS

Table 5-123 Fields for Register: T_ConnectionState

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	CS	R/W	Specifies the CPort connection status. #- 0: Closed. The CPort is in the Closed state #- 1: Connected. The CPort is in the Connected state or any other set field_value [create_register_field_value VALUE_OF_0 -field \$field -value 0x0] set_attribute \$field_value -attr DisplayName -value {Closed. The CPort is in the Closed state} set field_value [create_register_field_value VALUE_OF_1 -field \$field -value 0x1] set_attribute \$field_value -attr DisplayName -value {Connected. The CPort is in the Connected state or any other} State in which the data is transferred until receiving a FIN CM Extension or the T_CO_CLOSE.req primitive. Value After Reset: {(IS_UFS?1:(!IS_UFS && IS_UFS_DEVICE)?1:(!IS_UFS && !IS_UFS_DEVICE)?0:0)} Exists: Always

5.5.4 T_PeerDeviceID

- **Name:** T_PeerDeviceID
- **Description:** This attribute specifies the device ID of the connected peer CPort. Configure this attribute only when the CS field of T_ConnectionState attribute is 0. Write access to a connected CPort is rejected when the CS field of T_ConnectionState attribute is 1.
- **Size:** 32 bits
- **Offset:** 0x4021
- **Exists:** Always

31:7	6:0
Rsvd	PDID

Table 5-124 Fields for Register: T_PeerDeviceID

Bits	Name	Memory Access	Description
31:7			Reserved Field: Yes
6:0	PDID	R/W	This field specifies the device ID of the connected peer CPort. Value After Reset: 0x0 Exists: Always

5.5.5 T_PeerCPortID

- **Name:** T_PeerCPortID
- **Description:** This attribute specifies the CPort ID of the connected peer CPort. Configure this attribute only when the CS field of T_ConnectionState attribute is 0. Write access to a connected CPort is rejected when the CS field of T_ConnectionState attribute is 1.
- **Size:** 32 bits
- **Offset:** 0x4022
- **Exists:** Always

31:5	4:0
Rsvd	PCPID

Table 5-125 Fields for Register: T_PeerCPortID

Bits	Name	Memory Access	Description
31:5			Reserved Field: Yes
4:0	PCPID	R/W	This field specifies the CPort ID of the connected peer CPort Value After Reset: 0x0 Exists: Always

5.5.6 T_TrafficClass

- **Name:** T_TrafficClass
- **Description:** This attribute determines the used traffic class. Configure this attribute only when the CS field of T_ConnectionState attribute is 0. Write access to a connected CPort is rejected when the CS field of T_ConnectionState attribute is 1.
- **Size:** 32 bits
- **Offset:** 0x4023
- **Exists:** Always

31:1	0
Rsvd	TC

Table 5-126 Fields for Register: T_TrafficClass

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	TC	R/W	This bit specifies the traffic class of current connection, which determines the value of the TC field in the data link layer header. Value After Reset: 0x0 Exists: Always

5.5.7 T_ProtocolID

- **Name:** T_ProtocolID
- **Description:** This attribute specifies the connection protocol. Configure this attribute only when the CS field of T_ConnectionState attribute is 0. Write access to a connected CPort is rejected when the CS field of T_ConnectionState attribute is 1.
- **Size:** 32 bits
- **Offset:** 0x4024
- **Exists:** Always

31:16	15:0
Rsvd	PID

Table 5-127 Fields for Register: T_ProtocolID

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	PID	R/W	This field specifies the connection protocol Value After Reset: 0x0 Exists: Always

5.5.8 T_CPortFlags

- **Name:** T_CPortFlags
- **Description:** This attribute controls the CPort. Configure this attribute only when the CS field of T_ConnectionState attribute is 0. Write access to a connected CPort is rejected when the CS field of T_ConnectionState attribute is 1.
- **Size:** 32 bits
- **Offset:** 0x4025
- **Exists:** Always

31:3	2	1	0
Rsvd	CSV	CSD	E2EFC

Table 5-128 Fields for Register: T_CPortFlags

Bits	Name	Memory Access	Description
31:3			Reserved Field: Yes
2	CSV	R/W	This bit enables or disables the CPort Safety Valve. Values: ■ 0x1 (DISABLED): CPort Safety Valve is disabled ■ 0x0 (ENABLED): CPort Safety Valve is enabled Value After Reset: {(IS_UFS?1:(!IS_UFS && IS_UFS_DEVICE)?1:(!IS_UFS && !IS_UFS_DEVICE)?0:0)} Exists: Always
1	CSD	R/W	This bit enables or disables the Controlled Segment Dropping. Values: ■ 0x1 (DISABLED): Controlled Segment Dropping is disabled ■ 0x0 (ENABLED): Controlled Segment Dropping is enabled Value After Reset: {(IS_UFS?1:(!IS_UFS && IS_UFS_DEVICE)?1:(!IS_UFS && !IS_UFS_DEVICE)?0:0)} Exists: Always

Table 5-128 Fields for Register: T_CPortFlags (Continued)

Bits	Name	Memory Access	Description
0	E2EFC	R/W	This bit enables or disables the End-to-End Flow Control. Values: ■ 0x1 (ENABLED): End-to-End Flow Control is enabled ■ 0x0 (DISABLED): End-to-End Flow Control is disabled Value After Reset: {(IS_UFS?0:(!IS_UFS && IS_UFS_DEVICE)?0:(!IS_UFS && !IS_UFS_DEVICE)?1:1)} Exists: Always

5.5.9 T_TxTokenValue

- **Name:** T_TxTokenValue
- **Description:** This attribute determines the value a token is worth (in bytes) for a transmitted FCT. Configure this attribute only when the CS field of T_ConnectionState attribute is 0. Write access to a connected CPort is rejected when the CS field of T_ConnectionState attribute is 1.
- **Size:** 32 bits
- **Offset:** 0x4026
- **Exists:** Always

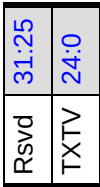


Table 5-129 Fields for Register: T_TxTokenValue

Bits	Name	Memory Access	Description
31:25			Reserved Field: Yes
24:0	TXTV	R/W	This field determines the value a token is worth (in bytes) for a transmitted FCT. Only one bit should be set at any given time. Value After Reset: 0x20 Exists: Always

5.5.10 T_RxTokenValue

- **Name:** T_RxTokenValue
- **Description:** This attribute determines the value a token is worth (in bytes) for a received FCT. Configure this attribute only when the CS bit of T_ConnectionState attribute is 0. Write access to a connected CPort is rejected when the CS bit of T_ConnectionState attribute is 1.
- **Size:** 32 bits
- **Offset:** 0x4027
- **Exists:** Always

31:25	24:0
Rsvd	RXTV

Table 5-130 Fields for Register: T_RxTokenValue

Bits	Name	Memory Access	Description
31:25			Reserved Field: Yes
24:0	RXTV	R/W	Determines the value a token is worth (in bytes) for a received FCT. Only one bit should be set at any given time. Value After Reset: 0x20 Exists: Always

5.5.11 T_LocalBufferSpace

- **Name:** T_LocalBufferSpace
- **Description:** This attribute specifies the amount of local buffer space as tracked by the local CPort. Configure this attribute only when the CS field of T_ConnectionState attribute is 0. Write access to a connected CPort is rejected when the CS field of T_ConnectionState attribute is 1.
- **Size:** 32 bits
- **Offset:** 0x4028
- **Exists:** Always



Table 5-131 Fields for Register: T_LocalBufferSpace

Bits	Name	Memory Access	Description
31:0	LBS	R/W	This field specifies the amount of local buffer space as tracked by the local CPort. Value After Reset: 0x0 Exists: Always

5.5.12 T_PeerBufferSpace

- **Name:** T_PeerBufferSpace
- **Description:** This attribute specifies the value of the amount of buffer space at the peer CPort. Configure this attribute only when the CS field of T_ConnectionState attribute is 0. Write access to a connected CPort is rejected when the CS field of T_ConnectionState attribute is 1.
- **Size:** 32 bits
- **Offset:** 0x4029
- **Exists:** Always



Table 5-132 Fields for Register: T_PeerBufferSpace

Bits	Name	Memory Access	Description
31:0	PBS	R/W	This field specifies the value of the amount of buffer space at the peer CPort Value After Reset: 0x0 Exists: Always

5.5.13 T_CreditsToSend

- **Name:** T_CreditsToSend
- **Description:** This attribute specifies the amount of space in the local buffer that has not been communicated to the remote destination port yet. Configure this attribute only when the CS field of T_ConnectionState attribute is 0. Write access to a connected CPort is rejected when the CS field of T_ConnectionState attribute is 1.
- **Size:** 32 bits
- **Offset:** 0x402a
- **Exists:** Always

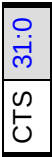


Table 5-133 Fields for Register: T_CreditsToSend

Bits	Name	Memory Access	Description
31:0	CTS	R/W	This field specifies the value of the amount of buffer space at the peer CPort Value After Reset: 0x0 Exists: Always

5.5.14 T_CPortMode

- **Name:** T_CPortMode
- **Description:** This attribute indicates the CPort Mode. Configure this attribute only when the CS field of T_ConnectionState attribute is 0. Write access to a connected CPort is rejected when the CS field of T_ConnectionState attribute is 1.
- **Size:** 32 bits
- **Offset:** 0x402b
- **Exists:** Always

31:2	1:0
Rsvd	CPM

Table 5-134 Fields for Register: T_CPortMode

Bits	Name	Memory Access	Description
31:2			Reserved Field: Yes
1:0	CPM	R/W	When set to CPORT_APPLICATION, the CPort is used by the application. When set to CPORT_UNDER_TEST, the CPort is used by the transport layer test feature #- 1: CPORT_APPLICATION #- 2: CPORT_UNDER_TEST Values: ■ 0x1 (VALUE_OF_1): CPORT_APPLICATION ■ 0x2 (VALUE_OF_2): CPORT_UNDER_TEST Value After Reset: 0x1 Exists: Always

5.5.15 T_TC0TxMaxSDUSize

- **Name:** T_TC0TxMaxSDUSize
- **Description:** This attribute specifies the maximum number of payload bytes per SDU for TC0.
- **Size:** 32 bits
- **Offset:** 0x4060
- **Exists:** Always



Table 5-135 Fields for Register: T_TC0TxMaxSDUSize

Bits	Name	Memory Access	Description
x:0	T_TC0TX_SDU	R	The maximum number of payload bytes per SDU for TC0 is 272. This value is fixed and cannot be controlled by parameter settings. Value After Reset: "(P_L2_BUF_CREDUNIT_MODE==1) ? \"1144\" : \"272\"" Exists: Always Range Variable[x]: "(P_L2_CREDUNIT_128==1) ? \"11\" : \"9\" - 1

5.5.16 T_TC1TxMaxSDUSize

- **Name:** T_TC1TxMaxSDUSize
- **Description:** This attribute specifies the maximum number of payload bytes per SDU for TC1.
- **Size:** 32 bits
- **Offset:** 0x4061
- **Exists:** Always



Table 5-136 Fields for Register: T_TC1TxMaxSDUSize

Bits	Name	Memory Access	Description
x:0	T_TC1TX_SDU	R	The maximum number of payload bytes per SDU for TC1 is 272. This value is fixed and cannot be controlled by parameter settings. Value After Reset: "(P_L2_BUF_CREDUNIT_MODE==1) ? \"1144\" : \"272\"" Exists: Always Range Variable[x]: "(P_L2_CREDUNIT_128==1) ? \"11\" : \"9\" - 1

5.5.17 T_MaxSegmentSize

- **Name:** T_MaxSegmentSize
- **Description:** This attribute represents the Maximum Segment Size (T_PDU) of Transport Layer.
- **Size:** 32 bits
- **Offset:** 0x4062
- **Exists:** Always



Table 5-137 Fields for Register: T_MaxSegmentSize

Bits	Name	Memory Access	Description
31:0	MaxSegmentSize	R	The field represents the maximum segment size of Transport layer. In DLCreditUnit32_Mode, MaxSegmentSize equals to 273. In DLCreditUnit128_Mode, MaxSegmentSize equals to 1145. Value After Reset: *Varies with the parameter P_L2_BUF_CREDUNIT_MODE Exists: Always

5.5.18 T_MTU

- **Name:** T_MTU
- **Description:** This attribute represents the Maximum Transport Unit of Transport Layer.
- **Size:** 32 bits
- **Offset:** 0x4063
- **Exists:** Always



Table 5-138 Fields for Register: T_MTU

Bits	Name	Memory Access	Description
31:0	MTU	R	The field represents the maximum transport unit of Transport layer. In DLCreditUnit32_Mode, MTU equals to 272. In DLCreditUnit128_Mode, MTU equals to 1144. Value After Reset: *Varies with the parameter P_L2_BUF_CREDUNIT_MODE Exists: Always

5.5.19 T_TstCPortID

- **Name:** T_TstCPortID
- **Description:** Test feature common attribute. This attribute specifies the ID of the CPort-under-test. Configure this attribute only when the TSO field of T_TstSrcOn attribute is 0. Write access to a connected CPort is rejected when the CS field of T_ConnectionState attribute is 1.
- **Size:** 32 bits
- **Offset:** 0x4080
- **Exists:** Always

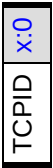


Table 5-139 Fields for Register: T_TstCPortID

Bits	Name	Memory Access	Description
x:0	TCPID	R/W	This field specifies the ID of the CPort-under-test. It defines the CPort for generating or receiving the data. Value After Reset: 0x0 Exists: Always Range Variable[x]: P_CA - 1

5.5.20 T_TstSrcOn

- **Name:** T_TstSrcOn
- **Description:** Test feature data generator attribute. This attribute specifies the message generation state. This is cleared automatically once the requested number of messages are transmitted. When the test feature is set to transmit infinite messages (T_TstSrcMessageCount = 0), then this bit has to be reset to stop the transmission.
- **Size:** 32 bits
- **Offset:** 0x4081
- **Exists:** Always

31:1	0
Rsvd	TSO

Table 5-140 Fields for Register: T_TstSrcOn

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	TSO	R/W	This field specifies Message generation state. Set this to 1 to start the pattern generator. Do not start the data generator before the peer data checker is configured and switched on; otherwise, test data is lost leading to errors. Value After Reset: 0x0 Exists: Always

5.5.21 T_TstSrcPattern

- **Name:** T_TstSrcPattern
- **Description:** Test feature data generator attribute. This attribute specifies the pattern used for message generation. Only value 0 is accepted.
- **Size:** 32 bits
- **Offset:** 0x4082
- **Exists:** Always

31:1	0
Rsvd	TSP

Table 5-141 Fields for Register: T_TstSrcPattern

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	TSP	R/W	Pattern used for Message generation. Only value 0 is accepted and only sawtooth is supported Value After Reset: 0x0 Exists: Always

5.5.22 T_TstSrcIncrement

- **Name:** T_TstSrcIncrement
- **Description:** Test feature data generator attribute. This attribute specifies the increment value between two consecutive bytes. Configure this attribute only when the TSO field of T_TstSrcOn attribute is 0. Write access to a connected CPort is rejected when the CS field of T_ConnectionState attribute is 1.
- **Size:** 32 bits
- **Offset:** 0x4083
- **Exists:** Always

31:8	7:0
Rsvd	TSI

Table 5-142 Fields for Register: T_TstSrcIncrement

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	TSI	R/W	This field specifies the increment value between two consecutive bytes. Value After Reset: 0x1 Exists: Always

5.5.23 T_TstSrcMessageSize

- **Name:** T_TstSrcMessageSize
- **Description:** Test feature data generator attribute. This attribute specifies the size of each generated message. Configure this attribute only when the TSO field of T_TstSrcOn attribute is 0. Write access to a connected CPort is rejected when the CS field of T_ConnectionState attribute is 1.
- **Size:** 32 bits
- **Offset:** 0x4084
- **Exists:** Always

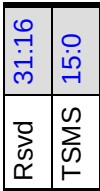


Table 5-143 Fields for Register: T_TstSrcMessageSize

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	TSMS	R/W	This field specifies the size of each generated message to send. Value After Reset: 0x100 Exists: Always

5.5.24 T_TstSrcMessageCount

- **Name:** T_TstSrcMessageCount
- **Description:** Test feature data generator attribute. This attribute specifies the number of messages generated. Configure this attribute only when the TSO field of T_TstSrcOn attribute is 0. Write access to a connected CPort is rejected when the CS field of T_ConnectionState attribute is 1.
- **Size:** 32 bits
- **Offset:** 0x4085
- **Exists:** Always

31:16	15:0
Rsvd	TSMC

Table 5-144 Fields for Register: T_TstSrcMessageCount

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	TSMC	R/W	This field specifies the number of messages generated to send: ■ Non-zero: Finite message stream ■ Zero: Infinite message stream Value After Reset: 0x100 Exists: Always

5.5.25 T_TstSrcInterMessageGap

- **Name:** T_TstSrcInterMessageGap
- **Description:** Test feature data generator attribute. This attribute specifies the gap time between two messages. Configure this attribute only when the TSO field of T_TstSrcOn attribute is 0. Write access to a connected CPort is rejected when the CS field of T_ConnectionState attribute is 1.
- **Size:** 32 bits
- **Offset:** 0x4086
- **Exists:** Always

31:16	15:0
Rsvd	TSMG

Table 5-145 Fields for Register: T_TstSrcInterMessageGap

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	TSMG	R/W	This field specifies the inter-message gap in microseconds. When non-zero, specifies the time gap between two messages. Value After Reset: 0x0 Exists: Always

5.5.26 T_TstDstOn

- **Name:** T_TstDstOn
- **Description:** Test feature data checker attribute. This attribute specifies the state of TstDst. This is cleared automatically when any of the received messages has an error as compared to the expected message. The error status is given in the T_TstDstErrorCode attribute.
- **Size:** 32 bits
- **Offset:** 0x40a1
- **Exists:** Always

31:1	0
Rsvd	TDO

Table 5-146 Fields for Register: T_TstDstOn

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	TDO	R/W	This bit specifies the state of TstDst. Set to 1 to start pattern checker. Start the data checker before the peer data generator, otherwise, the test data is lost leading to errors. When 0, pattern checker is deactivated. Value After Reset: 0x0 Exists: Always

5.5.27 T_TstDstErrorDetectionEnable

- **Name:** T_TstDstErrorDetectionEnable
- **Description:** Test feature data checker attribute. This attribute enables analyzing incoming messages. Configure this attribute only when the TDO field of T_TstDstOn attribute is 0. Write access to a connected CPort is rejected when the CS field of T_ConnectionState attribute is 1.
- **Size:** 32 bits
- **Offset:** 0x40a2
- **Exists:** Always

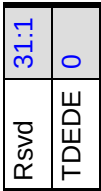


Table 5-147 Fields for Register: T_TstDstErrorDetectionEnable

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	TDEDE	R/W	This bit enables analyzing incoming messages. When set to 1, enables error detection. Value After Reset: 0x0 Exists: Always

5.5.28 T_TstDstPattern

- **Name:** T_TstDstPattern
- **Description:** Test feature data checker attribute. This attribute specifies the expected pattern in the incoming messages. Configure this attribute only when the TDO field of T_TstDstOn attribute is 0. Write access to a connected CPort is rejected when the CS field of T_ConnectionState attribute is 1.
- **Size:** 32 bits
- **Offset:** 0x40a3
- **Exists:** Always

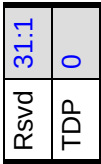


Table 5-148 Fields for Register: T_TstDstPattern

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	TDP	R/W	This bit specifies the expected pattern in the incoming messages. Only value 0 is accepted and only sawtooth is supported. Value After Reset: 0x0 Exists: Always

5.5.29 T_TstDstIncrement

- **Name:** T_TstDstIncrement
- **Description:** Test feature data checker attribute. This attribute specifies the increment value between two consecutive bytes. Configure this attribute only when the TDO field of T_TstDstOn attribute is 0. Write access to a connected CPort is rejected when the CS field of T_ConnectionState attribute is 1.
- **Size:** 32 bits
- **Offset:** 0x40a4
- **Exists:** Always

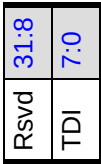


Table 5-149 Fields for Register: T_TstDstIncrement

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	TDI	R/W	This field specifies the expected increment value between two consecutive bytes. Value After Reset: 0x1 Exists: Always

5.5.30 T_TstDstMessageCount

- **Name:** T_TstDstMessageCount
- **Description:** Test feature data checker attribute. This attribute specifies the number of received messages. Configure this attribute only when the TDO field of T_TstDstOn attribute is 0. Write access to a connected CPort is rejected when the CS field of T_ConnectionState attribute is 1.
- **Size:** 32 bits
- **Offset:** 0x40a5
- **Exists:** Always

31:16	15:0
Rsvd	TDMC

Table 5-150 Fields for Register: T_TstDstMessageCount

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	TDMC	R/W	This field specifies the number of received messages.A Write to this attribute allows you to clear or initialize the attribute. Value After Reset: 0x0 Exists: Always

5.5.31 T_TstDstMessageOffset

- **Name:** T_TstDstMessageOffset
- **Description:** Test feature data checker attribute. This attribute specifies the position of the last detected error. Configure this attribute only when the TDO field of T_TstDstOn attribute is 0. Write access to a connected CPort is rejected when the CS field of T_ConnectionState attribute is 1.
- **Size:** 32 bits
- **Offset:** 0x40a6
- **Exists:** Always

31:16	15:0
Rsvd	TDMO

Table 5-151 Fields for Register: T_TstDstMessageOffset

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	TDMO	R/W	This field specifies the offset of error from the start of the message Value After Reset: 0x0 Exists: Always

5.5.32 T_TstDstMessageSize

- **Name:** T_TstDstMessageSize
- **Description:** Test feature data checker attribute. This attribute specifies the expected size of message to be received. Configure this attribute only when the TDO field of T_TstDstOn attribute is 0. Write access to a connected CPort is rejected when the CS field of T_ConnectionState attribute is 1.
- **Size:** 32 bits
- **Offset:** 0x40a7
- **Exists:** Always

31:16	15:0
Rsvd	TDMS

Table 5-152 Fields for Register: T_TstDstMessageSize

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	TDMS	R/W	This field specifies the expected size of incoming message. If there is a mismatch between the incoming message size and the value of this attribute and T_TstDstErrorDetectionEnable is set to TRUE, then Test Analyzer considers it as an error and overwrites the attribute value with the incoming message length. Value After Reset: 0x100 Exists: Always

5.5.33 T_TstDstFCCredits

- **Name:** T_TstDstFCCredits
- **Description:** Test feature data checker attribute. This attribute specifies the amount of credits passed to the flow control service primitive. This is optional, if flow control and CSD are enabled. Configure this attribute only when the TDO field of T_TstDstOn attribute is 0. Write access to a connected CPort is rejected when the CS field of T_ConnectionState attribute is 1.
- **Size:** 32 bits
- **Offset:** 0x40a8
- **Exists:** Always

31:17	Rsvd
16:0	TDFCC

Table 5-153 Fields for Register: T_TstDstFCCredits

Bits	Name	Memory Access	Description
31:17			Reserved Field: Yes
16:0	TDFCC	R/W	This field specifies the amount of credits passed to the flow control service primitive. This value is limited by the P_FC parameter. Width of the attribute is P_FC+1 Value After Reset: 0x100 Exists: Always

5.5.34 T_TstDstInterFCTokenGap

- **Name:** T_TstDstInterFCTokenGap
- **Description:** Test feature data checker attribute. This attribute specifies gap between two credit interface updates. This is optional, if flow control and CSD are enabled. Configure this attribute only when the TDO field of T_TstDstOn attribute is 0. Write access to a connected CPort is rejected when the CS field of T_ConnectionState attribute is 1.
- **Size:** 32 bits
- **Offset:** 0x40a9
- **Exists:** Always

31:16	Rsvd
15:0	TDL_FCTG

Table 5-154 Fields for Register: T_TstDstInterFCTokenGap

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	TDL_FCTG	R/W	This field specifies the interval in microseconds for which requesting T_CO_FLOWCONTROL.req is skipped. Value After Reset: 0x0 Exists: Always

5.5.35 T_TstDstInitialFCCredits

- **Name:** T_TstDstInitialFCCredits
- **Description:** Test feature data checker attribute. This attribute specifies the initial credit update when the data checker is enabled. This is optional, if flow control and CSD are enabled.

Configure this attribute only when the CS field of T_TstDstOn attribute is 0. Write access to a connected CPort is rejected when the CS field of T_ConnectionState attribute is 1.

- **Size:** 32 bits
- **Offset:** 0x40aa
- **Exists:** Always

Rsvd	31:17
TDI_FCC	16:0

Table 5-155 Fields for Register: T_TstDstInitialFCCredits

Bits	Name	Memory Access	Description
31:17			Reserved Field: Yes
16:0	TDI_FCC	R/W	This field specifies the initial number of credits passed by the TstDst to T_CO_FLOWCONTROL.req after enabling the TstDst. This value is limited by the P_FC parameter. Width of the attribute is P_FC+1 Value After Reset: 0x100 Exists: Always

5.5.36 T_TstDstErrorCode

- **Name:** T_TstDstErrorCode
- **Description:** Test feature data checker attribute. This attribute specifies the error code that generated the TstDst disconnection. Optional, if flow control and CSD are enabled.
- **Size:** 32 bits
- **Offset:** 0x40ab
- **Exists:** Always

31:2	1:0
Rsvd	TDEC

Table 5-156 Fields for Register: T_TstDstErrorCode

Bits	Name	Memory Access	Description
31:2			Reserved Field: Yes
1:0	TDEC	R/W	This field indicates the error-code generated in the test feature analyzer (TstDst). This attribute is valid when the T_TstDstOn attribute is cleared to 0 during the test. #- 0: NO_ERROR #- 1: FRAGMENT_CORRUPT #- 2: INVALID_MSG_SIZE #- 3: UNEXPECTED_BYTE_VALUE The non-zero status has to be cleared by writing 0 into this register. Unless this is cleared, the T_TstDstOn cannot be set to 1 again. Values: ■ 0x0 (VALUE_OF_0): NO_ERROR ■ 0x1 (VALUE_OF_1): FRAGMENT_CORRUPT ■ 0x2 (VALUE_OF_2): INVALID_MSG_SIZE ■ 0x3 (VALUE_OF_3): UNEXPECTED_BYTE_VALUE Value After Reset: 0x0 Exists: Always

5.6 DWC_mipi_unipro_map/DWC_mipi_unipro_ddb_block Registers

DDB attributes. Follow the link for the register to see a detailed description of the register.

Table 5-157 Registers for Address Block: DWC_mipi_unipro_map/DWC_mipi_unipro_ddb_block

Register	Offset	Description
DM_Revision on page 333	0x5000	DM_Revision
DM_Level on page 334	0x5001	DM_Level
DM_DeviceClass on page 335	0x5002	DM_DeviceClass
DM_ManufacturerID on page 336	0x5003	DM_ManufacturerID
DM_ProductID on page 337	0x5004	DM_ProductID
DM_Length on page 338	0x5005	DM_Length

5.6.1 DM_Revision

- **Name:** DM_Revision
- **Description:** This attribute specifies the DDB L1 revision ID. Once this attribute is written with a nonzero value, it becomes a Read-Only attribute.
- **Size:** 32 bits
- **Offset:** 0x5000
- **Exists:** Always

31:8	Rsvd
7:0	DM_REV

Table 5-158 Fields for Register: DM_Revision

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	DM_REV	R/W	This field specifies the DDB L1 revision ID. Value After Reset: P_DDB_REV Exists: Always

5.6.2 DM_Level

- **Name:** DM_Level
- **Description:** This attribute specifies the DDB level support. This attribute cannot be modified when the DM_Revision attribute has a non-zero value.
- **Size:** 32 bits
- **Offset:** 0x5001
- **Exists:** Always

Rsvd	31:8
DM_LEV	7:0

Table 5-159 Fields for Register: DM_Level

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	DM_LEV	R/W	This field specifies the DDB level support. Value After Reset: P_DDB_LVL Exists: Always

5.6.3 DM_DeviceClass

- **Name:** DM_DeviceClass
- **Description:** This attribute specifies the UniPro device class. This attribute cannot be modified when the DM_Revision attribute has a non-zero value.
- **Size:** 32 bits
- **Offset:** 0x5002
- **Exists:** Always

31:16	Rsvd
15:0	DM_DCL

Table 5-160 Fields for Register: DM_DeviceClass

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	DM_DCL	R/W	This field specifies the UniPro device class. Value After Reset: P_DDB_DCL Exists: Always

5.6.4 DM_ManufacturerID

- **Name:** DM_ManufacturerID
- **Description:** This attribute specifies the UniPro manufacturer ID. This attribute cannot be modified when the DM_Revision attribute has a non-zero value.
- **Size:** 32 bits
- **Offset:** 0x5003
- **Exists:** Always

Rsvd	31:16
DM_MID	15:0

Table 5-161 Fields for Register: DM_ManufacturerID

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	DM_MID	R/W	This field specifies the UniPro manufacturer ID. Value After Reset: P_DDB_MID Exists: Always

5.6.5 DM_ProductID

- **Name:** DM_ProductID
- **Description:** This attribute specifies the user-defined product ID. This attribute cannot be modified when the DM_Revision attribute has a non-zero value.
- **Size:** 32 bits
- **Offset:** 0x5004
- **Exists:** Always

31:16	Rsvd
15:0	DM_PID

Table 5-162 Fields for Register: DM_ProductID

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	DM_PID	R/W	This attribute specifies the user-defined product ID. Value After Reset: P_DDB_PID Exists: Always

5.6.6 DM_Length

- **Name:** DM_Length
- **Description:** This attribute specifies the length of DDB Level 2. For DDB L1, the value is equal to 0. This attribute cannot be modified when the DM_Revision attribute has a non-zero value.
- **Size:** 32 bits
- **Offset:** 0x5005
- **Exists:** Always

31:16	Rsvd
15:0	DM_LEN

Table 5-163 Fields for Register: DM_Length

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	DM_LEN	R/W	This field specifies the length of DDB Level 2. Value After Reset: P_DDB_LGT Exists: Always

5.7 DWC_mipi_unipro_map/DWC_mipi_unipro_dme_qos_block Registers

Quality of Service Attributes. Follow the link for the register to see a detailed description of the register.

Table 5-164 Registers for Address Block: DWC_mipi_unipro_map/DWC_mipi_unipro_dme_qos_block

Register	Offset	Description
DME_TX_DATA_OFL on page 340	0x5100	DME_TX_DATA_OFL
DME_TX_NAC_RECEIVED on page 341	0x5101	DME_TX_NAC_RECEIVED
DME_TX_QOS_COUNT on page 342	0x5102	DME_TX_QOS_COUNT
DME_TX_DL_LM_ERROR on page 343	0x5103	DME_TX_DL_LM_ERROR
DME_RX_DATA_OFL on page 344	0x5110	DME_RX_DATA_OFL
DME_RX_CRC_ERROR on page 345	0x5111	DME_RX_CRC_ERROR
DME_RX_QOS_COUNT on page 346	0x5112	DME_RX_QOS_COUNT
DME_RX_DL_LM_ERROR on page 347	0x5113	DME_RX_DL_LM_ERROR
DME_TXRX_DATA_OFL on page 348	0x5120	DME_TXRX_DATA_OFL
DME_TXRX_PA_INIT_REQUEST on page 349	0x5121	DME_TXRX_PA_INIT_REQUEST
DME_TXRX_QOS_COUNT on page 350	0x5122	DME_TXRX_QOS_COUNT
DME_TXRX_DL_LM_ERROR on page 351	0x5123	DME_TXRX_DL_LM_ERROR
DME_QOS_ENABLE on page 352	0x5130	DME_QOS_ENABLE : Enable QoS monitoring
DME_QOS_STATUS on page 353	0x5131	DME_QOS_STATUS

5.7.1 DME_TX_DATA_OFL

- **Name:** DME_TX_DATA_OFL
- **Description:** This attribute is used define QOS TX data overflow.
- **Size:** 32 bits
- **Offset:** 0x5100
- **Exists:** Always



Table 5-165 Fields for Register: DME_TX_DATA_OFL

Bits	Name	Memory Access	Description
31:5			Reserved Field: Yes
4:0	TX_DATA_OFL	R/W	This field defines upper count range for TX QoS counter counting transmitted PHY symbols to $2^{TX_DATA_OFL+10}$. Value After Reset: 0x0 Exists: Always

5.7.2 DME_TX_NAC_RECEIVED

- **Name:** DME_TX_NAC_RECEIVED
- **Description:** This attribute is used define QoS Threshold for NAC received.
- **Size:** 32 bits
- **Offset:** 0x5101
- **Exists:** Always

31:6	
5:0	TX_NAC_RECEIVED

Table 5-166 Fields for Register: DME_TX_NAC_RECEIVED

Bits	Name	Memory Access	Description
31:6			Reserved Field: Yes
5:0	TX_NAC_RECEIVED	R/W	This field defines threshold for number of NAC_RECEIVED to trigger DME_QoS.ind(TX) during TX QoS counter counts from 0 to $2^{(TX_DATA_OFL+10)}$. Value After Reset: 0x3f Exists: Always

5.7.3 DME_TX_QOS_COUNT

- **Name:** DME_TX_QOS_COUNT
- **Description:** This attribute contains the result value of TX QoS counter.
- **Size:** 32 bits
- **Offset:** 0x5102
- **Exists:** Always



Table 5-167 Fields for Register: DME_TX_QOS_COUNT

Bits	Name	Memory Access	Description
31:0	TX_QOS_COUNT	R	This field contains TX QoS counter value latched at time of last DME_QoS.ind(TX). Indicates the number of transmitted PHY symbols until QoS problem is flagged. Resolution is 2**10 Symbols, a value of 1 means that between 1024 and 2047 PHY symbols have been transmitted. Value After Reset: 0x0 Exists: Always

5.7.4 DME_TX_DL_LM_ERROR

- **Name:** DME_TX_DL_LM_ERROR
- **Description:** This attribute contains the actual counter value of NACs received.
- **Size:** 32 bits
- **Offset:** 0x5103
- **Exists:** Always

31:6	
5:0	
Rsvd	TX_DL_LM_ERROR

Table 5-168 Fields for Register: DME_TX_DL_LM_ERROR

Bits	Name	Memory Access	Description
31:6			Reserved Field: Yes
5:0	TX_DL_LM_ERROR	R	This field contains actual counter value of DL_LM_ERROR.ind (NAC_RECEIVED) events. Value After Reset: 0x0 Exists: Always

5.7.5 DME_RX_DATA_OFL

- **Name:** DME_RX_DATA_OFL
- **Description:** This attribute is used define QOS RX data overflow.
- **Size:** 32 bits
- **Offset:** 0x5110
- **Exists:** Always

31:5	
4:0	RX_DATA_OFL

Table 5-169 Fields for Register: DME_RX_DATA_OFL

Bits	Name	Memory Access	Description
31:5			Reserved Field: Yes
4:0	RX_DATA_OFL	R/W	This field defines upper count range for RX QoS counter counting transmitted PHY symbols to $2^{**}(\text{RX_DATA_OFL}+10)$. Value After Reset: 0x0 Exists: Always

5.7.6 DME_RX_CRC_ERROR

- **Name:** DME_RX_CRC_ERROR
- **Description:** This attribute is used define QoS Threshold for CRC Errors.
- **Size:** 32 bits
- **Offset:** 0x5111
- **Exists:** Always

31:6	
5:0	RX_CRC_ERROR
	Rsvd

Table 5-170 Fields for Register: DME_RX_CRC_ERROR

Bits	Name	Memory Access	Description
31:6			Reserved Field: Yes
5:0	RX_CRC_ERROR	R/W	This field defines threshold for number of CRC_ERRORS required to trigger DME_QoS.ind(RX) during RX QoS counter counts from 0 to RX_DATA_OFL. Value After Reset: 0x3f Exists: Always

5.7.7 DME_RX_QOS_COUNT

- **Name:** DME_RX_QOS_COUNT
- **Description:** This attribute contains the result value of RX QoS counter.
- **Size:** 32 bits
- **Offset:** 0x5112
- **Exists:** Always



Table 5-171 Fields for Register: DME_RX_QOS_COUNT

Bits	Name	Memory Access	Description
31:0	RX_QOS_COUNT	R	This field contains RX QoS counter value latched at time of last DME_QoS.ind(RX). Indicates the number of received PHY Symbols until QoS problem is flagged. Resolution is 2*10 Symbols, a value of 1 means that between 1024 and 2047 PHY symbols have been received. Value After Reset: 0x0 Exists: Always

5.7.8 DME_RX_DL_LM_ERROR

- **Name:** DME_RX_DL_LM_ERROR
- **Description:** This attribute contains the actual count value of CRC errors occurred.
- **Size:** 32 bits
- **Offset:** 0x5113
- **Exists:** Always

31:6	
5:0	RX_DL_LM_ERROR
	Rsvd

Table 5-172 Fields for Register: DME_RX_DL_LM_ERROR

Bits	Name	Memory Access	Description
31:6			Reserved Field: Yes
5:0	RX_DL_LM_ERROR	R	This field contains actual counter value of CRC_ERROR events occurred. Value After Reset: 0x0 Exists: Always

5.7.9 DME_TXRX_DATA_OFL

- **Name:** DME_TXRX_DATA_OFL
- **Description:** This attribute is used define QOS TX RX data overflow.
- **Size:** 32 bits
- **Offset:** 0x5120
- **Exists:** Always



Table 5-173 Fields for Register: DME_TXRX_DATA_OFL

Bits	Name	Memory Access	Description
31:5			Reserved Field: Yes
4:0	TXRX_DATA_OFL	R/W	This field defines upper count range for TXRX counter counting the sum of transmitted and received symbols to 2**(TXRX_DATA_OFL+10). Value After Reset: 0x0 Exists: Always

5.7.10 DME_TXRX_PA_INIT_REQUEST

- **Name:** DME_TXRX_PA_INIT_REQUEST
- **Description:** This attribute is used define QoS Threshold for PA_INIT received.
- **Size:** 32 bits
- **Offset:** 0x5121
- **Exists:** Always



Table 5-174 Fields for Register: DME_TXRX_PA_INIT_REQUEST

Bits	Name	Memory Access	Description
31:6			Reserved Field: Yes
5:0	TXRX_PA_INIT_REQUEST	R/W	This field defines threshold for number of PA_INIT requests required to trigger DME_QoS.ind(PA_INIT) during TXRX QoS counter counts from 0 to TXRX_DATA_OFL. Value After Reset: 0x3f Exists: Always

5.7.11 DME_TXRX_QOS_COUNT

- **Name:** DME_TXRX_QOS_COUNT
- **Description:** This attribute contains the result value of TXRX QoS counter.
- **Size:** 32 bits
- **Offset:** 0x5122
- **Exists:** Always

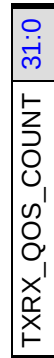


Table 5-175 Fields for Register: DME_TXRX_QOS_COUNT

Bits	Name	Memory Access	Description
31:0	TXRX_QOS_COUNT	R	This field contains TXRX QoS counter value latched at time of last DME_QoS.ind(PA_INIT). Indicates the sum of transmitted and received PHY symbols until QoS problem is flagged. Resolution is 2**10 PHY Symbols. Value After Reset: 0x0 Exists: Always

5.7.12 DME_TXRX_DL_LM_ERROR

- **Name:** DME_TXRX_DL_LM_ERROR
- **Description:** This attribute contains the actual counter value of PA_INIT requests/indications.
- **Size:** 32 bits
- **Offset:** 0x5123
- **Exists:** Always

31:6	
5:0	TXRX_DL_LM_ERROR
	Rsvd

Table 5-176 Fields for Register: DME_TXRX_DL_LM_ERROR

Bits	Name	Memory Access	Description
31:6			Reserved Field: Yes
5:0	TXRX_DL_LM_ERROR	R	This field contains actual counter value of DL_LM_ERROR.ind (PA_INIT) events occurred. Value After Reset: 0x0 Exists: Always

5.7.13 DME_QOS_ENABLE

- **Name:** DME_QOS_ENABLE : Enable QoS monitoring
- **Description:** This attribute is used to enable QoS monitoring.
- **Size:** 32 bits
- **Offset:** 0x5130
- **Exists:** Always

31:3	
2	PA_INIT_ENABLE
1	RX_QOS_ENABLE
0	TX_QOS_ENABLE

Table 5-177 Fields for Register: DME_QOS_ENABLE

Bits	Name	Memory Access	Description
31:3			Reserved Field: Yes
2	PA_INIT_ENABLE	R/W	This field used to enable PA_INIT QOS monitoring. Value After Reset: 0x0 Exists: Always
1	RX_QOS_ENABLE	R/W	This field used to enable RX QOS monitoring. Value After Reset: 0x0 Exists: Always
0	TX_QOS_ENABLE	R/W	This field used to enable TX QOS monitoring. Value After Reset: 0x0 Exists: Always

5.7.14 DME_QOS_STATUS

- **Name:** DME_QOS_STATUS
- **Description:** This attribute contains the status of QoS monitoring. It is a clear on read attribute.
- **Size:** 32 bits
- **Offset:** 0x5131
- **Exists:** Always

31:3	Rsvd
2	PA_INIT_DETECTED
1	RX_QOS_DETECTED
0	TX_QOS_DETECTED

Table 5-178 Fields for Register: DME_QOS_STATUS

Bits	Name	Memory Access	Description
31:3			Reserved Field: Yes
2	PA_INIT_DETECTED	R	This field contains the status of PA_INIT QoS monitoring. Value After Reset: 0x0 Exists: Always
1	RX_QOS_DETECTED	R	This field contains the status of RX QoS monitoring. Value After Reset: 0x0 Exists: Always
0	TX_QOS_DETECTED	R	This field contains the status of TX QoS monitoring. Value After Reset: 0x0 Exists: Always

5.8 DWC_mipi_unipro_map/DWC_mipi_unipro_vsl4_block Registers

Vendor Specific Layer 4 Attributes. Follow the link for the register to see a detailed description of the register.

Table 5-179 Registers for Address Block: DWC_mipi_unipro_map/DWC_mipi_unipro_vsl4_block

Register	Offset	Description
VST_CPortCfg on page 355	0xc000	VST_CPortCfg

5.8.1 VST_CPortCfg

- **Name:** VST_CPortCfg
- **Description:** CPort arbiter configuration attribute. This attribute is controlled by the P_CP top-level parameter.
- **Size:** 32 bits
- **Offset:** 0xc000
- **Exists:** Always

31:16	15:12	11:8	7:4	3:1	0
Rsvd	CTO	Rsvd	QW	Rsvd	PL

Table 5-180 Fields for Register: VST_CPortCfg

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:12	CTO	R/W	This field selects the number of clock cycles for which the arbiter waits when the currently selected/active CPort's request LA_TxDataValid signal is de-asserted without reaching a segment, fragment or message boundary. ■ 0000: 1 cycle ■ 0001: 2 cycles ■ 0010: 3 cycles ■ 1111: 16 cycles This attribute is controlled by the P_CP top-level parameter. Value After Reset: 0x0 Exists: Always
11:8			Reserved Field: Yes

Table 5-180 Fields for Register: VST_CPortCfg (Continued)

Bits	Name	Memory Access	Description
7:4	QW	R/W	This field decides the allocated weight of this CPort in terms of segments. The arbiter services this CPort for the programmed number of consecutive segments or fragments when the request is continuously high. ■ 0000: 1 Segment ■ 0001: 2 Segments ■ 0010: 3 Segments ■ 1111: 16 Segments This attribute is controlled by the P_CP top-level parameter. Value After Reset: 0x0 Exists: Always
3:1			Reserved Field: Yes
0	PL	R/W	This field specifies the priority level for quality of service. #- 0: Best Effort #- 1: Low Latency (High Priority) This attribute is controlled by the P_CP top-level parameter. Values: ■ 0x0 (VALUE_OF_0): Best Effort ■ 0x1 (VALUE_OF_1): Low Latency (High Priority) Value After Reset: 0x0 Exists: Always

5.9 DWC_mipi_unipro_map/DWC_mipi_unipro_Dme_n_VS_block Registers

Dme and Vendor Specific attributes. Follow the link for the register to see a detailed description of the register.

Table 5-181 Registers for Address Block: DWC_mipi_unipro_map/DWC_mipi_unipro_Dme_n_VS_block

Register	Offset	Description
DME_LayerEnable on page 363	0xd000	DME_LayerEnable
DME_Reset on page 364	0xd010	DME_Reset
DME_EndpointReset on page 365	0xd011	DME_EndpointReset
DME_EndpointResetInd on page 366	0xd012	DME_EndpointResetInd
DME_LinkStartup on page 367	0xd020	DME_LinkStartup
DME_LinkStartupInd on page 368	0xd021	DME_LinkStartupInd
DME_LinkLostInd on page 369	0xd022	DME_LinkLostInd
DME_HibernateEnter on page 370	0xd030	DME_HibernateEnter
DME_HibernateEnterInd on page 371	0xd031	DME_HibernateEnterInd
DME_HibernateExit on page 373	0xd032	DME_HibernateExit
DME_HibernateExitInd on page 374	0xd033	DME_HibernateExitInd
DME_PowerModeInd on page 375	0xd040	DME_PowerModeInd
DME_FC0ProtectionTimeOutVal on page 376	0xd041	DME_FC0ProtectionTimeOutVal
DME_TC0ReplayTimeOutVal on page 377	0xd042	DME_TC0ReplayTimeOutVal
DME_AFC0ReqTimeOutVal on page 378	0xd043	DME_AFC0ReqTimeOutVal
DME_FC1ProtectionTimeOutVal on page 379	0xd044	DME_FC1ProtectionTimeOutVal
DME_TC1ReplayTimeOutVal on page 380	0xd045	DME_TC1ReplayTimeOutVal
DME_AFC1ReqTimeOutVal on page 381	0xd046	DME_AFC1ReqTimeOutVal
DME_TestMode on page 382	0xd050	DME_TestMode
DME_TestModeInd on page 383	0xd051	DME_TestModeInd
DME_ErrorPHYInd on page 384	0xd060	DME_ErrorPHYInd
DME_ErrorPAInd on page 386	0xd061	DME_ErrorPAInd
DME_ErrorDInd on page 387	0xd062	DME_ErrorDInd
DME_ErrorNInd on page 389	0xd063	DME_ErrorNInd

Table 5-181 Registers for Address Block: DWC_mipi_unipro_map/DWC_mipi_unipro_Dme_n_VS_block

Register	Offset	Description
DME_ErrorTInd on page 390	0xd064	DME_ErrorTInd
VS_InterruptEnable on page 392	0xd080	VS_InterruptEnable
VS_InterruptStatus on page 395	0xd081	VS_InterruptStatus
VS_L2Status on page 398	0xd082	VS_L2Status
VS_PowerState on page 399	0xd083	VS_PowerState
VS_TxBurstClosureDelay on page 400	0xd084	VS_TxBurstClosureDelay
VS_MphyCfgUpdt on page 401	0xd085	VS_MphyCfgUpdt
VS_AdjustTrailingClocks on page 402	0xd086	VS_AdjustTrailingClocks
VS_SuppressReq on page 403	0xd087	VS_SuppressReq
VS_L2Timeout on page 404	0xd088	VS_L2Timeout
VS_PACapabilityUserData on page 405	0xd089	VS_PACapabilityUserData
VS_BackwardCompatibilityMode on page 406	0xd08a	VS_BackwardCompatibilityMode
VS_EarlySeqNumUpdate on page 407	0xd08b	VS_EarlySeqNumUpdate
VS_DebugAODomain on page 408	0xd08c	VS_DebugAODomain
VS_DebugAdapt on page 410	0xd08d	VS_DebugAdapt
VS_DebugL15ci on page 412	0xd08e	VS_DebugL15ci
VS_TBD on page 416	0xd090	VS_TBD : Depth of the layer 2 transmit buffers
VS_RBD on page 417	0xd091	VS_RBD
VS_DebugTxByteCount on page 418	0xd092	VS_DebugTxByteCount
VS_DebugRxByteCount on page 419	0xd093	VS_DebugRxByteCount
VS_DebugInvalidByteEnable on page 420	0xd094	VS_DebugInvalidByteEnable
VS_DebugLinkStartup on page 421	0xd095	VS_DebugLinkStartup
VS_DebugPwrChg on page 423	0xd096	VS_DebugPwrChg
VS_DebugStates on page 425	0xd097	VS_DebugStates
VS_DebugCounter0 on page 427	0xd098	VS_DebugCounter0
VS_DebugCounter1 on page 428	0xd099	VS_DebugCounter1
VS_DebugCounter0Mask on page 429	0xd09a	VS_DebugCounter0Mask
VS_DebugCounter1Mask on page 431	0xd09b	VS_DebugCounter1Mask

Table 5-181 Registers for Address Block: DWC_mipi_unipro_map/DWC_mipi_unipro_Dme_n_VS_block

Register	Offset	Description
VS_DebugCounterControl on page 433	0xd09c	VS_DebugCounterControl
VS_DebugCounterOverflow on page 434	0xd09d	VS_DebugCounterOverflow
VS_DebugOMC on page 435	0xd09e	VS_DebugOMC
VS_DebugCounterBMask on page 436	0xd09f	VS_DebugCounterBMask
VS_DebugSaveConfigTime on page 438	0xd0a0	VS_DebugSaveConfigTime
VS_DebugLoopback on page 440	0xd0a1	VS_DebugLoopback
VS_Debug_DisCBAutoWrite on page 441	0xd0a2	VS_Debug_DisCBAutoWrite
VS_Debug_Tx_Tst_RtControl on page 442	0xd0a3	VS_Debug_Tx_Tst_RtControl
VS_Debug_Rx_Tst_RtControl on page 443	0xd0a4	VS_Debug_Rx_Tst_RtControl
VS_Debug_CB_Tst_RtControl on page 444	0xd0a5	VS_Debug_CB_Tst_RtControl
VS_Debug_Tx_Tst_RtObserve on page 445	0xd0a6	VS_Debug_Tx_Tst_RtObserve
VS_Debug_Rx_Tst_RtObserve on page 446	0xd0a7	VS_Debug_Rx_Tst_RtObserve
VS_Debug_CB_Tst_RtObserve on page 447	0xd0a8	VS_Debug_CB_Tst_RtObserve
VS_Mk2ExtnSupport on page 448	0xd0ab	VS_Mk2ExtnSupport
VS_ULPH8_Cntrl on page 451	0xd0af	VS_ULPH8_Cntrl
VS_RX_SQ_Cntrl on page 454	0xd0b0	VS_RX_SQ_Cntrl
VS_OverrideAttrEnable on page 456	0xd0b1	VS_OverrideAttrEnable
VS_OverrideAdvTactivate on page 469	0xd0b2	VS_OverrideAdvTactivate
VS_OverrideAdvHiber8 on page 470	0xd0b3	VS_OverrideAdvHibn8
VS_OverrideAdvGran on page 471	0xd0b4	VS_OverrideAdvGranularity
VS_ForceLanealign on page 472	0xd0b5	VS_ForceLanealign
VS_OverridePwmBurstClosureLength on page 473	0xd0b7	VS_OverridePwmBurstClosureLength
VS_OverridePwmSyncLength on page 474	0xd0b8	VS_OverridePwmSyncLength

Table 5-181 Registers for Address Block: DWC_mipi_unipro_map/DWC_mipi_unipro_Dme_n_VS_block

Register	Offset	Description
VS_OverridePwmPrepareLength on page 475	0xd0b9	VS_OverridePwmPrepareLength
VS_SaveConfigTimeIncrement on page 476	0xd0ba	VS_SaveConfigTimeIncrement
VS_OverrideHSG3SyncLength on page 477	0xd0bb	VS_OverrideHSG3SyncLength
VS_OverrideHSG3PrepareLength on page 478	0xd0bc	VS_OverrideHSG3PrepareLength
VS_MphyCfgUpdtAckWait on page 479	0xd0bd	VS_MPhyCfgUpdtAckWait
VS_TxControlActTimer on page 480	0xd0c0	VS_SaveConfigTimeIncrement
VS_MphyDisable on page 481	0xd0c1	VS_MphyDisable
VS_Gen2MphyPIIOn on page 482	0xd0c2	VS_Gen2MphyPIIOn
VS_TxFlowControl on page 484	0xd0ca	VS_TxFlowControl
VS_RxFlowControl on page 485	0xd0cb	VS_RxFlowControl
VS_GPO_Ctrl_0 on page 486	0xd0d0	VS_GPO_Ctrl_0
VS_GPO_Ctrl_1 on page 487	0xd0d1	VS_GPO_Ctrl_1
VS_GPO_Ctrl_2 on page 488	0xd0d2	VS_GPO_Ctrl_2
VS_GPI_Status_0 on page 489	0xd0d3	VS_GPI_Status_0
VS_GPI_Status_1 on page 490	0xd0d4	VS_GPI_Status_1
VS_GPI_Status_2 on page 491	0xd0d5	VS_GPI_Status_2
VS_NonAutoModeBurstControl on page 492	0xd0e0	VS_NonAutoModeBurstControl
VS_OverrideRxMinSleepNoConfigTimeCa pability on page 494	0xd0e1	VS_OverrideRxMinSleepNoConfigTimeCapability
VS_OverrideTxHsUnterminatedLineDrive Capability on page 495	0xd0e2	VS_OverrideTxHsUnterminatedLineDriveCapability
VS_OverrideTxLsTerminatedLineDriveCa pability on page 496	0xd0e3	VS_OverrideTxLsTerminatedLineDriveCapability
VS_OverrideRxMinStallNoConfigTimeCap ability on page 497	0xd0e4	VS_OverrideRxMinStallNoConfigTimeCapability
VS_OverrideRxMinSaveConfigTimeCapab ility on page 498	0xd0e5	VS_OverrideRxMinSaveConfigTimeCapability

Table 5-181 Registers for Address Block: DWC_mipi_unipro_map/DWC_mipi_unipro_Dme_n_VS_block

Register	Offset	Description
VS_OverrideRxExtdMinSaveConfigTimeCapability on page 499	0xd0e6	VS_OverrideRxExtdMinSaveConfigTimeCapability
VS_OverrideRxHsG1PrepareLengthCapability on page 500	0xd0e7	VS_OverrideRxHsG1PrepareLengthCapability
VS_OverrideRxHsG1SyncLengthCapability on page 501	0xd0e8	VS_OverrideRxHsG1SyncLengthCapability
VS_OverrideRxHsG2PrepareLengthCapability on page 502	0xd0e9	VS_OverrideRxHsG2PrepareLengthCapability
VS_OverrideRxHsG2SyncLengthCapability on page 503	0xd0ea	VS_OverrideRxHsG2SyncLengthCapability
VS_OverrideRxHsG4SyncLengthCapability on page 504	0xd0eb	VS_OverrideRxHsG4SyncLengthCapability
VS_OverrideRxHsG4PrepareLengthCapability on page 505	0xd0ec	VS_OverrideRxHsG4PrepareLengthCapability
VS_OverrideRxHsAdaptRefreshCapability on page 506	0xd0ed	VS_OverrideRxHsAdaptRefreshCapability
VS_OverrideRxHsAdaptInitialCapability on page 507	0xd0ee	VS_OverrideRxHsAdaptInitialCapability
VS_DowngradedValueOverrideEnable on page 508	0xd0f0	VS_DowngradedValueOverrideEnable
VS_OverrideTxMinSleepNoConfigTimeDowngraded on page 511	0xd0f1	VS_OverrideTxMinSleepNoConfigTimeDowngraded
VS_OverrideTxMinStallNoConfigTimeDowngraded on page 512	0xd0f2	VS_OverrideTxMinStallNoConfigTimeDowngraded
VS_OverrideTxPwmG6G7SyncLengthDowngraded on page 513	0xd0f3	VS_OverrideTxPwmG6G7SyncLengthDowngraded
VS_OverrideTxLsPrepareLengthDowngraded on page 514	0xd0f4	VS_OverrideTxLsPrepareLengthDowngraded
VS_OverrideTxPwmBurstClosureExtensionDowngraded on page 515	0xd0f5	VS_OverrideTxPwmBurstClosureExtensionDowngraded
VS_LssBypass on page 516	0xd0f6	VS_LssBypass
VS_PmcInterventionEn on page 517	0xd0f7	VS_PmcInterventionEn
VS_PmcInterventionDone on page 519	0xd0f8	VS_PmcInterventionDone
VS_Ext1StatusCtrl on page 520	0xd0f9	VS_Ext1StatusCtrl
VS_MphyLaneCfgUpdt on page 521	0xd0fa	VS_MphyLaneCfgUpdt

Table 5-181 Registers for Address Block: DWC_mipi_unipro_map/DWC_mipi_unipro_Dme_n_VS_block

Register	Offset	Description
VS_ClkMuxSwitchingTimer on page 523	0xd0fb	VS_ClkMuxSwitchingTimer
VS_PMGGlobalCtrl on page 524	0xd100	VS_PMGGlobalCtrl
VS_Debug on page 525	0xd101	VS_Debug
VS_PMGGlobalCounter on page 526	0xd102	VS_PMGGlobalCounter
VS_OverrideRxHsG5SyncLengthCapability on page 527	0xd103	VS_OverrideRxHsG5SyncLengthCapability Register
VS_OverrideRxHsG5PrepareLengthCapability on page 528	0xd104	VS_OverrideRxHsG5PrepareLengthCapability Register
VS_OverrideRxHsG5AdaptRefreshCapability on page 529	0xd105	VS_OverrideRxHsG5AdaptRefreshCapability Register
VS_OverrideRxHsG5AdaptInitialCapability on page 530	0xd106	VS_OverrideRxHsG5AdaptInitialCapability
VS_PMCounter on page 531	0xd110-0xd117	VS_PMCounter
VS_PMMetricSelector on page 532	0xd140-0xd147	VS_PMMetricSelector

5.9.1 DME_LayerEnable

- **Name:** DME_LayerEnable
- **Description:** This attribute enables the boot process by enabling all layers one after the other starting with layer 1.5 and confirms the completion of the layer enable sequence.
- **Size:** 32 bits
- **Offset:** 0xd000
- **Exists:** Always

31:2	1	0
Rsvd	LED	LA

Table 5-182 Fields for Register: DME_LayerEnable

Bits	Name	Memory Access	Description
31:2			Reserved Field: Yes
1	LED	R	When this bit reads 1, it indicates that the LayerEnable request is not accepted as the DME is not in the RESET state. This is cleared after the attribute is read by the application. Value After Reset: 0x0 Exists: Always
0	LA	R/W	When set to 1, a request is given to enable the UniPro stack. When read, this bit confirms the completion of the layer enable sequence: Values: ■ 0x1 (ENABLED): Layer enable sequence is completed. This is cleared after the attribute is read by the application. ■ 0x0 (DISABLED): Layer enable sequence is in progress Value After Reset: 0x0 Exists: Always

5.9.2 DME_Reset

- **Name:** DME_Reset
- **Description:** This attribute activates the internal reset using the provided reset level. It confirms the completion of the reset procedure and provides the type of the ResetMode.
- **Size:** 32 bits
- **Offset:** 0xd010
- **Exists:** Always

31:3	2	1	0
Rsvd	RRRM	RRW	RRC

Table 5-183 Fields for Register: DME_Reset

Bits	Name	Memory Access	Description
31:3			Reserved Field: Yes
2	RRRM	R	When read as 1, HS_MODE is selected as ResetMode. when read as 0, LS_MODE is selected as ResetMode. Values: ■ 0x1 (HS_MODE): ResetMode is HS_MODE ■ 0x0 (LS_MODE): ResetMode is LS_MODE Value After Reset: 0x0 Exists: Always
1	RRW	W	When set to 1, this bit requests for Warm Reset Value After Reset: 0x0 Exists: Always
0	RRC	R/W	When set to 1, this bit requests for cold reset. When read, this bit indicates if the reset procedure (cold and warm) is completed. Values: ■ 0x1 (ENABLED): Reset sequence is not completed ■ 0x0 (DISABLED): Reset sequence is completed Value After Reset: 0x1 Exists: Always

5.9.3 DME_EndpointReset

- **Name:** DME_EndpointReset
- **Description:** This attribute sends an endpoint reset to the peer device and confirms the sending of an endpoint reset to the peer device.
- **Size:** 32 bits
- **Offset:** 0xd011
- **Exists:** Always

31:2	1	0
Rsvd	ENRC	ENR

Table 5-184 Fields for Register: DME_EndpointReset

Bits	Name	Memory Access	Description
31:2			Reserved Field: Yes
1	ENRC	R	When this bit reads 1, it indicates that an error occurred while sending the endpoint reset. This is cleared after the attribute is read by the application. Value After Reset: 0x0 Exists: Always
0	ENR	R/W	When set to 1, Send endpoint reset to peer When read, this bit indicates if the endpoint reset is sent successfully. This is cleared after the attribute is read by the application. Values: ■ 0x0 (DISABLED): Endpoint reset is not sent successfully ■ 0x1 (ENABLED): Endpoint reset is sent successfully Value After Reset: 0x0 Exists: Always

5.9.4 DME_EndpointResetInd

- **Name:** DME_EndpointResetInd
- **Description:** This attribute indicates the reception of an endpoint reset from the peer device.
- **Size:** 32 bits
- **Offset:** 0xd012
- **Exists:** Always

31:1	0
Rsvd	ERI

Table 5-185 Fields for Register: DME_EndpointResetInd

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	ERI	R	When set to 1, this bit indicates that an endpoint reset is received from peer device. This is cleared after the attribute is read by the application. Value After Reset: 0x0 Exists: Always

5.9.5 DME_LinkStartup

- **Name:** DME_LinkStartup
- **Description:** This attribute triggers the link startup procedure and confirms the completion of the last requested link startup sequence.
- **Size:** 32 bits
- **Offset:** 0xd020
- **Exists:** Always

31:2	1	0
Rsvd	LSRC	LSR

Table 5-186 Fields for Register: DME_LinkStartup

Bits	Name	Memory Access	Description
31:2			Reserved Field: Yes
1	LSRC	R	When this bit reads 1, it indicates that the link startup sequence has failed. This is cleared after the attribute is read by the application. Value After Reset: 0x0 Exists: Always
0	LSR	R/W	When set to 1, link startup sequence is initiated by the DWC_mipi_unipro. When this bit reads 1, it indicates that link startup sequence is successful. This is cleared after the attribute is read by the application. Value After Reset: 0x0 Exists: Always

5.9.6 DME_LinkStartupInd

- **Name:** DME_LinkStartupInd
- **Description:** This attribute indicates the detection of a link startup sequence from the remote device.
- **Size:** 32 bits
- **Offset:** 0xd021
- **Exists:** Always

31:1	
Rsvd	LSI

Table 5-187 Fields for Register: DME_LinkStartupInd

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	LSI	R	When this bit reads 1, the link startup sequence is detected. This is cleared after the attribute is read by the application. Value After Reset: 0x0 Exists: Always

5.9.7 DME_LinkLostInd

- **Name:** DME_LinkLostInd
- **Description:** This attribute indicates that the link is lost and the peer device tries a link startup procedure.
- **Size:** 32 bits
- **Offset:** 0xd022
- **Exists:** Always

31:1	0
Rsvd	LLI

Table 5-188 Fields for Register: DME_LinkLostInd

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	LLI	R	When this bit reads 1, it indicates that the link is lost. This is cleared after the attribute is read by the application Value After Reset: 0x0 Exists: Always

5.9.8 DME_HibernateEnter

- **Name:** DME_HibernateEnter
- **Description:** This attribute requests the link to go into hibernation state and indicates the successful reception of the hibernate enter request.
- **Size:** 32 bits
- **Offset:** 0xd030
- **Exists:** Always

31:2	1	0
Rsvd	EH8C	EH8

Table 5-189 Fields for Register: DME_HibernateEnter

Bits	Name	Memory Access	Description
31:2			Reserved Field: Yes
1	EH8C	R	When this bit reads 1, it indicates that the hibernate enter request is denied (for example, due to ongoing power mode change). This is cleared after the attribute is read by the application. Value After Reset: 0x0 Exists: Always
0	EH8	R/W	When set to 1, hibernation entry sequence is requested. When this bit reads 1, it indicates that the hibernate enter request is successfully received and can be processed. This is cleared after the attribute is read by the application. Value After Reset: 0x0 Exists: Always

5.9.9 DME_HibernateEnterInd

- **Name:** DME_HibernateEnterInd
- **Description:** This attribute indicates the completion of the hibernate enter sequence. This attribute is cleared after it is read by the application.
- **Size:** 32 bits
- **Offset:** 0xd031
- **Exists:** Always

31:6	5	4	3	2	1	0
Rsvd	HEN_RC	HEN_RJ	HEN_RA	HEN_RR	HEN_LR	Rsvd

Table 5-190 Fields for Register: DME_HibernateEnterInd

Bits	Name	Memory Access	Description
31:6			Reserved Field: Yes
5	HEN_RC	R	When this bit reads 1, the request is aborted due to a communication problem. The link may not be operable. Value After Reset: 0x0 Exists: Always
4	HEN_RJ	R	When this bit reads 1, the request is rejected due to an impossible configuration. Value After Reset: 0x0 Exists: Always
3	HEN_RA	R	When this bit reads 1, the request is aborted due to concurrent requests Value After Reset: 0x0 Exists: Always
2	HEN_RR	R	When this bit reads 1, the remote request is successfully applied Value After Reset: 0x0 Exists: Always

Table 5-190 Fields for Register: DME_HibernateEnterInd (Continued)

Bits	Name	Memory Access	Description
1	HEN_LR	R	When this bit reads 1, the local request is successfully applied Value After Reset: 0x0 Exists: Always
0			Reserved Field: Yes

5.9.10 DME_HibernateExit

- **Name:** DME_HibernateExit
- **Description:** This attribute requests the link to exit hibernation state and indicates the successful reception of the hibernate exit request. This attribute is cleared after the application reads the register.
- **Size:** 32 bits
- **Offset:** 0xd032
- **Exists:** Always

31:2	1	0
Rsvd	H8EC	H8E

Table 5-191 Fields for Register: DME_HibernateExit

Bits	Name	Memory Access	Description
31:2			Reserved Field: Yes
1	H8EC	R	When this bit reads 1, hibernate exit request is denied. Value After Reset: 0x0 Exists: Always
0	H8E	R/W	When set to 1, hibernation exit sequence is requested. When read 1, hibernate exit request is successfully received and can be processed. Value After Reset: 0x0 Exists: Always

5.9.11 DME_HibernateExitInd

- **Name:** DME_HibernateExitInd
- **Description:** This attribute indicates the completion of the hibernate exit sequence. This attribute is cleared after the application reads the register.
- **Size:** 32 bits
- **Offset:** 0xd033
- **Exists:** Always

31:6	5	4	3	2	1	0
Rsvd	HEX_RC	HEX_RJ	HEX_RA	HEX_RR	HEX_LR	Rsvd

Table 5-192 Fields for Register: DME_HibernateExitInd

Bits	Name	Memory Access	Description
31:6			Reserved Field: Yes
5	HEX_RC	R	The request is aborted due to a communication problem. The link may be inoperable Value After Reset: 0x0 Exists: Always
4	HEX_RJ	R	The request is rejected due to an impossible configuration. Value After Reset: 0x0 Exists: Always
3	HEX_RA	R	The request is aborted due to concurrent requests Value After Reset: 0x0 Exists: Always
2	HEX_RR	R	The remote request is successfully applied Value After Reset: 0x0 Exists: Always
1	HEX_LR	R	The local request is successfully applied Value After Reset: 0x0 Exists: Always
0			Reserved Field: Yes

5.9.12 DME_PowerModelInd

- **Name:** DME_PowerModelInd
- **Description:** This attribute indicates the result of the last power mode change. This attribute is cleared after the application reads the register.
- **Size:** 32 bits
- **Offset:** 0xd040
- **Exists:** Always

31:6	5	4	3	2	1	0
Rsvd	PWI_RC	PWI_RJ	PWI_RA	PWI_RR	PWI_LR	Rsvd

Table 5-193 Fields for Register: DME_PowerModelInd

Bits	Name	Memory Access	Description
31:6			Reserved Field: Yes
5	PWI_RC	R	The request is aborted due to a communication problem. The link may be inoperable Value After Reset: 0x0 Exists: Always
4	PWI_RJ	R	The request is rejected due to an impossible configuration. Value After Reset: 0x0 Exists: Always
3	PWI_RA	R	The request is aborted due to concurrent requests Value After Reset: 0x0 Exists: Always
2	PWI_RR	R	The remote request is successfully applied Value After Reset: 0x0 Exists: Always
1	PWI_LR	R	The local request is successfully applied Value After Reset: 0x0 Exists: Always
0			Reserved Field: Yes

5.9.13 DME_FC0ProtectionTimeOutVal

- **Name:** DME_FC0ProtectionTimeOutVal
- **Description:** This attribute stores the FC protection time out value (16 bit) for TC0 applied with the next power mode change.
- **Size:** 32 bits
- **Offset:** 0xd041
- **Exists:** Always

31:16	
15:0	
Rsvd	DME_FC0PTV

Table 5-194 Fields for Register: DME_FC0ProtectionTimeOutVal

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	DME_FC0PTV	R/W	This field specifies the next protection time out value for TC0 Value After Reset: 0x1fff Exists: Always

5.9.14 DME_TC0ReplayTimeOutVal

- **Name:** DME_TC0ReplayTimeOutVal
- **Description:** This attribute stores the replay time out value (16 bit) for TC0 applied with the next power mode change.
- **Size:** 32 bits
- **Offset:** 0xd042
- **Exists:** Always

31:16	Rsvd
15:0	DME_TC0_RTV

Table 5-195 Fields for Register: DME_TC0ReplayTimeOutVal

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	DME_TC0_RTV	R/W	This field specifies the next replay time out value for TC0 Value After Reset: 0xffff Exists: Always

5.9.15 DME_AFC0ReqTimeOutVal

- **Name:** DME_AFC0ReqTimeOutVal
- **Description:** This attribute stores the request time out value (16 bit) for TC0 applied with the next power mode change.
- **Size:** 32 bits
- **Offset:** 0xd043
- **Exists:** Always

31:16	
15:0	DME_AFC0_RQTV
Rsvd	

Table 5-196 Fields for Register: DME_AFC0ReqTimeOutVal

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	DME_AFC0_RQTV	R/W	This field specifies the next request time out value for TC0 Value After Reset: 0x7fff Exists: Always

5.9.16 DME_FC1ProtectionTimeOutVal

- **Name:** DME_FC1ProtectionTimeOutVal
- **Description:** This attribute stores the FC protection time out value (16 bit) for TC1 applied with the next power mode change.
- **Size:** 32 bits
- **Offset:** 0xd044
- **Exists:** Always

31:16	
15:0	
Rsvd	DME_FC1PTV

Table 5-197 Fields for Register: DME_FC1ProtectionTimeOutVal

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	DME_FC1PTV	R/W	This field specifies the next protection time out value for TC1 Value After Reset: 0x1fff Exists: Always

5.9.17 DME_TC1ReplayTimeOutVal

- **Name:** DME_TC1ReplayTimeOutVal
- **Description:** This attribute stores the replay time out value (16 bit) for TC1 applied with the next power mode change.
- **Size:** 32 bits
- **Offset:** 0xd045
- **Exists:** Always



Table 5-198 Fields for Register: DME_TC1ReplayTimeOutVal

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	DME_TC1_RTV	R/W	This fields specifies the next replay time out value for TC1 Value After Reset: 0xffff Exists: Always

5.9.18 DME_AFC1ReqTimeOutVal

- **Name:** DME_AFC1ReqTimeOutVal
- **Description:** This attribute stores the request time out value (16 bit) for TC1 applied with the next power mode change.
- **Size:** 32 bits
- **Offset:** 0xd046
- **Exists:** Always

31:16	Rsvd
15:0	DME_AFC1_RQTV

Table 5-199 Fields for Register: DME_AFC1ReqTimeOutVal

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15:0	DME_AFC1_RQTV	R/W	This fields specifies the next request time out value for TC1 Value After Reset: 0x7fff Exists: Always

5.9.19 DME_TestMode

- **Name:** DME_TestMode
- **Description:** This attribute triggers sending of a PACP frame to the peer device and indicates that the last test mode request is sent. This attribute is cleared after the application reads the register.
- **Size:** 32 bits
- **Offset:** 0xd050
- **Exists:** Always

31:2	1	0
Rsvd	TMC	TMR

Table 5-200 Fields for Register: DME_TestMode

Bits	Name	Memory Access	Description
31:2			Reserved Field: Yes
1	TMC	R	When this bit reads 1, it indicates that the test mode request is rejected. Value After Reset: 0x0 Exists: Always
0	TMR	R/W	When set to 1, a request is given to the DWC_mipi_unipro to enter test mode and to send test mode PACP frame to the peer device (last test mode request). When this bit reads 1, it indicates that the test mode request is accepted and a test mode PACP frame is sent. Value After Reset: 0x0 Exists: Always

5.9.20 DME_TestModeInd

- **Name:** DME_TestModeInd
- **Description:** This attribute indicates the reception of a test mode request from the peer device and the local device is set to test mode.
- **Size:** 32 bits
- **Offset:** 0xd051
- **Exists:** Always

31:1	0
Rsvd	TMI

Table 5-201 Fields for Register: DME_TestModeInd

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	TMI	R	When this bit reads 1, it indicates that the test mode is entered. Value After Reset: 0x0 Exists: Always

5.9.21 DME_ErrorPHYInd

- **Name:** DME_ErrorPHYInd
- **Description:** This attribute indicates the error event detected by L1 (M-PHY) showing one of the following:
 - Link error during a transmission
 - MPx_RX_DataNCtrl for lower byte is set
 - A wrong MK1 is detected (not part of TRG0, TRG1, TRG2, or PACP packet).

Once set, the bits get updated only after the attribute is read by the application.

- **Size:** 32 bits
- **Offset:** 0xd060
- **Exists:** Always

31:4	3	2	1	0
Rsvd	EPIL3	EPIL2	EPIL1	EPIL0

Table 5-202 Fields for Register: DME_ErrorPHYInd

Bits	Name	Memory Access	Description
31:4			Reserved Field: Yes
3	EPIL3	R	When this bit reads 1, it indicates that error detected on physical lane 3. Value After Reset: 0x0 Exists: Always
2	EPIL2	R	When this bit reads 1, it indicates that error is detected on physical lane 2. Value After Reset: 0x0 Exists: Always
1	EPIL1	R	When this bit reads 1, it indicates that error is detected on physical lane 1. Value After Reset: 0x0 Exists: Always

Table 5-202 Fields for Register: DME_ErrorPHYInd (Continued)

Bits	Name	Memory Access	Description
0	EPILO	R	When this bit reads 1, it indicates that error is detected on physical lane 0. Value After Reset: 0x0 Exists: Always

5.9.22 DME_ErrorPAInd

- **Name:** DME_ErrorPAInd
- **Description:** This attribute indicates the detection of a PA layer line reset. Once set, the bits gets updated only after the attribute is read by the application.
- **Size:** 32 bits
- **Offset:** 0xd061
- **Exists:** Always

31:2	1	0
Rsvd	LRRX	LRTX

Table 5-203 Fields for Register: DME_ErrorPAInd

Bits	Name	Memory Access	Description
31:2			Reserved Field: Yes
1	LRRX	R	When this bit reads 1, line reset is detected on inbound lanes (RX) Value After Reset: 0x0 Exists: Always
0	LRTX	R	When this bit reads 1, line reset is detected on output lanes (TX) Value After Reset: 0x0 Exists: Always

5.9.23 DME_ErrorDInd

- **Name:** DME_ErrorDInd
- **Description:** This attribute indicates the error event detected by the data link layer. Once set, the bits gets updated only after the attribute is read by the application.
- **Size:** 32 bits
- **Offset:** 0xd062
- **Exists:** Always

31:16	Rsvd
15	EDI_PAINIT
14	EDI_PEIR
13	EDI_PIE
12	EDI_BCST
11	EDI_FSE
10	EDI_ESE
9	EDI_NFSE
8	EDI_AFSE
7	EDI_WSN
6	EDI_MFLE
5	EDI_RBO
4	EDI_CE
3	EDI_FPTE
2	EDI_ARTE
1	EDI_TRTE
0	EDI_NR

Table 5-204 Fields for Register: DME_ErrorDInd

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15	EDI_PAINIT	R	When this bit reads 1, it indicates that the DL layer has issued a PA_INIT.req to the PA layer or that the DL layer has received a PA_INIT.ind from the PA layer. Value After Reset: 0x0 Exists: (UPRO_V18_EN)
14	EDI_PEIR	R	When this bit reads 1, it indicates that PA error indication is received. Value After Reset: 0x0 Exists: Always
13	EDI_PIE	R	When this bit reads 1, it indicates PA initialization error. Value After Reset: 0x0 Exists: Always
12	EDI_BCST	R	When this bit reads 1, it indicates bad control symbol type. Value After Reset: 0x0 Exists: Always
11	EDI_FSE	R	When this bit reads 1, it indicates frame syntax error. Value After Reset: 0x0 Exists: Always

Table 5-204 Fields for Register: DME_ErrorDInd (Continued)

Bits	Name	Memory Access	Description
10	EDI_ESE	R	When this bit reads 1, it indicates EOF syntax error. Value After Reset: 0x0 Exists: Always
9	EDI_NFSE	R	When this bit reads 1, it indicates NAC frame syntax error. Value After Reset: 0x0 Exists: Always
8	EDI_AFSE	R	When this bit reads 1, it indicates AFC frame syntax error. Value After Reset: 0x0 Exists: Always
7	EDI_WSN	R	When this bit reads 1, it indicates wrong sequence number. Value After Reset: 0x0 Exists: Always
6	EDI_MFLE	R	When this bit reads 1, it indicates that maximum frame length is exceeded. Value After Reset: 0x0 Exists: Always
5	EDI_RBO	R	When this bit reads 1, it indicates RX buffer overflow. Value After Reset: 0x0 Exists: Always
4	EDI_CE	R	When this bit reads 1, it indicates CRC error. Value After Reset: 0x0 Exists: Always
3	EDI_FPTE	R	When this bit reads 1, it indicates that FCx protection timer is expired. Value After Reset: 0x0 Exists: Always
2	EDI_ARTE	R	When this bit reads 1, it indicates that AFCx request timer is expired. Value After Reset: 0x0 Exists: Always
1	EDI_TRTE	R	When this bit reads 1, it indicates that TCx replay timer is expired. Value After Reset: 0x0 Exists: Always
0	EDI_NR	R	When this bit reads 1, it indicates that NAC is received. Value After Reset: 0x0 Exists: Always

5.9.24 DME_ErrorNInd

- **Name:** DME_ErrorNInd
- **Description:** This attribute indicates the error event detected by the network layer. Once set, the bits gets updated only after the attribute is read by the application.
- **Size:** 32 bits
- **Offset:** 0xd063
- **Exists:** Always

31:3	2	1	0
Rsvd	ENI_LPTD	ENI_BDE	ENI_UHT

Table 5-205 Fields for Register: DME_ErrorNInd

Bits	Name	Memory Access	Description
31:3			Reserved Field: Yes
2	ENI_LPTD	R	When this bit reads 1, it indicates LHDR trap packet dropping Value After Reset: 0x0 Exists: Always
1	ENI_BDE	R	When this bit reads 1, it indicates bad device ID Value After Reset: 0x0 Exists: Always
0	ENI_UHT	R	When this bit reads 1, it indicates unsupported header type Value After Reset: 0x0 Exists: Always

5.9.25 DME_ErrorTInd

- **Name:** DME_ErrorTInd
- **Description:** This attribute indicates the error event detected by the transport layer. Once set, the bits gets updated only after the attribute is read by the application.
- **Size:** 32 bits
- **Offset:** 0xd064
- **Exists:** Always

31:7	Rsvd
6	ETI_SVD
5	ETI_ECO
4	ETI_BT
3	ETI_CSD
2	ETI_NCR
1	ETI_UC
0	ETI_UHT

Table 5-206 Fields for Register: DME_ErrorTInd

Bits	Name	Memory Access	Description
31:7			Reserved Field: Yes
6	ETI_SVD	R	When this bit reads 1, it indicates that Safety value is dropping Value After Reset: 0x0 Exists: Always
5	ETI_ECO	R	When this bit reads 1, it indicates E2E credit overflow Value After Reset: 0x0 Exists: Always
4	ETI_BT	R	When this bit reads 1, it indicates Bad TC Value After Reset: 0x0 Exists: Always
3	ETI_CSD	R	When this bit reads 1, it indicates that controlled segment is dropping. Value After Reset: 0x0 Exists: Always
2	ETI_NCR	R	When this bit reads 1, it indicates no RX connection. Value After Reset: 0x0 Exists: Always

Table 5-206 Fields for Register: DME_ErrorTInd (Continued)

Bits	Name	Memory Access	Description
1	ETI_UC	R	When this bit reads 1, it indicates unknown CPort ID. Value After Reset: 0x0 Exists: Always
0	ETI_UHT	R	When this bit reads 1, it indicates unsupported header type. Value After Reset: 0x0 Exists: Always

5.9.26 VS_InterruptEnable

- **Name:** VS_InterruptEnable
- **Description:** Individual interrupt sources can be activated by writing 1 into the corresponding bit location. Initial value of each bit is 0 (disabled).
- **Size:** 32 bits
- **Offset:** 0xd080
- **Exists:** Always

31:16	Rsvd
15	IE_QI
14	IE_LSC
13	IE_DC
12	IE_IE
11	IE_ETIE
10	IE_EN
9	IE_ED
8	IE_EPA
7	IE_EP
6	IE_TMI
5	IE_PMI
4	IE_HEXI
3	IE_HENI
2	IE_LLI
1	IE_LSI
0	IE_ERI

Table 5-207 Fields for Register: VS_InterruptEnable

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15	IE_QI	R/W	When set to 1, this bit enables the PHY link degradation indication interrupt (for more information, see QosInd). Value After Reset: 0x0 Exists: Always
14	IE_LSC	R/W	When set to 1, this bit enables the link startup confirmation interrupt (for more information, see LinkStartupCnf). Value After Reset: 0x0 Exists: Always
13	IE_DC	R/W	When set to 1, this bit enables the debug counter overflow interrupt (for more information, see VS_DebugCounterOverflow). Value After Reset: 0x0 Exists: Always
12	IE_IE	R/W	When set to 1, this bit enables the PA initialization error (L2) interrupt (for more information, see PA Init Error) Value After Reset: 0x0 Exists: Always

Table 5-207 Fields for Register: VS_InterruptEnable (Continued)

Bits	Name	Memory Access	Description
11	IE_ETIE	R/W	When set to 1, this bit enables the transport layer error indication PA interrupt (for more information, see ErrorTInd) Value After Reset: 0x0 Exists: Always
10	IE_EN	R/W	When set to 1, this bit enables the network layer error indication interrupt (for more information, see ErrorNInd) Value After Reset: 0x0 Exists: Always
9	IE_ED	R/W	When set to 1, this bit enables the data link layer error indication interrupt (for more information, see ErrorDInd) Value After Reset: 0x0 Exists: Always
8	IE_EPA	R/W	When set to 1, this bit enables the PA layer line reset error indication interrupt (for more information, see ErrorPAInd) Value After Reset: 0x0 Exists: Always
7	IE_EP	R/W	When set to 1, this bit enables the PHY error indication interrupt (for more information, see ErrorPHYInd) Value After Reset: 0x0 Exists: Always
6	IE_TMI	R/W	When set to 1, this bit enables the test mode indication interrupt (for more information, see TestModeInd) Value After Reset: 0x0 Exists: Always
5	IE_PMI	R/W	When set to 1, this bit enables the power mode indication interrupt (for more information, see PowerModeInd) Value After Reset: 0x0 Exists: Always
4	IE_HEXI	R/W	When set to 1, this bit enables the hibernate exit indication interrupt (for more information, see HibernateExitInd) Value After Reset: 0x0 Exists: Always
3	IE_HENI	R/W	When set to 1, this bit enables the hibernate enter indication interrupt (for more information, see HibernateEnterInd) Value After Reset: 0x0 Exists: Always

Table 5-207 Fields for Register: VS_InterruptEnable (Continued)

Bits	Name	Memory Access	Description
2	IE_LLI	R/W	When set to 1, this bit enables the link loss indication interrupt (for more information, see LinkLostInd) Value After Reset: 0x0 Exists: Always
1	IE_LSI	R/W	When set to 1, this bit enables the link startup indication interrupt (for more information, see LinkStartupInd) Value After Reset: 0x0 Exists: Always
0	IE_ERI	R/W	When set to 1, this bit enables the endpoint reset indication interrupt (for more information, see EndpointResetInd) Value After Reset: 0x0 Exists: Always

5.9.27 VS_InterruptStatus

- **Name:** VS_InterruptStatus
- **Description:** This attribute shows the occurrence of any of the defined interrupt sources independent of the interrupt enable settings. Bit assignment is the same as the interrupt enable attribute.
- **Size:** 32 bits
- **Offset:** 0xd081
- **Exists:** Always

31:16	Rsvd
15	IS_QI
14	IS_LSC
13	IS_DC
12	IS_IE
11	IS_ETIE
10	IS_EN
9	IS_ED
8	IS_EPA
7	IS_EP
6	IS_TMI
5	IS_PMI
4	IS_HEXI
3	IS_HENI
2	IS_LLI
1	IS_LSI
0	IS_ERI

Table 5-208 Fields for Register: VS_InterruptStatus

Bits	Name	Memory Access	Description
31:16			Reserved Field: Yes
15	IS_QI	R	When this bit reads 1, it indicates PHY link degradation (for more information, see QosInd) Value After Reset: 0x0 Exists: Always
14	IS_LSC	R	When this bit reads 1, it indicates link startup confirmation (for more information, see LinkStartupCnf) Value After Reset: 0x0 Exists: Always
13	IS_DC	R	When this bit reads 1, it indicates debug counter overflow interrupt (for more information, see VS_DebugCounterOverflow) Value After Reset: 0x0 Exists: Always
12	IS_IE	R	When this bit reads 1, it indicates PA initialization error (L2) interrupt (for more information, see PA Init Error) Value After Reset: 0x0 Exists: Always

Table 5-208 Fields for Register: VS_InterruptStatus (Continued)

Bits	Name	Memory Access	Description
11	IS_ETIE	R	When this bit reads 1, it indicates transport layer error indication PA interrupt (for more information, see ErrorTInd) Value After Reset: 0x0 Exists: Always
10	IS_EN	R	When this bit reads 1, it indicates network layer error indication interrupt (for more information, see ErrorNInd) Value After Reset: 0x0 Exists: Always
9	IS_ED	R	When this bit reads 1, it indicates data link layer error indication interrupt (for more information, see ErrorDInd) Value After Reset: 0x0 Exists: Always
8	IS_EPA	R	When this bit reads 1, it indicates PA layer line reset error indication interrupt (for more information, see ErrorPAInd) Value After Reset: 0x0 Exists: Always
7	IS_EP	R	When this bit reads 1, it indicates PHY error indication interrupt (for more information, see ErrorPHYInd) Value After Reset: 0x0 Exists: Always
6	IS_TMI	R	When this bit reads 1, it indicates test mode indication interrupt (for more information, see TestModeInd) Value After Reset: 0x0 Exists: Always
5	IS_PMI	R	When this bit reads 1, it indicates power mode indication interrupt (for more information, see PowerModeInd) Value After Reset: 0x0 Exists: Always
4	IS_HEXI	R	When this bit reads 1, it indicates hibernate exit indication interrupt (for more information, see HibernateExitInd) Value After Reset: 0x0 Exists: Always
3	IS_HENI	R	When this bit reads 1, it indicates hibernate enter indication interrupt (for more information, see HibernateEnterInd) Value After Reset: 0x0 Exists: Always

Table 5-208 Fields for Register: VS_InterruptStatus (Continued)

Bits	Name	Memory Access	Description
2	IS_LLI	R	When this bit reads 1, it indicates link loss indication interrupt (for more information, see LinkLostInd) Value After Reset: 0x0 Exists: Always
1	IS_LSI	R	When this bit reads 1, it indicates link startup indication interrupt (for more information, see LinkStartupInd) Value After Reset: 0x0 Exists: Always
0	IS_ERI	R	When this bit reads 1, it indicates endpoint reset indication interrupt (for more information, see EndpointResetInd) Value After Reset: 0x0 Exists: Always

5.9.28 VS_L2Status

- **Name:** VS_L2Status
- **Description:** This attribute reflects the current status of L2; bits are not cleared when read. Due to the nature of the attribute, it does not trigger an interrupt.
- **Size:** 32 bits
- **Offset:** 0xd082
- **Exists:** Always

31:3	2	1	0
Rsvd	L2S_UTC0	L2S_UTC1	L2S_CFT

Table 5-209 Fields for Register: VS_L2Status

Bits	Name	Memory Access	Description
31:3			Reserved Field: Yes
2	L2S_UTC0	R	When this bit reads 1, it indicates that the Unacknowledged TC0 frames are queued Value After Reset: 0x0 Exists: Always
1	L2S_UTC1	R	When this bit reads 1, it indicates that the Unacknowledged TC1 frames are queued Value After Reset: 0x0 Exists: Always
0	L2S_CFT	R	When this bit reads 1, it indicates that the Control frames are queued for transmission Value After Reset: 0x0 Exists: Always

5.9.29 VS_PowerState

- **Name:** VS_PowerState
- **Description:** This attribute specifies the state of the link.
- **Size:** 32 bits
- **Offset:** 0xd083
- **Exists:** Always

31:3	2:0
Rsvd	PS

Table 5-210 Fields for Register: VS_PowerState

Bits	Name	Memory Access	Description
31:3			Reserved Field: Yes
2:0	PS	R	Indicates the link powerstate #- 0: DISABLE #- 1: LINKDOWN #- 2: LINKUP #- 3: HIBERNATE #- 4: LINKLOST #- 5: LINKCFG Values: ■ 0x0 (VALUE_OF_0): DISABLE ■ 0x1 (VALUE_OF_1): LINKDOWN ■ 0x2 (VALUE_OF_2): LINKUP ■ 0x3 (VALUE_OF_3): HIBERNATE ■ 0x4 (VALUE_OF_4): LINKLOST ■ 0x5 (VALUE_OF_5): LINKCFG Value After Reset: 0x0 Exists: Always

5.9.30 VS_TxBurstClosureDelay

- **Name:** VS_TxBurstClosureDelay
- **Description:** This attribute specifies the delay in TX clock cycles to wait before initiating a burst closure. When there is no more data to send, a timer is started with that delay value. When the timer expires, burst closure is initiated. The data arriving after the expiry of initiation of burst closure results in the start of a new burst including the prepare and sync periods. In Unipro V2.0, By default PA_EoBDealy attribute is used to delay the Tx Burst closure but Write to VS_TxBurstClosureDelay register will override the effect of PA_EoBDealy attribute.If PA_EoBDealy attribute is used, then usage of VS_TxBurstClosureDelay is discouraged
- **Size:** 32 bits
- **Offset:** 0xd084
- **Exists:** Always

31:8	7:0
Rsvd	BCD

Table 5-211 Fields for Register: VS_TxBurstClosureDelay

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	BCD	R/W	Burst closure delay (initially 0) Value After Reset: 0x8 Exists: Always

5.9.31 VS_MphyCfgUpdt

- **Name:** VS_MphyCfgUpdt
- **Description:** Any write access to this attribute triggers a configuration update of all connected TX and RX M-PHY modules.
- **Size:** 32 bits
- **Offset:** 0xd085
- **Exists:** Always



Table 5-212 Fields for Register: VS_MphyCfgUpdt

Bits	Name	Memory Access	Description
31:0	MCU	R/W	Any write access to this attribute triggers a configuration update of all connected TX and RX M-PHY modules. Before link startup, all lanes are updated. Reading this attribute always returns 0. Timing of the configuration update needs to be handled by the application, that is, by checking SaveConfigTime. Note: It is not recommended to use this mechanism to update M-PHY related attributes. A power mode change (even without changing link properties) should be used instead. Value After Reset: 0x0 Exists: Always

5.9.32 VS_AdjustTrailingClocks

- **Name:** VS_AdjustTrailingClocks
- **Description:** This attribute controls the value of the PA_MinRxTrailingClocks attribute.
- **Size:** 32 bits
- **Offset:** 0xd086
- **Exists:** Always



Table 5-213 Fields for Register: VS_AdjustTrailingClocks

Bits	Name	Memory Access	Description
x:0	ATC	R/W	Controls the value of the PA_MinRxTrailingClocks attribute. The initial value is defined by parameter P_RXTRAIL. Any write access overwrites this value. Note: After reset, the parameter value is valid again. Setting this attribute to a too low value results in reception errors. For information about the calculation of minimum requirement of PA_MinRxTrailingClocks, see the <i>Minimum RX Trailing Clocks</i> section in <i>DWC MIPI UniPro User Guide</i>.For backward compatibility purposes, the range is limited t 255. Value After Reset: P_RXTRAIL Exists: Always Range Variable[x]: "(UPRO_V2_EN) ? \"10\" : \"8\" - 1

5.9.33 VS_SuppressReq

- **Name:** VS_SuppressReq
- **Description:** This attribute configures the NAC frames.
- **Size:** 32 bits
- **Offset:** 0xd087
- **Exists:** Always

31:1	
Rsvd	SRNAC
	0

Table 5-214 Fields for Register: VS_SuppressReq

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	SRNAC	R/W	Configures the NAC frames: Values: ■ 0x0 (NAC_REQ): NAC frames with RReq bit are generated as defined by MIPI UniPro 1.41 specification (may lead to double link reinitialization) ■ 0x1 (AVOID_REQ): Avoids NAC frames with RReq bit. Set to avoid double link reinitialization Value After Reset: 0x1 Exists: Always

5.9.34 VS_L2Timeout

- **Name:** VS_L2Timeout
- **Description:** This attribute enables or disables the layer 2 time out.
- **Size:** 32 bits
- **Offset:** 0xd088
- **Exists:** Always

31:1	
Rsvd	L2TDS

Table 5-215 Fields for Register: VS_L2Timeout

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	L2TDS	R/W	Timeout control Values: ■ 0x1 (ENABLED): Timeout is enabled ■ 0x0 (DISABLED): Timeout is disabled (default) Value After Reset: 0x0 Exists: Always

5.9.35 VS_PACapabilityUserData

- **Name:** VS_PACapabilityUserData
- **Description:** This attribute allows the user to program the private capability data to the peer DMA.
- **Size:** 32 bits
- **Offset:** 0xd089
- **Exists:** Always

31:1	
Rsvd	DLCreditUnit_Mode
	0

Table 5-216 Fields for Register: VS_PACapabilityUserData

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	DLCreditUnit_Mode	* Varies	This field is used to program the CreditUnitMode of the controller if the configuration parameter P_L2_BUF_CREDUNIT_MODE is set to 1. - 0 - CreditUnitMode-32 - 1 - CreditUnitMode-128 This register is read-only and will be set to 0 (CreditUnitMode-32) if parameter P_L2_BUF_CREDUNIT_MODE is set to 0. NOTE: Please ensure L2 Tx and Rx buffer sizes configured are sufficient to meet the functionality and throughput based on the Maximum Payload Length of chosen CreditUnitMode. Values: ■ 0x0 (CREDIT_UNIT_32): Credit Unit 32 Mode is selected ■ 0x1 (CREDIT_UNIT_128): Credit Unit 128 Mode is selected Value After Reset: P_L2_BUF_CREDUNIT_MODE Exists: Always Memory Access: "(P_L2_BUF_CREDUNIT_MODE==1) ? \"read-write\" : \"read-only\""

5.9.36 VS_BackwardCompatibilityMode

- **Name:** VS_BackwardCompatibilityMode
- **Description:** This attribute allows the application to enable backward compatibility to V1.8 when UniPro is used in V2.0 mode.
- **Size:** 32 bits
- **Offset:** 0xd08a
- **Exists:** False

31:1	
Rsvd	BCKCOMP
	0

Table 5-217 Fields for Register: VS_BackwardCompatibilityMode

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	BCKCOMP	R/W	Backward Compatibility to V1.8 Values: ■ 0x1 (ENABLED): Enable the Backward Compatibility mode in Unipro v2.0 ■ 0x0 (DISABLED): Disable the Backward Compatibility mode in Unipro v2.0 Value After Reset: 0x0 Exists: Always

5.9.37 VS_EarlySeqNumUpdate

- **Name:** VS_EarlySeqNumUpdate
- **Description:** This attribute enables or disables early sequence number update
- **Size:** 32 bits
- **Offset:** 0xd08b
- **Exists:** Always

31:1	
0	EARLYSEQNUMEN
Rsvd	

Table 5-218 Fields for Register: VS_EarlySeqNumUpdate

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	EARLYSEQNUMEN	R/W	If enabled Sequence number is updated in AFC frame after CRC of received frame is correct, When disabled Sequence number is updated after application reads the data from buffer Values: ■ 0x1 (ENABLED): Early Sequence number update is enabled ■ 0x0 (DISABLED): Early Sequence number update is disabled (default) Value After Reset: P_EARLY_SEQ_NUM_RST_VAL Exists: Always

5.9.38 VS_DebugAODomain

- **Name:** VS_DebugAODomain
- **Description:** This attribute provides debug information for always on domain when UniPro is configured for V1.8 with LPM enabled.
- **Size:** 32 bits
- **Offset:** 0xd08c
- **Exists:** Always

Rsvd	31:18
DAO_RESETCOMPLETE	17
DAO_ISOEN	16
DAO_SYSCLOCKREQ	15
DAO_CLKGATEDIS	14
DAO_LPCLOCKSEL	13
DAO_LPH8EXIT	12
DAO_LPH8STATE	11
DAO_RXH8EXIT	10
DAO_RXH8EXITLANE	9:6
DAO_LPRESACK	5
DAO_LPSAVEACK	4
DAO_LPSAVEEN	3
DAO_LPRESEN	2
DAO_TXRXSAVE	1
DAO_H8EXIT	0

Table 5-219 Fields for Register: VS_DebugAODomain

Bits	Name	Memory Access	Description
31:18			Reserved Field: Yes
17	DAO_RESETCOMPLETE	R	LP reset release operation complete indication Value After Reset: 0x0 Exists: Always
16	DAO_ISOEN	R	LP isolation enable indication Value After Reset: 0x0 Exists: Always
15	DAO_SYSCLOCKREQ	R	LP sys Clock request indication Value After Reset: 0x1 Exists: Always
14	DAO_CLKGATEDIS	R	LP sys Clock gating disable Value After Reset: 0x0 Exists: Always

Table 5-219 Fields for Register: VS_DebugAODomain (Continued)

Bits	Name	Memory Access	Description
13	DAO_LPCLKSEL	R	LP Clock switching select indication Value After Reset: 0x0 Exists: Always
12	DAO_LPH8EXIT	R	LP hibernate exit request from L15 layer Value After Reset: 0x0 Exists: Always
11	DAO_LPH8STATE	R	LP hibernate state indication Value After Reset: 0x0 Exists: Always
10	DAO_RXH8EXIT	R	Rx interface hiberante exit indication on any lane Value After Reset: 0x0 Exists: Always
9:6	DAO_RXH8EXITLANE	R	Rx interface hiberante exit indication for indivisual lanes Value After Reset: 0x0 Exists: Always
5	DAO_LPRESACK	R	Low power restore complete Value After Reset: 0x0 Exists: Always
4	DAO_LPSAVEACK	R	Low power save complete Value After Reset: 0x0 Exists: Always
3	DAO_LPSAVEEN	R	Low power save enable Value After Reset: 0x0 Exists: Always
2	DAO_LPRESEN	R	Low power restrore enable Value After Reset: 0x0 Exists: Always
1	DAO_TXRXSAVE	R	TX and RX interfaces are into Save State Value After Reset: 0x0 Exists: Always
0	DAO_H8EXIT	R	Local hiberante exit operation request Value After Reset: 0x0 Exists: Always

5.9.39 VS_DebugAdapt

- **Name:** VS_DebugAdapt
- **Description:** This attribute provides debug information related to ADAPT variables at L15 layer when UniPro is configured for V1.8.
- **Size:** 32 bits
- **Offset:** 0xd08d
- **Exists:** Always

Rsvd	31:27
TESTSTATE	26:20
RXBURST	19:16
RXADAPTCOMPELETE	15:12
RXADAPTDONE	11
TXADAPTDONE	10
TXADAPTACTIVE	9:6
TXADAPTREQ	5:2
RXADAPTEN	1
TXADAPTEN	0

Table 5-220 Fields for Register: VS_DebugAdapt

Bits	Name	Memory Access	Description
31:27			Reserved Field: Yes
26:20	TESTSTATE	R	TestMode FSM state. Value After Reset: 0x0 Exists: Always
19:16	RXBURST	R	RX burst of lanes at L15 layer. Value After Reset: 0x0 Exists: Always
15:12	RXADAPTCOMPELETE	R	RX Adapt complete indication. Value After Reset: 0x0 Exists: Always
11	RXADAPTDONE	R	RX Adapt request complete indication. Value After Reset: 0x1 Exists: Always

Table 5-220 Fields for Register: VS_DebugAdapt (Continued)

Bits	Name	Memory Access	Description
10	TXADAPTDONE	R	TX Adapt request complete indication. Value After Reset: 0x1 Exists: Always
9:6	TXADAPTACTIVE	R	TX Adapt request start indication from MPHY Value After Reset: 0x0 Exists: Always
5:2	TXADAPTREQ	R	TX Adapt request for the TX lanes Value After Reset: 0x0 Exists: Always
1	RXADAPTEN	R	RX Adapt requested for last received PMC Value After Reset: 0x0 Exists: Always
0	TXADAPTEN	R	Tx adapt requested for last received PMC Value After Reset: 0x0 Exists: Always

5.9.40 VS_DebugL15ci

- **Name:** VS_DebugL15ci
- **Description:** This attribute provides debug information related to L15 ci module of the UniPro controller
- **Size:** 32 bits
- **Offset:** 0xd08e
- **Exists:** Always

31	Rsvd
30	TRG0TRANSTRG_HS
29	LNRSTPH0B_COMP
28	TRG0RCVD_HS
27	TRG0PH0B_COMP
26	TRG0PH0_COMP
25	SAVECFGEXP_HS
24	DMEL2WRITEACK
23	TXRXCFGREADYSTATUS
22	TXSTALLACK
21	TXSTALLREQ
20	LSSTIMEOUT
19	TXRXCFGSTATUS
18	RXCFGSTATUS
17	TXCFGSTATUS
16	TXCFGREADYSTATUS
15	TXCFGBUSY
14	RXCFGBUSY
13	TACTTIMERENLATCH
12	TACTTIMEREN
11	H8EXITDONE
10	RXCFGRDYSTATUS
9	SAVECONFIGREQ
8	SAVECONFIGEXPIRY
7	SAVECONFIGEXPIRYINT
6	DESKEWINSERT
5	DUMMYBURSTEN
4	TXSAVESTATUS
3	RXBURST
2	TXBURST
1	RXH8EXIT
0	PMCCOMPLT

Table 5-221 Fields for Register: VS_DebugL15ci

Bits	Name	Memory Access	Description
31			Reserved Field: Yes
30	TRG0TRANSTRG_HS	R	Transmit TRG_UPR0 messages in HS_MODE Value After Reset: 0x0 Exists: P_UPRO_LSS_HS=1
29	LNRSTPH0B_COMP	R	Line Reset before transmitting TRG_UPR0 for Phase 0b has been completed Value After Reset: 0x0 Exists: P_UPRO_LSS_HS=1
28	TRG0RCVD_HS	R	Valid, consecutive and identical TRG_UPR0 message has been detected after Save Config timer expired on transmit side Value After Reset: 0x0 Exists: P_UPRO_LSS_HS=1

Table 5-221 Fields for Register: VS_DebugL15ci (Continued)

Bits	Name	Memory Access	Description
27	TRG0PH0B_COMP	R	TRG0 transmission during phase 0b of LSS has been completed during HS_MODE. Value After Reset: 0x0 Exists: P_UPRO_LSS_HS=1
26	TRG0PH0_COMP	R	TRG0 transmission during phase 0 of LSS has been completed during HS_MODE. Value After Reset: 0x0 Exists: P_UPRO_LSS_HS=1
25	SAVECFGEXP_HS	R	Save Config Timer has expired during TRG0 transmission in LSS in HS Value After Reset: 0x0 Exists: P_UPRO_LSS_HS=1
24	DMEL2WRITEACK	R	Writing user data to L2 complete indication. Value After Reset: 0x0 Exists: Always
23	TXRXCFGREADYSTATUS	R	Tx and Rx Config RdyN status including all lanes. Value After Reset: 0x0 Exists: Always
22	TXSTALLACK	R	Tx Interface stall acknowledge. Value After Reset: 0x0 Exists: Always
21	TXSTALLREQ	R	Tx Interface stall request. Value After Reset: 0x0 Exists: Always
20	LSSTIMEOUT	R	LSS Timeout indication because of the capability packet issue. Value After Reset: 0x0 Exists: Always
19	TXRXCFGSTATUS	R	TX and Rx Interface config interface status. Value After Reset: 0x1 Exists: Always
18	RXCFGSTATUS	R	Rx Interface config interface status. Value After Reset: 0x1 Exists: Always
17	TXCFGSTATUS	R	Tx Interface config interface status. Value After Reset: 0x1 Exists: Always

Table 5-221 Fields for Register: VS_DebugL15ci (Continued)

Bits	Name	Memory Access	Description
16	TXCFGREADYSTATUS	R	Tx Config RdyN status including all lanes. Value After Reset: 0x0 Exists: Always
15	TXCFGGBUSY	R	Tx Config interface busy indication. Value After Reset: 0x0 Exists: Always
14	RXCFCGBUSY	R	Rx Config interface busy indication. Value After Reset: 0x0 Exists: Always
13	TACTTIMERENLATCH	R	Tactivate timer enable indication after successful MPHY configuration for hiberante exit. Value After Reset: 0x0 Exists: Always
12	TACTTIMEREN	R	Tactivate timer enable indication for hiberante exit of inactive lane. Value After Reset: 0x0 Exists: Always
11	H8EXITDONE	R	Indicates hiberante exit operation of the inactive TX lane. Value After Reset: 0x1 Exists: Always
10	RXCFCGRDYSTATUS	R	Rx Config RdyN status including all lanes. Value After Reset: 0x0 Exists: Always
9	SAVECONFIGREQ	R	Save config timer request signal. Value After Reset: 0x0 Exists: Always
8	SAVECONFIGEXPIRY	R	Save config timer expiry indication to TX module. Value After Reset: 0x0 Exists: Always
7	SAVECONFIGEXPIRYINT	R	Save config timer expiry internal indication. Value After Reset: 0x0 Exists: Always
6	DESKEWININSERT	R	Deskew pattern insertion indication. Value After Reset: 0x0 Exists: Always

Table 5-221 Fields for Register: VS_DebugL15ci (Continued)

Bits	Name	Memory Access	Description
5	DUMMYBURSTEN	R	Dummy presence indication. Value After Reset: 0x1 Exists: Always
4	TXSAVESTATUS	R	Tx Save Status indication. Value After Reset: 0x0 Exists: Always
3	RXBURST	R	Rx Burst status in sysclk domain. Value After Reset: 0x0 Exists: Always
2	TXBURST	R	Tx Burst status in sysclk domain. Value After Reset: 0x0 Exists: Always
1	RXH8EXIT	R	Hiberante exit indication for RX interface on logical lane 0. Value After Reset: 0x0 Exists: Always
0	PMCCOMPLT	R	Indicates proper closure of the TX and RX burst after PMC operation. Value After Reset: 0x1 Exists: Always

5.9.41 VS_TBD

- **Name:** VS_TBD : Depth of the layer 2 transmit buffers
- **Description:** This attribute returns the depth of the layer 2 transmit buffers.
- **Size:** 32 bits
- **Offset:** 0xd090
- **Exists:** Always

TBD1	31:16
TBD0	15:0

Table 5-222 Fields for Register: VS_TBD

Bits	Name	Memory Access	Description
31:16	TBD1	R	Parameter P_TBD1 (Traffic Class 1) Value After Reset: P_TBD1 Exists: Always
15:0	TBD0	R	Parameter P_TBD0 (Traffic Class 0) Value After Reset: P_TBD0 Exists: Always

5.9.42 VS_RBD

- **Name:** VS_RBD
- **Description:** This attribute returns the depth of the layer 2 receive buffers.
- **Size:** 32 bits
- **Offset:** 0xd091
- **Exists:** Always

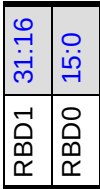


Table 5-223 Fields for Register: VS_RBD

Bits	Name	Memory Access	Description
31:16	RBD1	R	Parameter P_RBD1 (Traffic Class 1) Value After Reset: P_RBD1 Exists: Always
15:0	RBD0	R	Parameter P_RBD0 (Traffic Class 0) Value After Reset: P_RBD0 Exists: Always

5.9.43 VS_DebugTxByteCount

- **Name:** VS_DebugTxByteCount
- **Description:** This attribute returns the total number of bytes transmitted by layer 4.
- **Size:** 32 bits
- **Offset:** 0xd092
- **Exists:** Always



Table 5-224 Fields for Register: VS_DebugTxByteCount

Bits	Name	Memory Access	Description
31:0	DTXBC	R	Returns the total number of bytes transmitted by layer 4. It is read-only and rolls over when the maximum of 232 bytes is reached. Reset is only with a cold or warm reset. Value After Reset: 0x0 Exists: Always

5.9.44 VS_DebugRxByteCount

- **Name:** VS_DebugRxByteCount
- **Description:** This attribute returns the total number of bytes received by layer 4.
- **Size:** 32 bits
- **Offset:** 0xd093
- **Exists:** Always



Table 5-225 Fields for Register: VS_DebugRxByteCount

Bits	Name	Memory Access	Description
31:0	DRXBC	R	Returns the total number of bytes received by layer 4. It is read only and rolls over when the maximum of 2^32 bytes is reached. Reset is only with a cold or warm reset. Value After Reset: 0x0 Exists: Always

5.9.45 VS_DebugInvalidByteEnable

- **Name:** VS_DebugInvalidByteEnable
- **Description:** This attribute detects the byte enable errors on the application data interface. Reset only with a cold or warm reset.
- **Size:** 32 bits
- **Offset:** 0xd094
- **Exists:** Always

31:2	1	0
Rsvd	DIBE_RX	DIBE_TX

Table 5-226 Fields for Register: VS_DebugInvalidByteEnable

Bits	Name	Memory Access	Description
31:2			Reserved Field: Yes
1	DIBE_RX	R	Returns 1 if an invalid byte enable combination on the RX data. interface is detected Value After Reset: 0x0 Exists: Always
0	DIBE_TX	R	Returns 1 if an invalid byte enable combination on the TX data. interface is detected Value After Reset: 0x0 Exists: Always

5.9.46 VS_DebugLinkStartup

- **Name:** VS_DebugLinkStartup
- **Description:** This attribute contains information about the last link startup procedure.
- **Size:** 32 bits
- **Offset:** 0xd095
- **Exists:** Always

31:6	Rsvd
5	DLS_L2
4	DLS_PACP
3	DLS_EXT1_PACP
2	DLS_EXT2_PACP
1	DLS_UPR2
0	DLS_UPR1

Table 5-227 Fields for Register: VS_DebugLinkStartup

Bits	Name	Memory Access	Description
31:6			Reserved Field: Yes
5	DLS_L2	R	When this bit reads 1, it indicates that L2 link startup is requested Value After Reset: 0x0 Exists: Always
4	DLS_PACP	R	When this bit reads 1, it indicates that Capability PACP packet is received Value After Reset: 0x0 Exists: Always
3	DLS_EXT1_PACP	R	When this bit reads 1, it indicates that Ext1 Capability PACP packet is received Value After Reset: 0x0 Exists: Always
2	DLS_EXT2_PACP	R	When this bit reads 1, it indicates that Ext2 Capability PACP packet is received Value After Reset: 0x0 Exists: Always

Table 5-227 Fields for Register: VS_DebugLinkStartup (Continued)

Bits	Name	Memory Access	Description
1	DLS_UPR2	R	When this bit reads 1, it indicates that Trigger UPR2 is received Value After Reset: 0x0 Exists: Always
0	DLS_UPR1	R	When this bit reads 1, it indicates that Trigger UPR1 is received. Value After Reset: 0x0 Exists: Always

5.9.47 VS_DebugPwrChg

- **Name:** VS_DebugPwrChg
- **Description:** This attribute provides information about the last local and peer power mode change request.
- **Size:** 32 bits
- **Offset:** 0xd096
- **Exists:** Always

Rsvd	31:25
DPC_PR	24
DPC_TR	23
DPC_NL	22
DPC_GR	21
DPC_CC	20:18
DPC_PM	17
DPC_TL	16
DPC_LL	15
DPC_GL	14
DPC_WA	13
DPC_CS	12
DPC_RT	11:10
DPC_CL	9:6
DPC_CI	5:0

Table 5-228 Fields for Register: VS_DebugPwrChg

Bits	Name	Memory Access	Description
31:25			Reserved Field: Yes
24	DPC_PR	R	Result of the capability check for the power mode of the last peer request. Value After Reset: 0x0 Exists: Always
23	DPC_TR	R	Result of the capability check for the termination of the last peer request. Value After Reset: 0x1 Exists: Always
22	DPC_NL	R	Result of the capability check for the number of lanes of the last peer request. Value After Reset: 0x1 Exists: Always
21	DPC_GR	R	Result of the capability check for the gear of the last peer request. Value After Reset: 0x1 Exists: Always
20:18	DPC_CC	R	Last peer power mode change confirmation. Value After Reset: 0x0 Exists: Always

Table 5-228 Fields for Register: VS_DebugPwrChg (Continued)

Bits	Name	Memory Access	Description
17	DPC_PM	R	Result of the capability check for the power mode of the last local request. Value After Reset: 0x0 Exists: Always
16	DPC_TL	R	Result of the capability check for the termination of the last local request. Value After Reset: 0x0 Exists: Always
15	DPC_LL	R	Result of the capability check for the number of lanes of the last local request. Value After Reset: 0x0 Exists: Always
14	DPC_GL	R	Result of the capability check for the gear of the last local request. Value After Reset: 0x0 Exists: Always
13	DPC_WA	R	Result of the write access to the PwrMode attribute (0 = invalid access) Value After Reset: 0x0 Exists: Always
12	DPC_CS	R	Result if the last local request is performed in a correct state Value After Reset: 0x0 Exists: Always
11:10	DPC_RT	R	Number of retries of the last local power mode change request Value After Reset: 0x0 Exists: Always
9:6	DPC_CL	R	Last power mode change confirmation of a local request Value After Reset: 0x0 Exists: Always
5:0	DPC_CI	R	Last power mode change indication Value After Reset: 0x0 Exists: Always

5.9.48 VS_DebugStates

- **Name:** VS_DebugStates
- **Description:** This attribute contains LP fsm state, layer 1.5 and layer 2 main state machine states.
- **Size:** 32 bits
- **Offset:** 0xd097
- **Exists:** Always

31:27	26:24	23:19	18:16	15:0
Rsvd	DSLPPFSM	Rsvd	DSL2	DSL15

Table 5-229 Fields for Register: VS_DebugStates

Bits	Name	Memory Access	Description
31:27			Reserved Field: Yes
26:24	DSLPPFSM	R	LP controller FSM state. This register bits are available when UPRO_LPM_EN = 1: - 000: Idle - 001: Run - 010: Check Hibernate Enter Confirmation - 011: Save State - 100: Hibernate - 101: Check Hibernate Exit Confirmation - 110: Restore State - 111: Reserved Values: ■ 0x0 (VALUE_OF_0): Idle ■ 0x1 (VALUE_OF_1): Run ■ 0x2 (VALUE_OF_2): Check Hibernate Enter Confirmation ■ 0x3 (VALUE_OF_3): Save State ■ 0x4 (VALUE_OF_4): Hibernate ■ 0x5 (VALUE_OF_5): Check Hibernate Exit Confirmation ■ 0x6 (VALUE_OF_6): Restore State ■ 0x7 (VALUE_OF_7): Reserved Value After Reset: 0x0 Exists: Always
23:19			Reserved Field: Yes

Table 5-229 Fields for Register: VS_DebugStates (Continued)

Bits	Name	Memory Access	Description
18:16	DSL2	R	Layer 2 state Value After Reset: 0x1 Exists: Always
15:0	DSL15	R	Layer 1.5 state Value After Reset: 0x0 Exists: Always

5.9.49 VS_DebugCounter0

- **Name:** VS_DebugCounter0
- **Description:** This attribute has the debug counter 0 for counting various sources.
- **Size:** 32 bits
- **Offset:** 0xd098
- **Exists:** Always

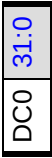


Table 5-230 Fields for Register: VS_DebugCounter0

Bits	Name	Memory Access	Description
31:0	DC0	R	Debug counter for counting various sources. For a description of all available sources, see <i>VS_DebugCounter0Mask</i> in the <i>Databook</i> . Value After Reset: 0x0 Exists: Always

5.9.50 VS_DebugCounter1

- **Name:** VS_DebugCounter1
- **Description:** This attribute has the debug counter 1 for counting various sources.
- **Size:** 32 bits
- **Offset:** 0xd099
- **Exists:** Always

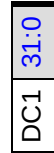


Table 5-231 Fields for Register: VS_DebugCounter1

Bits	Name	Memory Access	Description
31:0	DC1	R	Debug counter for counting various sources. For a description of all available sources, see <i>VS_DebugCounter1Mask</i> in the <i>Databook</i> . Value After Reset: 0x0 Exists: Always

5.9.51 VS_DebugCounter0Mask

- **Name:** VS_DebugCounter0Mask
- **Description:** This mask defines the sources for incrementing the debug counter 0. The mapping of each error indication field is the same as in the corresponding indication. Enabling the symbol count instructs the debug counter to count all received UniPro symbols (2 bytes) between the Sync and the EoB marker (including Fillers) divided by 1024. A value of 300 correlates to a value between $300 \times 2 \times 1024$ and $301 \times 2 \times 1024$ received bytes from the M-PHY interface.
- **Size:** 32 bits
- **Offset:** 0xd09a
- **Exists:** Always

31	DC0M_SCE
30:27	DC0M_EPH
26:25	DC0M_EPA
24:10	DC0M_ED
9:7	DC0M_EN
6:0	DC0M_ET

Table 5-232 Fields for Register: VS_DebugCounter0Mask

Bits	Name	Memory Access	Description
31	DC0M_SCE	R/W	Symbol count enable Value After Reset: 0x0 Exists: Always
30:27	DC0M_EPH	R/W	ErrorPHYInd Value After Reset: 0x0 Exists: Always
26:25	DC0M_EPA	R/W	ErrorPAInd Value After Reset: 0x0 Exists: Always
24:10	DC0M_ED	R/W	ErrorDInd Value After Reset: 0x0 Exists: Always
9:7	DC0M_EN	R/W	ErrorNInd Value After Reset: 0x0 Exists: Always

Table 5-232 Fields for Register: VS_DebugCounter0Mask (Continued)

Bits	Name	Memory Access	Description
6:0	DC0M_ET	R/W	ErrorTInd Value After Reset: 0x0 Exists: Always

5.9.52 VS_DebugCounter1Mask

- **Name:** VS_DebugCounter1Mask
- **Description:** This mask defines the sources for incrementing the debug counter 1. The mapping of each error indication field is the same as in the corresponding indication. Enabling the symbol count instructs the debug counter to count all received UniPro symbols (2 bytes) between the Sync and the EoB marker (including Fillers) divided by 1024. A value of 300 correlates to a value between $300 \times 2 \times 1024$ and $301 \times 2 \times 1024$ received bytes from the M-PHY interface.
- **Size:** 32 bits
- **Offset:** 0xd09b
- **Exists:** Always

31	DC1M_SCE
30:27	DC1M_EPH
26:25	DC1M_EPA
24:10	DC1M_ED
9:7	DC1M_EN
6:0	DC1M_ET

Table 5-233 Fields for Register: VS_DebugCounter1Mask

Bits	Name	Memory Access	Description
31	DC1M_SCE	R/W	Symbol count enable Value After Reset: 0x0 Exists: Always
30:27	DC1M_EPH	R/W	ErrorPHYInd Value After Reset: 0x0 Exists: Always
26:25	DC1M_EPA	R/W	ErrorPAInd Value After Reset: 0x0 Exists: Always
24:10	DC1M_ED	R/W	ErrorDInd Value After Reset: 0x0 Exists: Always
9:7	DC1M_EN	R/W	ErrorNInd Value After Reset: 0x0 Exists: Always

Table 5-233 Fields for Register: VS_DebugCounter1Mask (Continued)

Bits	Name	Memory Access	Description
6:0	DC1M_ET	R/W	ErrorTInd Value After Reset: 0x0 Exists: Always

5.9.53 VS_DebugCounterControl

- **Name:** VS_DebugCounterControl
- **Description:** Enables and resets the debug counters. Operations can be done simultaneously, for example, enable and clear.

For example, for calculating the bit error rate, one counter should enable all bits of the ErrorPHYInd and the other counter should enable the symbol count option. Then a single access to this attribute enables both counters simultaneously. When the measurement period has passed another access to this attribute, it disables the counters and both values can be read out. Another access clears and enables the counters again for further measurements.

- **Size:** 32 bits
- **Offset:** 0xd09c
- **Exists:** Always

31:4	3	2	1	0
Rsvd	DCC_CC1	DCC_CC0	DCC_EC1	DCC_EC0

Table 5-234 Fields for Register: VS_DebugCounterControl

Bits	Name	Memory Access	Description
31:4			Reserved Field: Yes
3	DCC_CC1	W	Clear counter 1. Read always returns 0. Value After Reset: 0x0 Exists: Always
2	DCC_CC0	W	Clear counter 0. Read always returns 0. Value After Reset: 0x0 Exists: Always
1	DCC_EC1	R/W	Enable counter 1 Value After Reset: 0x0 Exists: Always
0	DCC_EC0	R/W	Enable counter 0 Value After Reset: 0x0 Exists: Always

5.9.54 VS_DebugCounterOverflow

- **Name:** VS_DebugCounterOverflow
- **Description:** This attribute indicates the overflow in a counter. A read access clears the interrupt source.
- **Size:** 32 bits
- **Offset:** 0xd09d
- **Exists:** Always

31:2	1	0
Rsvd	DOC1	DOC0

Table 5-235 Fields for Register: VS_DebugCounterOverflow

Bits	Name	Memory Access	Description
31:2			Reserved Field: Yes
1	DOC1	R	When this bit reads 1, it indicates that the Overflow in counter 1 is detected. Value After Reset: 0x0 Exists: Always
0	DOC0	R	When this bit reads 1, it indicates that the Overflow in counter 0 is detected. Value After Reset: 0x0 Exists: Always

5.9.55 VS_DebugOMC

- **Name:** VS_DebugOMC
- **Description:** This attribute allows the application to override the setting of the P_OMC parameter. The parameter value is the reset value of this attribute.
- **Size:** 32 bits
- **Offset:** 0xd09e
- **Exists:** Always

31:1	0
Rsvd	DOMC

Table 5-236 Fields for Register: VS_DebugOMC

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	DOMC	R/W	LCC sequence Values: ■ 0x1 (ENABLED): Wait for the LCC sequence ■ 0x0 (DISABLED): Skip waiting for the LCC sequence Value After Reset: P_OMC Exists: (!UPRO_V2_EN)

5.9.56 VS_DebugCounterBMask

- **Name:** VS_DebugCounterBMask
- **Description:** This attribute controls the debug counters. This attribute enables counting of TX/RX bursts and the bad RX bursts. Bad RX bursts are bursts without any detected sync or end of burst.
- **Size:** 32 bits
- **Offset:** 0xd09f
- **Exists:** Always

31:19	Rsvd
18	BAD_RXDC1
17	RXDC1
16	TXDC1
15:3	Rsvd
2	BAD_RXDC0
1	RXDC0
0	TXDC0

Table 5-237 Fields for Register: VS_DebugCounterBMask

Bits	Name	Memory Access	Description
31:19			Reserved Field: Yes
18	BAD_RXDC1	R/W	When set to 1, this bit enables Bad RX burst count enable (DebugCounter1) Value After Reset: 0x0 Exists: Always
17	RXDC1	R/W	When set to 1, this bit enables RX burst count enable (DebugCounter1) Value After Reset: 0x0 Exists: Always
16	TXDC1	R/W	When set to 1, this bit enables TX burst count enable (DebugCounter1) Value After Reset: 0x0 Exists: Always
15:3			Reserved Field: Yes
2	BAD_RXDC0	R/W	When set to 1, this bit enables Bad RX burst count enable (DebugCounter0) Value After Reset: 0x0 Exists: Always

Table 5-237 Fields for Register: VS_DebugCounterBMask (Continued)

Bits	Name	Memory Access	Description
1	RXDC0	R/W	When set to 1, this bit enables RX burst count enable (DebugCounter0) Value After Reset: 0x0 Exists: Always
0	TXDC0	R/W	When set to 1, this bit enables TX burst count enable (DebugCounter0) Value After Reset: 0x0 Exists: Always

5.9.57 VS_DebugSaveConfigTime

- **Name:** VS_DebugSaveConfigTime
- **Description:** This attribute controls the start of the SaveConfigTime check as well as the time reference for counting SaveConfigTime to increase the maximum time for slower PLLs.
- **Size:** 32 bits
- **Offset:** 0xd0a0
- **Exists:** Always

31:5	
4:2	
1:0	
Rsvd	TREF
	ST_SCT

Table 5-238 Fields for Register: VS_DebugSaveConfigTime

Bits	Name	Memory Access	Description
31:5			Reserved Field: Yes

Table 5-238 Fields for Register: VS_DebugSaveConfigTime (Continued)

Bits	Name	Memory Access	Description
4:2	TREF	R/W	The value of the PA_SaveConfigTime attribute defines the wait period between bursts when a new configuration is performed. The default unit interval for this wait period is 1 us. Time reference: Values: 0x0 (VALUE_OF_0): Time reference for SaveConfigTime is 40 ns 0x1 (VALUE_OF_1): Time reference for SaveConfigTime is 50 ns 0x2 (VALUE_OF_2): Time reference for SaveConfigTime is 100 ns 0x3 (VALUE_OF_3): Time reference for SaveConfigTime is 125 ns 0x4 (VALUE_OF_4): Time reference for SaveConfigTime is 250 ns 0x5 (VALUE_OF_5): Time reference for SaveConfigTime is 500 ns 0x6 (VALUE_OF_6): Time reference for SaveConfigTime is 1 us (default, MIPI compliant) 0x7 (VALUE_OF_7): Reserved Value After Reset: 0x6 Exists: Always
1:0	ST_SCT	R/W	Control of RxCfgRdy signal to check the start of SaveConfigTime Timer: Values: 0x0 (VALUE_OF_0): Waits for SaveStateStatusN and then waits for RxCfgRdy (default, as defined by MIPI Specification) 0x1 (VALUE_OF_1): Does not wait for RxCfgRdy 0x2 (VALUE_OF_2): Checks RxCfgRdy only for local power mode change request 0x3 (VALUE_OF_3): Remembers every transition of RxCfgRdy to zero after the TX burst ends Value After Reset: 0x3 Exists: Always

5.9.58 VS_DebugLoopback

- **Name:** VS_DebugLoopback
- **Description:** This attribute allows the application to enable the debug loopback mode. Power on reset is required to exit from the loopback mode.
- **Size:** 32 bits
- **Offset:** 0xd0a1
- **Exists:** Always

31:1	0
Rsvd	DLPB

Table 5-239 Fields for Register: VS_DebugLoopback

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	DLPB	R/W	Debug loopback Values: ■ 0x1 (ENABLED): Enables the loopback mode ■ 0x0 (DISABLED): Disables the loopback mode Value After Reset: 0x0 Exists: Always

5.9.59 VS_Debug_DisCBAutoWrite

- **Name:** VS_Debug_DisCBAutoWrite
- **Description:** This attribute is used to disable the automatic TX and RX related attributes inside the CB block during power mode change. Application can write the required attributes inside the CB block. This attribute exists only when the SNPS_MPHY_EXT_EN parameter is enabled during the core configuration in the coreConsultant tool.
- **Size:** 32 bits
- **Offset:** 0xd0a2
- **Exists:** Always

31:1	
Rsvd	DisCBAutoWrite
	0

Table 5-240 Fields for Register: VS_Debug_DisCBAutoWrite

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	DisCBAutoWrite	R/W	When set to 1, this bit disables the auto attribute write operation inside the CB block through DWC_mipi_unipro. Value After Reset: 0x0 Exists: Always

5.9.60 VS_Debug_Tx_Tst_RtControl

- **Name:** VS_Debug_Tx_Tst_RtControl
- **Description:** This attribute allows the application to drive real time signals of the Synopsys GEN2 M-PHY TX control interface to control the implementation specific signals. In the Disable mode of Synopsys M-PHY, the continuity test is asserted through bits 0, 8, 16, and 24 of this attribute for individual lanes. This attribute exists only when the SNPS_MPHY_EXT_EN parameter is enabled during the core configuration in the coreConsultant tool.
- **Size:** 32 bits
- **Offset:** 0xd0a3
- **Exists:** Always



Table 5-241 Fields for Register: VS_Debug_Tx_Tst_RtControl

Bits	Name	Memory Access	Description
31:0	MPx_TxTstControl	R/W	Transmit test control signal is used to drive real time control signals of Synopsys GEN2 M-PHY. Each lane of the M-PHY has eight control signals. Value After Reset: 0x0 Exists: Always

5.9.61 VS_Debug_Rx_Tst_RtControl

- **Name:** VS_Debug_Rx_Tst_RtControl
- **Description:** This attribute allows the application to drive real time signals of the Synopsys GEN2 M-PHY RX control interface to control the implementation specific signals. In the Disable mode of Synopsys M-PHY, the continuity test is asserted through bits 0, 8, 16, and 24 of this attribute for individual lanes. This attribute exists only when the SNPS_MPHY_EXT_EN parameter is enabled during the core configuration in the coreConsultant tool.
- **Size:** 32 bits
- **Offset:** 0xd0a4
- **Exists:** Always



Table 5-242 Fields for Register: VS_Debug_Rx_Tst_RtControl

Bits	Name	Memory Access	Description
31:0	MPx_RxTstControl	R/W	Receive test control signal is used to drive real time control signals of Synopsys GEN2 M-PHY. Each lane of the M-PHY has eight control signals. Value After Reset: 0x0 Exists: Always

5.9.62 VS_Debug_CB_Tst_RtControl

- **Name:** VS_Debug_CB_Tst_RtControl
- **Description:** This attribute allows the application to drive real time signals of the Synopsys GEN2 M-PHY CB control interface to control the implementation specific signals. In the Disable mode of Synopsys M-PHY, the continuity test is asserted through bit 0 of this attribute. This attribute exists only when the SNPS_MPHY_EXT_EN parameter is enabled during the core configuration in the coreConsultant tool.
- **Size:** 32 bits
- **Offset:** 0xd0a5
- **Exists:** Always

31:8	Rsvd
7:0	MP_CBTstControl

Table 5-243 Fields for Register: VS_Debug_CB_Tst_RtControl

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	MP_CBTstControl	R/W	Common block test control signals are used to drive real time control signals of Synopsys GEN2 M-PHY CB block. Value After Reset: 0x0 Exists: Always

5.9.63 VS_Debug_Tx_Tst_RtObserve

- **Name:** VS_Debug_Tx_Tst_RtObserve
- **Description:** This attribute allows the application to observe real time signals of M-TX interface. If bit 0 of MPx_TX_Tst_RtObserve signal of each lane is asserted in the Disable mode, then this attribute is used to read the result of continuity test of M-PHY. This attribute exists only when the SNPS_MPHY_EXT_EN parameter is enabled during the core configuration in the coreConsultant tool.
- **Size:** 32 bits
- **Offset:** 0xd0a6
- **Exists:** Always



Table 5-244 Fields for Register: VS_Debug_Tx_Tst_RtObserve

Bits	Name	Memory Access	Description
31:0	MPx_TxTstObserve	R	Transmit interface real time observe signal. Four lanes are supported with each lane having eight observe pins. Value After Reset: 0x0 Exists: Always

5.9.64 VS_Debug_Rx_Tst_RtObserve

- **Name:** VS_Debug_Rx_Tst_RtObserve
- **Description:** This attribute allows the application to observe real time signals of M-RX interface. If bit 0 of MPx_RX_Tst_RtControl signal of each lane is asserted in the Disable mode, then this attribute is used to read the result of continuity test of M-PHY. This attribute exists only when the SNPS_MPHY_EXT_EN parameter is enabled during the core configuration in the coreConsultant tool.
- **Size:** 32 bits
- **Offset:** 0xd0a7
- **Exists:** Always



Table 5-245 Fields for Register: VS_Debug_Rx_Tst_RtObserve

Bits	Name	Memory Access	Description
31:0	MPx_RxTstObserve	R	Receive interface real time observe signal. Four lanes are supported with each lane having eight observe pins. Value After Reset: 0x0 Exists: Always

5.9.65 VS_Debug_CB_Tst_RtObserve

- **Name:** VS_Debug_CB_Tst_RtObserve
- **Description:** This attribute allows the application to observe real-time signals of M-CB interface. If bit 0 of MPx_CB_Tst_RtControl signal of each lane is asserted in the Disable mode, then this attribute is used to read the result of continuity test of M-PHY. This attribute exists only when the SNPS_MPHY_EXT_EN parameter is enabled during the core configuration in the coreConsultant tool.
- **Size:** 32 bits
- **Offset:** 0xd0a8
- **Exists:** Always

31:8	Rsvd
7:0	MP_CBTstObserve

Table 5-246 Fields for Register: VS_Debug_CB_Tst_RtObserve

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	MP_CBTstObserve	R	CB interface real time observe signal. Value After Reset: 0x0 Exists: Always

5.9.66 VS_Mk2ExtnSupport

- **Name:** VS_Mk2ExtnSupport
- **Description:** This attribute controls the MK2 features in DWC_mipi_unipro.
- **Size:** 32 bits
- **Offset:** 0xd0ab
- **Exists:** Always

31:6	Rsvd
5	CNCT_RXPHASE2TXEN
4	MK2_AFCSEND_EN
3	TX_DEEPSTALL_EXIT_EN
2	MK2_PLLMASK
1	MK2_EXTNSUPPORT
0	TX_DEEPSTALL_EN

Table 5-247 Fields for Register: VS_Mk2ExtnSupport

Bits	Name	Memory Access	Description
31:6			Reserved Field: Yes
5	CNCT_RXPHASE2TXEN	R/W	This bit is used to enable TX MK2 extension internally based Rx_Mk2ExterntionPhase or Tx_Mk2ExtensionEn signal: Values: ■ 0x0 (VALUE_OF_0): Tx MK2 Extension enable depends on Rx_Mk2ExterntionPhase signal. ■ 0x1 (VALUE_OF_1): Tx MK2 Extension enable depends on Tx_Mk2ExtensionEn signal. Value After Reset: 0x0 Exists: Always

Table 5-247 Fields for Register: VS_Mk2ExtnSupport (Continued)

Bits	Name	Memory Access	Description
4	MK2_AFCSEND_EN	R/W	This field controls the transfer of the outstanding AFC packets when <MK2,MK2> is received on RX logical lane 0: #- 0: DWC_mipi_unipro does not transfer immediately any outstanding AFCs before RX is put into Deep Stall. #AFCs are generated when either TCOAckThreshold is met or AFC request timer has timed out. #- 1: DWC_mipi_unipro forcibly transfers AFCs immediately after receiving <MK2,MK2> on RX logical lane 0. #It does not wait until AFC Request Timer timeout or TCOAckThreshold or other AFC generation conditions are met. Values: 0x0 (VALUE_OF_0): DWC_mipi_unipro does not transfer immediately any outstanding AFCs before RX is put into Deep Stall.AFCs are generated when either TCOAckThreshold is met or AFC request timer has timed out 0x1 (VALUE_OF_1): DWC_mipi_unipro forcibly transfers AFCs immediately after receiving <MK2,MK2> on RX logical lane 0.It does not wait until AFC Request Timer timeout or TCOAckThreshold or other AFC generation conditions are met Value After Reset: 0x1 Exists: Always
3	TX_DEEPSTALL_EXIT_EN	R/W	This field is used to indicate whether deep stall exit needs to be carried on TX lanes : #- 0: Disable the deep stall exit on Tx lanes. #- 1: Enable the deep stall exit on Tx lanes. Values: 0x0 (DISABLED): Disable the deep stall exit on Tx lanes 0x1 (ENABLED): Enable the deep stall exit on Tx lanes Value After Reset: 0x0 Exists: Always

Table 5-247 Fields for Register: VS_Mk2ExtnSupport (Continued)

Bits	Name	Memory Access	Description
2	MK2_PLLMASK	R/W	This field masks the 'Switching OFF' of the PLL when Synopsys GEN2 M-PHY is selected and MK2 extension feature is used: Values: 0x0 (VALUE_OF_0): After the transmission of <MK2,MK2> and reception of <MK2,MK2> , DWC_mipi_unipro autonomously programs M-PHY to enter deep stall state. This means, DWC_mipi_unipro turns off the PLL in M-PHY. 0x1 (VALUE_OF_1): M-PHY is not programmed to turn off the PLL. DWC_mipi_unipro only supports the transmission of <MK2,MK2> and reception of <MK2,MK2>. This is useful while debugging the hardware. Value After Reset: 0x0 Exists: Always
1	MK2_EXTNSUPPORT	R/W	This field enables the Mk2 extension support for the RX interface in EXT_1_Cap packet. During the link startup sequence, this bit is used to inform to the peer whether local UniPro supports <MK2,MK2> detection on RX logical lane '0' or not: Values: 0x0 (VALUE_OF_0): PA does not detect the EoB extension. When set to '0' before the link startup, UniPro does not detect <MK2,MK2> on RX logical lane 0 at EOB 0x1 (VALUE_OF_1): PA detects the EoB extension when set to '1' before link startup. This information is passed to the peer during the capability exchange and DWC_mipi_unipro detects <MK2,MK2> on RX logical lane 0 at EOB Value After Reset: 0x1 Exists: Always
0	TX_DEEPSTALL_EN	R/W	This field enables the Mk2 extension for TX interface: Values: 0x1 (ENABLED): MK2 extension is enabled. DWC_mipi_unipro transmits <MK2,MK2> at the end of burst 0x0 (DISABLED): MK2 extension is disabled. DWC_mipi_unipro does not transmit <MK2,MK2> at the end of burst Value After Reset: 0x0 Exists: Always

5.9.67 VS_ULPH8_Cntrl

- **Name:** VS_ULPH8_Cntrl
- **Description:** This attribute controls the support of Ultra Low Power Hibenate (ULPH8)feature of Synopsys GEN2 M-PHY.
- **Size:** 32 bits
- **Offset:** 0xd0af
- **Exists:** Always

31:4	Rsvd
3	Ulp_ModeEn
2	Ulp_SftrResComplete
1	Ulp_SftrResEn
0	Ulp_CtrlMode

Table 5-248 Fields for Register: VS_ULPH8_Cntrl

Bits	Name	Memory Access	Description
31:4			Reserved Field: Yes
3	Ulp_ModeEn	R/W	This filed enables the ULP H8 mode support: ■ 0: ULP H8 mode support is disabled. ■ 1: ULP H8 mode support is enabled. This bit should not be updated during hibernate. If done so, the behavior is undefined. Values: ■ 0x0 (ULP_H8_DIS): ULP H8 mode support is disabled ■ 0x1 (ULP_H8_EN): ULP H8 mode support is enabled Value After Reset: 0x1 Exists: Always

Table 5-248 Fields for Register: VS_ULPH8_Cntrl (Continued)

Bits	Name	Memory Access	Description
2	Ulp_SftrResComplete	R/W	This field indicates whether the software completed the M-PHY attribute restore: 0: software has yet not completed the restore of MPHY attributes. 1: software completed the restore of MPHY attributes. This bit is applicable when UlpCtrlMode bit is set to '0'. This bit should be set to '1' once software completes the M-PHY attribute restore operation. Setting this bit to '1' enables LowPower state machine to complete the ULP H8 exit operation. Otherwise, LP state machine remains in 'Restore State (3'b100)'. This bit is auto cleared once H8 exit is complete. Values: 0x0 (VALUE_OF_0): software has yet not completed the restore of MPHY attributes 0x1 (VALUE_OF_1): software completed the restore of MPHY attributes Value After Reset: 0x0 Exists: Always
1	Ulp_SftrResEn	R	This field indicates when software can perform M-PHY attribute restore. - 0: Software restore period has not yet occurred. - 1: Software restore window occurred. Software can start restoring the M-PHY attributes. This bit is applicable when UlpCtrlMode bit is set to '0'. This bit should be used during software restore operation. Values: 0x0 (VALUE_OF_0): Software restore period has not yet occurred 0x1 (VALUE_OF_1): Software restore window occurred Value After Reset: 0x0 Exists: Always

Table 5-248 Fields for Register: VS_ULPH8_Cntrl (Continued)

Bits	Name	Memory Access	Description
0	Ulp_CtrlMode	R/W	This field decides the Save/Restore of M-PHY attributes through hardware or software when ULP H8 mode is enabled. - 0: Save/Restore of M-PHY attributes through controller is disabled and it has to be done by software. - 1: Save/Restore of M-PHY attributes is performed by the controller. This field should not be updated during hibernate. If done so, the behavior is undefined. Values: ■ 0x0 (VALUE_OF_0): Save/Restore of M-PHY attributes through controller is disabled ■ 0x1 (VALUE_OF_1): Save/Restore of M-PHY attributes is performed by the controller Value After Reset: 0x1 Exists: Always

5.9.68 VS_RX_SQ_Cntrl

- **Name:** VS_RX_SQ_Cntrl
- **Description:** This attribute controls the squelch ports of Synopsys GEN2 M-PHY when ULPH8 feature is enabled.
- **Size:** 32 bits
- **Offset:** 0xd0b0
- **Exists:** Always

31:10	Rsvd
9	ulp_RxSq_WeakKeep
8:6	ulp_RxSq_CtrlThres
5:4	ulp_RxSq_CtrlResp
3	ulp_RxSq_CtrlSel
2	ulp_RxSqInv_Polarity
1	ulp_RxSq_On
0	ulp_RxSq_PhySoc

Table 5-249 Fields for Register: VS_RX_SQ_Cntrl

Bits	Name	Memory Access	Description
31:10			Reserved Field: Yes
9	ulp_RxSq_WeakKeep	R/W	If ulp_RxSq_CtrlSel is set to 0, then the value of this bit is directly driven on ULP_MPx_RX_SqWeakKeep port, irrespective of ulp_ModeEn bit value. If ulp_RxSq_CtrlSel is set to 1, then this bit is ignored. Value After Reset: 0x0 Exists: Always
8:6	ulp_RxSq_CtrlThres	R/W	The values that are set to these bits are used to drive the ULP_MPx_RX_SqCtrlThresh port. Value After Reset: 0x0 Exists: Always
5:4	ulp_RxSq_CtrlResp	R/W	The values that are set to these bits are used to drive the ULP_MPx_RX_SqCtrlResp port. Value After Reset: 0x0 Exists: Always

Table 5-249 Fields for Register: VS_RX_SQ_Cntrl (Continued)

Bits	Name	Memory Access	Description
3	ulp_RxSq_CtrlSel	R/W	By default, DWC_mipi_unipro drives the ULP_MPx_RX_SqPhySoc, ULP_MPx_RX_SqOnv, and ULP_MPx_RX_SqWeakKeep signals. When this bit is set to 0, the ulp_RxSq_PhySoc, ulp_RxSq_On, and ulp_RxSq_WeakKeep bits are directly connected to ULP_MPx_RX_SqPhySoc, ULP_MPx_RX_SqOn, and ULP_MPx_RX_SqWeakKeep signals, respectively. Value After Reset: 0x1 Exists: Always
2	ulp_RxSqInv_Polarity	R/W	The value that is set to this bit is used to drive the ULP_MPx_RX_SqInvPolarity port. Value After Reset: 0x0 Exists: Always
1	ulp_RxSq_On	R/W	If ulp_RxSq_CtrlSel is set to 0, then the value of this bit is directly driven on the ULP_MPx_RX_SqOn port, irrespective of ulp_ModeEn bit value. If ulp_RxSq_CtrlSel is set to 1, then the value of this bit is ignored. Value After Reset: 0x0 Exists: Always
0	ulp_RxSq_PhySoc	R/W	If the ulp_ModeEn, ulp_RxSq_CtrnSel, and ulp_RxSq_PhySoc bits are all set to 1, then ■ DWC_mipi_unipro drives ULP_MPx_RX_SqPhySoc with the value 1 during the entry and exit of ULP H8 ■ DWC_mipi_unipro drives ULP_MPx_RX_SqPhySoc with the value 0 during other times (other than entry and exit of ULP H8) If ulp_RxSq_CtrlSel is set to 0, then the value of this bit is directly driven on ULP_MPx_RX_PhySoc port, irrespective of the ulp_ModeEn bit value. For more information, see <i>Squelch Circuit Usage of M-PHY</i> section in the <i>Databook</i> Value After Reset: 0x0 Exists: Always

5.9.69 VS_OverrideAttrEnable

- **Name:** VS_OverrideAttrEnable
- **Description:** This attribute controls the override operation of the M-PHY advance capability attributes.
- **Size:** 32 bits
- **Offset:** 0xd0b1
- **Exists:** Always

Rsvd	31:26
RXEXTDMINSAVECONFIGTIMEOVREN	25
HSG5ADAPTINITIALOVREN	24
HSG5ADAPTREFRESHOVREN	23
HSG5PREPARELENGTHOVREN	22
HSG5SYNCLengthOVREN	21
HSADAPTINITIALOVREN	20
HSADAPTREFRESHOVREN	19
HSG4PREPARELENGTHOVREN	18
HSG4SYNCLengthOVREN	17
RXMINSAVECONFIGTIMEOVREN	16
RXMINSTALLNOCONFIGTIMEOVREN	15
TXLSTERMLINEDRIVEOVREN	14
TXHSUNTERMLINEDRIVEOVREN	13
RXMINSLEEPNOCONFIGTIMEOVREN	12
HSG1PREPARELENGTHOVREN	11
HSG1SYNCLengthOVREN	10
HSG2PREPARELENGTHOVREN	9
HSG2SYNCLengthOVREN	8
HSG3PREPARELENGTHOVREN	7
HSG3SYNCLengthOVREN	6
PWMPREPARELENGTHOVREN	5
PWMSYNCLengthOVREN	4
PWMBURSTCLOSURELENGTHOVREN	3
ADVGRANOVREN	2
ADVHIB8OVREN	1
ADVTACTOVREN	0

Table 5-250 Fields for Register: VS_OverrideAttrEnable

Bits	Name	Memory Access	Description
31:26			Reserved Field: Yes

Table 5-250 Fields for Register: VS_OverrideAttrEnable (Continued)

Bits	Name	Memory Access	Description
25	RXEXTDMINSAVECONFIGTIME OVREN	R/W	RX_Min_SAVE_Config_Time_Capability Override control 0: Disable the RX_Extended_Min_SAVE_Config_Time_Capability override 1: Enable the RX_Extended_Min_SAVE_Config_Time_Capability override The override value is specified by the VS_OverrideRxExtDMinSaveConfigTimeCapability register. Values: 0x0 (DISABLED): Disable the RX_Extended_Min_SAVE_Config_Time_Capability override 0x1 (ENABLED): Enable the RX_Extended_Min_SAVE_Config_Time_Capability override Value After Reset: 0x0 Exists: (P_EXTD_SAVE_CONFIG_EN)
24	HSG5ADAPTINITIALOVREN	R/W	RX_HS_G5_ADAPT_INITIAL_Capability Override Control: 0: Disable the RX_HS_G5_ADAPT_INITIAL_Capability override 1: Enable the RX_HS_G5_ADAPT_INITIAL_Capability override The override value is specified by the VS_OverrideRxHsG5AdaptInitialCapability register. Values: 0x0 (DISABLED): Disable the RX_HS_G5_ADAPT_INITIAL_Capability override 0x1 (ENABLED): Enable the RX_HS_G5_ADAPT_INITIAL_Capability override Value After Reset: 0x0 Exists: (UPRO_V2_EN)

Table 5-250 Fields for Register: VS_OverrideAttrEnable (Continued)

Bits	Name	Memory Access	Description
23	HSG5ADAPTREFRESHOVREN	R/W	RX_HS_G5_ADAPT_REFRESH_Capability Override control: 0: Disable the RX_HS_G5_ADAPT_REFRESH_Capability override 1: Enable the RX_HS_G5_ADAPT_REFRESH_Capability override The override value is specified by the VS_OverrideRxHsG5AdaptRefreshCapability register. Values: 0x0 (DISABLED): Disable the RX_HS_G5_ADAPT_REFRESH_Capability override 0x1 (ENABLED): Enable the RX_HS_G5_ADAPT_REFRESH_Capability override Value After Reset: 0x0 Exists: (UPRO_V2_EN)
22	HSG5PREPARELENGTHOVREN	R/W	RX_HS_G5_PREPARE_LENGTH_Capability Override Control: 0: Disable the RX_HS_G5_PREPARE_LENGTH_Capability override 1: Enable the RX_HS_G5_PREPARE_LENGTH_Capability override The override value is specified by the VS_OverrideRxHsG5PrepareLengthCapability register. Values: 0x0 (DISABLED): Disable the RX_HS_G5_PREPARE_LENGTH_Capability override 0x1 (ENABLED): Enable the RX_HS_G5_PREPARE_LENGTH_Capability override Value After Reset: 0x0 Exists: (UPRO_V2_EN)

Table 5-250 Fields for Register: VS_OverrideAttrEnable (Continued)

Bits	Name	Memory Access	Description
21	HSG5SYNCLENGTHOVREN	R/W	RX_HS_G5_SYNC_LENGTH_Capability Override Control: 0: Disable the RX_HS_G5_SYNC_LENGTH_Capability override 1: Enable the RX_HS_G5_SYNC_LENGTH_Capability override The override value is specified by the VS_OverrideRxHsG5SyncLengthCapability register. Values: 0x0 (DISABLED): Disable the RX_HS_G5_SYNC_LENGTH_Capability override 0x1 (ENABLED): Enable the RX_HS_G5_SYNC_LENGTH_Capability override Value After Reset: 0x0 Exists: (UPRO_V2_EN)
20	HSADAPTINITIALOVREN	R/W	RX_HS_ADAPT_INITIAL_Capability Override Control: 0: Disable the RX_HS_ADAPT_INITIAL_Capability override 1: Enable the RX_HS_ADAPT_INITIAL_Capability override The override value is specified by the VS_OverrideRxHsAdaptInitialCapability register. Values: 0x0 (DISABLED): Disable the RX_HS_ADAPT_INITIAL_Capability override 0x1 (ENABLED): Enable the RX_HS_ADAPT_INITIAL_Capability override Value After Reset: 0x0 Exists: (UPRO_V18_EN)

Table 5-250 Fields for Register: VS_OverrideAttrEnable (Continued)

Bits	Name	Memory Access	Description
19	HSADAPTREFRESHOVREN	R/W	RX_HS_ADAPT_REFRESH_Capability Override control 0: Disable the RX_HS_ADAPT_REFRESH_Capability override 1: Enable the RX_HS_ADAPT_REFRESH_Capability override The override value is specified by the VS_OverrideRxHsAdaptRefreshCapability register. Values: 0x0 (DISABLED): Disable the RX_HS_ADAPT_REFRESH_Capability override 0x1 (ENABLED): Enable the RX_HS_ADAPT_REFRESH_Capability override Value After Reset: 0x0 Exists: (UPRO_V18_EN)
18	HSG4PREPARELENGTHOVREN	R/W	RX_HS_G4_PREPARE_LENGTH_Capability Override Control: 0: Disable the RX_HS_G4_PREPARE_LENGTH_Capability override 1: Enable the RX_HS_G4_PREPARE_LENGTH_Capability override The override value is specified by the VS_OverrideRxHsG4PrepareLengthCapability register. Values: 0x0 (DISABLED): Disable the RX_HS_G4_PREPARE_LENGTH_Capability override 0x1 (ENABLED): Enable the RX_HS_G4_PREPARE_LENGTH_Capability override Value After Reset: 0x0 Exists: (UPRO_V18_EN)

Table 5-250 Fields for Register: VS_OverrideAttrEnable (Continued)

Bits	Name	Memory Access	Description
17	HSG4SYNCLENGTHOVREN	R/W	RX_HS_G4_SYNC_LENGTH_Capability Override Control: 0: Disable the RX_HS_G4_SYNC_LENGTH_Capability override 1: Enable the RX_HS_G4_SYNC_LENGTH_Capability override The override value is specified by the VS_OverrideRxHsG4SyncLengthCapability register. Values: 0x0 (DISABLED): Disable the RX_HS_G4_SYNC_LENGTH_Capability override 0x1 (ENABLED): Enable the RX_HS_G4_SYNC_LENGTH_Capability override Value After Reset: 0x0 Exists: (UPRO_V18_EN)
16	RXMINSAVECONFIGTIMEOVREN	R/W	RX_Min_SAVE_Config_Time_Capability Override control 0: Disable the RX_Min_SAVE_Config_Time_Capability override 1: Enable the RX_Min_SAVE_Config_Time_Capability override The override value is specified by the VS_OverrideRxMinSaveConfigTimeCapability register. Values: 0x0 (DISABLED): Disable the RX_Min_SAVE_Config_Time_Capability override 0x1 (ENABLED): Enable the RX_Min_SAVE_Config_Time_Capability override Value After Reset: 0x0 Exists: Always

Table 5-250 Fields for Register: VS_OverrideAttrEnable (Continued)

Bits	Name	Memory Access	Description
15	RXMINSTALLNOCONFIGTIMEOVREN	R/W	RX_Min_STALL_NoConfig_Time_Capability Override control 0: Disable the RX_Min_STALL_NoConfig_Time_Capability override 1: Enable the RX_Min_STALL_NoConfig_Time_Capability override The override value is specified by the VS_OverrideRxMinStallNoConfigTimeCapability register. Values: 0x0 (DISABLED): Disable the RX_Min_STALL_NoConfig_Time_Capability override 0x1 (ENABLED): Enable the RX_Min_STALL_NoConfig_Time_Capability override Value After Reset: 0x0 Exists: Always
14	TXLSSTERMLINEDRIVEOVREN	R/W	TX_LS_Terminated_LINE_Drive_Capability Override control 0: Disable the TX_LS_Terminated_LINE_Drive_Capability override 1: Enable the TX_LS_Terminated_LINE_Drive_Capability override The override value is specified by the VS_OverrideTxLsTerminatedLineDriveCapability register. Values: 0x0 (DISABLED): Disable the TX_LS_Terminated_LINE_Drive_Capability override 0x1 (ENABLED): Enable the TX_LS_Terminated_LINE_Drive_Capability override Value After Reset: 0x0 Exists: Always

Table 5-250 Fields for Register: VS_OverrideAttrEnable (Continued)

Bits	Name	Memory Access	Description
13	TXHSUNTERMLINEDRIVEOVREN	R/W	TX_HS_Unterminated_LINE_Drive_Capability Override Control: 0: Disable the TX_HS_Unterminated_LINE_Drive_Capability override 1: Enable the TX_HS_Unterminated_LINE_Drive_Capability override The override value is specified by the VS_OverrideTxHsUnterminatedLineDriveCapability register. Values: 0x0 (DISABLED): Disable the TX_HS_Unterminated_LINE_Drive_Capability override 0x1 (ENABLED): Enable the TX_HS_Unterminated_LINE_Drive_Capability override Value After Reset: 0x0 Exists: Always
12	RXMINSLLEEPNOCONFIGTIMEOVREN	R/W	RX_Min_SLEEP_NoConfig_Time_Capability Override Control: 0: Disable the RX_Min_SLEEP_NoConfig_Time_Capability override 1: Enable the RX_Min_SLEEP_NoConfig_Time_Capability override The override value is specified by the VS_OverrideRxMinSleepNoConfigTimeCapability register. Values: 0x0 (DISABLED): Disable the RX_Min_SLEEP_NoConfig_Time_Capability override 0x1 (ENABLED): Enable the RX_Min_SLEEP_NoConfig_Time_Capability override Value After Reset: 0x0 Exists: Always

Table 5-250 Fields for Register: VS_OverrideAttrEnable (Continued)

Bits	Name	Memory Access	Description
11	HSG1PREPARELENGTHOVREN	R/W	RX_HS_G1_PREPARE_LENGTH_Capability Override Control: 0: Disable the RX_HS_G1_PREPARE_LENGTH_Capability override 1: Enable the RX_HS_G1_PREPARE_LENGTH_Capability override The override value is specified by the VS_OverrideRxHsG1PrepareLengthCapability register. Values: 0x0 (DISABLED): Disable the RX_HS_G1_PREPARE_LENGTH_Capability override 0x1 (ENABLED): Enable the RX_HS_G1_PREPARE_LENGTH_Capability override Value After Reset: 0x0 Exists: Always
10	HSG1SYNCLengthOVREN	R/W	RX_HS_G1_SYNC_LENGTH_Capability Override Control: 0: Disable the RX_HS_G1_SYNC_LENGTH_Capability override 1: Enable the RX_HS_G1_SYNC_LENGTH_Capability override The override value is specified by the VS_OverrideRxHsG1SyncLengthCapability register. Values: 0x0 (DISABLED): Disable the RX_HS_G1_SYNC_LENGTH_Capability override 0x1 (ENABLED): Enable the RX_HS_G1_SYNC_LENGTH_Capability override Value After Reset: 0x0 Exists: Always

Table 5-250 Fields for Register: VS_OverrideAttrEnable (Continued)

Bits	Name	Memory Access	Description
9	HSG2PREPARELENGTHOVREN	R/W	RX_HS_G2_PREPARE_LENGTH_Capability Override Control: 0: Disable the RX_HS_G2_PREPARE_LENGTH_Capability override 1: Enable the RX_HS_G2_PREPARE_LENGTH_Capability override The override value is specified by the VS_OverrideRxHsG2PrepareLengthCapability register. Values: 0x0 (DISABLED): Disable the RX_HS_G2_PREPARE_LENGTH_Capability override 0x1 (ENABLED): Enable the RX_HS_G2_PREPARE_LENGTH_Capability override Value After Reset: 0x0 Exists: Always
8	HSG2SYNCLengthOVREN	R/W	RX_HS_G2_SYNC_LENGTH_Capability Override Control: 0: Disable the RX_HS_G2_SYNC_LENGTH_Capability override 1: Enable the RX_HS_G2_SYNC_LENGTH_Capability override The override value is specified by the VS_OverrideRxHsG2SyncLengthCapability register. Values: 0x0 (DISABLED): Disable the RX_HS_G2_SYNC_LENGTH_Capability override 0x1 (ENABLED): Enable the RX_HS_G2_SYNC_LENGTH_Capability override Value After Reset: 0x0 Exists: Always

Table 5-250 Fields for Register: VS_OverrideAttrEnable (Continued)

Bits	Name	Memory Access	Description
7	HSG3PREPARELENGTHOVREN	R/W	RX_HS_G3_PREPARE_LENGTH_Capability Override Control: 0: Disable the RX_HS_G3_PREPARE_LENGTH_Capability override 1: Enable the RX_HS_G3_PREPARE_LENGTH_Capability override The override value is specified by the VS_OverrideHSG3PrepareLength register. Values: 0x0 (DISABLED): Disable the RX_HS_G3_PREPARE_LENGTH_Capability override 0x1 (ENABLED): Enable the RX_HS_G3_PREPARE_LENGTH_Capability override Value After Reset: 0x0 Exists: Always
6	HSG3SYNCLengthOVREN	R/W	RX_HS_G3_SYNC_LENGTH_Capability Override Control: 0: Disable the RX_HS_G3_SYNC_LENGTH_Capability override 1: Enable the RX_HS_G3_SYNC_LENGTH_Capability override The override value is specified by the VS_OverrideHSG3SyncLength register. Values: 0x0 (DISABLED): Disable the RX_HS_G3_SYNC_LENGTH_Capability override 0x1 (ENABLED): Enable the RX_HS_G3_SYNC_LENGTH_Capability override Value After Reset: 0x0 Exists: Always
5	PWMPREPARELENGTHOVREN	R/W	PWM prepare length override control 0: Disable the prepare length override 1: Enable the prepare length override The override value is specified by the VS_OverridePwmPrepareLength register. Values: 0x0 (DISABLED): Disable the prepare length override 0x1 (ENABLED): Enable the prepare length override Value After Reset: 0x0 Exists: Always

Table 5-250 Fields for Register: VS_OverrideAttrEnable (Continued)

Bits	Name	Memory Access	Description
4	PWMSYNCLENGTHOVREN	R/W	PWM sync length override control 0: Disable the sync length override 1: Enable the sync length override The override value is specified by the VS_OverridePwmSyncLength register. Values: 0x0 (DISABLED): Disable the sync length override 0x1 (ENABLED): Enable the sync length override Value After Reset: 0x0 Exists: Always
3	PWMBURSTCLOSURELENGTHOVREN	R/W	PWM burst closure length override control 0: Disable the burst closure length override 1: Enable the burst closure length override The override value is specified by the VS_OverridePwmBurstClosureLength register. Values: 0x0 (DISABLED): Disable the burst closure length override 0x1 (ENABLED): Enable the burst closure length override Value After Reset: 0x0 Exists: Always
2	ADVGRANOVREN	R/W	Advance granularity override control 0: Disable the advance granularity override 1: Enable the advance granularity override The override value is specified by the VS_OverrideAdvGranularity register. Note: When this bit is enabled, set the VS_OverrideAdvTactivate and VS_OverrideAdvGran with proper value. Values: 0x0 (DISABLED): Disable the advance granularity override 0x1 (ENABLED): Enable the advance granularity override Value After Reset: 0x0 Exists: Always

Table 5-250 Fields for Register: VS_OverrideAttrEnable (Continued)

Bits	Name	Memory Access	Description
1	ADVHIB8OVREN	R/W	Advance Hibern8 override control 0: Disable the advance Hibern8 override 1: Enable the advance Hibern8 override The override value is specified by the VS_OverrideAdvHibn8 register. Values: 0x0 (DISABLED): Disable the advance Hibern8 override 0x1 (ENABLED): Enable the advance Hibern8 override Value After Reset: 0x0 Exists: Always
0	ADVTACTOVREN	R/W	Advance TActivate override control 0: Disable the advance TActivate override 1: Enable the advance TActivate override The override value is specified by the VS_OverrideAdvTactivate register. Values: 0x0 (DISABLED): Disable the advance TActivate override 0x1 (ENABLED): Enable the advance TActivate override Value After Reset: 0x0 Exists: Always

5.9.70 VS_OverrideAdvTactivate

- **Name:** VS_OverrideAdvTactivate
- **Description:** This attribute configures the override value for advance TActivate capability of M-PHY which(RX_Advanced_Min_ActivateTime_Capability) gets transferred inside the EXT1 packet during the link startup process.
- **Size:** 8 bits
- **Offset:** 0xd0b2
- **Exists:** Always

7:4	Rsvd
3:0	ADVTACTOVRVAL

Table 5-251 Fields for Register: VS_OverrideAdvTactivate

Bits	Name	Memory Access	Description
7:4			Reserved Field: Yes
3:0	ADVTACTOVRVAL	R/W	Advance TActivate capability override value. Note: Value should be in multiples of two only. Value After Reset: 0x7 Exists: Always

5.9.71 VS_OverrideAdvHiber8

- **Name:** VS_OverrideAdvHibn8
- **Description:** This attribute configures the override value for advance Hibern8 capability of M-PHY(RX_Advanced_Hibern8Time_Capability) which gets transferred inside the EXT1 packet during the link startup process.
- **Size:** 32 bits
- **Offset:** 0xd0b3
- **Exists:** Always



Table 5-252 Fields for Register: VS_OverrideAdvHiber8

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	ADVHBR8OVRVAL	R/W	Advance hibernate capability override value. Note: The value should be in multiples of two only. Value After Reset: 0x4 Exists: Always

5.9.72 VS_OverrideAdvGran

- **Name:** VS_OverrideAdvGranularity
- **Description:** This attribute configures the override value for advance granularity capability of M-PHY(RX_Advanced_Granularity_Capability), which gets transferred inside the EXT1 packet during the link startup process.
- **Size:** 32 bits
- **Offset:** 0xd0b4
- **Exists:** Always

31:4		
	ADVGRAOVRVAL	3:1
	ADVFINEGRANEN	0

Table 5-253 Fields for Register: VS_OverrideAdvGran

Bits	Name	Memory Access	Description
31:4			Reserved Field: Yes
3:1	ADVGRAOVRVAL	R/W	Advance granularity capability override value. - 000: 4 us - 001: 8 us - 010: 16 us - 011: 32 us Values: ■ 0x0 (VALUE_OF_0): 4 us ■ 0x1 (VALUE_OF_1): 8 us ■ 0x2 (VALUE_OF_2): 16 us ■ 0x3 (VALUE_OF_3): 32 us Value After Reset: 0x0 Exists: Always
0	ADVFINEGRANEN	R/W	Fine granularity control. Values: ■ 0x0 (DISABLED): Disable fine granularity ■ 0x1 (ENABLED): Enable fine granularity Value After Reset: 0x1 Exists: Always

5.9.73 VS_ForceLanealign

- **Name:** VS_ForceLanealign
- **Description:** This attribute controls the behavior of the RMMI TX_PHYDORDY port in multilane configuration.
- **Size:** 32 bits
- **Offset:** 0xd0b5
- **Exists:** Always



Table 5-254 Fields for Register: VS_ForceLanealign

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	VSFORCELANEALIGN	R/W	This attribute controls the TX_PHYDORDY assertion in multilane configuration. #- 0: Controller drives TX_PHYDORDY without waiting for TX_PHYDIRDY #- 1: controller drives TX_PHYDORDY on all lane after TX_PHYDIRDY goes high for all lanes Values: ■ 0x1 (ENABLED): Controller drives TX_PHYDORDY on all lane after TX_PHYDIRDY goes high for all lanes ■ 0x0 (DISABLED): Controller drives TX_PHYDORDY without waiting for TX_PHYDIRDY Value After Reset: VS_FORCE_LANE_ALIGN Exists: Always

5.9.74 VS_OverridePwmBurstClosureLength

- **Name:** VS_OverridePwmBurstClosureLength
- **Description:** This attribute configures the override value for pwm burst closure length capability of M-PHY, which gets transferred inside the EXT1 packet during the link startup process.
- **Size:** 32 bits
- **Offset:** 0xd0b7
- **Exists:** Always

31:8	
7:0	PWMBURSTCLOSURELENGTH
	Rsvd

Table 5-255 Fields for Register: VS_OverridePwmBurstClosureLength

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	PWMBURSTCLOSURELENGTH	R/W	Pwm burst closure length override value Value After Reset: 0x5 Exists: Always

5.9.75 VS_OverridePwmSyncLength

- **Name:** VS_OverridePwmSyncLength
- **Description:** This attribute configures the override value for pwm sync length capability of M-PHY, which gets transferred inside the EXT1 packet during the link startup process.
- **Size:** 32 bits
- **Offset:** 0xd0b8
- **Exists:** Always



Table 5-256 Fields for Register: VS_OverridePwmSyncLength

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	PWMSYNCLENGTH	R/W	Pwm sync length override value Value After Reset: 0x5 Exists: Always

5.9.76 VS_OverridePwmPrepareLength

- **Name:** VS_OverridePwmPrepareLength
- **Description:** This attribute configures the override value for pwm prepare length capability of MPHY, which gets transferred inside the EXT1 packet during the link startup process.
- **Size:** 32 bits
- **Offset:** 0xd0b9
- **Exists:** Always

31:8	Rsvd
7:0	PWMPREPARELENGTH

Table 5-257 Fields for Register: VS_OverridePwmPrepareLength

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	PWMPREPARELENGTH	R/W	Pwm prepare length override value Value After Reset: 0xf Exists: Always

5.9.77 VS_SaveConfigTimeIncrement

- **Name:** VS_SaveConfigTimeIncrement
- **Description:** This attribute holds the value which is added to the SaveConfigTime capability of M-PHY before the value gets transferred to the remote side.
- **Size:** 32 bits
- **Offset:** 0xd0ba
- **Exists:** Always



Table 5-258 Fields for Register: VS_SaveConfigTimeIncrement

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	SAVECNFTIMEINC	R/W	Save config time capability value increment register. Note: Ensure that the sum of M-PHY capability value and this attribute value does not go beyond 250. Value After Reset: 0x0 Exists: Always

5.9.78 VS_OverrideHSG3SyncLength

- **Name:** VS_OverrideHSG3SyncLength
- **Description:** This attribute overrides the value for HS G3 Sync Length of the M-PHY capability which gets transferred inside the EXT1 packet during the link start-up process.
- **Size:** 32 bits
- **Offset:** 0xd0bb
- **Exists:** Always

31:8	Rsvd
7:0	HSG3SYNCLNGTH

Table 5-259 Fields for Register: VS_OverrideHSG3SyncLength

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	HSG3SYNCLNGTH	R/W	HS G3 Sync Length Override value. Value After Reset: 0xf Exists: Always

5.9.79 VS_OverrideHSG3PrepareLength

- **Name:** VS_OverrideHSG3PrepareLength
- **Description:** This attribute overrides the value for HS G3 Prepare Length of the M-PHY capability which gets transferred inside the EXT1 packet during the link start-up process.
- **Size:** 32 bits
- **Offset:** 0xd0bc
- **Exists:** Always



Table 5-260 Fields for Register: VS_OverrideHSG3PrepareLength

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	HSG3PREPARELENGTH	R/W	HS G3 Prepare Length Override value. Value After Reset: 0xf Exists: Always

5.9.80 VS_MphyCfgUpdtAckWait

- **Name:** VS_MPhyCfgUpdtAckWait
- **Description:** This attribute controls the VS_MPhyCfgUpdt acknowledgement. Using this attribute, VS_MphyCfgUpdt acknowledgment can be bypassed before actual update of the effective bank inside the M-PHY.
- **Size:** 32 bits
- **Offset:** 0xd0bd
- **Exists:** Always

31:1	
Rsvd	CFGUPDTACKWAIT
0	

Table 5-261 Fields for Register: VS_MphyCfgUpdtAckWait

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	CFGUPDTACKWAIT	R/W	This bit controls the acknowledgment wait for the vendor specific update from the application: #- 0: Bypass the wait for the acknowledgement for VS_MphyCfgUpdt. #- 1: Wait for the acknowledgement for VS_MphyCfgUpdt till actual update within a M-PHY. Values: ■ 0x0 (VALUE_OF_0): Bypass the wait for the acknowledgement for VS_MphyCfgUpdt ■ 0x1 (VALUE_OF_1): Wait for the acknowledgement for VS_MphyCfgUpdt till actual update within a M-PHY Value After Reset: 0x1 Exists: Always

5.9.81 VS_TxControlActTimer

- **Name:** VS_SaveConfigTimeIncrement
- **Description:** This attribute controls the drive value of the port MPx_TX_Control_ActTimer of the M-PHY.
- **Size:** 32 bits
- **Offset:** 0xd0c0
- **Exists:** Always



Table 5-262 Fields for Register: VS_TxControlActTimer

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	TXCNTACTTIMER	R/W	MPx_Tx_Control_actTimer Port Control: ■ 0 : Drives MPx_Tx_Control_ActTimer to value 0. ■ 1 : Drives MPx_Tx_Control_ActTimer to value 1. Note: TX_DIFNDRIVE remains 0 when value of this attribute is set to 1. Values: ■ 0x0 (DISABLED): Drives MPx_Tx_Control_ActTimer to value 0 ■ 0x1 (ENABLED): Drives MPx_Tx_Control_ActTimer to value 1 Value After Reset: 0x0 Exists: Always

5.9.82 VS_MphyDisable

- **Name:** VS_MphyDisable
- **Description:** This attribute keeps the M-PHY into disable state after reset release.
- **Size:** 32 bits
- **Offset:** 0xd0c1
- **Exists:** SNPS_GEN3_MPHY_EXT_EN=1

31:1	
Rsvd	MPHYDIS
	0

Table 5-263 Fields for Register: VS_MphyDisable

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	MPHYDIS	R/W	mphy_disabe port control. #- 0 : M-PHY is enabled after reset release. #- 1 : M-PHY is in disable state after reset release. Values: ■ 0x0 (DISABLED): M-PHY is enabled after reset release ■ 0x1 (ENABLED): M-PHY is in disable state after reset release Value After Reset: 0x1 Exists: Always

5.9.83 VS_Gen2MphyPllOn

- **Name:** VS_Gen2MphyPllOn
- **Description:** This attribute controls configuration of Synopsys GEN2 M-PHY based on GEN2 M-PHY PLL usage.
- **Size:** 32 bits
- **Offset:** 0xd0c2
- **Exists:** Always

31:2	1	0
Rsvd	GEN2MPHYREFCLK	GEN2MPHYPLLON

Table 5-264 Fields for Register: VS_Gen2MphyPllOn

Bits	Name	Memory Access	Description
31:2			Reserved Field: Yes
1	GEN2MPHYREFCLK	R/W	Synopsys GEN2 M-PHY REFCLK. #- 0 : REFCLK always available to GEN2 M-PHY. #- 1 : REFCLK available into HS mode only. Values: ■ 0x0 (VALUE_OF_0): REFCLK always available to GEN2 M-PHY ■ 0x1 (VALUE_OF_1): REFCLK available into HS mode only Value After Reset: 0x0 Exists: Always

Table 5-264 Fields for Register: VS_Gen2MphyPllOn (Continued)

Bits	Name	Memory Access	Description
0	GEN2MPHYPLLON	R/W	Synopsys GEN2 M-PHY configuration control port. - 0 : M-PHY configuration with PLL configuration. - 1 : M-PHY configuration for PLL divider only. Setting this field to 1 reduces the SaveConfigTime requirement (the defined limit of 10us in UniPro Specification) for M-RX. However, it is required that the M-PHY PLL is configured in HS-G3 during M-PHY initialization process if refclk is always present. Values: ■ 0x0 (VALUE_OF_0): M-PHY configuration with PLL configuration ■ 0x1 (VALUE_OF_1): M-PHY configuration for PLL divider only Value After Reset: 0x0 Exists: Always

5.9.84 VS_TxFlowControl

- **Name:** VS_TxFlowControl
- **Description:** This attribute provides the status of Flow Control Registers on Tx side.
- **Size:** 32 bits
- **Offset:** 0xd0ca
- **Exists:** Always

31:30	29:20	19:10	9:0
Rsvd	FLOWCPARTUTC0	FLOWCUTC0	FLOWCRTC0

Table 5-265 Fields for Register: VS_TxFlowControl

Bits	Name	Memory Access	Description
31:30			Reserved Field: Yes
29:20	FLOWCPARTUTC0	R	Stores the number of partial credits of U (Part_U). The maximum value is 31 for credit unit of 32 bytes and 127 for credit unit of 128 bytes. Value After Reset: 0x0 Exists: Always
19:10	FLOWCUTC0	R	Stores the number of credits used (U). The maximum value for R is 128. Value After Reset: 0x0 Exists: Always
9:0	FLOWCRTC0	R	Stores the number of credits received (R). The maximum value for R is 128. Value After Reset: 0x0 Exists: Always

5.9.85 VS_RxFlowControl

- **Name:** VS_RxFlowControl
- **Description:** This attribute provides the status of Flow Control Registers on Rx side.
- **Size:** 32 bits
- **Offset:** 0xd0cb
- **Exists:** Always

31:30	29:20	19:10	9:0
Rsvd	FLOWCPARTATC0	FLOWCATC0	FLOWCSTC0

Table 5-266 Fields for Register: VS_RxFlowControl

Bits	Name	Memory Access	Description
31:30			Reserved Field: Yes
29:20	FLOWCPARTATC0	R	Stores the number of partial credits of U (Part_U). The maximum value is 31 for credit unit of 32 bytes and 127 for credit unit of 128 bytes. Value After Reset: 0x0 Exists: Always
19:10	FLOWCATC0	R	Stores the number of credits Available (U). The maximum value for R is 128. Value After Reset: 0x0 Exists: Always
9:0	FLOWCSTC0	R	Stores the number of credits received (R). The maximum value for R is 128. Value After Reset: 0x0 Exists: Always

5.9.86 VS_GPO_Ctrl_0

- **Name:** VS_GPO_Ctrl_0
- **Description:** General Purpose Output. The value written to this field gets reflected on the LA_GPO output ports. The number of bits available in this field depends on parameter P_GPIO. Other bits are not used (reserved and always reset).**Note:** This register gets implemented when the parameter P_GPIO is set to a non-zero value.
- **Size:** 32 bits
- **Offset:** 0xd0d0
- **Exists:** P_GPIO>0



Table 5-267 Fields for Register: VS_GPO_Ctrl_0

Bits	Name	Memory Access	Description
31:0	GPO_0	R/W	General Purpose Output. The value written to this field gets reflected on the LA_GPO output ports. Unused bits are reserved and always reset to zero. Value After Reset: 0x0 Exists: Always

5.9.87 VS_GPO_Ctrl_1

- **Name:** VS_GPO_Ctrl_1
- **Description:** General Purpose Output. The value written to this field gets reflected on the LA_GPO output ports. The number of bits available in this field depends on parameter P_GPIO width option. Other bits are not used (reserved and always reset).
- **Note:** This register gets implemented when P_GPIO is greater than 32.
- **Size:** 32 bits
- **Offset:** 0xd0d1
- **Exists:** P_GPIO>32



Table 5-268 Fields for Register: VS_GPO_Ctrl_1

Bits	Name	Memory Access	Description
31:0	GPO_1	R/W	General Purpose Output. The value written to this field gets reflected on the LA_GPO output ports. Unused bits are reserved and always reset to zero. Value After Reset: 0x0 Exists: Always

5.9.88 VS_GPO_Ctrl_2

- **Name:** VS_GPO_Ctrl_2
- **Description:** General Purpose Output. The value written to this field gets reflected on the LA_GPO output ports. The number of bits available in this field depends on parameter P_GPIO width option. Other bits are not used (reserved and always reset).
- **Note:** This register gets implemented when P_GPIO is greater than 64.
- **Size:** 32 bits
- **Offset:** 0xd0d2
- **Exists:** P_GPIO>64



Table 5-269 Fields for Register: VS_GPO_Ctrl_2

Bits	Name	Memory Access	Description
31:0	GPO_1	R/W	General Purpose Output. The value written to this field gets reflected on the LA_GPO output ports. Unused bits are reserved and always reset to zero. Value After Reset: 0x0 Exists: Always

5.9.89 VS_GPI_Status_0

- **Name:** VS_GPI_Status_0
- **Description:** General Purpose Input status. This register reflects the status of signals connected to LA_GPI ports. The number of bits available in this field depends on parameter P_GPIO width option. Other bits are not used (reserved and always reset).

Note: This register gets implemented when the parameter P_GPIO is set to a non-zero value.

- **Size:** 32 bits
- **Offset:** 0xd0d3
- **Exists:** P_GPIO>0



Table 5-270 Fields for Register: VS_GPI_Status_0

Bits	Name	Memory Access	Description
31:0	GPI_0	R	General Purpose Input. This register reflects the status of signals connected to LA_GPI ports. Value After Reset: *Varies with the value of the input port LA_Gpi Exists: Always

5.9.90 VS_GPI_Status_1

- **Name:** VS_GPI_Status_1
- **Description:** General Purpose Input status. This register reflects the status of signals connected to LA_GPI ports. The number of bits available in this field depends on parameter P_GPIO width option. Other bits are not used (reserved and always reset).

Note: This register gets implemented when P_GPIO is greater than 32.
- **Size:** 32 bits
- **Offset:** 0xd0d4
- **Exists:** P_GPIO>32



Table 5-271 Fields for Register: VS_GPI_Status_1

Bits	Name	Memory Access	Description
31:0	GPI_0	R	General Purpose Input.This register reflects the status of signals connected to LA_GPI ports. Value After Reset: *Varies with the value of the input port LA_Gpi Exists: Always

5.9.91 VS_GPI_Status_2

- **Name:** VS_GPI_Status_2
- **Description:** General Purpose Input status. This register reflects the status of signals connected to LA_GPI ports. The number of bits available in this field depends on parameter P_GPIO width option. Other bits are not used (reserved and always reset).
Note: This register gets implemented when P_GPIO is greater than 64.
- **Size:** 32 bits
- **Offset:** 0xd0d5
- **Exists:** P_GPIO>64



Table 5-272 Fields for Register: VS_GPI_Status_2

Bits	Name	Memory Access	Description
31:0	GPI_0	R	General Purpose Input.This register reflects the status of signals connected to LA_GPI ports. Value After Reset: *Varies with the value of the input port LA_Gpi Exists: Always

5.9.92 VS_NonAutoModeBurstControl

- **Name:** VS_NonAutoModeBurstControl
- **Description:** This attribute controls the condition at which the TX burst gets asserted after power mode is changed to Fast_Mode or Slow_Mode.

The value in this attribute comes into effect when power mode change request with Power_Mode set to Fast_Mode or Slow_Mode on outbound link is issued.
- **Size:** 32 bits
- **Offset:** 0xd0e0
- **Exists:** Always



Table 5-273 Fields for Register: VS_NonAutoModeBurstControl

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes

Table 5-273 Fields for Register: VS_NonAutoModeBurstControl (Continued)

Bits	Name	Memory Access	Description
0	ENABLE_BURST_CONTROL	R/W	This bit controls the timing of TX burst assertion after a PMC to Fast_Mode or Slow_Mode: 0: This is the default core behavior. When the core enters Fast_Mode or Slow_Mode, the TX burst is asserted just after PA_SaveConfigTime is lapsed. 1: When the core enters Fast_Mode or Slow_Mode, the assertion of TX burst just after PA_SaveConfigTime depends on data being available for transmission. If the data is already available for transmission, then the TX burst is asserted just after PA_SaveConfigTime. If there is no data to be transmitted, then the link remains in Save state and TX burst is asserted only when data is queued for transmission. Values: 0x0 (VALUE_OF_0): Default core behavior 0x1 (VALUE_OF_1): Core Behavior based on data availability Value After Reset: 0x0 Exists: Always

5.9.93 VS_OverrideRxMinSleepNoConfigTimeCapability

- **Name:** VS_OverrideRxMinSleepNoConfigTimeCapability
- **Description:** This attribute configures the override value for RX_Min_SLEEP_NoConfig_Time_Capability attribute of M-PHY, which gets transferred inside the PACP_CAP packet during the link startup process.
- **Size:** 32 bits
- **Offset:** 0xd0e1
- **Exists:** Always



Table 5-274 Fields for Register: VS_OverrideRxMinSleepNoConfigTimeCapability

Bits	Name	Memory Access	Description
31:4			Reserved Field: Yes
3:0	RXMINSLEEPNOCONFIGTIMEOVR	R/W	RX_Min_SLEEP_NoConfig_Time_Capability override value. Value After Reset: 0xf Exists: Always

5.9.94 VS_OverrideTxHsUnterminatedLineDriveCapability

- **Name:** VS_OverrideTxHsUnterminatedLineDriveCapability
- **Description:** This attribute configures the override value for TX_HS_Unterminated_LINE_Drive_Capability attribute of M-PHY, which gets transferred inside the PACP_CAP packet during the link startup process.
- **Size:** 32 bits
- **Offset:** 0xd0e2
- **Exists:** Always

31:1	Rsvd
0	TXHSUNTERMLINEDRIVEOVR

Table 5-275 Fields for Register: VS_OverrideTxHsUnterminatedLineDriveCapability

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	TXHSUNTERMLINEDRIVEOVR	R/W	TX_HS_Unterminated_LINE_Drive_Capability override value. Value After Reset: 0x0 Exists: Always

5.9.95 VS_OverrideTxLsTerminatedLineDriveCapability

- **Name:** VS_OverrideTxLsTerminatedLineDriveCapability
- **Description:** This attribute configures the override value for TX_LS_Terminated_LINE_Drive_Capability attribute of M-PHY, which gets transferred inside the PACP_CAP packet during the link startup process.
- **Size:** 32 bits
- **Offset:** 0xd0e3
- **Exists:** Always



Table 5-276 Fields for Register: VS_OverrideTxLsTerminatedLineDriveCapability

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	TXLSTERMLINEDRIVEOVR	R/W	TX_LS_Terminated_LINE_Drive_Capability override value Value After Reset: 0x0 Exists: Always

5.9.96 VS_OverrideRxMinStallNoConfigTimeCapability

- **Name:** VS_OverrideRxMinStallNoConfigTimeCapability
- **Description:** This attribute configures the override value for RX_Min_STALL_NoConfig_Time_Capability attribute of M-PHY, which gets transferred inside the PACP_CAP packet during the link startup process.
- **Size:** 32 bits
- **Offset:** 0xd0e4
- **Exists:** Always

31:8	Rsvd
7:0	RXMINSTALLNOCONFIGTIMEOVR

Table 5-277 Fields for Register: VS_OverrideRxMinStallNoConfigTimeCapability

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	RXMINSTALLNOCONFIGTIMEOVR	R/W	RX_Min_STALL_NoConfig_Time_Capability override value Value After Reset: 0xff Exists: Always

5.9.97 VS_OverrideRxMinSaveConfigTimeCapability

- **Name:** VS_OverrideRxMinSaveConfigTimeCapability
- **Description:** This attribute configures the override value for RX_Min_SAVE_Config_Time_Capability attribute of M-PHY, which gets transferred inside the PACP_CAP packet during the link startup process.
- **Size:** 32 bits
- **Offset:** 0xd0e5
- **Exists:** Always



Table 5-278 Fields for Register: VS_OverrideRxMinSaveConfigTimeCapability

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	RXMINSAVECONFIGTIMEOVR	R/W	RX_Min_SAVE_Config_Time_Capability override value Value After Reset: 0xf0 Exists: Always

5.9.98 VS_OverrideRxExtdMinSaveConfigTimeCapability

- **Name:** VS_OverrideRxExtdMinSaveConfigTimeCapability
- **Description:** This attribute configures the override value for RX_Extended_Min_SAVE_Config_Time_Capability attribute of M-PHY, which gets transferred via TExtSaveConf field in PACP_CAP_EXT2 packet during the link start-up process.
- **Size:** 32 bits
- **Offset:** 0xd0e6
- **Exists:** Always



Table 5-279 Fields for Register: VS_OverrideRxExtdMinSaveConfigTimeCapability

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	RXMINEXTDSAVECONFIGTIMEOVR	R/W	RX_Extended_Min_SAVE_Config_Time_Capability override value Value After Reset: 0x0 Exists: Always

5.9.99 VS_OverrideRxHsG1PrepareLengthCapability

- **Name:** VS_OverrideRxHsG1PrepareLengthCapability
- **Description:** This attribute configures the override value for RX_HS_G1_PREPARE_LENGTH_Capability attribute of M-PHY, which gets transferred inside the PACP_CAP_EXT1 packet during the link startup process.
- **Size:** 32 bits
- **Offset:** 0xd0e7
- **Exists:** Always



Table 5-280 Fields for Register: VS_OverrideRxHsG1PrepareLengthCapability

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	HSG1PREPARELENGTHOVR	R/W	RX_HS_G1_PREPARE_LENGTH_Capability override value Value After Reset: 0xf Exists: Always

5.9.100 VS_OverrideRxHsG1SyncLengthCapability

- **Name:** VS_OverrideRxHsG1SyncLengthCapability
- **Description:** This attribute configures the override value for RX_HS_G1_SYNC_LENGTH_Capability attribute of M-PHY, which gets transferred inside the PACP_CAP_EXT1 packet during the link startup process.
- **Size:** 32 bits
- **Offset:** 0xd0e8
- **Exists:** Always

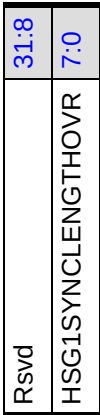


Table 5-281 Fields for Register: VS_OverrideRxHsG1SyncLengthCapability

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	HSG1SYNCLENGTHOVR	R/W	RX_HS_G1_SYNC_LENGTH_Capability override value Value After Reset: 0xf Exists: Always

5.9.101 VS_OverrideRxHsG2PrepareLengthCapability

- **Name:** VS_OverrideRxHsG2PrepareLengthCapability
- **Description:** This attribute configures the override value for RX_HS_G2_PREPARE_LENGTH_Capability attribute of M-PHY, which gets transferred inside the PACP_CAP_EXT1 packet during the link startup process.
- **Size:** 32 bits
- **Offset:** 0xd0e9
- **Exists:** Always

31:8	
7:0	HSG2PREPARELENGTHOVR

Table 5-282 Fields for Register: VS_OverrideRxHsG2PrepareLengthCapability

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	HSG2PREPARELENGTHOVR	R/W	RX_HS_G2_PREPARE_LENGTH_Capability override value Value After Reset: 0xf Exists: Always

5.9.102 VS_OverrideRxHsG2SyncLengthCapability

- **Name:** VS_OverrideRxHsG2SyncLengthCapability
- **Description:** This attribute configures the override value for RX_HS_G2_SYNC_LENGTH_Capability attribute of M-PHY, which gets transferred inside the PACP_CAP_EXT1 packet during the link startup process.
- **Size:** 32 bits
- **Offset:** 0xd0ea
- **Exists:** Always

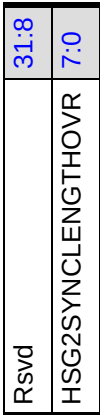


Table 5-283 Fields for Register: VS_OverrideRxHsG2SyncLengthCapability

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	HSG2SYNCLENGTHOVR	R/W	RX_HS_G2_SYNC_LENGTH_Capability override value Value After Reset: 0xf Exists: Always

5.9.103 VS_OverrideRxHsG4SyncLengthCapability

- **Name:** VS_OverrideRxHsG4SyncLengthCapability
- **Description:** This attribute configures the override value for RX_HS_G4_SYNC_LENGTH_Capability attribute of M-PHY, which gets transferred inside the PACP_CAP_EXT2 packet during the link startup process.
- **Size:** 32 bits
- **Offset:** 0xd0eb
- **Exists:** Always

31:8	Rsvd
7:0	HSG4SYNCLNGTHOVR

Table 5-284 Fields for Register: VS_OverrideRxHsG4SyncLengthCapability

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	HSG4SYNCLNGTHOVR	R/W	RX_HS_G4_SYNC_LENGTH_Capability override value Value After Reset: 0xff Exists: Always

5.9.104 VS_OverrideRxHsG4PrepareLengthCapability

- **Name:** VS_OverrideRxHsG4PrepareLengthCapability
- **Description:** This attribute configures the override value for RX_HS_G4_PREPARE_LENGTH_Capability attribute of M-PHY, which gets transferred inside the PACP_CAP_EXT2 packet during the link startup process.
- **Size:** 32 bits
- **Offset:** 0xd0ec
- **Exists:** Always

31:8	Rsvd
7:0	HSG4PREPARELENGTHOVR

Table 5-285 Fields for Register: VS_OverrideRxHsG4PrepareLengthCapability

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	HSG4PREPARELENGTHOVR	R/W	RX_HS_G4_PREPARE_LENGTH_Capability override value Value After Reset: 0xff Exists: Always

5.9.105 VS_OverrideRxHsAdaptRefreshCapability

- **Name:** VS_OverrideRxHsAdaptRefreshCapability
- **Description:** This attribute configures the override value for RX_HS_ADAPT_REFRESH_Capability attribute of M-PHY, which gets transferred inside the PACP_CAP_EXT2 packet during the link startup process.
- **Size:** 32 bits
- **Offset:** 0xd0ed
- **Exists:** Always



Table 5-286 Fields for Register: VS_OverrideRxHsAdaptRefreshCapability

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	HSADAPTREFRESHOVR	R/W	RX_HS_ADAPT_REFRESH_Capability override value Value After Reset: 0x7f Exists: Always

5.9.106 VS_OverrideRxHsAdaptInitialCapability

- **Name:** VS_OverrideRxHsAdaptInitialCapability
- **Description:** This attribute configures the override value for RX_HS_ADAPT_INITIAL_Capability attribute of M-PHY, which gets transferred inside the PACP_CAP_EXT2 packet during the link startup process.
- **Size:** 32 bits
- **Offset:** 0xd0ee
- **Exists:** Always

31:8	Rsvd
7:0	HSADAPTINITIALOVR

Table 5-287 Fields for Register: VS_OverrideRxHsAdaptInitialCapability

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	HSADAPTINITIALOVR	R/W	RX_HS_ADAPT_INITIAL_Capability override value Value After Reset: 0x7f Exists: Always

5.9.107 VS_DowngradedValueOverrideEnable

- **Name:** VS_DowngradedValueOverrideEnable
- **Description:** This attribute controls the override operation of the Downgraded values of M-PHY attributes after capability exchange.
- **Size:** 32 bits
- **Offset:** 0xd0f0
- **Exists:** Always

31:5	Rsvd
4	TXPWMBURSTCLOSUREEXTENSIONOVREN
3	TXLSPREPARELENGTHHOVREN
2	TXPWMG6G7SYNCLNGTHOVREN
1	TXMINSTALLNOCONFIGTIMEOVREN
0	TXMINSLEEPNOCONFIGTIMEOVREN

Table 5-288 Fields for Register: VS_DowngradedValueOverrideEnable

Bits	Name	Memory Access	Description
31:5			Reserved Field: Yes

Table 5-288 Fields for Register: VS_DowngradedValueOverrideEnable (Continued)

Bits	Name	Memory Access	Description
4	TXPWMBURSTCLOSUREEXTENSIONOVREN	R/W	TX_PWM_BURST_Closure_Extension downgraded value override enable: #- 0: Disable TX_PWM_BURST_Closure_Extension override #- 1: Enable TX_PWM_BURST_Closure_Extension override Values: 0x1 (ENABLED): Enable TX_PWM_BURST_Closure_Extension override 0x0 (DISABLED): Disable TX_PWM_BURST_Closure_Extension override Value After Reset: 0x0 Exists: Always
3	TXLSPREPARELENGTHOVREN	R/W	TX_LS_PREPARE_LENGTH downgraded value override enable #- 0: Disable TX_LS_PREPARE_LENGTH override #- 1: Enable TX_LS_PREPARE_LENGTH override Values: 0x1 (ENABLED): Enable TX_LS_PREPARE_LENGTH override 0x0 (DISABLED): Disable TX_LS_PREPARE_LENGTH override Value After Reset: 0x0 Exists: Always
2	TXPWMG6G7SYNCLengthOVREN	R/W	TX_PWM_G6_G7_SYNC_LENGTH downgraded value override enable #- 0: Disable TX_PWM_G6_G7_SYNC_LENGTH override #- 1: Enable TX_PWM_G6_G7_SYNC_LENGTH override NOTE: Enabling this bit does not have any effect when UniPro is used as V2.0 Values: 0x1 (ENABLED): Enable TX_PWM_G6_G7_SYNC_LENGTH override 0x0 (DISABLED): Disable TX_PWM_G6_G7_SYNC_LENGTH override Value After Reset: 0x0 Exists: Always

Table 5-288 Fields for Register: VS_DowngradedValueOverrideEnable (Continued)

Bits	Name	Memory Access	Description
1	TXMINSTALLNOCONFIGTIMEOVREN	R/W	TX_Min_STALL_NoConfig_Time downgraded value override enable #- 0: Disable TX_Min_STALL_NoConfig_Time override #- 1: Enable TX_Min_STALL_NoConfig_Time override Values: 0x1 (ENABLED): Enable TX_Min_STALL_NoConfig_Time override 0x0 (DISABLED): Disable TX_Min_STALL_NoConfig_Time override Value After Reset: 0x0 Exists: Always
0	TXMINSLEEPNOCONFIGTIMEOVREN	R/W	TX_Min_SLEEP_NoConfig_Time downgraded value override enable #- 0: Disable TX_Min_SLEEP_NoConfig_Time override #- 1: Enable TX_Min_SLEEP_NoConfig_Time override Values: 0x1 (ENABLED): Enable TX_Min_SLEEP_NoConfig_Time override 0x0 (DISABLED): Disable TX_Min_SLEEP_NoConfig_Time override Value After Reset: 0x0 Exists: Always

5.9.108 VS_OverrideTxMinSleepNoConfigTimeDowngraded

- **Name:** VS_OverrideTxMinSleepNoConfigTimeDowngraded
- **Description:** This attribute configures the override value for TX_Min_SLEEP_NoConfig_Time attribute of M-PHY, which gets written after Capability downgrading.
- **Size:** 32 bits
- **Offset:** 0xd0f1
- **Exists:** Always

31:4	
3:0	TXMINSLEEPNOCONFIGTIMEOVR
	Rsvd

Table 5-289 Fields for Register: VS_OverrideTxMinSleepNoConfigTimeDowngraded

Bits	Name	Memory Access	Description
31:4			Reserved Field: Yes
3:0	TXMINSLEEPNOCONFIGTIMEOVR	R/W	TX_Min_SLEEP_NoConfig_Time override value Value After Reset: 0xf Exists: Always

5.9.109 VS_OverrideTxMinStallNoConfigTimeDowngraded

- **Name:** VS_OverrideTxMinStallNoConfigTimeDowngraded
- **Description:** This attribute configures the override value for TX_Min_STALL_NoConfig_Time attribute of M-PHY, which gets written after Capability downgrading.
- **Size:** 32 bits
- **Offset:** 0xd0f2
- **Exists:** Always

31:8	
7:0	TXMINSTALLNOCONFIGTIMEOVR

Table 5-290 Fields for Register: VS_OverrideTxMinStallNoConfigTimeDowngraded

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	TXMINSTALLNOCONFIGTIMEOVR	R/W	TX_Min_STALL_NoConfig_Time override value Value After Reset: 0xff Exists: Always

5.9.110 VS_OverrideTxPwmG6G7SyncLengthDowngraded

- **Name:** VS_OverrideTxPwmG6G7SyncLengthDowngraded
- **Description:** This attribute configures the override value for TX_PWM_G6_G7_SYNC_LENGTH attribute of M-PHY, which gets written after Capability downgrading.
- **Size:** 32 bits
- **Offset:** 0xd0f3
- **Exists:** Always



Table 5-291 Fields for Register: VS_OverrideTxPwmG6G7SyncLengthDowngraded

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	TXPWMG6G7SYNCLNGTHOVR	R/W	TX_PWM_G6_G7_SYNC_LENGTH override value Value After Reset: 0xf Exists: Always

5.9.111 VS_OverrideTxLsPrepareLengthDowngraded

- **Name:** VS_OverrideTxLsPrepareLengthDowngraded
- **Description:** This attribute configures the override value for TX_LS_PREPARE_LENGTH attribute of M-PHY, which gets written after Capability downgrading.
- **Size:** 32 bits
- **Offset:** 0xd0f4
- **Exists:** Always



Table 5-292 Fields for Register: VS_OverrideTxLsPrepareLengthDowngraded

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	TXLSPREPARELENGTHOVR	R/W	TX_LS_PREPARE_LENGTH override value Value After Reset: 0xf Exists: Always

5.9.112 VS_OverrideTxPwmBurstClosureExtensionDowngraded

- **Name:** VS_OverrideTxPwmBurstClosureExtensionDowngraded
- **Description:** This attribute configures the override value for TX_PWM_BURST_Closure_Extension attribute of M-PHY, which gets written after Capability downgrading.
- **Size:** 32 bits
- **Offset:** 0xd0f5
- **Exists:** Always

31:8	
7:0	TXPWMBURSTCLOSUREEXTN
	Rsvd

Table 5-293 Fields for Register: VS_OverrideTxPwmBurstClosureExtensionDowngraded

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	TXPWMBURSTCLOSUREEXTN	R/W	TX_PWM_BURST_Closure_Extension override value Value After Reset: 0xff Exists: Always

5.9.113 VS_LssBypass

- **Name:** VS_LssBypass
- **Description:** This attribute controls the bypassing of linerreset and TRG pattern communication during the LSS process.
- **Size:** 32 bits
- **Offset:** 0xd0f6
- **Exists:** Always

31:1	
Rsvd	LSSBYPASS
	0

Table 5-294 Fields for Register: VS_LssBypass

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	LSSBYPASS	R/W	LSS Bypass Control Signal: #- 0: Linerreset and TRG communication present during LSS process. #- 1: Linerreset and TRG communication skipped during LSS process. Values: ■ 0x0 (VALUE_OF_0): Linerreset and TRG communication present during LSS process ■ 0x1 (VALUE_OF_1): Linerreset and TRG communication skipped during LSS process Value After Reset: 0x0 Exists: Always

5.9.114 VS_PmcInterventionEn

- **Name:** VS_PmcInterventionEn
- **Description:** This attribute enables the PMC intervention feature for desired housekeeping activities by application during power mode change communication.
- **Size:** 32 bits
- **Offset:** 0xd0f7
- **Exists:** Always

31:2	1	0
Rsvd	PMCINTADPTEN	PMCINTEN

Table 5-295 Fields for Register: VS_PmcInterventionEn

Bits	Name	Memory Access	Description
31:2			Reserved Field: Yes
1	PMCINTADPTEN	R/W	PMC Intervention Feature Control Signal: ■ 0: PMC Intervention with Adapt for PMC request to G4/G5 is disabled. ■ 1: PMC Intervention with Adapt for PMC request to G4/G5 is enabled. Note: When this feature is enabled, UniPro generates a pulse on LA_PowerModeAdptRecd port when it receives PMC request for G4/G5 with adapt enabled from remote side. Values: ■ 0x0 (DISABLED): PMC Intervention with adapt is disabled ■ 0x1 (ENABLED): PMC Intervention with adapt is enabled Value After Reset: 0x0 Exists: Always

Table 5-295 Fields for Register: VS_PmcInterventionEn (Continued)

Bits	Name	Memory Access	Description
0	PMCINTEN	R/W	PMC Intervention Feature Control Signal: ■ 0: PMC Intervention is disabled. ■ 1: PMC Intervention is enabled. Note: When this feature is enabled, UniPro generates a pulse on LA_PowerModeRecd port when it receives PMC request from remote side. Values: ■ 0x0 (DISABLED): PMC Intervention is disabled ■ 0x1 (ENABLED): PMC Intervention is enabled Value After Reset: 0x0 Exists: Always

5.9.115 VS_PmcInterventionDone

- **Name:** VS_PmcInterventionDone
- **Description:** This attribute gives conformation about the PMC Intervention completion.
- **Size:** 32 bits
- **Offset:** 0xd0f8
- **Exists:** Always



Table 5-296 Fields for Register: VS_PmcInterventionDone

Bits	Name	Memory Access	Description
31:1			Reserved Field: Yes
0	PMCINTDONE	R/W	PMC Intervention Done indication signal: ■ 0: PMC Intervention is pending. ■ 1: PMC Intervention is done. Note: Attribute gets auto clear once PMC operation is completed. Values: ■ 0x0 (DISABLED): PMC Intervention is pending ■ 0x1 (ENABLED): PMC Intervention is done Value After Reset: 0x0 Exists: Always

5.9.116 VS_Ext1StatusCtrl

- **Name:** VS_Ext1StatusCtrl
- **Description:** This attribute controls the scrambling and skip support status bits of EXT1 capability packet.
- **Size:** 32 bits
- **Offset:** 0xd0f9
- **Exists:** Always

31:2	1	0
Rsvd	SKIPEN	SCREN

Table 5-297 Fields for Register: VS_Ext1StatusCtrl

Bits	Name	Memory Access	Description
31:2			Reserved Field: Yes
1	SKIPEN	R/W	Skip support of EXT1 status field. #- 0: PA RX does not require skip symbol insertion. #- 1: PA RX requires skip symbol insertion. Values: ■ 0x1 (ENABLED): PA RX requires skip symbol insertion. ■ 0x0 (DISABLED): PA RX does not require skip symbol insertion. Value After Reset: RX_SKIP_REQUEST Exists: Always
0	SCREN	R/W	Scrambling support of EXT1 status field. #- 0: PA RX does not implement scrambling. #- 1: PA RX implements scrambling. Values: ■ 0x1 (ENABLED): PA RX implements scrambling ■ 0x0 (DISABLED): PA RX does not implement scrambling Value After Reset: 0x1 Exists: Always

5.9.117 VS_MphyLaneCfgUpdt

- **Name:** VS_MphyLaneCfgUpdt
- **Description:** Write access to this attribute generates Mphy Lane specific Cfg updates with or without wait of Cfg update within M-PHY.
- **Size:** 32 bits
- **Offset:** 0xd0fa
- **Exists:** Always

31:9	Rsvd
8	LANECFGUPDTACKWAIT
7	RXUPDTLANE3
6	RXUPDTLANE2
5	RXUPDTLANE1
4	RXUPDTLANE0
3	TXUPDTLANE3
2	TXUPDTLANE2
1	TXUPDTLANE1
0	TXUPDTLANE0

Table 5-298 Fields for Register: VS_MphyLaneCfgUpdt

Bits	Name	Memory Access	Description
31:9			Reserved Field: Yes
8	LANECFGUPDTACKWAIT	R/W	This bit controls the acknowledgement wait for the vendor specific updates requested by the lower 8 bits. #- 0 : Bypass the wait for the acknowledgement for VS_MphyLaneCfgUpdt #- 1 : Wait for the acknowledgement for VS_MphyLaneCfgUpdt till actual update within a M-PHY Values: ■ 0x0 (DISABLED): Bypass the wait for the acknowledgement for VS_MphyLaneCfgUpdt ■ 0x1 (ENABLED): Wait for the acknowledgement for VS_MphyLaneCfgUpdt till actual update within a M-PHY Value After Reset: 0x0 Exists: Always

Table 5-298 Fields for Register: VS_MphyLaneCfgUpdt (Continued)

Bits	Name	Memory Access	Description
7	RXUPDTLANE3	R/W	RX Lane 3 Cfg Update request. Note: Bit gets auto clear once MPHY update is generated. Value After Reset: 0x0 Exists: Always
6	RXUPDTLANE2	R/W	RX Lane 2 Cfg Update request. Note: Bit gets auto clear once MPHY update is generated. Value After Reset: 0x0 Exists: Always
5	RXUPDTLANE1	R/W	RX Lane 1 Cfg Update request. Note: Bit gets auto clear once MPHY update is generated. Value After Reset: 0x0 Exists: Always
4	RXUPDTLANE0	R/W	RX Lane 0 Cfg Update request. Note: Bit gets auto clear once MPHY update is generated. Value After Reset: 0x0 Exists: Always
3	TXUPDTLANE3	R/W	TX Lane 3 Cfg Update request. Note: Bit gets auto clear once MPHY update is generated. Value After Reset: 0x0 Exists: Always
2	TXUPDTLANE2	R/W	TX Lane 2 Cfg Update request. Note: Bit gets auto clear once MPHY update is generated. Value After Reset: 0x0 Exists: Always
1	TXUPDTLANE1	R/W	TX Lane 1 Cfg Update request. Note: Bit gets auto clear once MPHY update is generated. Value After Reset: 0x0 Exists: Always
0	TXUPDTLANE0	R/W	TX Lane 0 Cfg Update request. Note: Bit gets auto clear once MPHY update is generated. Value After Reset: 0x0 Exists: Always

5.9.118 VS_ClkMuxSwitchingTimer

- **Name:** VS_ClkMuxSwitchingTimer
- **Description:** This attribute provides timer value for LP_ClkMuxSelect port toggling after TX and RX interfaces are into Save state during hibernate entry process.
- **Size:** 32 bits
- **Offset:** 0xd0fb
- **Exists:** Always

31:8	
7:0	CLKMUXTIMER
	Rsvd

Table 5-299 Fields for Register: VS_ClkMuxSwitchingTimer

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	CLKMUXTIMER	R/W	LP_ClkMuxSelect port toggling timing delay. Value After Reset: 0xa Exists: Always

5.9.119 VS_PMGlobalCtrl

- **Name:** VS_PMGlobalCtrl
- **Description:** Global control to performance monitoring metric counters.
- **Size:** 32 bits
- **Offset:** 0xd100
- **Exists:** Always

31	30:2	1	0
MRST	Rsvd	EN	FRZ

Table 5-300 Fields for Register: VS_PMGlobalCtrl

Bits	Name	Memory Access	Description
31	MRST	R/W	Master reset control to all performance counters. Values: ■ 0x0 (DISABLED): Metric counters are not under reset ■ 0x1 (ENABLED): Resets all metric counters Value After Reset: 0x0 Exists: Always
30:2			Reserved Field: Yes
1	EN	R/W	Enable performance counter control. Values: ■ 0x0 (DISABLED): Disable Performance Monitor ■ 0x1 (ENABLED): Enable Performance Monitor Value After Reset: 0x0 Exists: Always
0	FRZ	R/W	Metric counter freeze control. Values: ■ 0x0 (DISABLED): Metric counters are not in freeze state ■ 0x1 (ENABLED): Freeze all performance monitoring metric counters to collect profile Value After Reset: 0x0 Exists: Always

5.9.120 VS_Debug

- **Name:** VS_Debug
- **Description:** Cport arbiter debug register.
- **Size:** 32 bits
- **Offset:** 0xd101
- **Exists:** Always

31	30:9	8	7:0
ARST	Rsvd	ARSTSTAT	ACID

Table 5-301 Fields for Register: VS_Debug

Bits	Name	Memory Access	Description
31	ARST	W	Write this bit to 1 results in re-initialization of arbiter. Note: ARST re-initializes the flow control arbiter interface. Note: ARST has no effect on Rx CPort interfaces. Value After Reset: 0x0 Exists: Always
30:9			Reserved Field: Yes
8	ARSTSTAT	R	This bit reflects status of arbiter reset. #- 0: Arbiter reset is completed. #- 1: Arbiter reset is ongoing. Values: ■ 0x0 (VALUE_OF_0): Arbiter reset is completed ■ 0x1 (VALUE_OF_1): Arbiter reset is ongoing Value After Reset: 0x0 Exists: Always
7:0	ACID	R	Current active Cport ID being serviced by the arbiter. This is a read-only register. Value After Reset: 0xf Exists: Always

5.9.121 VS_PMGlobalCounter

- **Name:** VS_PMGlobalCounter
- **Description:** Counts the total number of segment transfers performed by the TX CPort arbiter.
- **Size:** 32 bits
- **Offset:** 0xd102
- **Exists:** Always

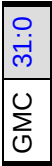


Table 5-302 Fields for Register: VS_PMGlobalCounter

Bits	Name	Memory Access	Description
31:0	GMC	R	Global Metric Counter Value. Value After Reset: 0x0 Exists: Always

5.9.122 VS_OverrideRxHsG5SyncLengthCapability

- **Name:** VS_OverrideRxHsG5SyncLengthCapability Register
- **Description:** This attribute configures the override value for RX_HS_G5_SYNC_LENGTH_Capability attribute of M-PHY, which gets transferred inside the PACP_CAP_EXT2 packet during the link startup process.
- **Size:** 32 bits
- **Offset:** 0xd103
- **Exists:** Always

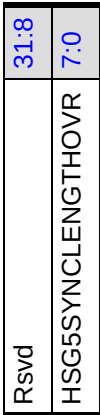


Table 5-303 Fields for Register: VS_OverrideRxHsG5SyncLengthCapability

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	HSG5SYNCLNGTHOVR	R/W	RX_HS_G5_SYNC_LENGTH_Capability override value Value After Reset: 0xff Exists: Always

5.9.123 VS_OverrideRxHsG5PrepareLengthCapability

- **Name:** VS_OverrideRxHsG5PrepareLengthCapability Register
- **Description:** This attribute configures the override value for RX_HS_G5_PREPARE_LENGTH_Capability attribute of M-PHY, which gets transferred inside the PACP_CAP_EXT2 packet during the link startup process.
- **Size:** 32 bits
- **Offset:** 0xd104
- **Exists:** Always



Table 5-304 Fields for Register: VS_OverrideRxHsG5PrepareLengthCapability

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	HSG5PREPARELENGTHOVR	R/W	RX_HS_G5_PREPARE_LENGTH_Capability override value Value After Reset: 0xff Exists: Always

5.9.124 VS_OverrideRxHsG5AdaptRefreshCapability

- **Name:** VS_OverrideRxHsG5AdaptRefreshCapability Register
- **Description:** This attribute configures the override value for RX_HS_G5_ADAPT_REFRESH_Capability attribute of M-PHY, which gets transferred inside the PACP_CAP_EXT2 packet during the link startup process.
- **Size:** 32 bits
- **Offset:** 0xd105
- **Exists:** Always

31:8	
7:0	HSG5ADAPTREFRESHOVR

Table 5-305 Fields for Register: VS_OverrideRxHsG5AdaptRefreshCapability

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	HSG5ADAPTREFRESHOVR	R/W	RX_HS_G5_ADAPT_REFRESH_Capability override value Value After Reset: 0x7f Exists: Always

5.9.125 VS_OverrideRxHsG5AdaptInitialCapability

- **Name:** VS_OverrideRxHsG5AdaptInitialCapability
- **Description:** This attribute configures the override value for RX_HS_G5_ADAPT_INITIAL_Capability attribute of M-PHY, which gets transferred inside the PACP_CAP_EXT2 packet during the link startup process.
- **Size:** 32 bits
- **Offset:** 0xd106
- **Exists:** Always

31:8	Rsvd
7:0	HSG5ADAPTINITIALOVR

Table 5-306 Fields for Register: VS_OverrideRxHsG5AdaptInitialCapability

Bits	Name	Memory Access	Description
31:8			Reserved Field: Yes
7:0	HSG5ADAPTINITIALOVR	R/W	RX_HS_G5_ADAPT_INITIAL_Capability override value Value After Reset: 0x7f Exists: Always

5.9.126 VS_PMCCounter

- **Name:** VS_PMCCounter
- **Description:** Performance metric counter status read register selected by register VS_PMMetricSelector[RMCS[31:30]].
- **Size:** 32 bits
- **Offset:** 0xd110-0xd117
- **Exists:** Always

31:20	19:0
Rsvd	SMC

Table 5-307 Fields for Register: VS_PMCCounter

Bits	Name	Memory Access	Description
31:20			Reserved Field: Yes
19:0	SMC	R	Selected Metric Counter Value. Value After Reset: 0x0 Exists: Always

5.9.127 VS_PMMetricSelector

- **Name:** VS_PMMetricSelector
- **Description:** This control register can be used to select any of the multiple CPort events to be routed to a performance monitor counter or to reset a selected performance counter.
- **Size:** 32 bits
- **Offset:** 0xd140-0xd147
- **Exists:** Always

31:30	29:17	16	15:12	11:8	7:4	3:0
RMCS	Rsvd	RST	MS3	MS2	MS1	MS0

Table 5-308 Fields for Register: VS_PMMetricSelector

Bits	Name	Memory Access	Description
31:30	RMCS	R/W	Read Metric Counter Select Selects any of the four Performance Monitor counters to read count value: ■ 00: PM Counter 0 selected for read ■ 01: PM Counter 1 selected for read ■ 10: PM Counter 2 selected for read ■ 11: PM Counter 3 selected for read Values: ■ 0x0 (VALUE_OF_0): PM Counter 0 selected for read ■ 0x1 (VALUE_OF_1): PM Counter 1 selected for read ■ 0x2 (VALUE_OF_2): PM Counter 2 selected for read ■ 0x3 (VALUE_OF_3): PM Counter 3 selected for read Value After Reset: 0x0 Exists: Always
29:17			Reserved Field: Yes

Table 5-308 Fields for Register: VS_PMMetricSelector (Continued)

Bits	Name	Memory Access	Description
16	RST	R/W	Reset Metric counter. #- 0: Metric counter is not under reset #- 1: Resets only the four metric counters Values: 0x0 (DISABLED): Metric counter is not under reset 0x1 (ENABLED): Resets only the four metric counters Value After Reset: 0x0 Exists: Always
15:12	MS3	R/W	Selects Metric Counter 3 event. 0000: CPORT0 end of segment event 0001: CPORT1 end of segment event 1111: CPORT15 end of segment event Value After Reset: 0x0 Exists: Always
11:8	MS2	R/W	Selects Metric Counter 2 event. 0000: CPORT0 end of segment event 0001: CPORT1 end of segment event 1111: CPORT15 end of segment event Value After Reset: 0x0 Exists: Always
7:4	MS1	R/W	Selects Metric Counter 1 event. 0000: CPORT0 end of segment event 0001: CPORT1 end of segment event 1111: CPORT15 end of segment event Value After Reset: 0x0 Exists: Always

Table 5-308 Fields for Register: VS_PMMetricSelector (Continued)

Bits	Name	Memory Access	Description
3:0	MS0	R/W	Selects Metric Counter 0 event. ■ 0000: CPORT0 end of segment event ■ 0001: CPORT1 end of segment event ■ 1111: CPORT15 end of segment event Value After Reset: 0x0 Exists: Always

6

Application Description

This chapter describes the usage of the DWC_mipi_unipro_20 with some examples for the most critical and complex parts of the UniPro protocol.

**Note**

All descriptions are examples for handling specific tasks in a general application scenario. These examples only provide a guidelines. You should adapt these examples to individual system requirements.

This chapter contains the following sections:

- [“Initial Configuration” on page 536](#)
- [“Link Startup” on page 536](#)
- [“Layer 1.5 Configuration” on page 540](#)
- [“Layer 2 Configuration” on page 546](#)
- [“CPort Connection Management” on page 556](#)
- [“Data Transfer using Arbitration and Segmentation when Using Multiplexed CPort Interface” on page 558](#)
- [“Application Flow Control” on page 560](#)
- [“Hibernation Mode” on page 561](#)
- [“Handling of Indications” on page 562](#)
- [“UniPro DDB Configuration” on page 564](#)
- [“L2 Buffer Size Requirements” on page 564](#)
- [“DWC_mipi_unipro_20 Capabilities” on page 564](#)
- [“CPort Connectivity” on page 565](#)
- [“L2 TX Buffer Size Calculation” on page 566](#)
- [“L2 RX Buffer Size Calculation” on page 568](#)
- [“Guidelines for Dimensioning the Layer 2 Buffers” on page 570](#)

- [“Simulation Time Reduction during LinkStartup Process”](#) on page 575

6.1 Initial Configuration

The DWC_mipi_unipro_20 does not need special configuration before the link is started. All required configuration accesses can be done when the link is up.

Perform the following steps for the link startup and the initial configuration. Additional attributes may be set depending on the needs of individual applications.

1. Initiate module enable and configure M-PHY (non-power mode related attributes).
2. Initiate link startup (see [“Link Startup”](#) on page 536).
3. Set the power mode related attributes and start the power mode change procedure (section [“Layer 1.5 Configuration”](#) on page 540).
4. Set the local and peer layer 2 attributes depending on the application requirements (section [“Layer 2 Configuration”](#) on page 546).
5. Set the local and peer layer 3 device ID.
6. Configure the local and the peer CPorts, and enable the connections (section [“CPort Connection Management”](#) on page 556).

When these steps are successfully performed, the DWC_mipi_unipro_20 and the M-PHY link are fully configured and can be used for data transmissions in both directions.

6.2 Link Startup

The link startup procedure is the DWC_mipi_unipro_20 sequence to bring the M-PHY link and the UniPro stack to a defined initial state, where simple packets can be exchanged between the two attached devices. The minimum sequence to start the link is:

1. Reset the DWC_mipi_unipro_20 by applying either the global reset, LA_ResetReqCold, or LA_ResetReqWarm, or by setting the DME_Reset attribute (if not done already).
Verify that the reset has completed successfully through the DME_Reset attribute or the LA_ResetCnf signal.
2. Request enable using the DME_LayerEnable attribute or the LA_EnableReq signal.
Verify that the enable has completed successfully through the DME_LayerEnable attribute or the LA_EnableCnf signal.
3. Request the link startup using the DME_LinkStartup attribute or the LA_LinkStartupReq signal.
Verify that the link startup has completed successfully through the DME_LinkStartup attribute or the LA_LinkStartupCnf signal.

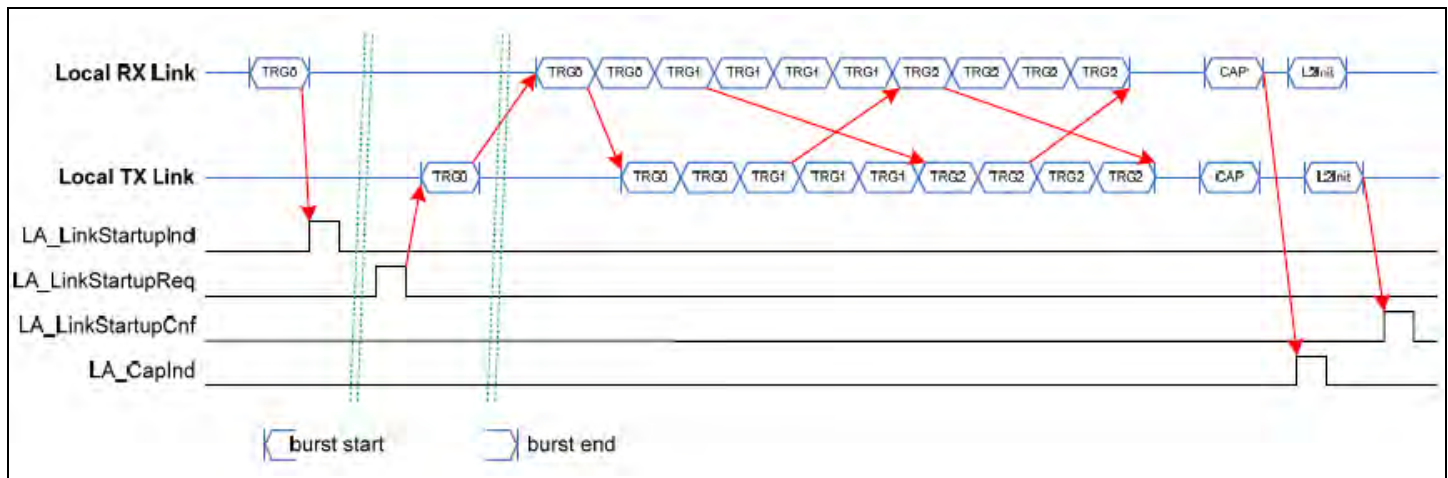


Note

If the link startup fails with a fatal error, the application must reset the device, enable it again (including configuration, if required), and restart the link startup procedure.
If the link startup procedure is not completed within 100 ms, it results in fatal error.

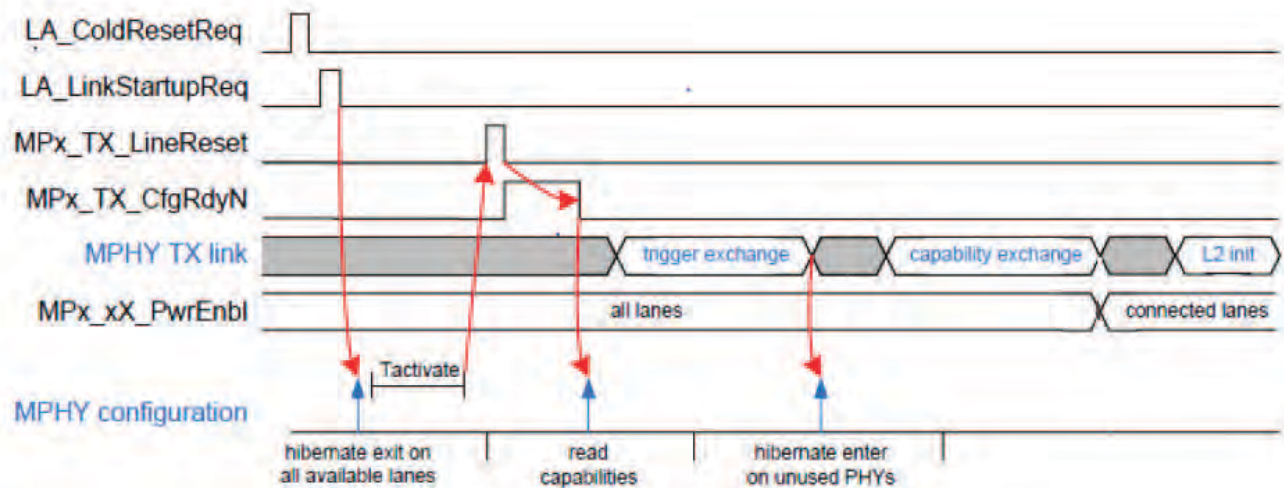
Figure 6-1 shows the individual M-PHY trigger sequences passed over the link during the link startup sequence. Here, the local device reacts on the link startup indication of the peer device.

Figure 6-1 Link Startup Sequence



The M-PHY configuration interface is accessed during link startup several times. Figure 6-2 shows the sequence of accesses after the link startup is requested until the M-PHYs are configured and the UniPro link is activated.

Figure 6-2 Link Startup Sequence (M-PHY)



A new link startup request re-powers all available M-PHY lanes to reinitialize the link configuration and lane mapping.

All timing requirements of the local M-PHY modules (for example, Tactivate) must be programmed before the link startup request to speed up the link startup procedure.

6.2.1 Link Startup in High Speed

DWC_mipi_unipro_20 supports LSS in High Speed when it is configured as follows:

- Set the parameter P_UPRO_LSS_HS to 1
- Drive the port LA_ResetModeReq with value 1

LA_ResetModeReq must be selected at Power On Reset and must stay with same value throughout till next Power On Reset is applied. This port should have a valid value before RG_ResetN and RG_AOResetN is released. The value assigned to this port is driven on MPx_TX_ResetMode and MPx_RX_ResetMode ports. DME_Reset[2] (0xD010) is a READ_ONLY bit. When read, it returns Link Startup type, where value of 1 indicates HS_MODE and value of 0 indicates LS_MODE.

In HS_MODE, the UniPro controller transmits TRG_UPRn(n=0,1and 2) patterns, PACP_CAP_EXT2_ind packet, PACP_CAP_EXT1_ind packet, PACP_CAP_ind packet, and initial AFCs with the following BURST functionality:

- DWC_mipi_unipro_20 operates in FAST_AUTO mode. It is expected that M-PHY is in HS-G1 Series A in HS_MODE.
- Data Scrambling is disabled
- Skip Transmission is disabled

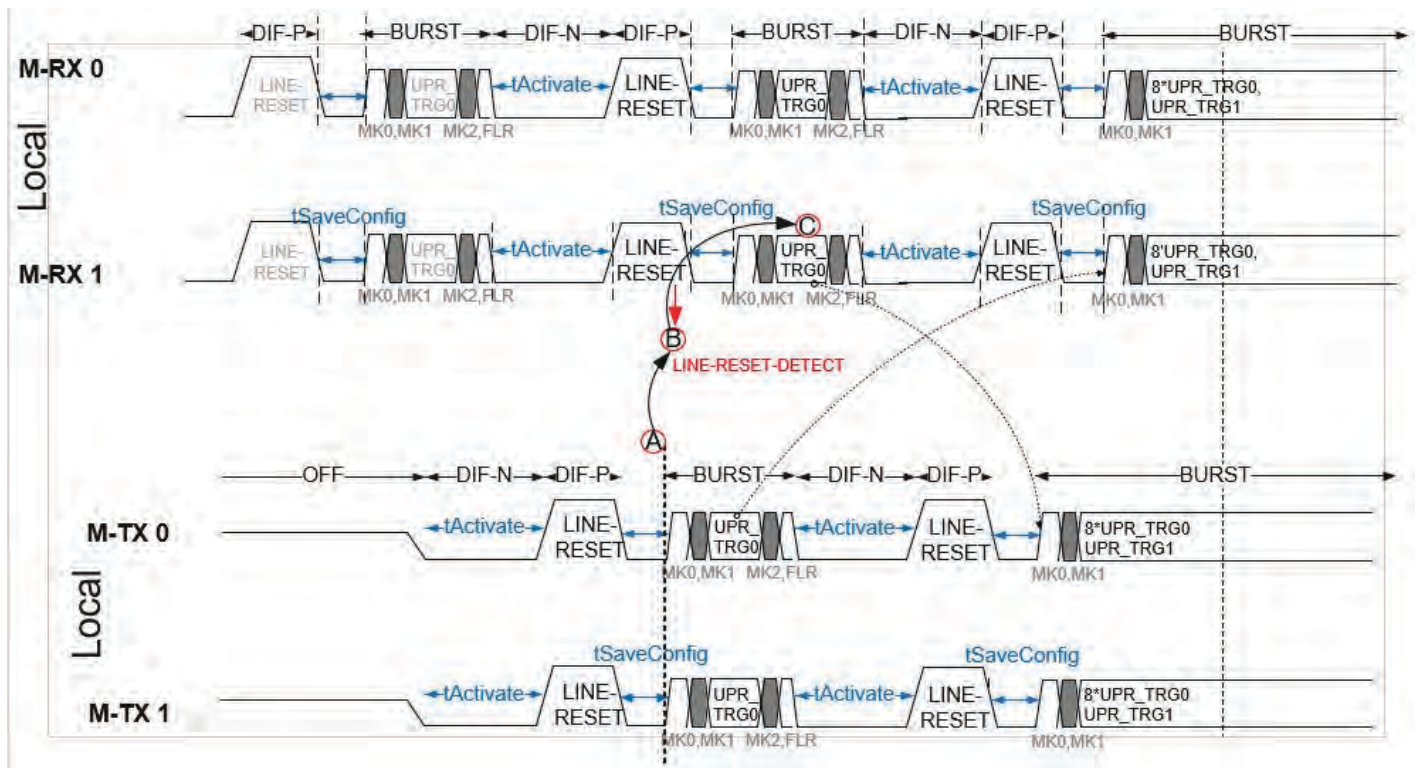
Scrambling is disabled in Link Startup process. Scrambling is enabled after completion of a successful Power Mode Change in which scrambling enable is requested in Flag[5] of PACP_PWR_req frame and peer has advertised scrambling support in Status[1] field of PACP_CAP_EXT1_ind frame during LSS.

Skip symbols are not inserted before capabilities are exchanged and PA Layer Link Startup is complete. However, they are inserted later based on PA_TxSkip(0x155C) value.

DWC_mipi_unipro_20 controller starts transmission of TRG_UPR1 messages only when LINE-RESET is detected on Rx RMMI and LINE-RESET followed by PA_SaveConfig timer expired on the Tx RMMI failing which UniPro controller stays in phase0.

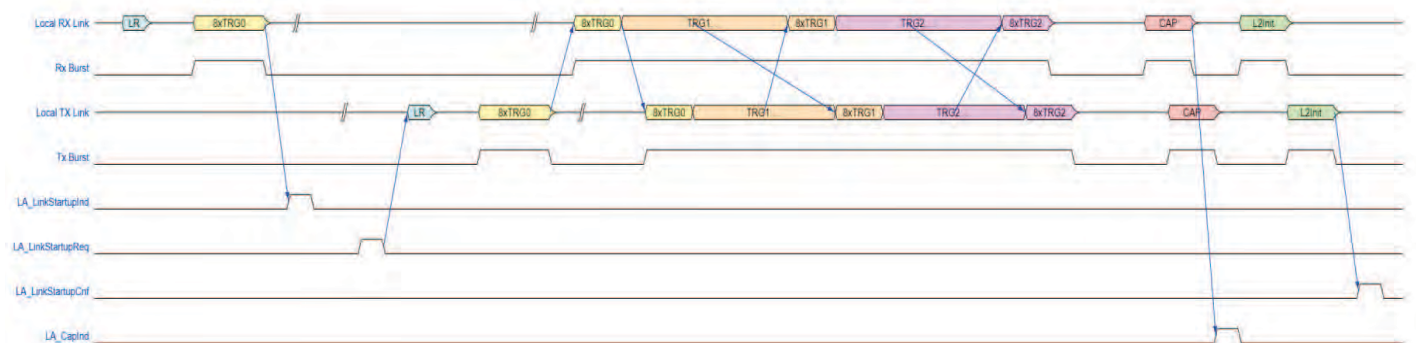
In [Figure 6-3](#),

- At time A, the first generated LINE-RESET on TX RMMI is complete and PA_SaveConfigTime has expired.
- At time B, the rising edge of LINE-RESET on RX is detected.
- At time C, first valid TRG_UPR0 Messages are received.

Figure 6-3 Start Timing for TRG_UPR0 Detection

DWC_mipi_unipro_20 asserts bit [0] of DME_LinkLostInd(0xD022) register and port LA_LinkLostInd when bits [2:0] field of VS_PowerState indicate either LINKUP state(0x2) or Hibernate (0x3) state.

Figure 6-4 shows individual M-PHY trigger sequences passed over the link during the link startup sequence in High Speed. Here, the local device reacts on the link startup indication of the peer device.

Figure 6-4 Link Startup Sequence in High-Speed Mode

DWC_mipi_unipro_20 times out the LSS procedure under the following conditions

- If either of valid TRG_UPR0 messages in phase0 or TRG_UPR1 messages in phase 1 or TRG_UPR2 messages in phase 3 are not received

- If PACP_CAP_EXT2_ind packet is not received, whereas, PACP_CAP_EXT1_ind packet and PACP_CAP_ind packets are received if peer device is V1.8 or higher.
- PACP_CAP_ind packet is received first followed by PACP_CAP_EXT2_ind and PACP_CAP_EXT1_ind packet if peer device is V1.8 or higher.
- PACP_CAP_ind packet is received first followed by PACP_CAP_EXT1_ind and PACP_CAP_EXT2_ind packet if peer device is V1.8 or higher.

DWC_mipi_unipro_20 progresses with LSS procedure if PACP_CAP_EXT1_ind packet is received first followed by PACP_CAP_EXT2_ind and PACP_CAP_ind is received.

After finishing Phase 4 of the Link Startup Sequence, new configuration is applied on the link and PA Layer starts TxBurst only on logical Lane 0. Hence, when ResetMode is HS_MODE, PA_SaveConfigTimer is run after TRG_UPR2 burst closure before starting a TxBurst on Logical Lane 0 for capability exchange.

6.3 Layer 1.5 Configuration

Layer 1.5 attributes are mostly configured by conducting a power mode change procedure. This procedure consists of several write accesses to the layer 1.5 attributes, which are locally and remotely checked for correctness before they are applied to the M-PHY modules.

In addition to a power mode change, several minor configuration accesses allow tuning and optimization of the interface.

6.3.1 Power Mode Change

A power mode change is an automated sequence defined in the UniPro protocol to change the link speed or transfer mode. In addition, several settings are applied to either side of the link. Each side is responsible for checking the requested link properties against its own capabilities and reporting a problem back to the initiator of the power mode change request.

A power mode change is structured in several accesses to the DWC_mipi_unipro_20. As the first step, several local attributes are set. These attributes are stored in specific registers as they are not directly applied to the current configuration. These attributes are:

- PA_TxGear
- PA_TxTermination
- PA_HSSeries
- PA_ActiveTxDataLanes
- PA_RxGear
- PA_RxTermination
- PA_ActiveRxDataLanes

These attribute accesses are locally stored in layer 1.5 but not forwarded to the M-PHY layer until the power mode change procedure is successful.

Reading these attributes returns the values of the currently active link configuration. This may also be the configuration done by the peer device.

After initializing layer 1.5, the layer 2 timer values for the local device and the peer device applied after the successful power mode change need to be programmed by accessing:

- Local DME_FC0ProtectionTimeOutVal
- Local DME_TC0ReplayTimeOutVal
- Local DME_AFC0ReqTimeOutVal
- Local DME_FC1ProtectionTimeOutVal
- Local DME_TC1ReplayTimeOutVal
- Local DME_AFC1ReqTimeOutVal
- PA_PWRModeUserData0 with the value used for peer DL_FC0ProtectionTimeOutVal
- PA_PWRModeUserData1 with the value used for peer DL_TC0ReplayTimeOutVal
- PA_PWRModeUserData2 with the value used for peer DL_AFC0ReqTimeOutVal
- PA_PWRModeUserData3 with the value used for peer DL_FC1ProtectionTimeOutVal
- PA_PWRModeUserData4 with the value used for peer DL_TC1ReplayTimeOutVal
- PA_PWRModeUserData5 with the value used for peer DL_AFC1ReqTimeOutVal

If only a single traffic class is used, it is not necessary to set the following attributes:

- Local DME_FC1ProtectionTimeOutVal
- Local DME_TC1ReplayTimeOutVal
- Local DME_AFC1ReqTimeOutVal
- PA_PWRModeUserData3 with the value used for peer DL_FC1ProtectionTimeOutVal
- PA_PWRModeUserData4 with the value used for peer DL_TC1ReplayTimeOutVal
- PA_PWRModeUserData5 with the value used for peer DL_AFC1ReqTimeOutVal

Layer 2 timer value calculation is described in detail in [“Layer 2 Configuration”](#) on page 546.

Finally, the power mode change request is started by accessing the power mode attribute PA_PWRMode inside layer 1.5. This attribute contains the requested transmission mode (fast or slow).



Hint

Clear the power mode indication attribute (DME_PowerModeInd) before writing this attribute to get the correct result of the request.

The return code of this last attribute access to the power mode attribute indicates if the request is locally accepted, that is, all the attributes are within the local capabilities. If they are rejected (return code INVALID_MIB_ATTRIBUTE_VALUE), the application must identify the problem by comparing the local capabilities to its own settings.

The access to the PA_PWRMode attribute triggers the power mode change procedure, which enables the DWC_mipi_unipro_20 to exchange the PACP frames with the peer device. When the peer device accepts or rejects the power mode change request, an indication (LA_PowerModeInd) is generated and the return

value is set to the attribute DME_PowerModeInd. If the power mode indication is configured as an interrupt source, the interrupt is generated as well.

All power mode settings including local layer 2 timer values are applied only if the peer device accepts the power mode change request.

The indication informs the application about the status of the power mode change request.

While DWC_mipi_unipro_20 allows direct access to M-PHY (L1 layer) attributes, it restricts the access to some attributes because they affect the link configuration. Access to these attribute is allowed through the power mode change protocol only. The attributes to which the access is restricted are:

- TX_MODE (0x21)
- TX_HSRATE_Series (0x22)
- TX_HSGEAR (0x23)
- TX_PWMGEAR (0x24)
- TX_HIBERN8_Control (0x2b)
- TX_HS_Unterminated_LINE_Drive_Enable (0x30)
- TX_LS_Terminated_LINE_Drive_Enable (0x31)
- MC_HS_Unterminated_Enable (0x62)
- MC_LS_Terminated_Enable (0x63)
- MC_HS_Unterminated_LINE_Drive_Enable (0x64)
- MC_LS_Terminated_LINE_Drive_Enable (0x65)
- RX_MODE (0xA1)
- RX_HSRATE_Serie (0xA2)
- RX_HSGEAR (0xA3)
- RX_PWMGEAR (0xA4)
- RX_LS_Terminated_Enable (0xA5)
- RX_HS_Unterminated_Enable (0xA6)
- RX_Enter_HIBERN8 (0xA7)

Figure 6-5 shows the power mode change flow.

Figure 6-5 Power Mode Change Flow

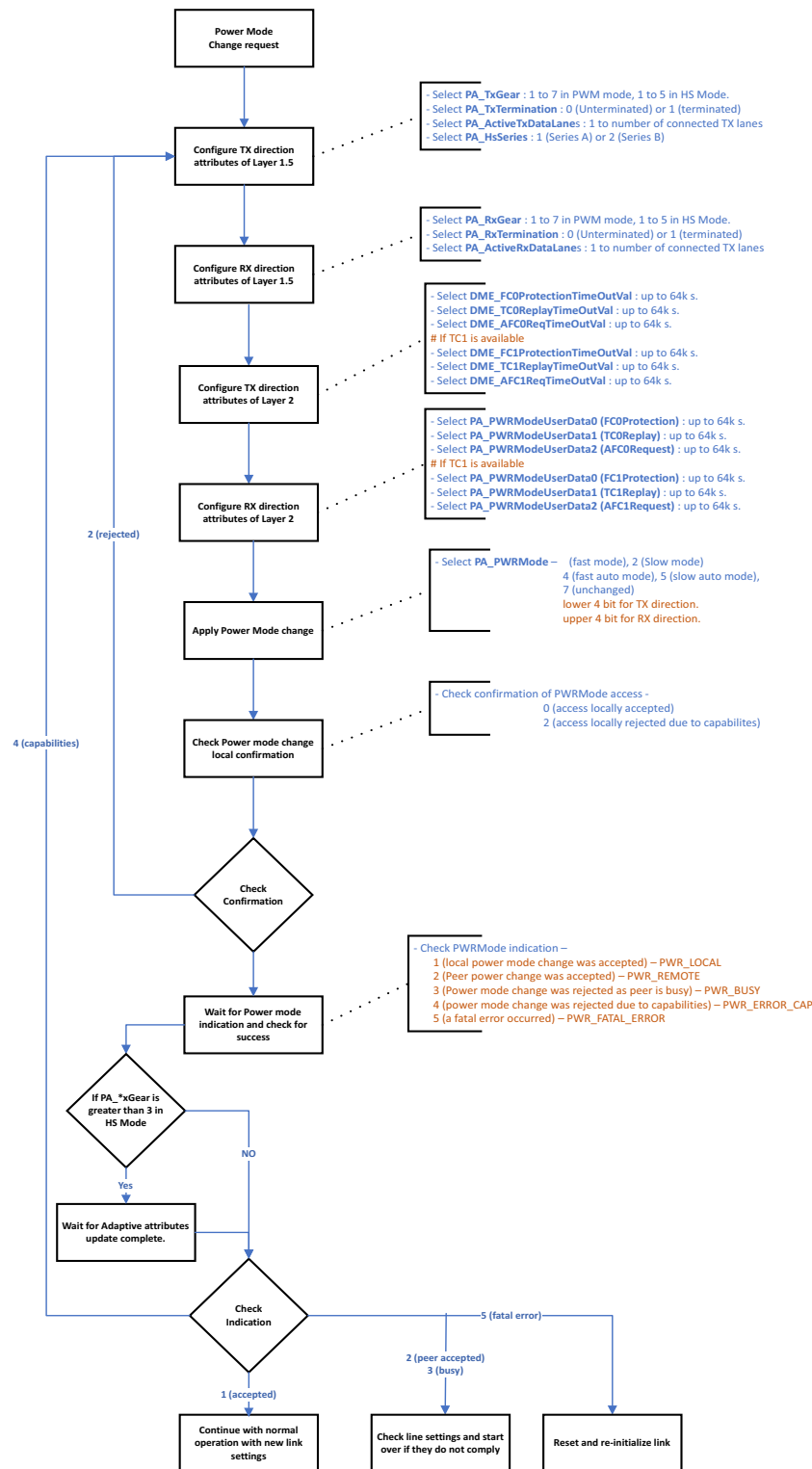


Table 6-1 Power Mode Return Codes

Power Mode Return Code	Bit	Description	Handling
PWR_OK	0	Request accepted -> not used	None
PWR_LOCAL	1	The local request is successfully applied	None
PWR_REMOTE	2	A remote request is successfully applied	Application can check new settings
PWR_BUSY	3	A concurrent request is detected and the request is not applied	Wait for the peer device to change the power mode or resend the request
PWR_ERROR_CAP	4	Capability conflict is detected and the request is not applied	Check the remote capabilities and correct the request before resending
PWR_FATAL_ERROR	5	A communication problem is detected and the link cannot be operated	Send endpoint reset request and reset local device

The VS_PowerState attribute indicates the current status of the link.

6.3.2 Link Optimization

By default, several UniPro attributes are set to a conservative value to secure initial transfers. However, a more realistic value can help in improving the data throughput and therefore optimize the link usage.

Set the following attributes after starting up the link:

- PA_TActivate: Automatically set during link startup
- PA_TxTrailingClocks: Automatically set during link startup
- PA_SleepNoConfigTime: Automatically set during link startup
- PA_StallNoConfigTime: Automatically set during link startup
- PA_SaveConfigTime: Automatically set during link startup
- VS_TxBurstClosureDelay: Can be optimized depending on the application requirements
- PA_PACPRReqTimeout: Must be larger than the PA_TActivate time

The value of the PA_TxTrailingClocks attribute is automatically set during Ext1 capability exchange. This value should not be set to a smaller value than received MinRxTrailingClocks of the peer device plus the process delay of the local M-PHY TX. This is because these additional filler symbols which are sent at the end of a burst allow the peer device to completely process its data.

Setting VS_TxBurstClosureDelay to a higher value gives the application more time to provide more data before an active burst is closed in AutoMode. However, a higher value also keeps the burst open for additional cycles, when no more data is available.

PA_SaveConfigTime Calculation

PA_SaveConfigTime is computed by addressing the Extended Save Config time attributes of MPHY. The MPHY timing information (RX_Extended_Min_SAVE_Config_Time_Capability) is transmitted in TextSaveConf field of PACP_CAP_EXT2_ind packet.

If VS_OverrideAttrEnable[25] is programmed to 1, the value in VS_OverrideRxExtdMinSaveConfigTimeCapability(0xd0e6) attribute is transmitted in TextSaveConf field of PACP_CAP_EXT2_ind packet rather than value read from RX_Extended_Min_SAVE_Config_Time_Capability of MPHY.

After capability management step, the downgraded value is stored in PA_SaveConfigTime(0x15a4) attribute and step size is microseconds(us). The local application can overwrite the computed value of PA_SaveConfigTime by programming the PA_SaveConfigTime(0x15a4) attribute and the downgraded value is ignored.

SYNC and ADAPT length PA Layer Attribute

The range or width of Sync and Adapt length PA Layer attribute is increased beyond the specification-defined value:

- SYNLEN field of all PA_TxHsGxSyncLength registers are updated to allow write access up to 6-bit value irrespective of value of SYN RANGE field (Fine/coarse selection). This applies to all high-speed gear from G1 to G5.
- ADAPTREF field of PA_PeerRxHsG5AdaptRefresh, PA_PeerRxHsG5AdaptInitial, PA_PeerRxHsAdaptRefresh, and PA_PeerRxHsAdaptInitial registers are updated to allow write access up to 7-bit value irrespective of Fine/coarse selection.

6.3.3 PA_INIT and Error Recovery During PMC When ResetMode is HS_Mode

When ResetMode is HS_MODE, DWC_mipi_unipro_20 issues LINE-RESET only at the beginning of Phase 0 of Link startup sequence. Unlike LS_MODE, LINE-RESET is not issued during PA-INIT and Error recovery during Power Mode Change.

For PA_INIT or Error recovery process, the local PA layer retries the link configuration by transmitting the PACP_PWR_req frame. The final retry is performed after transmitting a Deskew pattern, while LINE-RESET request flag is set to 0 in all the retries.

If PACP_REQUEST_TIMER expires before reception of correct PACP_PWR_cnf frame in all the trails, that is, if all retries are unsuccessful, the PA Layer aborts the recovery or PA_INIT process and pass PA_LM_PWR_MODE_CHANGED.ind(PWR_FATAL_ERROR) to the DME in case of Error recovery while PA_INIT.cnf_L primitive is sent to DL layer and application in case of PA-INIT process.

If PACP_PWR_cnf frame is received successfully and Status field is PWR_BUSY or PWR_ERROR_CAP, then DME_PWR_MODE_CHANGE.ind(LA_PowerModeInd port and DME_PowerModeInd attribute) is generated with the result code as PWR_BUSY or PWR_ERROR_CAP respectively. In such cases, DWC_mipi_unipro_20 controller continues to operate in the current settings (Lane, Gear and Mode) before the PACP_PWR_req packet is initiated and closes its MPx_TX_Burst.

When ResetMode is HS_MODE, DWC_mipi_unipro_20 controller issues DME_LINKLOST.ind upon reception of LINE-RESET if Link Startup process is complete. Hence, when LINE-RESET is received during the ongoing Power Mode change process, the value of PA_LM_PWR_MODE_CHANGED.ind/PA_LM_HIBERNATE_ENTER.ind is unknown.

6.4 Layer 2 Configuration

Layer 2 configuration is divided into two parts: thresholds and timers. Calculation and selection of these attribute values can have major influence on the performance of a UniPro system as they program the generation of AFC frames. Careful selection of these values is therefore very important.



Note

AFC frames for acknowledge generation are generated independently from AFC frames for credit exchange.

6.4.1 Layer 2 Threshold Value Calculation

Layer 2 defines the following configurable threshold values per traffic class (x is 0 or 1):

- `DL_TCxTXFCThreshold` (`DL_TC0TXFCThreshold` and `DL_TC1TXFCThreshold`)
- `DL_AFCxCreditThreshold` (`DL_AFC0CreditThreshold` and `DL_AFC1CreditThreshold`)
- `DL_TCxOutAckThreshold` (`DL_TC0OutAckThreshold` and `DL_TC1OutAckThreshold`)

These threshold values are used to control the AFC frame and the credit transmission between the two devices on the link.

The following values are recommended for these thresholds (x is 0 or 1):

- $DL_TCxTxFCThreshold = DL_PeerTCxRxInitCreditVal[credits] - 1$
- $DL_AFCxCreditThreshold = MaxFrameSizeTCx[credits] - 1$
- $DL_TCxOutAckThreshold = 0$

The following paragraphs describe the usage of the individual threshold values in detail and give some advice if more optimization is required.

`DL_TCxTXFCThreshold`

A change in `DL_TCxTXFCThreshold` influences the required level of available layer 2 credits (1 credit = 32 bytes) in a transmitter before the protection timer is started to request an update of credit information. For example, if this threshold value for TC0 is locally set to 10 credits, the local TX starts the protection timer when less than 10 credits are available for sending the data to the peer device. If the protection timer expires, the local TX sends an AFC0 frame to the peer device with the CReq set to 1 to request an update of the peer RX credit information. When the peer device answers with an AFC0 frame at any time, the protection timer is reset and restarted if there are still less than 10 credits available or if the timer is stopped.

This mechanism should prevent the local TX running out of credits and stalling its data transmissions, when for any reason the credit update is lost or not sent on time.

As a general guideline, this threshold value can be set to its maximum value (1 less than the maximum number of credits of the receive buffer for the corresponding TC, which is stored in the attributes `DL_PeerTC0RxInitCreditVal` or `DL_PeerTC1RxInitCreditVal`). With this setting, the local TX tries to get an update of the local credit information after the expiry of the protection timer as long as it does not have all possible credits available. The overhead for sending the AFC frames depends on the timeout value of the protection timer, but should not have a negative impact on the link performance.

DL_AFCxCreditThreshold

This threshold value is the counterpart of the DL_TCxTXFCThreshold attribute in the receiver. Layer 2 provides a mechanism to ensure that during normal operation, the peer layer 2 transmitter gets sufficient credit updates to continuously send data.

When RX buffer space is freed, that is, the data is passed to the upper layer, layer 2 keeps track of the amount of data read from the RX buffer since the last time flow control credits are communicated to the peer device through an AFC frame. If this amount of data (measured in L2 credits) exceeds the value of DL_AFCxCreditThreshold, L2 initiates the generation of an AFC frame in order to communicate new credits to the peer device. For example, if DL_AFCxCreditThreshold is set to 5 credits, the receiver sends an AFC frame after at least 6x32 bytes (six credits) have been read from the RX buffer of the corresponding traffic class since the last credit update.

Changing this attribute to a lower value increases the number of AFC frames sent because of flow control. A higher value for this parameter decreases the number of AFC frames sent because of flow control.

However, to prevent a credit stall, DL_AFCxCreditThreshold must not exceed an upper limit, which depends on the RX buffer size. If the parameter is set to a value higher than this upper limit, the peer transmitter may not always receive enough credits to continuously send the data, which leads to a timeout of the corresponding protection timer at the peer layer 2, and possibly long periods of inactivity.

The upper limit for DL_AFCxCreditThreshold to prevent a credit stall situation is defined as follows:

$$DL_AFCxCreditThreshold \leq LocalRxBufferSize_{TCx}[credits] - MaxFrameSize_{TCx}[credits]$$

For maximum performance, layer 2 must be able to send the data continuously without gaps or stall times. For this purpose, new frames that are to be transmitted must be already stored entirely in the TX buffer when layer 2 receives its credit updates. This can be ensured by setting the value for DL_AFCxCreditThreshold at the peer layer 2 receiver such that the local device gets credit updates (along with the corresponding data frame acknowledgments in the AFC frames). This allows the local layer 2 to delete all acknowledged frames from the TX buffer and thus accept new data frames from its upper layer. Setting DL_AFCxCreditThreshold at the peer device so that the layer 2 TX buffer at the local device can store two additional unacknowledged frames should give reasonable results.

The following equation reflects this:

$$DL_AFCxCreditThreshold \leq PeerTxBufferSize_{TCx}[credits] - 2 \times MaxFrameSize_{TCx}[credits] - 1$$

As a guideline, or as a starting point for further optimizations, setting the DL_AFCxCreditThreshold attribute to the credit value corresponding to the maximum size of a frame for this traffic class (MaxFrameSizeTCx), should give reasonable results. In this case, the receiver sends its credit update when at least one additional frame of size MaxFrameSizeTCx can be stored in its buffer:

$$DL_AFCxCreditThreshold = MaxFrameSize_{TCx}[credits] - 1$$



This setting is conservative because an AFC is sent after each maximum sized frame. Optimizing the value for DL_AFCxCreditThreshold depends on many system parameters as such as the TX and RX buffer sizes and the frame lengths. However, when optimizing the value of DL_AFCxCreditThreshold, you must ensure that the value does not exceed any of the upper limits defined earlier in this section. In any case, the impact of this parameter on the system performance is quite limited.

DL_TCxOutAckThreshold

This attribute defines the number of frames (+1) to be received before an AFC frame is sent to the peer device. For example, if this attribute is set to 2, the device sends an AFC frame back when three frames are received for the corresponding traffic class.

A lower value for this attribute increases the number of AFC frames sent from the receiver of the data frames to the transmitter. If the value for this parameter is too high, the transmitter may have to stop data transfer because it has 16 unacknowledged frames and is still waiting for the reception of the AFC frame. This scenario may occur if the transmitter sends many short frames in quick succession.

If the link usage is mostly in one direction (as in most camera or display systems), this value can be set to 0 because additional AFC frames have little impact on the reverse bandwidth.

If both link directions carry high bandwidth data traffic, the performance benefit by increasing the value for DL_TCxOutAckThreshold may be slightly higher.

Finding an optimized value for DL_TCxOutAckThreshold depends on many system parameters, such as the average frame lengths, power modes, and so on. When optimizing the L2 parameters, it is recommended to start with a value of 0 for DL_TCxOutAckThreshold, which is the most conservative choice, but provides a good performance. Subsequently, higher values for this parameter may be evaluated by appropriate system simulations.

6.4.2 Layer 2 Timeout Value Calculation

Layer 2 uses three timers per traffic class to keep the link operational in case of wrong configuration or unstable communication with the peer device. These timers can be separately configured using the following attributes per traffic class:

- “DL_FCxProtectionTimeOutVal” on page 550
- “DL_TCxReplayTimeOutVal and DL_AFCxReqTimeOutVal” on page 551



Attention

The [DL_FCxProtectionTimeOutVal](#) and [DL_TCxReplayTimeOutVal](#) and [DL_AFCxReqTimeOutVal](#) sections provide formulas for local device only. However, during a power mode change, the timer values for both local and peer device need to be programmed. To determine the timeout values for the peer device, the same formulas can be used with exchanged local and peer values.

In the following subsections, formulas are derived for estimating processing delays relevant for the layer 2 timers. Later, based on these calculated delay values, guidelines are provided for optimizing the timeout values in the system. Options with different implementation efforts are discussed, that may be used by a UniPro device driver to calculate layer 2 timeout values. Finally, these guidelines are illustrated by an example.



Note

Proper timer settings are heavily system-dependent and need to be validated carefully by system simulations.

Determination of TX Processing Delays

For correct operation of the system, you must set up the layer 2 timeout values so that no blocking occurs. For example, the protection timer must be set up so that the peer device has got sufficient time to return an AFC frame before the timer expires again. This must be ensured even under worst case conditions (for example, when the inbound link is operating in Auto Mode and the burst has just been closed). The AFC request timer must be programmed so that an AFC frame can be transferred to the peer device before the peer replay timer expires.

For timeout value calculation, TX processing delays must be taken into account. The following formulas are derived to calculate these delay values, which are finally used for determining the layer 2 timeout values.

TX processing delays depend on the following parameters:

- TX_SI_period: Symbol Interval (SI) period in μs of the M-TX in the power mode to be applied in the next power mode change
- TX_Hibern8Time_Capability: Value of TX_Hibern8Time_Capability of the M-TX
- RX_Hibern8Time_Capability: Value of RX_Hibern8Time_Capability of the M-RX
- PA_MinRxTrailingClocks: Minimum number of trailing RX clocks for the L1.5 RX
- RX_PWM_BURST_Closure_Length_Capability: Minimum burst closure time (in SI) at the M-RX
- TX_HS_PREPARE_LENGTH: High-Speed PREPARE length multiplier for the M-TX
- TX_HS_SYNC_LENGTH: High-speed synchronization pattern length of the M-TX in SI
- PA_StallNoConfigTime
- PA_SleepNoConfigTime



Hint

You can use the `_local` and `_peer` suffixes to distinguish between the parameter values at the local and the peer device.

If at least one of the links in the inbound or the outbound direction incorporates more than one connected lane, the hibernate time has to be taken into account. The hibernate time (in μs) used for timeout value calculation is determined from the maximum of all capabilities on the inbound and outbound links:

$$\text{HibernateTime}[\mu\text{s}] = 100\mu\text{s} \times \text{Max} \{ \text{TX_Hibern8Time_Capability_local}, \\ \text{TX_Hibern8Time_Capability_peer}, \\ \text{RX_Hibern8Time_Capability_local}, \\ \text{RX_Hibern8Time_Capability_peer} \}$$

The hibernate time must be set to 0 if both inbound and outbound links incorporate only one connected lane.

The delays (in SI) required after a burst has been closed on the outbound lanes of the local device in high-speed or low-speed modes are calculated as follows:

$$\text{TX_BurstEndDelay_HS_local}[\text{SI}] = \text{PA_MinRxTrailingClocks_peer}$$

$$\text{TX_BurstEndDelay_LS_local[SI]} = \text{PA_MinRxTrailingClocks_peer} + \text{RX_PWM_BURST_Closure_Length_Capability_peer}$$

The delays (in SI) required before a burst is started on the outbound lanes of the local device in high-speed or low-speed modes are calculated as follows:

$$\text{TX_BurstStartDelay_HS[SI]} = T_{\text{HS_PREPARE}}[\text{SI}] + T_{\text{HS_SYNC}}[\text{SI}]$$

$$\text{TX_BurstStartDelay_LS[SI]} = T_{\text{LS_PREPARE}}[\text{SI}]$$

The PREPARE and SYNC times (in SI) need to be calculated from the corresponding TX_HS_PREPARE_LENGTH and TX_HS_SYNC_LENGTH M-PHY parameters. For more information on calculating these parameters, see *MIPI M-PHY Specification*.

From these parameters, the M-PHY based worst case processing delays (in microseconds) for the outbound lanes in high-speed or low-speed modes are as follows:

$$\begin{aligned} \text{TX_MaxProcessingDelay_HS}[\mu\text{s}] &= (\text{PA_StallNoConfigTime} \\ &+ \text{TX_BurstStartDelay_HS} \\ &+ \text{TX_BurstEndDelay_HS}) \times \text{TX_SI_period} \end{aligned}$$

$$\begin{aligned} \text{TX_MaxProcessingDelay_LS}[\mu\text{s}] &= (\text{PA_SleepNoConfigTime} \\ &+ \text{TX_BurstStartDelay_LS} \\ &+ \text{TX_BurstEndDelay_LS}) \times \text{TX_SI_period} \end{aligned}$$

The actual values for the M-PHY based processing delays depend on the current power mode of the corresponding link:

$$\text{TX_MaxProcessingDelay} = \begin{cases} \text{TX_MaxProcessingDelay_HS, Link in HS mode} \\ \text{TX_MaxProcessingDelay, Link in LS mode} \end{cases}$$

In addition to the M-PHY based processing delay calculated in the previous example, a time of 1000 SI is assumed as a conservative estimate for internal Data Link layer processing delays at the local transmitter and the peer receiver:

$$\text{TX_MaxFrameTransferTime}[\mu\text{s}] = 1000 \times \text{TX_SI_period}$$


Attention

The “[DL_FCxProtectionTimeOutVal](#)” on page 550 and “[DL_TCxReplayTimeOutVal](#) and [DL_AFCxReqTimeOutVal](#)” on page 551 sections provide formulas for local device only. However, during a power mode change, the timer values for both local and peer device need to be programmed. To determine the timeout values for the peer device, the same formulas can be used with exchanged local and peer values.

DL_FCxProtectionTimeOutVal

The protection timer is used in conjunction with DL_TCxTXFCThreshold as described in “[Layer 2 Threshold Value Calculation](#)” on page 546. Its purpose is to protect the link against the loss of an AFC frame and the resulting stall in communication, if the required credits are not delivered.

This timer runs whenever lesser credits than specified by DL_TCxTXFCThreshold are available at the local device. After expiry, the local UniPro sends a promoted AFC frame to the peer device requesting a credit

information update (CReq set to one). When this AFC request is sent, the timer is restarted. After receiving the AFC frame, the peer device issues a promoted AFC with up-to-date credit information. If the AFC frame sent by the peer device is not received by the local device before the local protection timer expires again, layer 2 at the local device requests that the link be re-initialized by issuing a PA_INIT.req to the PA layer.

Configuration of the protection timer has minimum impact on system performance. If the timeout value is set too high, longer periods of inactivity may occur if AFC frames are lost because of errors on the link.

In any case, the timeout values for the TC0 and TC1 protection timers must be set so that the peer device has sufficient time to return an AFC frame after receiving the credit request, even under bad conditions (for example, when a burst has just been closed on the peer TX link).

The following equation takes account of possible delays on the peer TX link, as well as the maximum timeout tolerance of 10% defined by UniPro. It also provides a guideline for optimization of the timeout value of the protection timer:

$$\begin{aligned} &DL_FCxProtectionTimeOutVal_local > 1.1 \times (\text{HibernateTime} \\ &+ TX_MaxProcessingDelay_local \\ &+ TX_MaxFrameTransferTime_local \\ &+ TX_MaxProcessingDelay_peer \\ &+ TX_MaxFrameTransferTime_peer) \end{aligned}$$

If the timeout value calculated using the previous formula exceeds the maximum value of 65535 μ s allowed for the layer 2 timers, the value must be set to the maximum.

DL_TCxReplayTimeOutVal and DL_AFCxReqTimeOutVal

The replay timer running at the local device is closely related to the AFC request timer running on the peer device of the link. In general, the replay timer is running when transmitted frames are not acknowledged yet and is reset every time a new frame is sent. It is also reset after the reception of an AFC frame that acknowledges at least one unacknowledged data frame. The replay timer protects the protocol against lost AFC or NAC frames. The timer expires when no AFC is received for a period of DL_TCxReplayTimeOutVal microseconds, and no new frame is sent for the corresponding traffic class. After the timer expires, layer 2 reinitializes the link by issuing PA_INIT.req to the PA layer, and finally retransmits all data frames that have not been acknowledged.

The AFC request timer at the receiver is used to trigger the AFC generation before the replay timer at the transmitter expires, which can cause reinitialization and retransmission.

To ensure proper operation, the AFC request timer on the receiving device must be programmed to a significantly smaller value than the replay timer on the transmitting device. The configuring application must ensure that the replay timer value includes the AFC request value, internal processing delays while generating new AFC frames, and transmission over the link. On the other hand, the timeout value for the AFC request timer must not be so low that it generates more AFC frames than is actually required.

The AFC request timer is reset and restarted each time a data frame for the corresponding traffic class is received correctly. That is, the timeout value must be set up in such a way that at least one maximum-sized frame (DL_MTU) can be received in the current power mode before the timer expires. An additional Data Link layer processing delay, as well as the maximum allowed timeout tolerance of 10% must be taken into account:

$$DL_AFCxReqTimeOutVal_local > 1.1 \times TX_MaxFrameTransferTime_peer$$

An optimized timeout value for the AFC request timer depends on the maximum lengths of the data frames transferred over the link as well as the estimated average gaps between two consecutive frames. You must perform system simulations to fine-tune the timeout values for the AFC request timers. This may result in different optimized values for TC0 and TC1.

For proper operation, the timeout value of the replay timer at the local device must include the timeout value of the AFC request timer at the peer device, plus additional processing delays and frame transfer times. Additionally, this timeout value must take into account the worst case deviation according to timeout tolerances, when the local replay timer is running 10% faster, and the peer AFC request timer is running 10% slower than actually set:

$$\begin{aligned} &DL_TCxReplayTimeOutVal_local > 1.1 \times (1.1 \times DL_AFCxReqTimeOutVal_peer \\ &+ HibernateTime \\ &+ TX_MaxProcessingDelay_local \\ &+ TX_MaxFrameTransferTime_local \\ &+ TX_MaxProcessingDelay_peer \\ &+ TX_MaxFrameTransferTime_peer \end{aligned}$$

If a timeout value calculated using one of the previous formulas exceeds the maximum value of 65535 μ s allowed for the layer 2 timers, the value must be set to the maximum.

Layer 2 Timeout Value Optimization

A UniPro device driver can choose one of the following methods for setting up the timeout values for the local and peer layer 2 timers:

- Use the UniPro layer 2 reset values. The reset values form a pessimistic setup, which should work in most situations. However, if the replay timers are set to their maximum allowed values, longer periods of inactivity may occur in case AFC frames get lost.
- Calculate and set timeout values only once after link startup. In this case, the device driver calculates the timeout values for all layer 2 timers for PWM-G1 as well as for HS-G1, using the formulas given in the previous section. The actual timeout value for a specific timer is determined as the maximum of the timeout values for PWM-G1 and for HS-G1. This value is used as the timeout value in all power modes that can be applied while the system is running.
- Calculate and set new timeout values for each power mode change. In this case, the device driver calculates the timeout values for all layer 2 timers optimized depending on the power mode to be applied. This method comes with the highest effort, but provides optimized timeout values for all power modes.



Note

Always set the timeout values through a power mode change procedure using the DME and PA layer attributes.

Example of Layer 2 Timeout Calculation

In this example, layer 2 timeout values are calculated for the following configuration with two connected UniPro devices (device A and device B).

Device A Configuration	Device B Configuration
GEAR_A: HS_G1, RATE A-series (1.248 Gbps)	GEAR_B: PWM-G1, 0.004 Gbps
TX_SI_period_A: $10 \cdot (1/1.248) \text{ ns} = 8.01 \text{ ns} = 0.00801 \mu\text{s}$	TX_SI_period_B: $10 \cdot (1/0.004) \text{ ns} = 2500 \text{ ns} = 2.5 \mu\text{s}$
2 connected lanes	2 connected lanes
TX_Hibern8Time_Capability_A: 1	TX_Hibern8Time_Capability_B: 5
RX_Hibern8Time_Capability_A: 1	RX_Hibern8Time_Capability_B: 5
PA_MinRxTrailingClocks_A: 64	PA_MinRxTrailingClocks_B: 128
RX_PWM_BURST_Closure_Length_Capability_A: 31	RX_PWM_BURST_Closure_Length_Capability_B: 16
TX_HS_PREPARE_LENGTH_A: 15	TX_LS_PREPARE_LENGTH_B: 15
TX_HS_SYNC_LENGTH_A: FINE, 15	
PA_StallNoConfigTime_A: 226 SI	PA_StallNoConfigTime_B: 168 SI
PA_SleepNoConfigTime_A: 15 SI	PA_SleepNoConfigTime_B: 9 SI

In the following formulas, the parameters for the two devices are distinguished by the suffixes _A and _B.

The HibernateTime must be taken into account because two lanes are connected:

$$\text{HibernateTime} = \text{Max}\{1,1,5,5\} \times 100 \mu\text{s} = 500 \mu\text{s}$$



Note

In a single lane configuration (in both directions), hibernate time is not required because no unused data lanes are in hibernate mode.

The delays (in SI) required after a burst is closed on the outbound lanes of a device are calculated as follows with device A operating in high-speed mode, and device B operating in low-speed mode:

$$\begin{aligned} \text{TX_BurstEndDelay_HS_A} &= \text{PA_MinRxTrailingClocks_B} \\ &= 128 \text{ SI} \end{aligned}$$

$$\begin{aligned} \text{TX_BurstEndDelay_LS_B} &= \text{PA_MinRxTrailingClocks_A} \\ &+ \text{RX_PWM_BURST_Closure_Length_Capability_A} \\ &= 64 \text{ SI} + 31 \text{ SI} = 95 \text{ SI} \end{aligned}$$

PREPARE and SYNC times (in SI) for device A are calculated as follows from the corresponding TX_HS_PREPARE_LENGTH and TX_HS_SYNC_LENGTH M-PHY parameters. For more information on calculating these parameters, see *MIPI M-PHY Specification*.

$$\begin{aligned} T_{\text{HS_PREPARE_A}} &= \text{TX_HS_PREPARE_LENGTH_A} \times 2^{(\text{GEAR_A}-1)} \\ &= 15 \times 2^{(1-1)} \text{ SI} = 15 \text{ SI} \end{aligned}$$

$$T_{\text{HS_SYNC_A}} = \text{TX_HS_SYNC_LENGTH_A} = 15 \text{ SI}$$

The PREPARE time (in SI) for device B is calculated as follows from the corresponding TX_LS_PREPARE_LENGTH M-PHY parameter:

$$\begin{aligned} T_{LS_PREPARE_B} &= \max(2^{TX_LS_PREPARE_LENGTH_B + GEAR_B - 7}, 1) \\ &= \max\{2^{15+1-7}, 1\} \\ &= 512 \text{ SI} \end{aligned}$$

The delays (in SI) before a burst is started on the outbound lanes of Devices A and B are determined as follows:

$$TX_BurstStartDelay_HS_A = T_{HS_PREPARE_A} + T_{HS_SYNC_A} = 30 \text{ SI}$$

$$TX_BurstStartDelay_LS_B = T_{LS_PREPARE_B} = 512 \text{ SI}$$

These delays result in the following values for worst case processing delays in the respective outbound lanes of the two devices:

$$\begin{aligned} TX_MaxProcessingDelay_HS_A &= (PA_StallNoConfigTime_A \\ &+ TX_BurstStartDelay_HS_A \\ &+ TX_BurstEndDelay_HS_A) \times TX_SI_period_A \\ &= (226 \text{ SI} + 30 \text{ SI} + 128 \text{ SI}) \times 0.00801 \mu\text{s} \\ &= 3.08 \mu\text{s} \end{aligned}$$

$$\begin{aligned} TX_MaxProcessingDelay_LS_B &= (PA_SleepNoConfigTime_B \\ &+ TX_BurstStartDelay_LS_B \\ &+ TX_BurstEndDelay_LS_B) \times TX_SI_period_B \\ &= (9 \text{ SI} + 512 \text{ SI} + 95 \text{ SI}) \times 2.5 \mu\text{s} \\ &= 1540 \mu\text{s} \end{aligned}$$

The maximum times for transferring frames over the link are estimated as:

$$\begin{aligned} TX_MaxFrameTransferTime_A &= 1000 \times TX_SI_period_A \\ &= 1000 \times 0.00801 \mu\text{s} = 8.01 \mu\text{s} \end{aligned}$$

$$\begin{aligned} TX_MaxFrameTransferTime_B &= 1000 \times TX_SI_period_B \\ &= 1000 \times 2.5 \mu\text{s} = 2500 \mu\text{s} \end{aligned}$$

From these values, the lower bounds for the protection timers to ensure proper operation are determined as:

$$DL_FCxProtectionTimeOutVal_A$$

and

$$\begin{aligned} DL_FCxProtectionTimeOutVal_B &> 1.1 \times (HibernateTime \\ &+ TX_MaxProcessingDelay_A \\ &+ TX_MaxFrameTransferTime_A \\ &+ TX_MaxProcessingDelay_B \\ &+ TX_MaxFrameTransferTime_B) \\ &= 1.1 \times (500 \mu\text{s} + 3.08 \mu\text{s} + 8.01 \mu\text{s} + 1540 \mu\text{s} + 2500 \mu\text{s}) \\ &= 5006 \mu\text{s} \end{aligned}$$

The lower bounds for the AFC request timers are calculated as follows:

$$\begin{aligned}
 \text{DL_AFCxReqTimeOutVal_A} &> 1.1 \times \text{TX_MaxFrameTransferTime_B} \\
 &= 1.1 \times 2500 \mu\text{s} \\
 &= 2750 \mu\text{s}
 \end{aligned}$$

$$\begin{aligned}
 \text{DL_AFCxReqTimeOutVal_B} &> 1.1 \times \text{TX_MaxFrameTransferTime_A} \\
 &= 1.1 \times 8.01 \mu\text{s} \\
 &= 8.811 \mu\text{s}
 \end{aligned}$$

The minimum timeout values for the Replay Timers are determined as:

$$\begin{aligned}
 \text{DL_TCxReplayTimeOutVal_A} &> 1.1 \times (1.1 \times \text{DL_AFCxReqTimeOutVal_B} \\
 &+ \text{HibernateTime} \\
 &+ \text{TX_MaxProcessingDelay_A} \\
 &+ \text{TX_MaxFrameTransferTime_A} \\
 &+ \text{TX_MaxProcessingDelay_B} \\
 &+ \text{TX_MaxFrameTransferTime_B} \\
 &= 1.1 \times (1.1 \times 8.811 \mu\text{s} + 500 \mu\text{s} + 3.08 \mu\text{s} + 8.01 \mu\text{s} + 1540 \mu\text{s} + 2500 \mu\text{s}) \\
 &= 5017 \mu\text{s}
 \end{aligned}$$

$$\begin{aligned}
 \text{DL_TCxReplayTimeOutVal_B} &> 1.1 \times (1.1 \times \text{DL_AFCxReqTimeOutVal_A} \\
 &+ \text{HibernateTime} \\
 &+ \text{TX_MaxProcessingDelay_B} \\
 &+ \text{TX_MaxFrameTransferTime_B} \\
 &+ \text{TX_MaxProcessingDelay_A} \\
 &+ \text{TX_MaxFrameTransferTime_A} \\
 &= 1.1 \times (1.1 \times 2750 \mu\text{s} + 500 \mu\text{s} + 1540 \mu\text{s} + 2500 \mu\text{s} + 3.08 \mu\text{s} + 8.01 \mu\text{s}) \\
 &= 8333 \mu\text{s}
 \end{aligned}$$

If system simulations show that the AFC request timer expires too often in one device (for example, when there are large gaps between subsequent frames), the timeout value may be increased in the course of an optimization. In this case, the timeout value of the corresponding Replay timer at the peer device also has to be increased accordingly.

6.4.3 Layer 2 Credit Unit Mode

The UniPro V2.0 protocol defines the following two modes concerning the maximum transfer units (T_MTU, N_MTU, and DL_MTU):

- **DLCreditUnit32_Mode:** In this mode, the maximum transfer units defined for UniPro up to version 1.8 are used.
- **DLCreditUnit128_Mode:** This mode defines extended maximum transfer units

The optional **DLCreditUnit128_Mode** is used to support extended maximum transfer units. In **DLCreditUnit128_Mode**, the UniPro device should support a credit unit size of 128 bytes and a maximum payload length of 1160 bytes. If the local device supports **DLCreditUnit128_Mode**, it is indicated to the peer device through **PACapabilityUserData** field in the **PACP_CAP_ind** frame during Link Startup. In **DLCreditUnit128_Mode**, **PACapabilityUserData[0]** is set to 1 and in **DLCreditUnit32_Mode**, it is set to 0.

The **DLCreditUnitMode** supported by the peer device is determined from **PACapabilityUserData** field in the **PACP_CAP_ind** frame received from the peer device. Only if both the local and peer device support **DLCreditUnit128_Mode**, local and peer device can operate in **DLCreditUnit128_Mode** after Link Startup. Otherwise, both local and peer device should operate in **DLCreditUnit32_Mode** after Link Startup by

downgrading the capabilities. If the controller is programmed to act as a UniPro v1.8 device, program PACapabilityUserData[0] to 0.

The following features are supported in DLCreditUnit128_Mode feature:

- The maximum payload length is increased from 288 bytes to 1160 bytes.
- L2 flow control credit granularity is increased from 32 bytes to 128 bytes.
- PACapabilityUserData attribute is supported to include new user capabilities like DLCreditUnitMode.

Programming Flow for DLCreditUnit128_Mode

If P_L2_BUF_CREDUNIT_MODE is set to '1', then by default, DWC_mipi_unipro_20 uses 128-bit mode, which means that DLCreditUnit128_Mode is enabled. If this parameter is set to '0', then by default, DWC_mipi_unipro_20 uses 32-bit mode, which means that DLCreditUnit32_Mode is enabled.

VS_PACapabilityUserData (0xD089) can be used to select the credit unit mode between DLCreditUnit128_Mode or DLCreditUnit32_Mode. However, it is possible only if P_L2_BUF_CREDUNIT_MODE = 1.

When VS_PACapabilityUserData (0xD089) is set to 0x1, it enables DLCreditUnit128_Mode. When VS_PACapabilityUserData (0xD089) is set to 0x0, it enables DLCreditUnit32_Mode.

Effective Credit Unit Mode is decided based on the downgraded value during link startup which can be read from DLCreditUnitSize (0x2009) register.

Value present in VS_PACapabilityUserData (0xD089) attribute is used as PA_CapabilityUserData field of transmitted PACP_CAP_ind packet.



Note

If DWC_mipi_unipro_20 is configured as a v1.8 device, then ensure that credit unit is set to DLCreditUnit32_Mode only. Else, the functionality of the controller is unknown.

6.4.4 Flow Control Registers Mapped to VS Registers

The flow control registers used for the credit calculation are mapped to Vendor Specific (VS) registers for debug purpose. For this purpose, two new registers – VS_TxFlowControl (0xD0CA) and VS_RxFlowControl (0xD0CB) are introduced for Tx and Rx respectively. The details of the registers are documented in the sections VS_TxFlowControl and VS_RxFlowControl.

6.5 CPort Connection Management

A UniPro 1.41 device uses manual connection management to set up individual CPort connections between two attached devices. One or more applications set up all CPort related attributes on either side of the link and enable or disable individual connections.

A UniPro connection is defined as a pair of CPorts (one on each device) configured and enabled in the same way. The UniPro device has no means to check and refuse the wrongly configured connections, for example, flow control enabled on one side but disabled on the other side.

It is therefore important, that the application configures both devices properly so that the corresponding attributes match. For example, the PeerDeviceID programmed in the local module must be identical to the Device ID of the peer device. [Table 6-2](#) shows all the implications of connection related attributes:

Table 6-2 Connection Related Attributes

Local Attribute	Peer Attribute
T_PeerCPortID	CPortID
T_PeerDeviceID	N_DeviceID
T_TrafficClass	T_TrafficClass
T_CPortFlags	T_CPortFlags
T_TxTokenValue	T_RxTokenValue
T_RxTokenValue	T_TxTokenValue
T_ConnectionState	T_ConnectionState
Note: If end-to-end flow control is used, additional attributes must be set as described in “Setting up End-to-End Flow Control” on page 560.	

Individual CPorts are accessed locally and on the remote side using the upper bits of the **LA_AttrID** bus (for more information, see [“Attribute Mapping”](#) on page 29). When these attributes are set up for a CPort on both sides of the link, the CPorts are active and can transmit data in both directions. Activate the CPort by configuring the T_ConnectionState attribute to 1.

Example:

Application X in device A (device ID 7) wants to connect the local CPort 1 to the peer CPort 3 (device ID 12) using the traffic class 0 and end-to-end flow control. The flow control token size in the forward direction is set to the minimum value of 5 (relates to 2^5 bytes) and in reverse direction to a value of 8.

[Table 6-3](#) lists the values that are set in local and peer attributes.

Table 6-3 Connection Management Attribute Example

Attribute	Local Value (Select = 1)	Peer Value (Select = 3)
T_PeerCPortID	3	1
T_PeerDeviceID	12	7
T_TrafficClass	0	0
T_CPortFlags	1	1
T_TxTokenValue	5	8
T_RxTokenValue	8	5
T_ConnectionState	1	1

The last access of these attribute accesses should be to the `T_ConnectionState` attribute for both devices as this access activates the connection. All following accesses to any connection related attribute (except for `T_ConnectionState`) fail as UniPro does not allow reprogramming of active connections. To reprogram a connection, the `T_ConnectionState` attribute should be reset to 0.

An application can set up any number of CPorts and connections as long as all CPorts of one of the devices are not used. Data sent to any CPort is delivered in the same sequence to the target application of the other device. However, the data sent to different CPorts may be in a different order depending on the configured traffic classes and local arbitration (see [“Data Transfer using Arbitration and Segmentation when Using Multiplexed CPort Interface”](#)).

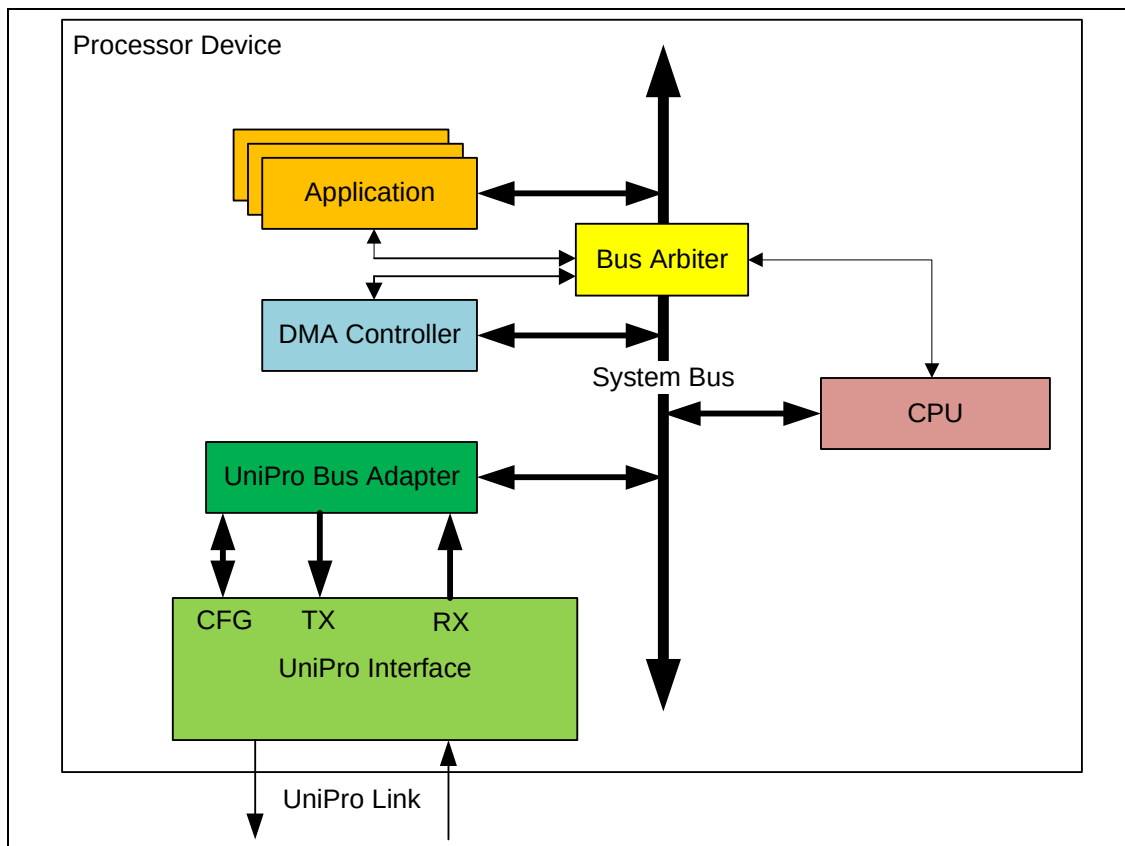
6.6 Data Transfer using Arbitration and Segmentation when Using Multiplexed CPort Interface

6.6.1 CPort Arbitration

The `DWC_mipi_unipro_20` uses a single CPort interface in each direction to transfer the data between any number of applications and the UniPro link.

The RX CPort interface transfers the received data in the same sequence as they are transferred by the peer device with the exception of the frames using a higher traffic class. These frames have a higher priority and may overtake frames of a lower traffic class either on the link or while reading out from the receive buffers of layer 2. All frames using the same traffic class are in the same order as they are sent by the peer device.

For the data to be transmitted, the MIPI UniPro Specification defines that the multiple CPorts need to be arbitrated according to a specified arbitration scheme. If multiplexed CPort is selected, and because the data transmission interface of `DWC_mipi_unipro_20` does not handle arbitration, this functionality must be provided by the application layer. In most systems where arbitration of different application sources is necessary, this functionality is anyway available in the form of a bus arbiter or a DMA controller being programmed by the application. [Figure 6-6](#) shows an example for different applications in a system using the `DWC_mipi_unipro_20` as the interface to a memory device.

Figure 6-6 Arbitration Example

In [Figure 6-6](#), each application (for example, graphics processor, audio processor, and so on) as well as the CPU may use one or more CPorts to transfer the data to and receive data from an attached memory device over the UniPro link. In the peer memory device, each CPort corresponds to a specific memory area, making the memory partitioning transparent to the applications.

The bus arbiter of the system is responsible for the arbitration of the individual bus masters and performs the required task of CPort arbitration. Even in a less complex device like a camera system, the arbitration of CPorts needs to take place, if more than one CPort is used for transmitting, for example, control and image data.

6.6.2 CPort Segmentation

If a CPort is configured to use end-to-end flow control, the application should take care of checking the available buffer space in the receiver before sending a message to the peer device. If multiplexed CPort is selected, the `DWC_mipi_unipro_20` supports this functionality by a dedicated signal

LA_TxSaFcMaxSegment of the status interface. An active level on the bits corresponding to the selected CPort flags, that the receiver has enough buffer space for a maximum sized segment (272 bytes). This information allows the transmitter to send any number of bytes up to 272 to the receiver before checking this signal again. For example, if a message of 1024 bytes should be sent to the peer device, the transmitter should check the status at least four times during the data transmission for the buffer space of the receiving CPort. The transmitter may check this status also with a higher frequency than every 272 bytes (for example, every 256 bytes).

The `DWC_mipi_unipro_20` performs the segmentation to keep the overhead for the header data as low as possible. The data is only sent, if a segment of 272 bytes is full or the EoM (End of Message) or EoF (End of Fragment) flag is set or the data is sent to another CPort.

6.7 Application Flow Control

6.7.1 Setting up End-to-End Flow Control

Application flow control can be activated for each connection independently. Flow control is activated by configuring the `E2EFC` field of the `T_CPortFlags` attribute with a value of 1. If this attribute is set, the local and peer `T_LocalBufferSpace`, `T_PeerBufferSpace`, and `T_CreditsToSend` attributes need to be configured as well. These attributes must not be written once CPorts are activated and receiving credit information, or if the flow control interface is active.

The `T_LocalBufferSpace` of the local device needs to be set to the same value as the `T_PeerBufferSpace` of the peer device and vice versa. They define the initial amount of bytes available for the application on the local and peer CPort. If these values are not initialized, the available buffer space needs to be reported using the flow control tokens later on.

The `T_CreditsToSend` attribute should always be initialized to zero before activating the connection. A write access with any other value is denied and results in the `INVALID_MIB_ATTRIBUTE_VALUE` error code.

[Table 6-4](#) provides an example where the local buffer space is 4096 bytes and the peer buffer space is 1024 bytes.

Table 6-4 Flow Control Attributes Example

Attribute	Local Value	Peer Value
<code>T_LocalBufferSpace</code>	4096	1024
<code>T_PeerBufferSpace</code>	1024	4096
<code>T_CreditsToSend</code>	0	0

After these steps are performed and the CPorts are activated (connected), the `DWC_mipi_unipro_20` handles the calculation of buffer spaces autonomously. An application can always read out the current status by accessing the buffer space attributes. If new buffer space is available (that is, when data is consumed by an application and the memory is free) the application can inform the `DWC_mipi_unipro_20` using an access on the flow control interface (see [“End-to-End Flow Control Support”](#) on page 27).

The received credits are reported to the peer device by sending a flow control token in a data frame or a separate frame, if no data is to be sent. The `T_TxTokenValue` defines the amount of credits reported in a single frame. For example, if this attribute is set to 5, a single token reports the availability of the additional $2^5 = 32$ bytes to the peer device. To reduce the number of credit frames, a larger token size can be selected by the application.

A good token size depends on the number of bytes (credits) provided by the application in parallel. A too low value results in a lot of additional frames for sending credit information only. A very high value may lead to long wait times on the peer device if not enough new credits are available.

As a general recommendation, the local $T_TxTokenValue$ and the peer $T_RxTokenValue$ must be set according to the following formula to prevent stall conditions on the link:

$$2^5 \leq T_TxTokenValue_{local} = T_RxTokenValue_{peer} \leq (LocalBufferSize/2) [bytes]$$

6.7.2 Local Buffer Protection (Controlled Segment Dropping)

Because flow control is not a mandatory feature for every application, the MIPI UniPro Specification defines an additional protection mechanism to avoid the buffer overflow for a local receiving application. This mechanism is called Controlled Segment Dropping. It is activated by configuring a value of 0 in the CSD field of the $T_CPortFlags$ attribute of the connected CPorts of both devices.

If CSD is activated, the receiver keeps track of the receive buffer space on its own. It receives the flow control credits from the application just like in E2EFC case. These credits however are not forwarded to the peer device but just locally stored in the $T_LocalBufferSpace$ attribute of the CPort. If the data is received, the local buffer space is decremented. If new credits are provided by the application, the buffer space is incremented accordingly.

If the local buffer space is less than a newly received segment, the segment is discarded internally and an error condition is flagged to the application by asserting the $LA_RxFragmentStatus$ and $LA_RxMsgStatus$ error condition. The first cycle where the error occurs is marked by a value of 01b on the $LA_RxFragmentStatus$ interface. If the error condition is valid for more cycles, which may occur when the credits are missing, a value of 11b shows that the data is discarded in the previous as well as the current cycle. The $LA_RxMsgStatus$ signal stays active until the $LA_RxDataEom$ signal is set and the message is completed to inform the application, that the corrupt message has ended.

Both fragment and message status signals are CPort dependent. This means that the message and fragment status are kept for each CPort individually. If a fragment for a different CPort is received before the end of the corrupt message of another CPort, the status is stored and flagged again, when the corrupt message is continued.

The transmitter is not informed about the incomplete message automatically. The receiving application needs to take care of informing the transmitter about this issue, which may start a retransmission of the complete message on application level.

6.8 Hibernation Mode

Hibernation is a special low-power mode of the M-PHY, which can be controlled by the application layers on each side of the link. $DWC_mipi_unipro_20$ offers the following methods for entering and exiting the hibernation mode:

- Through accesses to specific DME attributes
- Through special signals to the DME
- Through request from the peer device

All methods perform the same actions first on all connected TX lanes and afterwards on all connected RX lanes simultaneously. An enter hibernate request puts the connected M-PHY lanes to hibernation mode, while the exit hibernate request returns all connected M-PHYs to normal operation mode. Hibernation state is always set to the M-PHY lanes in both directions. $DWC_mipi_unipro_20$ communicates the enter request through the special PACP frames to the peer side, which takes care of the hibernation entry for its local M-PHY lanes.

The exit procedure can be executed by either side of the link, not necessarily the one that starts hibernate entry. This procedure is communicated on the link through a special state which is detected by the receiver. After the detecting that state, the receiver forwards this status to its own transmission path, which performs the hibernation exit on the local TX M-PHY lanes as well.

During the hibernation state, local attribute accesses to `DWC_mipi_unipro_20` and the M-PHY modules are still possible but may not be effective because (for example, all Layer 4 attributes are reset during hibernation).

Applications may be informed about a hibernate entry or exit detection by the following different sources:

- A dedicated signal for entry (`LA_HibernateEnterInd`) and exit (`LA_HibernateExitInd`)
- Reading the `DME_HibernateEnterInd` or `DME_HibernateExitInd` attribute
- An interrupt on signal `LA_Intr`, if the hibernate enter/exit indication is selected as interrupt source

Before entering the hibernate mode, the application should take care to stop sending all data and flow control information. By checking layer 2 frame indication attributes, the application can confirm that all frames are sent to and acknowledged from the peer device. After returning from hibernation, any data or flow control information, as well as layer 4 CPort setup is lost and needs to be re-sent.

The `DWC_mipi_unipro_20` does not check or control the required timing of hibernation mode (`T_HIBERN8` timing). The application should take care not to violate this minimum timing.



Note

In hibernation mode, the `PA_PWRMode` attribute returns a value of 33h.

6.9 Handling of Indications

`DWC_mipi_unipro_20` provides several indications of its processing state to the application through the dedicated signals (ending with “Ind”) or a configurable interrupt signal in combination with status attributes. The application should react on those indications because they report the occurrence of an error or a specific link procedure, which may require application interaction with the module.

Depending on the type of indication, the application must perform different actions. The following sections describe some examples of indication handling.

6.9.1 Status Indication Handling

Although `DWC_mipi_unipro_20` performs some operations automatically (for example, retransmission of wrong data frames and flow control handling), some actions need to be performed by the application to control further data processing. [Table 6-5](#) shows some indications of `DWC_mipi_unipro_20` and the action an application should perform on their detection.

Table 6-5 Indication Handling

Indication	Description	Proposed Application Reaction
Link Startup	The peer device wants to activate the link	If the system is ready, request a link startup to enable the link.

Table 6-5 Indication Handling (Continued)

Indication	Description	Proposed Application Reaction
Link Lost	The local device received a link startup although the link is already active	Reset and enable the local UniPro stack and re-configure all attributes. Generate a link startup request afterwards to bring the link in operational state again.
Endpoint Reset	The peer device requests to reset the local application	Determine if the reset request is applicable and perform a reset sequence on the local application.
Hibernate Enter	Either local or peer device requested to enter hibernate mode	Depending on the result code, the link may be in hibernate mode and no more data should be sent. The application may power-down certain parts of its system.
Hibernate Exit	Either local or peer device requested to exit hibernate mode	Depending on the result code, the link may be operational again, allowing the application to continue sending the data.
Powermode	Either local or peer device requested a power mode change	Depending on the result code, the link may be in a different power state now. The application should check that the current power mode complies with its current link requirements.
Error	Local device detected an error on the link or while processing requests	Check the error source and determine if the problem causes a loss of data. Many error indications may not inform that there is a loss of data.

6.9.2 Error Handling

DWC_mipi_unipro_20 reports errors back to the application in many ways. The following types of errors are reported:

- Configuration errors
- Data transmission errors
- Data reception errors
- Protocol errors
- Flow-control errors

Errors are reported on multiple interfaces:

- TX Data Interface
- RX Data Interface
- Configuration Interface
- Flow-Control Interface
- Interrupt signal
- Error indication interface signals

In general, all error types can be classified as critical errors and non-critical errors. While critical errors like a dropped segment may need interaction of the corresponding applications on both sides of the link, non-critical errors like CRC_ERROR (layer 2 error indication) just inform about a detected problem on the link, which is automatically solved by the protocol.

An application should take care of handling the critical error conditions on its own. These actions may include informing the connected application about the error up to the re-initializing the link, for example, if a fatal error occurred during power mode change. The MIPI UniPro Specification defines the individual reactions on error conditions in detail.

6.10 UniPro DDB Configuration

DDB attributes are configured in two possible ways:

- Through the selection of defined parameter values during synthesis
- Through the re-configuration of the DDB attributes using configuration accesses

The UniPro Specification defines the DDB attributes as read only, if the DDB revision attribute is set to a value other than zero. If this attribute is pre-configured through the synthesis parameter as non-zero, all the DDB attributes are read only and can never be set to a different value.

If the DDB revision attribute parameter is zero, all write accesses to the other DDB attributes are processed until the DDB revision attribute is set to non-zero. At that time all attributes are read only as well.

6.11 L2 Buffer Size Requirements

Three buffers are located inside layer 2 to store the incoming and outgoing frames. For transmission of the frames, all outgoing frames are stored in a single memory. While incoming frames are stored in different memories based on traffic class. The definition of the memory sizes depends on the number of symbols processed in parallel and the number of bytes to be stored. As the number of frames that can be stored in these buffers influence the system performance, the memory size needs to be selected carefully.

The physical buffer sizes of the layer 2 RX buffers are different from the logical buffer sizes, which are seen by the remote UniPro device. Receive buffers may contain gaps if the final bytes of a frame are not aligned to the buffer width as the next frame is always stored in a new line. Layer 2 takes care of this offset and transmits a lower initial credit value to the peer device in the initial credit exchange procedure. A total of 16 lines (one for each frame) is decremented from the total physical buffer size because every received frame can generate this gap.

It is important for the system calculations, that these 16 lines are added when defining the RX buffer sizes for layer 2. This may also affect other attributes like the AFCxCreditThreshold values.



Note

If TC1 is not used, the P_RBD1 and P_TBD1 parameters should be set to zero.

6.12 DWC_mipi_unipro_20 Capabilities

Table 6-6 shows all capabilities of the DWC_mipi_unipro_20 and their respective values or configuration options.

Table 6-6 UniPro Capabilities

Capability	Layer	Type	Description
PA_PhyType	L1.5	Constant	Value set to 1 (M-PHY)
PA_AvailTxDataLanes	L1.5	Parameter	Value set by the parameter P_NT
PA_AvailRxDataLanes	L1.5	Parameter	Value set by the parameter P_NR
PA_MinRxTrailingClocks	L1.5	Parameter	Value set by the parameter P_RXTRAIL
DL_TxPreemptionCap	L2	Constant	Value set to 1 (enabled)
DL_TC0TxMaxSDUSize	L2	Constant	Value set to 580(maximum) in DLCreditUnit128_Mode and set to 144(maximum) in DLCreditUnit32_Mode
DL_TC0RxInitCreditVal	L2	Parameter	Value set by the parameter P_RBD0
DL_TC1TxMaxSDUSize	L2	Constant	Value set to 580(maximum) in DLCreditUnit128_Mode and set to 144(maximum) in DLCreditUnit32_Mode
DL_TC1RxInitCreditVal	L2	Parameter	Value set by the parameter P_RBD1
DL_TC0TxBufferSize	L2	Parameter	Value set by the parameter P_TBD0
DL_TC1TxBufferSize	L2	Parameter	Value set by the parameter P_TBD1
N_TC0TxMaxSDUSize	L3	Constant	Value set to 1145(maximum) in DLCreditUnit128_Mode and set to 273(maximum) in DLCreditUnit32_Mode
N_TC1TxMaxSDUSize	L3	Constant	Value set to 1145(maximum) in DLCreditUnit128_Mode and set to 273(maximum) in DLCreditUnit32_Mode
T_NumCPorts	L4	Parameter	Value set by the parameter P_CP
T_NumTestFeatures	L4	Parameter	Value set by the parameter P_TF
T_TC0TxMaxSDUSize	L4	Constant	Value set to 1144(maximum) in DLCreditUnit128_Mode and set to 272(maximum) in DLCreditUnit32_Mode
T_TC1TxMaxSDUSize	L4	Constant	Value set to 1144(maximum) in DLCreditUnit128_Mode and set to 272(maximum) in DLCreditUnit32_Mode

6.13 CPort Connectivity

All data interfaces accessing the CPorts use a parameterized width for selecting the target CPort. The following list shows some examples of connections depending on the number of CPorts used:

- P_CP = 4 -> P_CA = 2 -> width of the CPort attribute ID: 2 bits
- P_CP = 3 -> P_CA = 2 -> width of the CPort attribute ID: 2 bits
- P_CP = 1 -> P_CA = 1 -> width of the CPort attribute ID: 1 bit

The last example is a special case. Although there is only one CPort available and no selection required, the CPort attribute ID interface should have at least a single bit of width. Since there is only one CPort as target, this bit can be tied to a constant value of zero.

6.14 L2 TX Buffer Size Calculation

Layer 2 TX buffer is implemented as one single physical memory, which stores data for TC0 and TC1 frames received from the upper layers (layer 3 and layer 4). The amount of buffer space reserved for TC0 and TC1 payload data to be transmitted to the peer device is determined by the P_TBD0 and P_TBD1 configuration parameters, which need to be chosen carefully in order to achieve the desired system performance.

The following system parameters are required to calculate the values for P_TBD0 and P_TBD1:

- P_NT_SYM: Number of parallel 16-bit symbols available in TX M-PHY lanes
- P_ST: Application layer TX interface width
Number of parallel 16-bit symbols at the CPort interface ($P_{ST} = P_{DWTX}/16$)
- $N_{Tx_frames_TCx}$: Number of frames to be stored in the TX buffer for TC0 and TC1
- $N_{Tx_payload_bytes_per_frame_TCx}$: Number of payload bytes per TX frame for TC0 and TC1. In general, the value for these parameters must be set to the number of payload bytes in an MTU sized frame, i.e., 272 bytes for DLCreditUnitMode 32, while 1144 bytes for DLCreditUnitMode 128. Only if a special purpose application layer is used, which divides all messages into fragments significantly smaller than 272 bytes or 1144 bytes (depending on DLCreditUnitMode), these parameters can be set to smaller values.

The P_TBD0 and P_TBD1 configuration parameters are calculated according to the following formula:

$$P_TBDx = N_{Tx_frames_TCx} \times \left\lceil \frac{N_{Tx_payload_bytes_frame_TCx} + 4}{2 \times \text{Max}(P_NT_SYM, P_ST)} \right\rceil + 8$$



Note

- The operator $\lceil \rceil$ denotes ceiling or rounding up to the next integer.
- The Max() function selects the maximum value of the two parameters.

The resulting total depth ($N_{Tx_buffer_depth}$ and $N_{Tx_buffer_depth_TC0_only}$, respectively) of the instantiated physical memory is determined by the following two equations.

If both traffic classes are present (TC0 and TC1), the depth of the physical memory instantiated as the L2 TX buffer is determined as:

$$Tx_buffer_depth = P_TBD0 + P_TBD1$$

Data Frames of TC0 are placed in address range from 0 to P_TBD0, whereas the Data Frame of TC1 are placed in address range from P_TBD0+1 to P_TBD0 + P_TBD1.

For TC0-only implementations (TC1 not implemented), the depth of the physical memory instantiated as the layer 2 TX buffer is determined as:

$$Tx_buffer_depth = P_TBD0$$

The width of the TX buffer is determined by the following equation:

$$Tx_buffer_width = \text{Max}(P_NT_SYM, P_ST) \times 16 \text{ bits}$$



Note

- The size of the instantiated physical memory must be in line with the P_TBD0 and P_TBD1 synthesis parameters. In other words, the number of rows of the instantiated physical memory for the L2 TX buffer must be equal to or greater than the number of rows calculated as $N_{Tx_buffer_depth}$ or $N_{Tx_buffer_depth_TC0_only}$, respectively, using the formulas. If a smaller physical memory is implemented by mistake, out-of-range RAM accesses occur and the UniPro device becomes inoperable.
- The TX buffer space reserved for TC0 and TC1 payload is stored in the layer 2 attributes DL_TC0TxBufferSize and DL_TC1TxBufferSize, measured in L2 credits (32 bytes or 128 bytes). The values read from these attributes may be up to four credits smaller than the physical buffer space for TC0/TC1 payload determined by P_TBD0 and P_TBD1. This is because a dedicated number of TX buffer rows are not usable for storing payload. However, only the amount of buffer space that is actually usable is communicated to a host reading these attributes.

The usage of these formulas is illustrated in the following examples:

Example TX (1)

A TX buffer storing eight symbols in parallel from the application (P_ST set to 8) and delivering eight symbols in parallel to the two M-PHY TX lanes (P_NT_SYM set to 8) has got a buffer width of eight symbols. It is assumed that a total of eight maximum-sized TC0 frames (TC0 MTUs), and five maximum-sized TC1 frames (TC1 MTUs) are stored in the TX buffer. DLCreditUnitMode is set to 128 then the P_TBD0 and P_TBD1 parameters are calculated as follows:

$$P_TBD0 = 8 \times \left\lceil \frac{1144 + 4}{2 \times \text{Max}(8, 8)} \right\rceil + 8 = 8 \times \left\lceil \frac{1148}{2 \times 8} \right\rceil + 8 = 8 \times 72 + 8 = 584$$

$$P_TBD1 = 5 \times \left\lceil \frac{1144 + 4}{2 \times \text{Max}(8, 8)} \right\rceil + 8 = 5 \times \left\lceil \frac{1148}{2 \times 8} \right\rceil + 8 = 5 \times 72 + 8 = 368$$

With P_TBD0 and P_TBD1 parameters calculated as shown, the (minimum) depth of the physical memory instantiated as the L2 TX buffer is determined as follows:

$$N_{Tx_buffer_depth} = P_TBD0 + P_TBD1 = 584 + 368 = 952$$

This results in a physical memory with 128 bits x 952 rows for the TX buffer.

Example TX (2)

TX buffer stores four symbols in parallel for both application and M-PHY ($P_ST=P_NT_SYM=4$). Four MTU-sized frames are stored for both TC0 and TC1. DLCreditUnitMode is set to 32 then the P_TBD0 and P_TBD1 parameters are calculated as follows:

$$P_TBD0 = P_TBD1 = 4 \times \left\lceil \frac{272 + 4}{2 \times \text{Max}(4, 4)} \right\rceil + 8 = 4 \times \left\lceil \frac{276}{2 \times 4} \right\rceil + 8 = 8 \times 35 + 8 = 288$$

With P_TBD0 and P_TBD1 calculated as shown, the minimum depth of the physical memory instantiated as the L2 TX buffer is determined as follows:

$$N_{Tx_buffer_depth} = P_TBD0 + P_TBD1 = 576$$

This results in a physical memory with 64 bits x 576 rows for the TX buffer.

6.15 L2 RX Buffer Size Calculation

In contrast to the TX buffer, which is a shared buffer for TC0 data, TC1 data and TC0 or TC1 frame addresses, separate buffers are implemented for storing received TC0 and TC1 data frames. The amount of buffer space reserved for TC0 and TC1 payload data received from the peer device is determined by the P_RBD0 and P_RBD1 synthesis parameters, which need to be chosen carefully to achieve the desired system performance.

The following system parameters are required in order to calculate the values for P_RBD0 and P_RBD1 :

- P_NR_SYM : Number of 16-bit parallel symbols available in RX PHY lanes
- P_SR : Application layer RX interface width
Number of parallel 16-bit symbols at the CPort interface ($P_SR=P_DWRX/16$)
- $N_{Rx_frames_TCx}$: Number of frames to be stored in the RX buffer for TCx (TC0 or TC1).
- $N_{Rx_payload_bytes_per_frame_TCx}$: Number of payload bytes per RX frame for TC0 and TC1. In general, the value for these parameters must be set to the number of payload bytes in an MTU-sized frame i.e., 272 bytes for DLCreditUnitMode 32 while 1144 bytes for DLCreditUnitMode 128. Only if the peer device uses a special-purpose application layer is used, which divides all messages into fragments significantly smaller than 272 bytes or 1144 bytes (depending on DLCreditUnitMode), these parameters can also be set to smaller values.

The P_RBD0 and P_RBD1 parameters are calculated according to the following formula:

$$P_RBDx = \left\lceil \frac{N_{Rx_frames_TCx} \times \left\lceil \frac{N_{Rx_payload_bytes_frame_TCx} + 2}{2 \times \text{Max}(P_NR_SYM, P_SR)} \right\rceil \times \text{Max}(P_NR_SYM, P_SR) \times \frac{1}{16}}{\text{Max}(P_NR_SYM, P_SR)} \times 16 \right\rceil + 16$$

**Note**

- The operator $\lceil \rceil$ denotes ceiling or rounding up to the next integer.
- The $\text{Max}()$ function selects the maximum value of the two parameters.

At the RX side, the number of physical memory rows ($N_{\text{Rx_buffer_depth_TC0}}$ and $N_{\text{Rx_buffer_depth_TC1}}$, respectively) required for proper operation equals the calculated values for P_{RBD0} and P_{RBD1} :

$$N_{\text{Rx_buffer_depth_TCx}} = P_{\text{RBDx}}$$

The width of the RX buffers is determined by the following equation:

$$N_{\text{Rx_buffer_width_TCx}} = \text{Max}(P_{\text{NR_SYM}}, P_{\text{SR}}) \times 16 \text{ bits}$$

**Note**

- The size of the instantiated physical memories for TC0 and TC1 RX data must be in line with the P_{RBD0} and P_{RBD1} parameters of the synthesized UniPro controller. In other words, the number of rows of the instantiated physical memory for the L2 RX buffers must be equal to or greater than the number of rows calculated as $N_{\text{Rx_buffer_depth_TC0}}$ and $N_{\text{Rx_buffer_depth_TC1}}$, respectively, using the formulas. If smaller physical memories are implemented by mistake, out-of-range RAM accesses occur and the UniPro device becomes inoperable.
- The RX buffer space of the TC0 and TC1 RX buffers communicated to the peer UniPro device in the Layer 2 Link Startup phase is calculated directly from the synthesis parameters P_{RBD0} and P_{RBD1} , and can be determined by reading the layer 2 attributes $\text{DL_TC0RxInitCreditVal}$ and $\text{DL_TC1RxInitCreditVal}$. The unit for these attributes is L2 credits (32 bytes or 128 bytes). The values read from these attributes are up to eight credits lesser than the physical buffer space for TC0/TC1 payload determined by P_{RBD0} and P_{RBD1} . This is due to the fact that for internal reasons a dedicated number of RX buffer rows is not usable for storing payload. However, only the amount of buffer space that is actually usable is communicated to the peer device during Link Startup.
- If P_{RBD0} is calculated by considering $N_{\text{Rx_payload_bytes_per_frame_TCx}}$ as 1144 bytes, then the resultant buffer size is more than 4KB if $N_{\text{Rx_frames_TC0}} > 3$. In such cases, if DL_CreditUnitSize is set to 32, either by configuring or by downgrading, then Credit Value in the initial AFC frame is advertised as 128 rather than the actual Credit Value.

The use of these formulas is illustrated by the following example:

RX buffer for TC0 using two symbols in parallel to the application layer ($P_{\text{SR}} = 2$) and four symbols in parallel to the M-PHY ($P_{\text{NR_SYM}} = 4$). Six frames can be stored in the buffer at the same time.

$$P_{\text{RBD0}} = \left\lceil \frac{\left(6 \times \left\lceil \frac{1144 + 2}{2 \times \text{Max}(4,2)} \right\rceil \times \text{Max}(4,2) \times \frac{1}{16} \right) \times 16}{\text{Max}(4,2)} \right\rceil + 16 = 880$$

This results in a physical memory with 64 bits x 880 rows for the TC0 RX buffer.

6.16 Guidelines for Dimensioning the Layer 2 Buffers

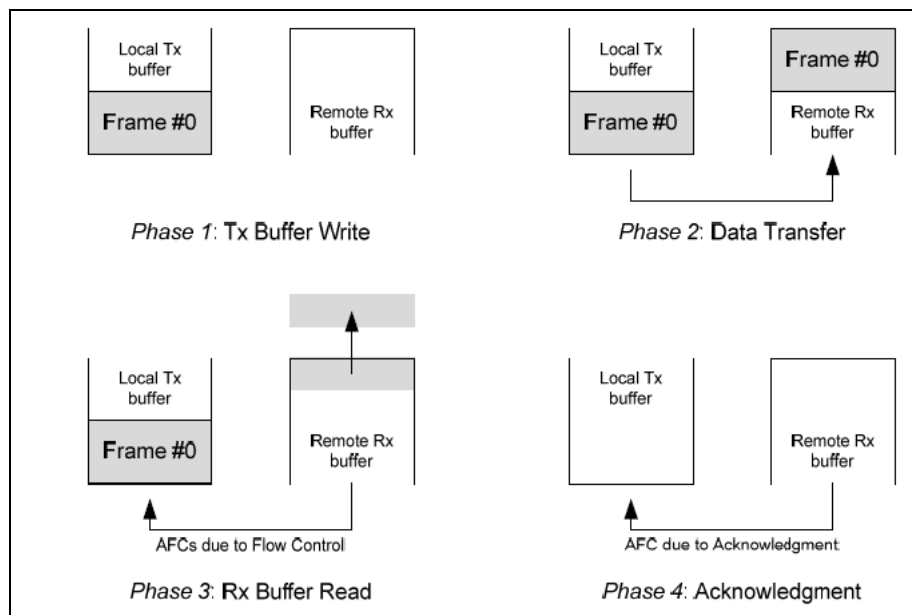
To achieve system performance goals, selecting proper sizes for the layer 2 TX and RX buffers is of utmost importance. For example, if the size of the RX buffer is too small, unintended gaps may occur during the data transmission. On the other hand, oversized buffers lead to unnecessary area and power consumption.

This section aims to enable a system integrator to choose proper dimensions for the layer 2 buffers. For this purpose, a basic knowledge of how data transfers are handled by DWC_mipi_unipro_20 is required.

6.16.1 Layer 2 Data Transfer Basics

Figure 6-7 shows the phases of a data frame transfer through the local TX buffer over the link and finally through the remote RX buffer. It is assumed, that Frame #0 in Figure 6-7 is an MTU-sized TC0 frame (272/1144 payload bytes). The DL_AFC0CreditThreshold is set to 4 at the remote device, while the DL_TC0OutAckThreshold is set to 0.

Figure 6-7 Four Phases of Data Frame Transfer



From a layer 2 perspective, a basic UniPro data frame transfer can be divided into the following phases:

- **TX Buffer Write (Phase 1):** The application layer at the local device has data to send, and writes the data to the local TX CPort interface. In general, the application layer messages are segmented into data chunks by Layer 4 with a maximum payload size of 272 or 1144 bytes. Additionally, Layer 4 and Layer 3 each add one header byte. In this example, it is assumed, that the application layer has exactly 272 bytes to send. In this phase, the data is only written to the layer 2 TX buffer. Data transfer to the remote device does not start yet.
- **Data Transfer (Phase 2):** Before starting the data transfer over the link to the peer device, the local layer 2 checks if enough credits are available for the remote receiver to store the entire frame in its RX buffer. If this is the case, the transmitter starts transferring the data.

Concurrently, the remote device stores the received data into its receive buffer. The transmitter keeps all data belonging to this frame in its TX buffer, that is, at the end of this phase, the frame data is stored in both the local TX and the remote RX buffer.

- **RX Buffer Read (Phase 3):** After the remote layer 2 has received and stored the entire frame, it checks the frame for correctness (CRC, frame length, syntactical correctness, and so on). If the frame is OK, the remote device starts transferring the frame from its RX buffer to the upper layers. Each time, a certain amount of data is read out of the RX buffer, the receiver sends an AFC frame due to flow control to give a credit update to the peer transmitter. The amount of data to be read from the buffer before an AFC is sent due to flow control is defined by `DL_AFCxCreditThreshold`. In this example, because `DL_AFC0CreditThreshold` is equal to 4, an AFC0 is sent each time 4*32 bytes are transferred to the upper layers. The local transmitter still keeps this frame in its TX buffer, because it did receive the flow control updates, but not the acknowledgment for the frame yet.
- **Acknowledgment (Phase 4):** After the remote layer 2 has transferred the entire frame to the upper layers, the frame is considered deleted from the RX buffer. In this example, the parameter `DL_TC0OutAckThreshold` is 0, so the remote device sends an AFC0 frame after each frame that has been transferred entirely to the upper layers. Only after the local device has received and processed this AFC containing the acknowledgment information, that is, the new updated sequence number, the local device is allowed to delete the frame also from its TX buffer.

6.16.2 Impact of Incorrectly Configured Buffer Sizes

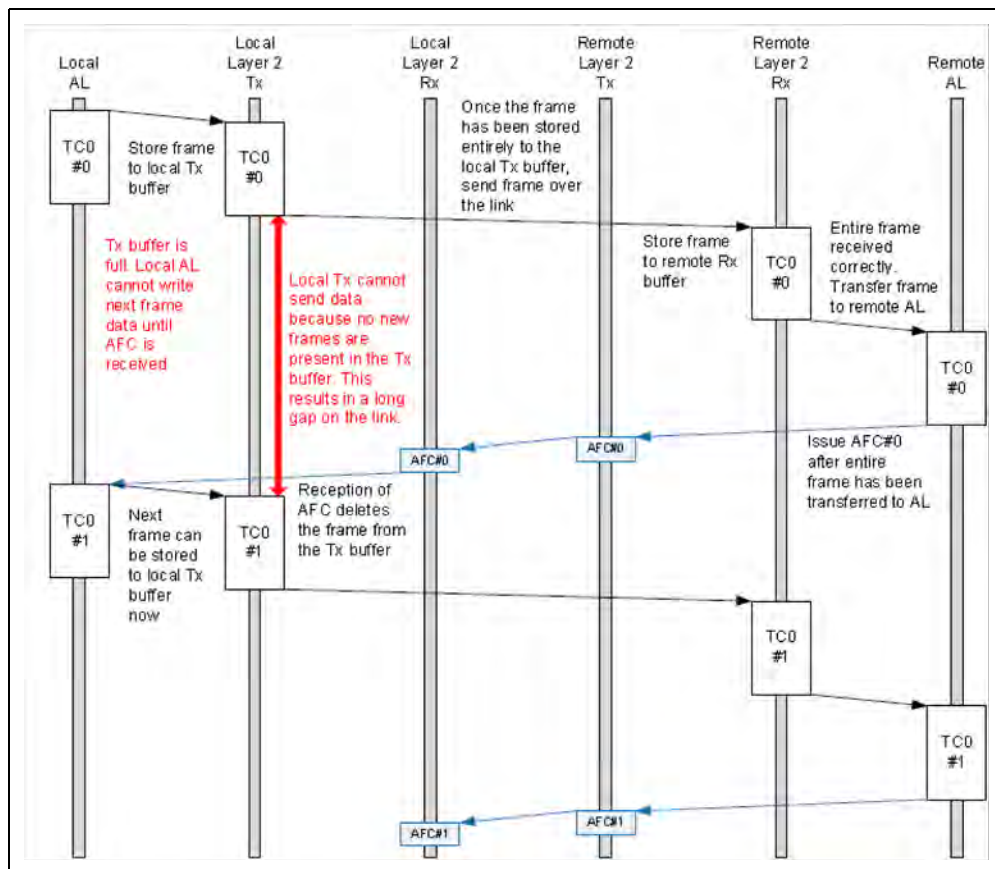
The importance of carefully selected buffer sizes can be best illustrated by an example for a system with poorly dimensioned TX and RX buffers. For example, if both the TX buffer and the peer RX buffer can store only one MTU-sized frame, data is streamed as follows:

1. Iteration 1, phase 1: One MTU-sized frame (frame #0) is stored in the local TX buffer. TX buffer is now full.
2. Iteration 1, phase 2: Frame #0 is read out of the local TX buffer, sent to the link and concurrently stored in the remote RX buffer. Both local TX buffer and remote RX buffer are now full.
3. Iteration 1, phase 3: Once the entire frame is received and checked for correctness, the frame is read out of the RX buffer by the upper layers in the remote device.
4. Iteration 1, phase 4: Once the entire frame is transferred to the upper layers at the remote device, an AFC is sent.
5. Iteration 2, phase 1: After the local transmitter receives the AFC and deleted frame #0 from its TX buffer, the local AL starts writing the next frame (frame #1) to the local TX buffer.
6. Iteration 2, phase 2: Frame #1 is transferred over the link.

The data is actually transferred over the link only in Steps 2 and 6, while in all other steps data transfer is stalled. In Step 3, the TX buffer at the local device is already full, such that the transmitter cannot send more data, although the local application layer still has data to transmit. Additionally, the remote RX buffer is also full, that is, the local transmitter may not have any more credits. Only after Step 4 is completed, the local transmitter is allowed to delete the frame from the TX buffer. However, the next frame to be sent must also be stored entirely into the transmit buffer before it can be sent to the link.

Gaps on the link are filled with FLR symbols as long as the burst is still active. For example, when the link is not operating in auto mode or when the number of TX symbol clock cycles defined by the `VS_TxBurstClosureDelay` attribute are not reached yet. When operating in auto mode, and the burst is closed between frames, the performance degrades further.

Figure 6-8 shows the time behavior of a system with TX and RX buffers dimensioned for only one MTU-sized frame.

Figure 6-8 Illustration of a System with Incorrectly Configured Buffer Sizes

6.16.3 General Guidelines and Common Pitfalls

To achieve maximum performance in a high-speed data transmission, the layer 2 TX and RX buffers need to be dimensioned correctly. High-speed transmission can only be achieved if the system is set up such that no gaps occur during data transmission on the link, that is, no FLR symbols are inserted, and the burst is not closed if the transmitter is in auto mode.

The system integrator should keep in mind that the size of the RX buffer and the corresponding layer 2 attribute configuration and the size of the TX buffer at the peer device must be adjusted to each other.

The RX buffer should be dimensioned such that the peer transmitter always has enough credits to send the data, even in a continuous data stream. Therefore, it must be taken into account that AFCs containing credit updates are only sent after the received data has been read out of the buffer from the upper layers, which is always delayed in time.

As a rule of thumb, an RX buffer for storing four MTU-sized frames can be considered a starting point for optimizations. Such a buffer enables the peer transmitter to send up to four frames before receiving the first credit update. The transfer time of four frames should also enable the receiver to process the received frames and return AFCs back to the transmitter.

The time required for reading out the received frames correctly from the RX buffer takes influence on the optimal buffer sizes. All system parameters affecting this time must be considered. For example, if the

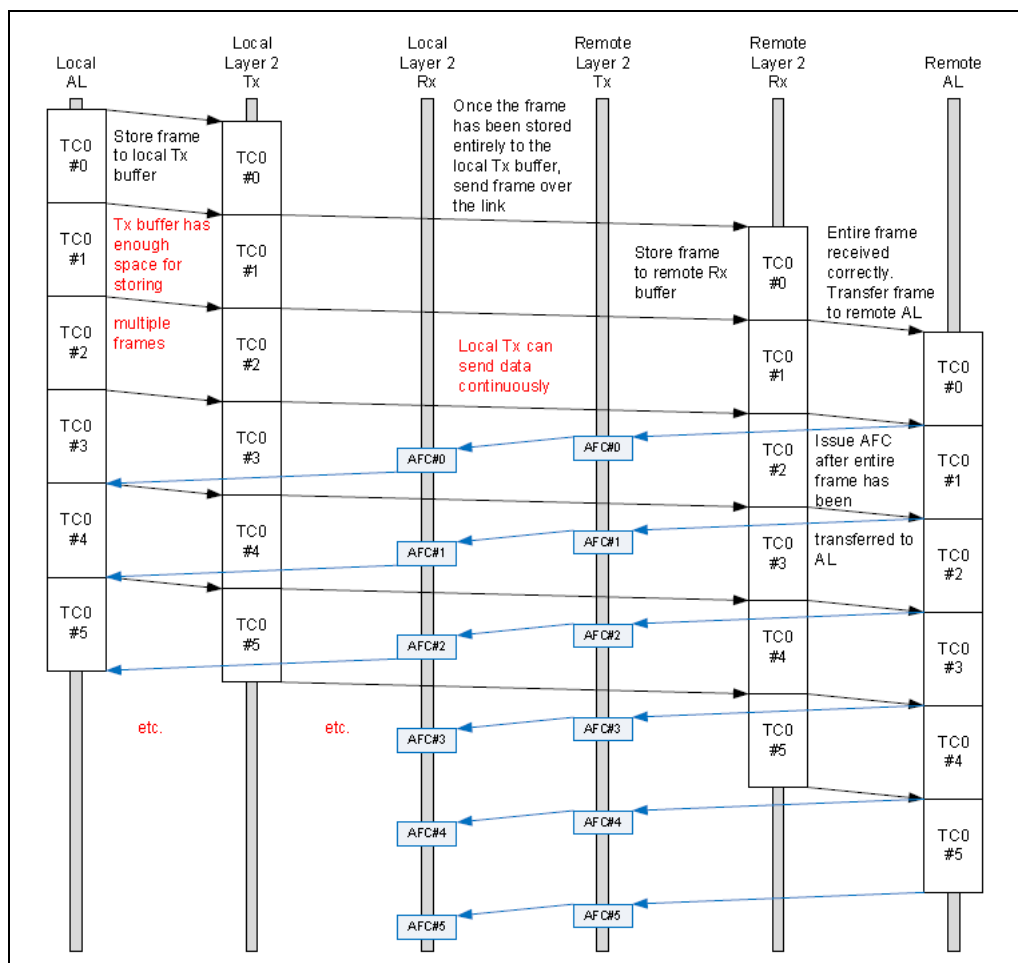
CG_Sysclk at the receiver is running at a lower speed, it takes longer until the frame has been transferred to the upper layer, and the corresponding AFC frame is sent later.

In addition, the transfer time of an AFC frame from the receiver to the transmitter must be taken into account. This time also depends on various system parameters, for example, the receiver CG_Sysclk frequency and the current power mode. The transfer time is affected, if the RX link operates in auto mode. In this case, a new burst has to be started before each AFC frame. The RX buffer has to be bigger, if the AFC frame return time is longer.

For the TX buffer, it must be taken into account that any data frame can only be deleted from the buffer after it has received its acknowledgment. For example, if the TX buffer can store four MTUs, and the peer receiver is configured to return an AFC after four frames, the transmitter has to wait for this AFC before the local application layer can start writing new frame data to the TX buffer. To enable continuous data transmission, the TX buffer should be larger than the RX buffer at the peer, which enables the local application layer to write the data to the TX buffer even before the AFC is received from the peer. As a rule of thumb, the peer RX buffer size plus two MTUs can be a starting point for TX buffer size optimizations.

In some cases, the size of the peer RX buffer is unknown prior to system integration, for example, if the peer is part of a removable memory card. In this case, a device driver has to take care of setting the local and peer attributes `DL_AFCxCreditThreshold` and `DL_TCxOutAckThreshold` according to the actual local and peer buffer sizes such that no gaps can occur.

In contrast to the counterexample with incorrectly configured buffer sizes shown in [Figure 6-8](#) and [Figure 6-9](#) depicts the time behavior of a system with adequately sized TX and RX buffers. The remote RX buffer is dimensioned such that the local TX always has enough credits to send data. In addition, the local TX buffer is big enough such that the application layer can continuously write new data to the buffer.

Figure 6-9 Illustration of a System with Properly Configured Buffer Sizes

In any case, the layer 2 buffer sizes must be validated before the integration by extensive system simulations.

6.17 Simulation Time Reduction during LinkStartup Process

LinkStartup procedure consists of LINE-RESET, TRG_UPRx communication (phase 0 to phase 4), and Capability Exchange (Phase5). Since LinkStartup process may take several ms of time to complete, it may be considered unnecessary simulation time overhead when simulation is not focused on LinkStartup verification. To reduce simulation time in LinkStartup process, a new register named as VS_LssBypass (0xd0f6) is implemented.

When VS_LssBypass[LSSBYPASS] is enabled, after hibernate exit, controller bypasses LINE-RESET and TRG_UPRx communication and directly enters to Capability Exchange Phase.

To use this feature, ensure the following:

- Set the value of VS_LssBypass to 0x0000_0001.
- Since no lane mapping happens, the lanes are fixed.
- Skipping line-reset and TRG_UPRx communication are also expected from the peer.



Note

This feature is provided only for speeding up simulation.

A

Area and Power

A.1 Area, Power, and DFT Coverage

A.1.1 Area and Power Numbers

Table A-1 Power Numbers

Configuration	Library	Area	Gate Count (Area/0.0768)	Idle Power	Hibernate Power	Active Power	Full Power
Configuration-1	Industry- standard 7nm	19535.67	264K gates	2.93mW	0.66μW	5.95mW	4.53mW
Configuration-2	Industry- standard 7nm	19610.26	255K gates	2.98mW	0.66μW	5.80mW	4.57mW
Configuration-3	Industry- standard 7nm	19692.46	256K gates	2.99mW	0.69μW	5.46mW	4.59mW

The clock frequencies used to generate the area and power are CG_sysclk = 400 MHz, CG_TxSymbolClk = 303 MHz, CG_RxSymbolClk = 303 MHz.

For synthesis, 7 nm library technology is used.

A.1.2 Spyglass DFT coverage

The SpyGlass DFT coverage is 97%.

**Note**

This information is provided for a selected configuration. Important parameters are listed in [Table A-2](#). Contact Synopsys for exact details.

Table A-2

Configuration Name	Configurable Parameter Values
Configuration-1	■ APB3_CLOCK_EN 0 ■ APB3_EN 0 ■ COMMON_RX_SYM_CLK 0 ■ DISBL_MCFG_CLK 0 ■ DISBL_TC1 1 ■ IS_UFS_DEVICE 1 ■ MK2_EXTN_SUPPORT 0 ■ NMUX_CPORT_SEL 0 ■ P_CP 1 ■ P_DDB_DCL 0x2 ■ P_DDB_LGT 0x0 ■ P_DDB_LVL 0x1 ■ P_DDB_MID 0x24a ■ P_DDB_PID 0xa00 ■ P_DDB_REV 0 ■ P_DEBUG_PORTS_ENABLE 1 ■ P_DWRX 128 ■ P_DWTX 128 ■ P_EARLY_SEQ_NUM_UPDATE 1 ■ P_FC 0 ■ P_GPIO 96 ■ P_L2_BUF_CREDUNIT_MODE 1 ■ P_NR 2 ■ P_NT 2 ■ P_PHYRESET 66 ■ P_RBD0 512 ■ P_RBD1 0 ■ P_RMMI_DATAWIDTH 80 ■ P_RXTRAIL 216 ■ P_TBD0 480 ■ P_TBD1 0 ■ P_TF 1 ■ P_UPRO_LSS_HS 1 ■ P_UPRO_VER 2 ■ RX_SKIP_REQUEST 0

Table A-2 (Continued)

Configuration Name	Configurable Parameter Values
Configuration-1	■ SNPS_MPHY 4 ■ UPRO_LPM_EN 1 ■ VS_RD_DATABYTE_PACK 0
Configuration-2	■ APB3_CLOCK_EN 0 ■ APB3_EN 1 ■ COMMON_RX_SYM_CLK 0 ■ DISBL_MCFG_CLK 0 ■ DISBL_TC1 1 ■ IS_UFS_DEVICE 1 ■ MK2_EXTN_SUPPORT 0 ■ NMUX_CPORT_SEL 0 ■ P_CP 1 ■ P_DDB_DCL 0 ■ P_DDB_LVL 1 ■ P_DDB_MID 0 ■ P_DDB_PID 0 ■ P_DDB_REV 0 ■ P_DEBUG_PORTS_ENABLE 1 ■ P_DELAY_TACTIVATE 1 ■ P_DWRX 128 ■ P_DWTX 128 ■ P_EARLY_SEQ_NUM_UPDATE 1 ■ P_FC 0 ■ P_GPIO 96 ■ P_L2_BUF_CREDUNIT_MODE 1 ■ P_NR 2 ■ P_NT 2 ■ P_PHYRESET 40 ■ P_RBD0 304 ■ P_RBD1 0 ■ P_RMMI_DATAWIDTH 80 ■ P_RXTRAIL 216 ■ P_SELF_LSS 0 ■ P_TBD0 296 ■ P_TBD1 0

Table A-2 (Continued)

Configuration Name	Configurable Parameter Values
Configuration-2	■ P_TF 1 ■ P_UPRO_B2B_H8_HANDLE 1 ■ P_UPRO_LSS_HS 1 ■ P_UPRO_VER 2 ■ RX_SKIP_REQUEST 0 ■ SNPS_MPHY 0 ■ UPRO_LPM_EN 1 ■ VS_RD_DATABYTE_PACK 0
Configuration-3	■ APB3_CLOCK_EN 0 ■ APB3_EN 1 ■ COMMON_RX_SYM_CLK 0 ■ DISBL_MCFG_CLK 0 ■ DISBL_TC1 1 ■ IS_UFS_DEVICE 1 ■ MK2_EXTN_SUPPORT 0 ■ NMUX_CPORT_SEL 0 ■ P_CP 1 ■ P_DDB_DCL 0 ■ P_DDB_LVL 1 ■ P_DDB_MID 0 ■ P_DDB_PID 0 ■ P_DDB_REV 0 ■ P_DEBUG_PORTS_ENABLE 1 ■ P_DWRX 128 ■ P_DWTX 128 ■ P_EARLY_SEQ_NUM_UPDATE 1 ■ P_FC 0 ■ P_GPIO 0 ■ P_L2_BUF_CREDUNIT_MODE 1 ■ P_NR 2 ■ P_NT 2 ■ P_PHYRESET 40 ■ P_RBD0 1168 ■ P_RBD1 0 ■ P_RETAIN_L4 1 ■ P_RMMI_DATAWIDTH 80

Table A-2 (Continued)

Configuration Name	Configurable Parameter Values
Configuration-3	■ P_RXTRAIL 216 ■ P_TBD0 1160 ■ P_TBD1 0 ■ P_TF 1 ■ P_UPRO_LSS_HS 1 ■ P_UPRO_VER 2 ■ RX_SKIP_REQUEST 0 ■ SNPS_MPHY 4 ■ UPRO_LPM_EN 1 ■ VS_RD_DATABYTE_PACK 0

B

Differences from the Specification

This appendix describes the differences between the DWC_mipi_unipro_20 and the MIPI UniPro Specification. It contains the following section:

- [“Differences between DWC_mipi_unipro_20 and MIPI UniPro Specification”](#) on page 583

B.1 Differences between DWC_mipi_unipro_20 and MIPI UniPro Specification

- Layer 3 Long Header Trap (Software Trap) is not supported.
- DME configuration accesses are accepted in every power mode.
- Attribute TL_CreditsToSend returns the result code INVALID_MIB_ATTRIBUTE_VALUE if written with a value other than zero.
- Signal naming convention on the CPort application interface is different than described in *Appendix D of UniPro Specification*. [Table B-1](#) shows the assignments.

Table B-1 Application Interface Signal Assignment

MIPI UniPro Signal Name	DWC_mipi_unipro_20 Signal Name
Common Group	
t_clk	CG_Sysclk
T_CO_DATA.req Group	
t_tx_valid	LA_TxDataValid
t_tx_accept	LA_TxDataAccept
t_tx_cportid	LA_TxCportID
t_tx_data	LA_TxData
t_tx_byte_en	LA_TxDataByteEn
t_tx_eom	LA_TxDataEndOfMessage
t_tx_eof	LA_TxDataEndOfFragment

Table B-1 Application Interface Signal Assignment (Continued)

MIPI UniPro Signal Name	DWC_mipi_unipro_20 Signal Name
t_tx_result	LA_TxResultCode
T_CO_DATA.ind Group	
t_rx_valid	LA_RxDataValid
t_rx_accept	LA_RxDataAccept
t_rx_cportid	LA_RxCportID
t_rx_data	LA_RxData
t_rx_byte_en	LA_RxDataByteEn
t_rx_eof	Not implemented. It is recommended to tie this to 1.
t_rx_eom	LA_RxEom
t_rx_som	LA_RxSom
t_rx_fragment_status	LA_RxFragmentStatus
	LA_RxMsgStatus
T_CO_FLOWCONTROL.req Group	
t_fc_valid	LA_FcValid
t_fc_accept	LA_FcAccept
t_fc_credits	LA_FcCredits
t_fc_credits_accepted	LA_FcCreditsAccepted
t_fc_result	LA_FcResultCode
TX Status/Arbitration Group	
t_txsa_cport_status	LA_TxSaCportStatus
t_txsa_fc_for_maxsegment	LA_TxSaFcMaxSegment
t_txsa_end_segment	LA_TxSaEndSegment

C

Synchronizer Methods

This appendix describes the synchronizer methods (blocks of synchronizer functionality) that are used in `DWC_mipi_unipro_20` to cross clock boundaries.

This appendix contains the following sections:

- “Synchronizer 1: Simple Double Register Synchronizer (`DWC_mipi_unipro_20_bcm21.v`)” on page 586
- “Synchronizer 2: Pulse Synchronizer (`DWC_mipi_unipro_20_bcm22.v`)” on page 591
- “Synchronizer 3: Data Bus Synchronizer without Acknowledge (`DWC_mipi_unipro_20_bcm26.v`)” on page 592

**Note**

If you use your customized synchronizer, replace the `DWC_mipi_unipro_20_bcm21`, `DWC_mipi_unipro_20_bcm22`, and `DWC_mipi_unipro_20_bcm26` modules with the appropriate synchronization cells. While constraining for synthesis, waive off the synchronizer paths as the false path.

C.1 Synchronizer 1: Simple Double Register Synchronizer (DWC_mipi_unipro_20_bcm21.v)

This is a single clock data bus synchronizer for synchronizing data that crosses asynchronous clock boundaries. Your controller may have parameters that allow you to configure for the number of synchronizing stages (2 or 3), the style of first-stage capturing flip-flop needed (negative or positive edge-triggered), and insertion of hold latches to facilitate scan testing.

Figure C-1 Synchronizer 1: Synchronization Type 1

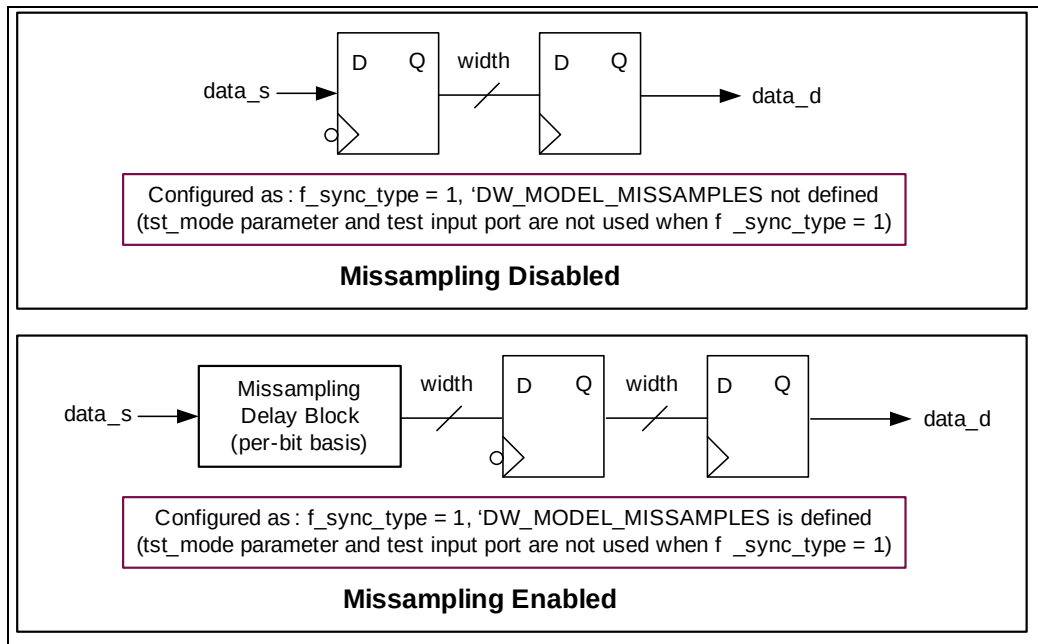


Figure C-2 Synchronizer 1: Synchronization Type 2

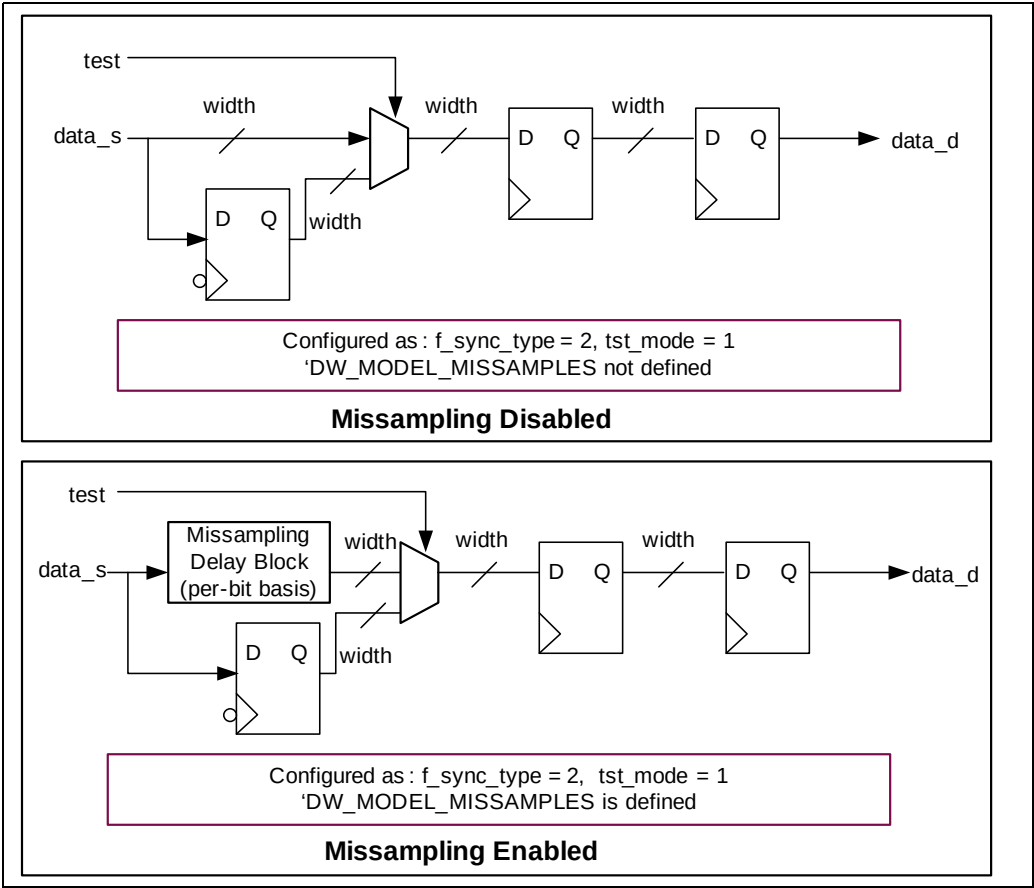


Figure C-3 Synchronizer 1: Synchronization Type 3

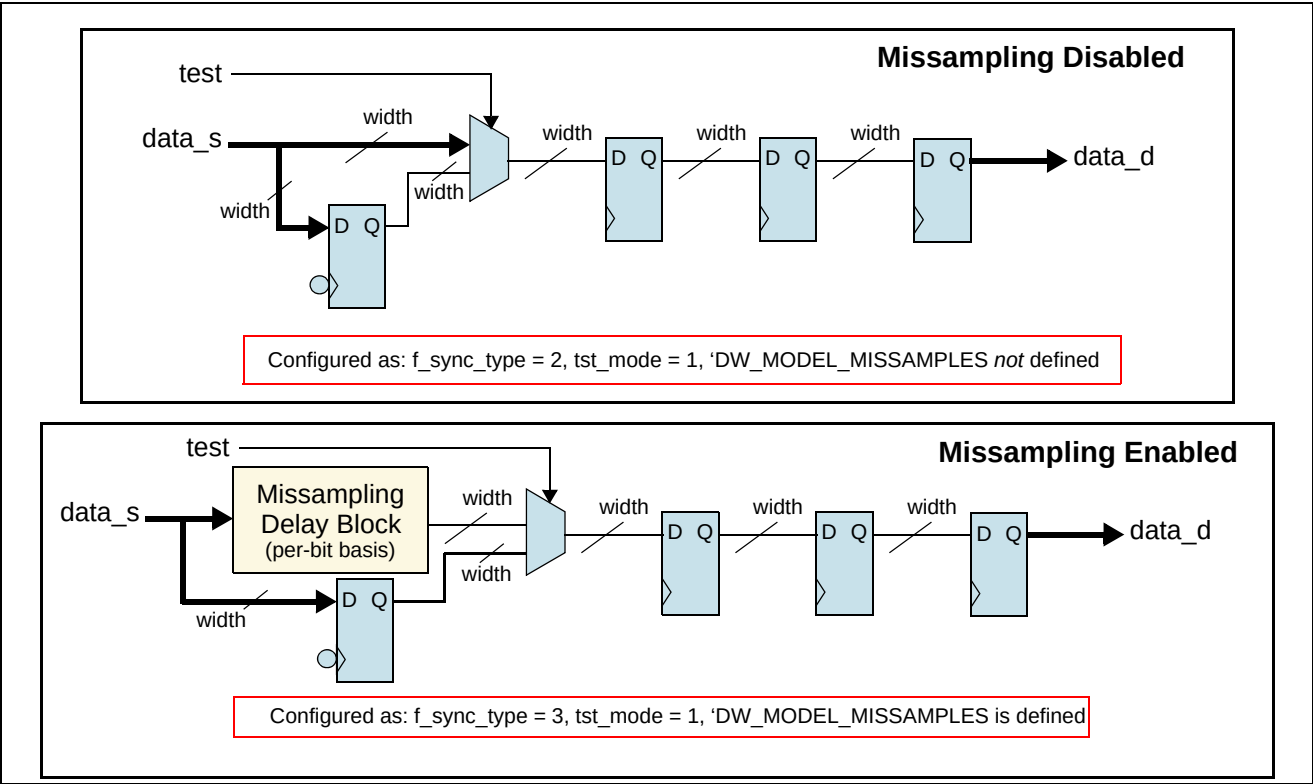


Figure C-4 Synchronizer 1: Missampling Model Type 0

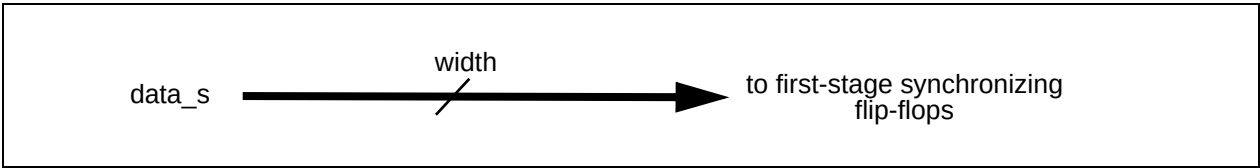


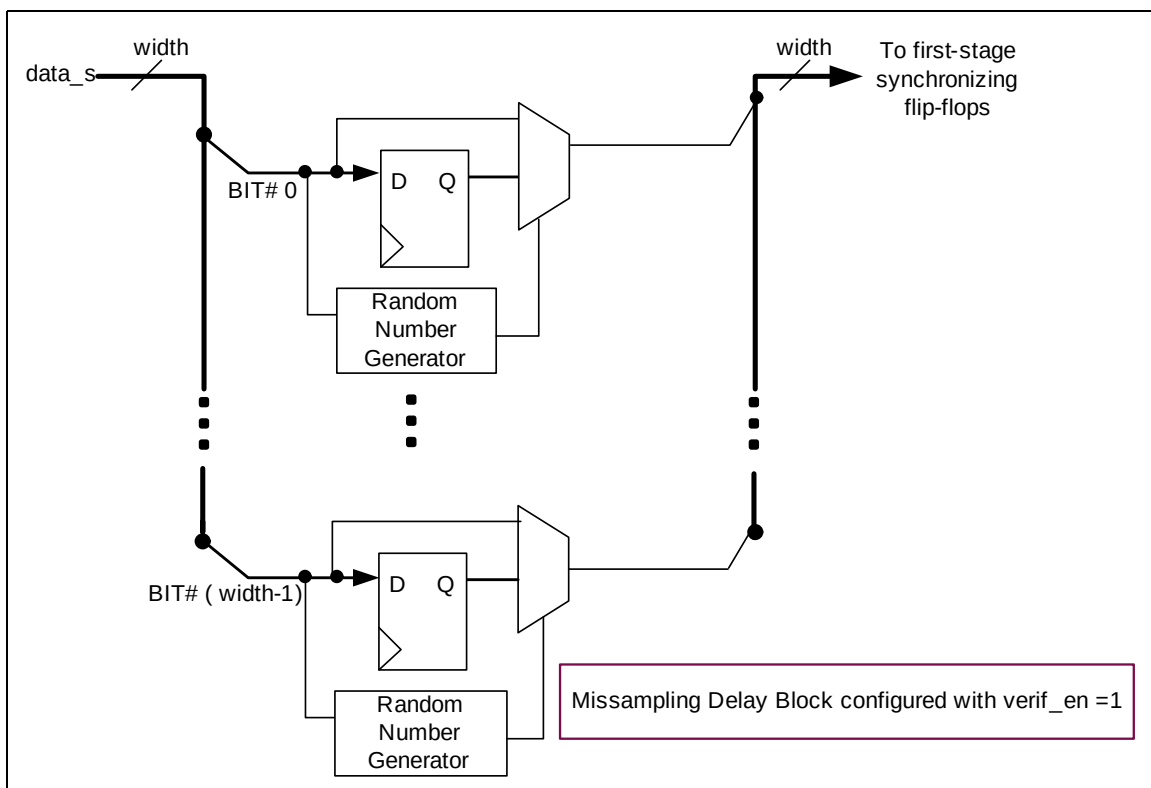
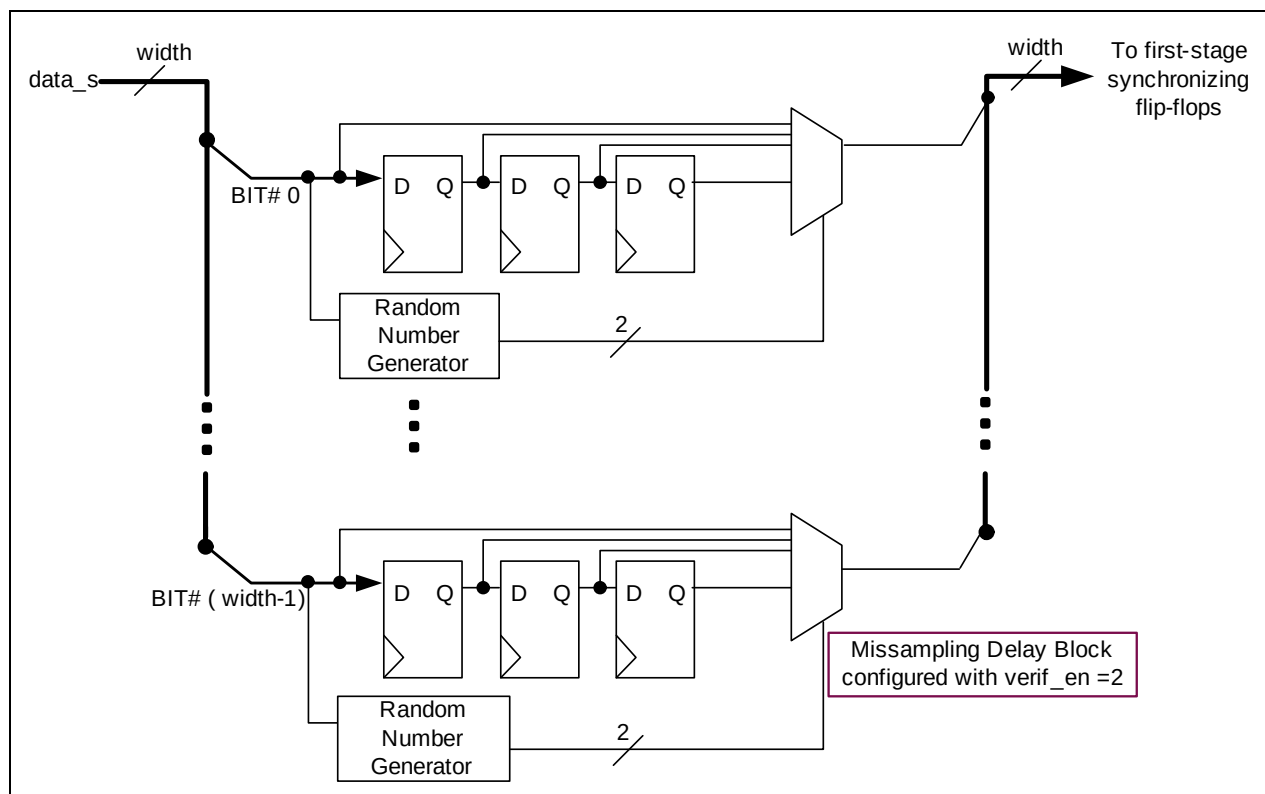
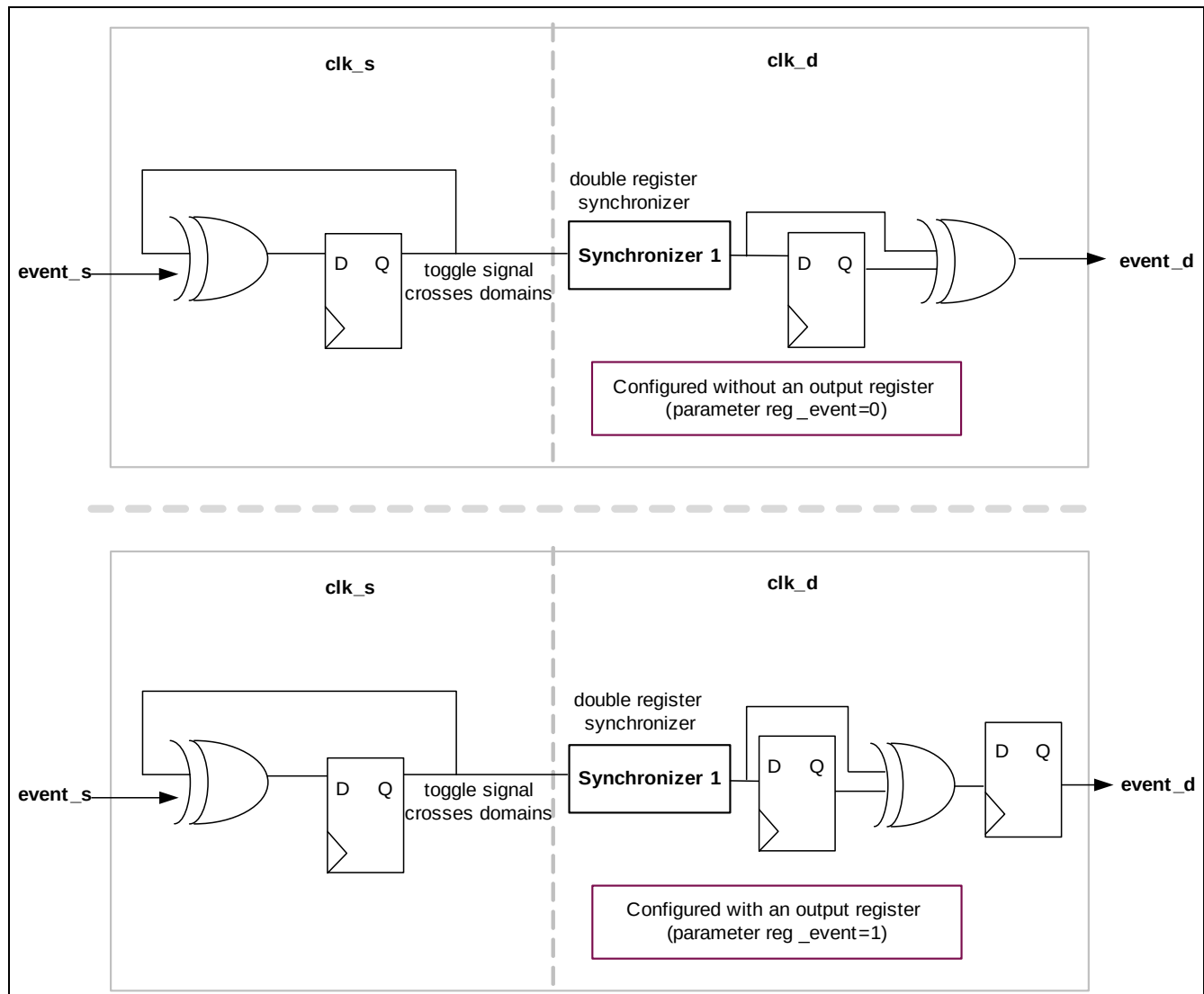
Figure C-5 Synchronizer 1: Missampling Model Type 1

Figure C-6 Synchronizer 1: Missampling Model Type 2

C.2 Synchronizer 2: Pulse Synchronizer (DWC_mipi_unipro_20_bcm22.v)

This is a dual clock pulse synchronizer for transmitting the single clock cycle pulses between two different clock domains. This method uses clock domain crossing techniques to safely transfer pulses between logic operating on different clocks.

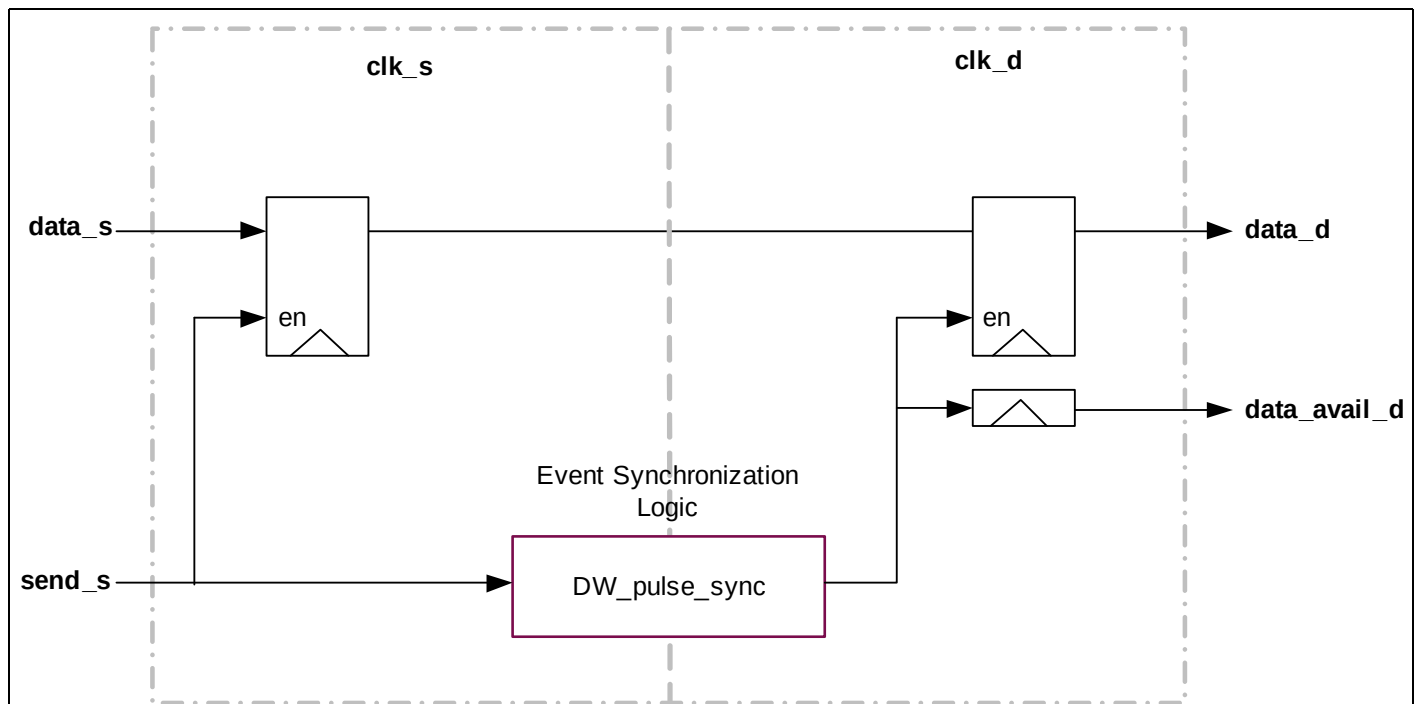
Figure C-7 Synchronizer 2 Block Diagram



C.3 Synchronizer 3: Data Bus Synchronizer without Acknowledge (DWC_mipi_unipro_20_bcm26.v)

DWC_mipi_unipro_20_bcm26 passes data values from the source domain to the destination domain. Full feedback handshake is not used. Therefore there is no Busy or Done status in the source domain. This synchronizer is useful in situations where the destination domain is known to always have a significantly faster clock rate than the source domain-greater than three times. This means that every transaction started by the source domain is guaranteed to be completed in the destination domain before another one can be started in the source domain.

Figure C-8 Synchronizer 3 Block Diagram



Glossary

This glossary defines the terms used in this databook.

Term	Definition
Application (Layer)	Logic or functionality within a device that uses the services provided by the transport layer and the configuration layer as provided by the UniPro stack.
Attribute	Atomic unit of information that can be read or modified using Get or Set commands. Attributes can be used to configure the behavior of DWC_mipi_unipro, to determine the current state of the interface or to determine the availability of interface options and interface capabilities.
Clock Lane	A unidirectional interconnect between two UniPorts which carries clock information needed for decoding the data on the data lanes.
Composition	The process of adding header, and possibly trailer, information to the protocol data unit of an upper layer to build a lower layer protocol data unit.
Connection	A bi-directional, logical communication channel between two CPorts. It is a concept used for connection-oriented data transmission, whereby two devices need to be connected before data can be exchanged.
Connection-oriented data transmission	Transfer of a service data unit from a source service access point to a destination service access point within the context of a connection that has previously been established.
CPort	A service access point on the transport layer (L4) within a device that is used for connection-oriented data transmission.
CPort Safety Valve	A mechanism by which a CPort discards received data whenever the CPort cannot deliver segments at the rate at which they arrive. This mechanism is used to manage network congestion and avoid some types of system-level deadlocks.
D-PHY	High-speed serial interconnect technology, based on source synchronous signaling, that is defined in MIPI® D-PHY Specification.
Data Lane	A physical interconnect between two UniPorts, which carries data. A link can consist of 1, 2, 3 or 4 data lanes per direction. A data lane in MIPI UniPro is used unidirectionally. A lane with embedded clocking information is also considered as a data lane.
Data Link Layer (L2)	A protocol layer. The data link layer provides the functional and procedural means to transfer data between two adjacent L2 instances. The data link layer also detects, and sometimes corrects, errors that occur in the physical layer. In addition, the data link layer maintains flow control between the L2 entities.

Term	Definition
Decomposition	The process of removing header, and possibly trailer, information from the protocol data unit of a lower layer to extract an upper layer protocol data unit.
Device	An addressable node on the network. All devices must comply with layers L3 and L4 as specified in the MIPI® UniPro Specification. Off-chip links must also comply to the UniPro protocol L1.5 and L2 and physical layer (L1) as specified in the MIPI® M-PHY Specification.
DeviceID	A number that uniquely identifies a device on the network.
Device Management Entity (DME)	The Device Management Entity is a layer-independent entity interfacing with all protocol layers and the application layer. The DME may set or get layer attributes and signal and control the resetting of individual layers.
End-to-end flow control	A credit-based flow control method that is implemented in the transport layer (L4) on a connection.
Fragment	A portion of a message that can be passed to, or received by, a CPort received fragments are not generally identical to transmitted fragments.
Flow Control	The process of managing the flow of data from one entity to another to ensure that the receiving entity can handle all of the incoming data.
Frame	A layer 2 protocol data unit for point-to-point transmission across an individual link. Frames can be either data frames or control frames.
Gear	A mechanism defined in the MIPI® M-PHY Specification that provides control over the speed of the link in steps of 2x. Higher values of high-speed gear or PWM gear imply higher frequencies and higher bandwidths.
Link	A bi-directional interconnection between two devices or switches.
Link Startup Sequence	An L1.5 handshake to establish initial link communication between two directly attached UniPorts.
Message	The Protocol Data Unit (PDU) of the application layer.
M-PHY	High-speed serial interconnect technology, based on clock embedding.
Network Layer (L3)	A protocol layer. The part of the communication protocol that manages the network and routes packets to their destinations.
PA Symbol	The Protocol Data Unit (PDU) of the PHY adapter layer (L1.5).
Packet	The Protocol Data Unit (PDU) of the network layer (L3).
PACP Frame	A Protocol Data Unit (PDU) used by PHY Adapter Control Protocol (PACP) for power control purposes.
PHY Adapter Control Protocol (PACP)	A control protocol within the PHY adapter layer (L1.5) used for setting power modes on individual links.
PHY Adapter Layer (L1.5)	A protocol layer. The PHY adapter layer presents a common interface to the data link layer for the supported physical layer.

Term	Definition
Pin Count	Number of pins required to implement a physical port. Typically includes signal, power, and control pins.
Power Mode	A configurable attribute used to control the UniPro power state. Some power modes have a one-to-one mapping to power states while other power modes give the PHY adapter layer (L1.5) the freedom to switch between two pre-determined power states autonomously.
Power State	An abstract representation within the PHY adapter layer (L1.5) of the current power state of the underlying physical layer. All active data lanes in a given direction of a link have the same power state. The distinction between UniPro power states and PHY power states serves to unify state names across alternative physical layers and to hide any transient PHY power states.
Segment	The Protocol Data Unit (PDU) of the transport layer (L4).
Segmentation	The function of splitting a message or fragment into segments within the transport layer transmitter.
Switch	A switch routes packets through the network. It contains a UniPro-compliant L3 with routing functionality.
SYNC	An 8b10b symbol sequence with high edge-density intended for fast phase alignment.
Synchronization	A mechanism used by the PHY adapter layer (L1.5) to detect and compensate for timing skew between multiple M-PHY data lanes.
Test Feature	A traffic generator (TstSrc) and traffic analyzer (TstDst) pair that may be connected to a CPort.
Tester	A device running the UniPro test suite and used for testing a remote UniPro Device-Under-Test (DUT).
TstDst	A programmable analyzer of incoming messages with the capability to check a stream of incoming messages against the expected values and lengths and report errors.
TstSrc	A programmable traffic generator that creates sequences of messages containing well-defined byte sequences of well-defined lengths.
Traffic Class	A priority equivalence class of messages, packets, or frames. Frames from a higher-priority traffic class are allowed to preempt lower-priority traffic classes.
Transport Layer (L4)	A protocol layer that can transfer segments between devices. The layer's functionality and concepts include message segmentation, connections, end-to-end flow control, and inorder data delivery per connection.